

A **Routledge** and **Guilford** FreeBook

CHANGE THE WAY YOU FEEL BY CHANGING THE WAY YOU THINK





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INTRODUCTION

Feel happier, calmer, and gain the confidence to make changes that can help you live the life you want with a new FreeBook from Routledge and Guilford Press. This highly practical resource contains a multitude of tips and exercises from bestselling Cognitive Behavioral Therapy (CBT) and positive psychology titles, including *Mind Over Mood*; *Worry Less, Live More*; and *Overcoming Depression and Low Mood Workbook*. Featuring easy-to-implement strategies and advice to change negative thought patterns and behaviour, this FreeBook will provide valuable help to those dealing with stress, anxiety, and depression.

Guilford Press is an independent publisher of books, periodicals, software, and DVDs in mental health, education, geography, and research methods. Guilford Press is distributed in the UK and Europe by Taylor & Francis. To order in other countries, visit www.guilford.com.

CHP 1: IT'S THE THOUGHT THAT COUNTS:

In this chapter from the bestseller, *Mind Over Mood*, Dr. Dennis Greenberger and Dr. Christine Padesky explore the connection between our thoughts and how they influence our mood and behaviour.

"Most people who are anxious, depressed, or angry can tell you that "just thinking positive thoughts" is not that easy...Looking at a situation from all sides and considering a wide range of information – positive, negative, and neutral – can lead to more helpful ways of understanding things and new solutions to difficulties you face."

CHP 2: STARTING OUT...AND HOW TO KEEP GOING IF YOU FEEL STUCK

How do you help yourself to plan for and overcome blocks to change? Chris Williams provides valuable tips, exercises and advice for using self-help resources in this chapter from *Overcoming Depression and Low Mood*.

"You can't expect to be able to swim immediately. You may need to start at the shallow end and practice at first. Pace what you do and don't jump straight away into the deep end"

INTRODUCTION

CHP 3: UNDERSTANDING FEAR AND ANXIETY

Written by Susan M. Orsillo and Lizabeth Roemer, this insightful chapter from *Worry Less, Live More* unpacks fear, anxiety, stress and worry, providing the reader with practical exercises and tips to help explore and control emotional responses.

“Although we are saying that fear and anxiety are a natural part of life, we are not suggesting that you “grin and bear it.” We have found that by understanding anxiety, noticing its many components as it evolves, learning to relate to anxiety differently...and clarifying what is important, people can make significant changes.”

CHP 4: DECIDE IF CBT IS FOR YOU

Author Windy Dryden outlines the main features of cognitive behavioural therapy and explains how and why it can be a beneficial form of therapy. This chapter from *How to Get the Most Out of CBT* also provides invaluable advice about choosing the right therapist.

“It is often thought that CBT therapists are not interested in your past. This is not correct, and in CBT you may talk about whatever it is you are bothered about, be it your past, your present or your future.”

CHP 5: DEPRESSION, UNHAPPINESS AND EMOTIONAL DISTRESS

Why do we get stuck in a negative cycle? This chapter from *The Mindful Way Workbook* will help readers gain a better understanding of how unhappiness and our state of mind can contribute to depression, as well as advice on how to break free from this cycle.

“But, somehow, most of us don’t feel able to let things take their natural course—when we feel sad or unhappy, we feel we have to do something, even if it’s only trying to understand what’s going on. Paradoxically, it is those very attempts to get rid of unwanted unhappy feelings that get us stuck in ever-deepening unhappiness.”

INTRODUCTION

CHP 6: CONNECTING TO OTHERS AND MANAGING YOUR EMOTIONS

In *Letting Go of Self-Destructive Behaviors*, Lisa Ferentz offers creative resources and behavioural techniques that calm the body, manage negative thoughts, and address tension and anxiety. The purpose of this chapter is to focus on the impact of negative experiences and relationships and how to manage negative emotions and create positive connections

“Attaching to people you love lets you feel protected, safe, physically comforted and soothed, and connected to others.”

CHP 7: THE INVENTION OF THE EXPRESSIVE WRITING APPROACH

Leading experts James W. Pennebaker and Joshua M. Smyth talk about the origin and development of the expressive writing approach. This chapter from *Opening Up by Writing It Down* explores the health benefits of this therapy and includes instructions on how to practice expressive writing.

“...people who wrote about their deepest thoughts and feelings related to stressful or traumatic experiences had reliable improvements in health in the two to three months after writing. Although there were also improvements in people’s self-reports of their health, there were equally large effects on people’s physiological functioning.”

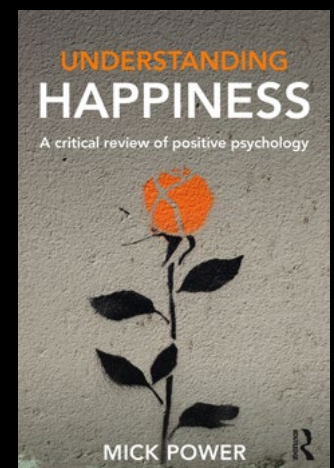
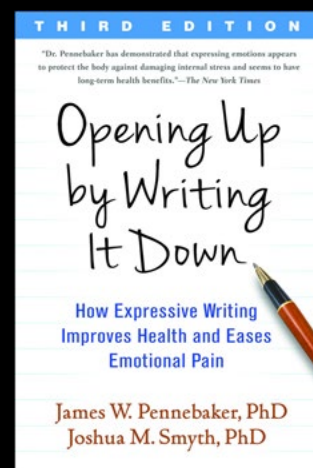
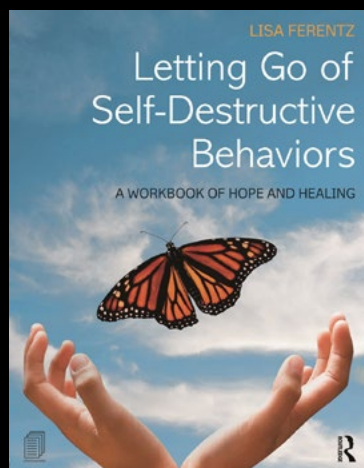
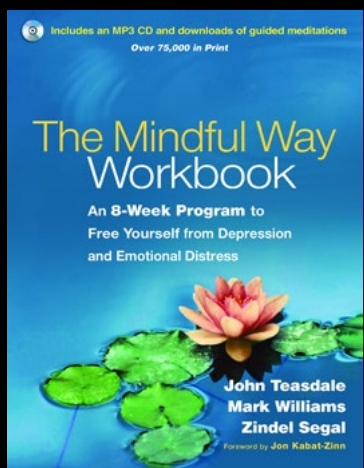
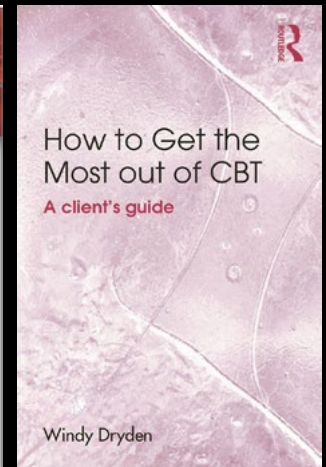
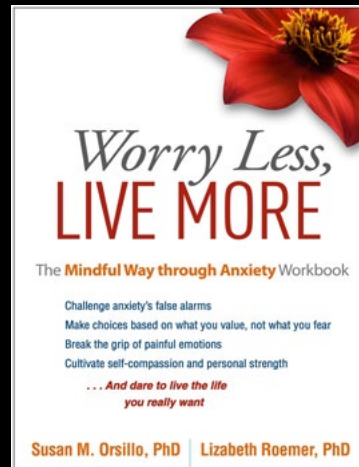
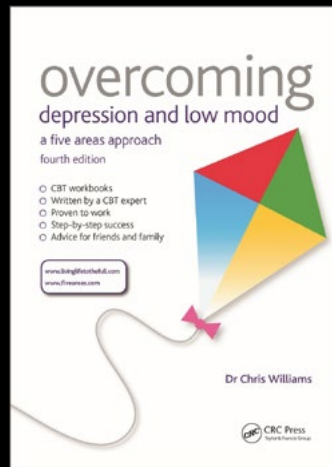
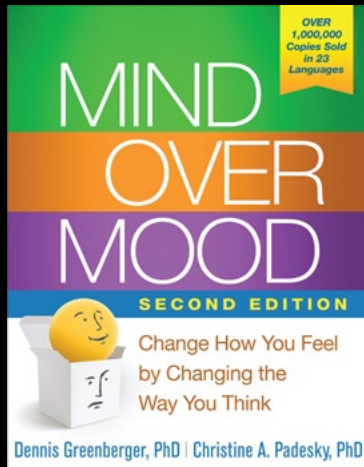
CHP 8: HAPPINESS: AN OVERVIEW

Is happiness really something we can actively aim for, or is it simply a by-product of how we live our lives more widely? In this opening chapter from *Understanding Happiness*, Dr. Mick Power provides an insightful critical assessment of how we define happiness and how it has been measured.

“Those of us who are a little sceptical of such high ideals might suggest that Americans have been more preoccupied with the pursuit of wealth than the pursuit of happiness, given that more than half of the world’s billionaires live in the US.”

*As you read through this FreeBook you will notice that some excerpts reference previous or further chapters. Please note that these are references to the original text and not the FreeBook.

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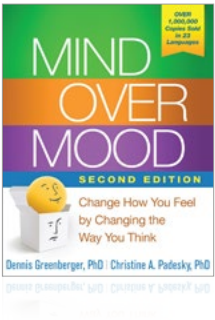
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IT'S THE THOUGHT THAT COUNTS

*From Mind Over Mood: Change How You
Feel by Changing the Way You Think,
2nd Edition*

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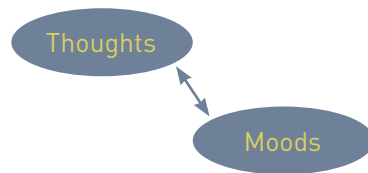
The following is excerpted from *Mind Over Mood* by Dennis Greenberger and Christine A. Padesky. © Dennis Greenberger and Christine A. Padesky, 2016. All rights reserved.

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In Chapter 2, you learned how thinking, mood, behavior, physical reactions, and environment/life situations all affect each other. In this chapter, you learn that when you want to feel better, your thoughts are often the place to start. This chapter describes how learning more about your thoughts can help you in many areas of your life.

WHAT IS THE THOUGHT–MOOD CONNECTION?



Whenever we experience a mood, there is a thought connected to it that helps define the mood. For example, suppose you are at a party, and a friend introduces you to Alex. As you talk, Alex never looks at you; in fact, throughout your brief conversation,

he looks over your shoulder across the room. Following are three different thoughts you might have in this situation. Four moods are listed below each thought. Mark the mood that you believe you would have with each thought:

Thought: Alex is rude. He is insulting me by ignoring me.

Possible moods (mark one): Irritated Sad Nervous Caring

Thought: Alex doesn't find me interesting. I bore everybody.

Possible moods (mark one): Irritated Sad Nervous Caring

Thought: Alex seems shy. He's probably too uncomfortable to look at me.

Possible moods (mark one): Irritated Sad Nervous Caring

This example illustrates that the moods we experience often depend upon our thoughts. Different interpretations of an event can lead to different moods. Since moods are often distressing or may lead to behavior with consequences (such as telling Alex he is rude), it is important to identify what you are thinking and to check out the accuracy of your thoughts before acting. For instance, if Alex is shy, it would be inaccurate to think of him as rude, and you may regret it later if you respond with anger or irritation.

Even situations you might think would create the same mood for everyone – such as losing a job – may, in fact, lead to different moods because of different personal beliefs and meanings. For example, one person facing a job loss might think, “I’m a failure,” and feel depressed. Another person might think, “They have no right to fire

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me; this is discrimination,” and feel angry. A third person might think, “I don’t like this, but now is my chance to try out a new job,” and feel a mixture of nervousness and anticipation.

Thoughts help determine which mood we experience in a given situation. Once a mood is present, we often begin thinking additional thoughts that support and strengthen the mood. For example, angry people think about ways they have been hurt, depressed people think about all the negative aspects of their lives, and anxious people think about danger. This does not mean that our thinking is wrong when we experience an intense mood. But when we feel intense moods, we are more likely to distort, discount, or disregard information that contradicts the validity of our moods and beliefs. In fact, the stronger our moods, the more extreme our thinking is likely to be.

For example, if we are mildly anxious before a party, we might have a thought: “I won’t know what to say when I meet new people, and I’ll feel really awkward.” However, if we are highly anxious, our thought may be “I won’t know what to say. I’ll blush as red as a beet, and I’ll make a complete fool of myself.” In addition, we won’t remember in this moment that we have been to many parties before, and usually we do think of something to say to new people and generally have a good time. All of us think like this sometimes. This is why it is helpful to be aware of our thoughts when we are most distressed. When we are aware of our thoughts, we more easily see how they are influencing our mood. The following example shows how Marissa’s thinking makes her depression worse.

MARISSA: *THE THOUGHT–MOOD CONNECTION.*

Marissa thought she was unlovable. This belief seemed absolutely true to her. Given her negative experiences with men, she couldn’t even imagine that someone could truly love her. This belief, coupled with her desire to be in a relationship, led her to feel depressed. When a colleague, Julio, began to be attracted to her, she had the following experiences:

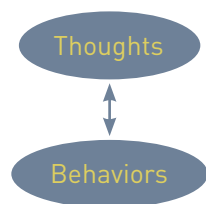
- A friend teased her about the frequent phone calls she received at work from Julio, saying, “I think you have an admirer, Marissa!” Marissa replied, “What do you mean? He doesn’t call that often.” (*Not noticing positive information*)
- Julio complimented Marissa, and she thought, “He is just saying this to keep up a good work relationship.” (*Discounting positive information*)

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- When Julio asked to meet her for lunch, Marissa thought, "I'm probably explaining the work project so poorly that he resents the extra time the project is taking." (*Jumping to a negative conclusion*)
- At lunch, Julio told Marissa he thought that they had both been very creative on the project, and said he had really enjoyed spending the extra time with her. He went on to tell her that he found her attractive. Marissa thought, "Oh, he probably says that to everyone and doesn't really mean it." (*Discounting positive experiences*)

Since Marissa was convinced that she was unlovable, she ignored or distorted information that was not consistent with her belief. Because she was very depressed, she had trouble believing the positive things people said that could help her feel better. Ignoring information that doesn't fit with our beliefs is something we can learn to change. For Marissa, learning to take in positive information about her attractiveness and lovability could be the start of something wonderful.

WHAT IS THE THOUGHT-BEHAVIOR CONNECTION?



Our thoughts and behaviors are usually closely connected. For example, we are more likely to try to do something if we believe it is possible. For many years, athletes believed it was impossible to run a four-minute mile. In track events around the world, top runners ran a mile in just over four minutes. Then a British miler, Roger Bannister, identified changes he could make in his running style and strategy to break the four-minute barrier. He believed it was possible to run faster and put many months of effort into changing his running technique to reach this goal. In 1954, Roger Bannister became the first man to run a mile in less than four minutes. His belief that he could succeed contributed to behavior change.

Remarkably, once Bannister broke the record, the best milers from around the world also began to run the mile in under four minutes. Unlike Bannister, these runners had not substantially changed their running techniques. What had changed were their beliefs; they now thought it was possible to run this fast, and their behavior followed this thought. Of course, just knowing it is possible to run fast does not mean that everyone can do this. Thinking is not the same as doing. But the more strongly we believe that something is possible, the more likely we are to attempt it and maybe succeed at it.

On a daily basis, we all have "automatic thoughts" that influence our behavior. These are the words and images that pop into our heads throughout the day. For example,

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imagine that you are at a family reunion. The food has just been laid out, and some family members go over to the buffet tables to fill their plates, while others remain seated and talking. You have been talking with your cousin for 10 minutes. Consider each of the following thoughts and write what behavior you would probably do if you had this thought.

Thought	Behavior
If I don't go now, they'll run out of food.	
It's rude to rush to the buffet tables when we're in the middle of a conversation.	
My grandfather looks too unsteady to carry a plate.	
My cousin and I are having such a wonderful conversation – I've never met anyone so interesting.	

Did your behavior change, depending on the thought you had?

Sometimes we are not aware of the thoughts that affect our behavior. Thoughts often occur rapidly, automatically, and just out of our awareness. We sometimes act out of habit, and the original thoughts that led to these habits have been forgotten. For example, perhaps we always give in when someone disagrees with us. This habit may have started with a belief such as "If we disagree, then it is best to just let it go, because otherwise our relationship won't last." We often are not aware of the thoughts guiding our behavior when our actions have become routine. An example from Ben's life illustrates the thought-behavior connection.

BEN: *THE THOUGHT-BEHAVIOR CONNECTION.*

After his friend Louie died, Ben cut back on meeting his friends for lunch and other activities he used to enjoy. At first, his family thought that avoiding his friends was part of Ben's grief over Louie's death. But as the months passed and Ben still refused to get together with friends, his wife, Sylvie, began to suspect there might be other reasons Ben was staying at home.

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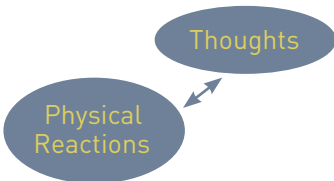
One morning, Sylvie sat down with Ben and asked him why he was not returning his friends' telephone calls. Ben shrugged and said, "What's the point? We're at that age where we're all just dying anyhow." Sylvie felt exasperated. "But you're alive now – do the things you enjoy!" Ben shook his head and thought, "Sylvie just doesn't understand."

Sylvie really didn't understand, because Ben was not aware of the thoughts guiding his behavior, and he couldn't fully explain to her why he had stopped doing activities he used to enjoy. As Ben learned to identify his thoughts, he realized that he had a series of thoughts: "Everyone is dying. What's the use in doing things when I'm just going to lose everyone anyhow? If I don't feel like doing something, then I won't enjoy myself."

When Louie died, Ben decided he had reached the age where death was close at hand. This awareness influenced his thoughts and his willingness to do the things he used to enjoy.

By contrast, Sylvie, who was only a little younger than Ben, thought she should do as many enjoyable activities as possible and enjoy life to the fullest. She frequently saw her friends and stayed quite active. As you can see, Sylvie's and Ben's different thoughts about growing older had a big impact on their behavior.

WHAT IS THE THOUGHT–PHYSICAL REACTIONS CONNECTION?



Thoughts also affect our physical reactions. Think about watching a really good movie. When you watch movies, you often anticipate what is coming next. If you think something scary or violent is about to happen, your body reacts as well. Your heart might start to beat more rapidly, and your breathing may actually change as your muscles get tight. If you anticipate a romantic scene, your body may feel warm or even sexually aroused.

Athletes are trained to use the powerful link between thoughts and physical reactions. Good coaches give their teams inspirational speeches, which they hope will "fire up" the team members, get adrenaline flowing, and lead to top performance. Olympic athletes are often taught to imagine in detail their performance in an event.

Research shows that athletes who do this type of vivid imagining actually experience small muscle contractions that reflect the bigger muscle movements they make in their event. This thought–muscle connection improves the athlete's performance.

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Research has also discovered that our thoughts, beliefs, and attitudes have an impact on our health. For example, you have probably heard that many medications and health treatments benefit from the “placebo effect.” What this means is that our expectation that a medication or treatment will help, increases the likelihood that it does help: Our belief that a pill will help us can itself lead to improvement, even if the pill is just a sugar pill. Modern brain research has found that the placebo effect comes about partly because our beliefs are a type of brain activity and can lead to real changes in physical responses.

LINDA: THE THOUGHT-PHYSICAL REACTIONS CONNECTION.

Just as our thoughts affect our physical reactions, our physical reactions can trigger thoughts. For example, after climbing up a set of stairs, Linda noticed that her heart was beating faster. Because Linda worried about her heart, when her heart rate went up, she had the thought “I’m having a heart attack” (Figure 3.1). This terrifying thought put her whole body on alert, and she experienced a series of physical changes, including quick, shallow breathing and profuse sweating. As Linda’s breathing became shallower, she took in less oxygen, which caused her heart to beat even faster. Her brain also temporarily received less oxygen, which caused a sensation of dizziness and light-headedness.

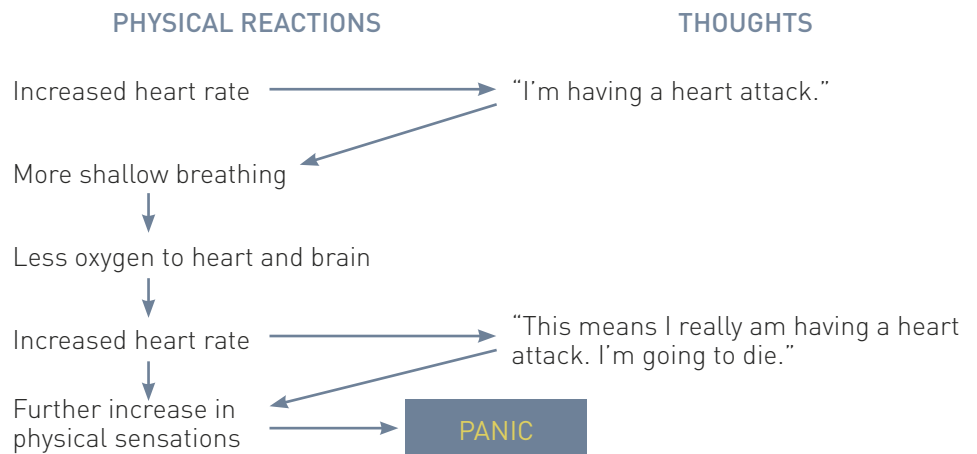


FIGURE 3.1. Linda's panic.

Linda's thought that she was having a heart attack increased her physical reactions and led her to believe she was in immediate danger of dying. Her physical responses to the idea that she was dying intensified until Linda experienced a full-blown panic

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attack. After a while, Linda realized that she was not having a heart attack. As she began to think this way, her physical symptoms gradually disappeared.

WHAT IS THE THOUGHT-ENVIRONMENT CONNECTION?

Environment



Thoughts

At the beginning of this chapter, you learned how thoughts influence the moods we experience. You may be wondering why some people are more prone to certain thoughts and moods rather than others. Some portion of these differences may be biological or genetically inherited. But we also know that our environment and life experiences can powerfully shape the beliefs and moods that color our lives. We use the words “environment” and “life experiences” to describe anything outside of us, including our families, our communities, the places we live, interactions with other people, and even our culture. We can be influenced by both present and past experiences that stretch over time from our childhoods to this moment.

Recall that Marissa was sexually and physically abused throughout her childhood and early adult years. These experiences shaped her beliefs that she was worthless, unacceptable, and unlovable, and that men were dangerous, abusive, and uncaring. It is understandable that Marissa’s earliest attempts to make sense of her experiences led her to devalue herself and be on the lookout for the negative reactions of others.

It doesn’t take traumatic environmental events to influence beliefs. The way we think about ourselves and our lives is influenced by culture, family, neighborhood, gender, religion, and the mass media. As an example of how culture influences beliefs, consider the messages we are given as children. In many cultures, girls are complimented for being pretty, and boys are rewarded for being strong and athletic. A girl might conclude that being pretty is the key to being well liked, and she might value herself for her appearance only. A boy might believe that he should be strong and athletic, and similarly judge himself on his athletic success or failure.

There is nothing inherently more likable about beauty or strength, but some cultures teach us to make these connections. Once these beliefs are formed, they can be difficult to change. Therefore, many girls who are athletic find it difficult to value their skills, and boys with musical or artistic talents but no strong athletic skills may feel cursed rather than blessed.

Vic was raised in a suburban community of educated professionals who valued achievement for themselves and their children. His family and school reflected these community values, emphasizing achievement and excellence. When Vic’s

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performance in school or on the athletic field was not superior, his family, teachers, and friends were disappointed and reacted as if Vic had failed.

From these reactions, Vic concluded that he was inadequate, even though his performance was generally very good. Since Vic believed he was inadequate, it is not surprising that he felt anxious in situations that required him to perform. He dreaded athletic events because there was a risk that he would not win or perform well. To him, those outcomes would mean that he was inadequate.

As you can see, Vic's childhood was not as traumatic as Marissa's. However, the environment he grew up in had a powerful impact on his thoughts that persisted into adulthood.

EXERCISE: THE THOUGHT CONNECTIONS

Worksheet 3.1 provides practice in recognizing the connections between thoughts and mood, behavior, and physical reactions.

WORKSHEET 3.1. THE THOUGHT CONNECTIONS

Sarah, a 34-year-old woman, sat in the back row of the auditorium during a school meeting for parents. She had concerns and questions regarding how her 8-year-old son was being taught, as well as questions about classroom security. As Sarah was about to raise her hand to voice her concerns and questions, she thought, "What if other people think my questions are stupid? Maybe I shouldn't ask these questions in front of the whole group. Someone may disagree with me and this could lead to a public argument. I could be humiliated."

THOUGHT-MOOD CONNECTION

Based on Sarah's thoughts, which of the following moods is she likely to experience? (Mark all that apply.)

- ☐ 1. Anxiety/nervousness
- ☐ 2. Sadness
- ☐ 3. Happiness
- ☐ 4. Anger
- ☐ 5. Enthusiasm

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THOUGHT-BEHAVIOR CONNECTION

Based on Sarah's thoughts, how do you predict she will behave?

- ☐ 1. She will speak loudly and voice her concerns.
- ☐ 2. She will remain silent.
- ☐ 3. She will openly disagree with what other people say.

THOUGHT-PHYSICAL REACTIONS CONNECTION

Based on Sarah's thoughts, which of the following physical changes might she notice? (Mark all that apply.)

- ☐ 1. Rapid heart rate
- ☐ 2. Sweaty palms
- ☐ 3. Breathing changes
- ☐ 4. Dizziness

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When Sarah had these thoughts, she felt anxious and nervous, remained silent, and experienced a rapid heart rate, sweaty palms, and breathing changes. Were these the reactions you anticipated Sarah would have? Not everyone experiences the same reactions to particular thoughts. However, it is important to recognize that thoughts influence our mood, behavior, and physical reactions.

IS POSITIVE THINKING THE SOLUTION?

Although our thoughts affect our moods, behavior, and physical reactions, positive thinking is not a solution to life's problems. Most people who are anxious, depressed, or angry can tell you that "just thinking positive thoughts" is not that easy. In fact, thinking only positive thoughts is overly simplistic, usually does not lead to lasting change, and can lead us to overlook information that might be important.

Mind Over Mood instead teaches you to consider all information and many different angles on a problem. Looking at a situation from all sides and considering a wide

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range of information – positive, negative, and neutral – can lead to more helpful ways of understanding things and new solutions to difficulties you face.

If Linda was planning a business trip that required her to fly on an airplane, simply thinking positive thoughts, such as “I won’t have a panic attack. Everything will be fine,” would not prepare her for the anxiety she might feel. In fact, with positive thinking, Linda might feel like a failure if she felt even a small amount of anxiety. A better solution for Linda would be to anticipate that she might feel anxious and to have a plan for how she will cope with her anxiety in flight. If we only think about the positive, we may not be able to accurately predict and cope with events that are worse than we expect.

IS CHANGING THE WAY YOU THINK THE ONLY WAY TO FEEL BETTER?

Even though the process of identifying, testing, and considering alternative thoughts is a central part of CBT and *Mind Over Mood*, it is often equally important to make changes in your physical reactions and/or your behavior. For example, if you have been anxious for a long time, you probably avoid things that make you anxious. Part of dealing with anxiety may be accepting your anxiety (cognitive shift), learning to relax (physical change), and approaching what frightens you so you can learn to cope with it (behavioral change). People do not usually overcome anxiety until they change their thoughts and overcome avoidance.

Making changes in your environment/life situations can also help you feel better. Reducing stress, learning to say no to unreasonable demands made by others, spending more time with supportive people, working with neighbors to increase neighborhood safety, and taking action to reduce discrimination or harassment on the job are all environmental/life changes that can help you feel better.

Some life situations are so challenging that simply thinking differently about things is not a wise idea. For example, someone who is being abused needs help either to change or to leave the situation. Just changing thoughts is not an adequate solution for abuse: The goal is to stop the abuse. Thought changes might help someone in this situation feel motivated to get help, but simply changing thoughts to permit acceptance of abuse is not a helpful solution.

As you complete the worksheets in this book, you will learn how to identify and change your thoughts, moods, behaviors, physical responses, and environment/life situations.

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CHAPTER 3 SUMMARY

- Thoughts help define the moods we experience.
- Thoughts influence how we behave and what we choose to do and not to do.
- Thoughts and beliefs affect our physical responses.
- Life experiences (environment) help determine the attitudes, beliefs, and thoughts that develop in childhood and often persist into adulthood.
- *Mind Over Mood* helps you look at all the information available; it is not simply positive thinking.
- While changes in thinking are often central, mood improvement may also require changes in behavior, physical reactions, and home or work situations/ environments.

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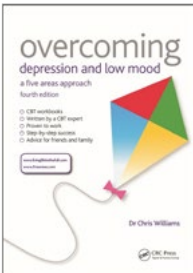
STARTING OUT

... AND HOW TO KEEP GOING IF YOU
FEEL STUCK

*From Overcoming Depression and Low
Mood : A Five Areas Approach,
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2 :: STARTING OUT

... AND HOW TO KEEP GOING IF YOU FEEL STUCK



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OVERCOMING DEPRESSION AND LOW MOOD – A FIVE AREAS APPROACH



Are you feeling like this?
If so... this course is for you.

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... AND HOW TO KEEP GOING IF YOU FEEL STUCK

WHAT YOU WILL LEARN:

- How to get the most out of this course.
- Make a clear but flexible plan of when to use this and the other workbooks.
- Discover how to overcome common blocks to change.

ABOUT THE COURSE

The workbooks in this course aim to help you understand why you feel as you do. They will teach you important life skills that will help you to turn the corner.

WHY SHOULD YOU USE THESE WORKBOOKS?

People use these workbooks because they want to make changes in their lives. **You, the reader, are in control** – and you can work on things at a time that suits you. Time and time again, people are surprised to see the amount of change they can make themselves using a structured approach.

These workbooks use an approach called cognitive behavioural therapy (CBT, a kind of talking treatment). Don't worry though – there won't be any more jargon like that in the rest of the course. Research has shown that self-help materials based on the CBT approach work well for problems such as depression and anxiety. Research on this book has also shown that it works very well as a treatment for depression. People using the book felt less depressed and were more able to live their lives as they wanted. This is why this book is one of only 30 books to be chosen to be included in every public library in England as part of the national Books on Prescription scheme.

REFERENCE

www.plosone.org/article/info%3Adoi%2F10.1371%2Fjournal.pone.0052735

www.bbc.co.uk/news/health-21083458

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GETTING GOING

Well done! You've done something that quite a few people struggle to do – **you're still reading**.

It can sometimes seem really hard starting to change. Have you ever bought or been given a book or a DVD and never even opened it or even taken the wrapper off? Using this course is no different. In fact, in some ways it's harder because it's not a book that's there for entertainment. Instead these are workbooks which aim to help you to change how you feel.

What should I read first?

People usually start by working through these two workbooks:

- This one – *Starting out... and how to keep going if you feel stuck*
- And then *Understanding why you feel as you do*.

They will help you work out how your low mood is affecting you and help you decide which other workbooks to use.

KEY POINT

Choose the workbooks **you** want to work on – making sure they deal with the problems/difficulties **you** are facing.

DEVELOPING A ROUTINE

Routines can be powerful. For example, if you are used to having a snack while you watch television, sometimes just sitting in the same chair can make you feel hungry!

In the same way, you might wish to set aside a particular place to complete the work-books. For example, sit on a chair at the kitchen table with a pen and some blank paper to jot down ideas as you read. Plan enough time so that you can get really involved in the workbook – preferably half an hour or so, if you have sufficient energy and concentration for this. If you have others around you, like young children who might interrupt, try to choose a time when they are asleep or away at nursery or school. Or see if a friend or relative could take them for a time while you work on the course.

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Getting into the mood: doing something physical can help you get started

You may feel physically and mentally sluggish when you feel low or when you aren't sleeping well. A good start to using the workbooks is to do something physical first. For example, get up and walk around the room and – if you have them – up and down the stairs. Then sit down in a chair like a kitchen chair that is upright and forces you to sit straight rather than slump back. Now start reading the workbook. Have some pens and paper handy so you can make notes and write in the book.

If you feel tired halfway through, stop, get physically going again, stretch, have some cold water, then get back to it.

But my life is too busy/unpredictable to use the course

Wouldn't it be great if we could always just sit down and plan a time to work? But sometimes life is busy, unpredictable or hard. If so, just take the time you can.

Here are some suggestions of how to build on this first step during the rest of the course.

Some dos and don'ts for getting the most out of the course

Do:

- Slow it down. Focus on using and applying just one workbook at a time.
- **Get a pen.** Writing things down means you are thinking and learning. In fact it's more than that. Sometimes **you actually work out what you really think** about something when you write it down.
 - Write it down. Things can look different when we write them down. We can notice patterns and habits we might otherwise miss. Sometimes we can start to really work out what is happening when we see it on a page in black and white. Therefore, answer the questions straight away rather than thinking you will come back to it later.
 - You might not wish to write in the book – it may be something you don't like to do or you may have borrowed the book from a library or someone else. If so, you can write your answers on a separate sheet of paper.
 - Many of the worksheets in the course are also available for download from www.lltff.com/odlm
- Answer all the questions – and do try to stop, think and reflect as you read.
- Ask: How might I use this in my life?

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- Try out what you read in the workbooks. You'll find a *Planner sheet* at the end of each workbook to help you to decide how to do this.
- Be realistic. You are more likely to succeed if you try changing things one step at a time rather than throwing yourself into things and then running out of steam.
- Re-read the workbooks. You may find that different parts become clearer or seem more useful on reading a second time.
- Use the workbooks to build up the help you receive in other ways, such as talking to friends, or from self-help organizations and support groups.
- Keep it organized and easy to handle. Create your own resource pack of key worksheets and other resources you use to improve how you feel.
- Log on to www.lltff.com to get access to free modules and resources.

Don't:

- Push and read through the entire workbook in one go. You need time to take it in and time to practice.
- Expect a sudden miracle cure. Change takes time and practice.
- Try to do this completely on your own. Supportive encouragement from a health professional or social care worker can really help you keep on track.
- Try to read the workbook when you are distracted, such as times when you are trying to do other tasks.
- Cut yourself off from other useful supports. You can do this course alongside other treatments, such as seeing your doctor or taking an antidepressant. These approaches can all be helpful parts of moving forward.

Finding extra support



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Having someone around who can offer *support and encouragement* can help. This is especially important if you are struggling or feel stuck. Sometimes just the act of telling someone – a family member, friend or health worker – that you are working on something, or plan to do a certain activity on a particular day can really help. Just knowing that someone else may ask you how it's going could help spur you to action. You might go through your answers to the questions in the workbooks with them, or keep your answers private and only discuss some of the course content.

BUILDING YOUR MOTIVATION TO CHANGE

Imagine it is 10 years in the future. You have made important changes in your life and things are much better. Write yourself an **encouraging letter** about why you need to make changes now.

Dear (your name)

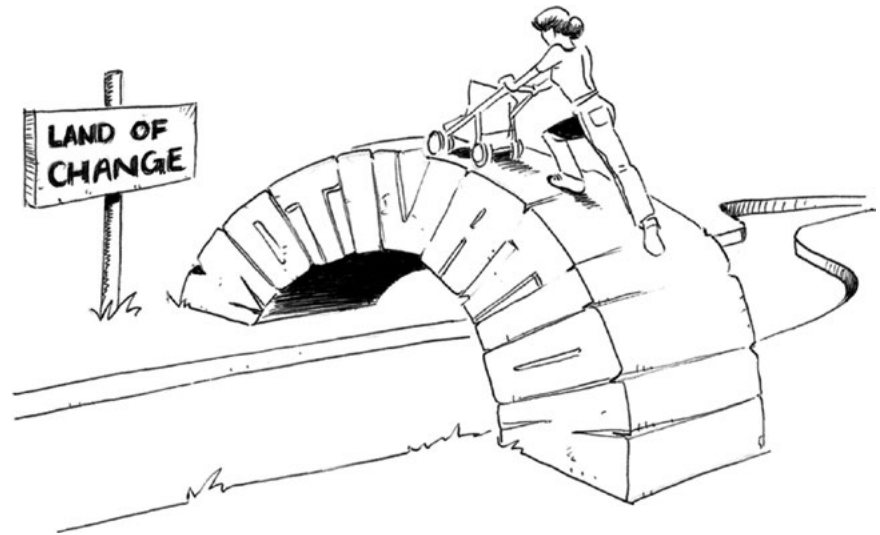
Signed:

Me

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CHANGE TAKES TIME AND EFFORT



Sometimes it's easy to forget how hard it is to learn new information or skills that you now take for granted. Think about some of the skills you have learned over the years. For example, if you can drive or swim or ride a bike, think back to your first driving/swimming lesson or attempt to cycle without stabilizers. You probably weren't very good at it that first time, yet with practice you developed the skills needed to do it. In the same way, you can help overcome how you feel by practicing what you learn, even if it may seem very hard at first.

Write down some other things you have learned that took time:

KEY POINT

You can't expect to be able to swim immediately. You may need to start at the shallow end and practice at first. Pace what you do and don't jump straight away into the deep end.

BE REALISTIC

It's important not to approach this course either far too positively or far too negatively. It would be untrue to claim that if you use this course you are guaranteed results. What we can guarantee is that this approach has helped many thousands of people –

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and that the workbooks teach clinically proved approaches that have been a help for many. Hopefully, at the very least you will learn some interesting and helpful things along the way.

COMMON PROBLEMS IN USING THE COURSE

I've no time

Life can be busy and complicated, especially if you have family or work commitments. There will be many demands on your time. But ...

Task

Imagine you have a close friend who is struggling. They don't like how they feel, and you knew that it is affecting them in lots of different ways. What helpful advice would you give them if they said 'I don't have time'?

Write down your encouraging advice here:

...if you would give your friend advice to make some time, could you use that same advice yourself?

I feel too down to do this now

Sometimes in really severe depression, it might not be the right time to use these workbooks. But you can always come back to them later if you are finding that things are too much now. If you can't concentrate for long just go at a pace you can manage. You should also discuss your treatment options with your doctor.

I'll never change

Perhaps the biggest block to getting better is not believing change can happen. Many people find that they gain much more from the course than they first thought they would. Could this be true for you?

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Task

What words of encouragement would you say to a friend who needed help but believed that change is impossible? Write them down here:

If you would offer helpful and positive advice to a friend, then why not also offer it to yourself?

EXPERIMENT

In this course, I'm going to ask you to experiment and try things out. Even if you have doubts about the course, or about your ability to use it effectively, try to give it a go and test out what happens. If you still find it doesn't help at all after you've given it a good go, it would be a sensible time to try something different. But perhaps you'll notice some positive things, even if it's just one or two things. Perhaps that is worth keeping going for.

SUMMARY

Well done – you've got to the last section – and you're still reading! That's a very important achievement. So many people who want to change find it hard starting out.

Let's review what you have learned in this workbook. You have covered:

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- How to get the most out of the course.
- How to write a clear but flexible plan of when to use the workbooks.
- How to overcome common blocks to change.

BEFORE YOU GO

Q What have I learned from this workbook?

Q What do I want to try *next*?

PUTTING INTO PRACTICE WHAT YOU HAVE LEARNED

You are likely to make the most progress if you can put into practice what you have learned in the workbook. Each workbook will encourage you to do this by suggesting certain tasks for you to do in the following days.

Suggested reading

The *Understanding why you feel as you do* is often a good workbook to read next. It will help you decide which other course workbooks you might need to use.

MAKING PLANS

The best plans say:

- **What** you are going to do.
- **When** you are going to do it.

and

- **Predict** things that might block or get in the way of you doing this so you can tackle problems head on.

You may find the following **Planning task** helpful. You can use it to plan any activities or tasks you want – picking a child up from school, getting a repeat prescription, planning to read the next workbook, painting the kitchen – anything.

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Here's an example plan that looks at when and how to start the *Understanding why you feel as you do* workbook. If this is something you would like to do, write your own plan into the Plan below. If you prefer to choose something else, then there are blank versions of this sheet available at the end of the workbook and for download from www.lltff.com.

1. *What am I going to do?*

Suggestion: Start to read the *Why do you feel as you do* workbook

If you decide to do this, think through in more detail exactly what you will do. Do you need to break it down into smaller parts (e.g. get a pen and paper, find the copy of the book or workbook, sit at the table with the door shut and radio off...)? Try to be clear exactly what you will do – so you will be clear when you've done it. Be realistic. Will you plan to read the whole workbook, or just a few pages at a time? Each workbook is also split up into sections that you can work through in smaller sections.

Write what you will do here:

2. *When am I going to do it?*

What date and time will suit? Many people with low mood notice they feel at their worst first thing in the morning. So you might find that the best time for you to read the workbook is after lunch, in the late afternoon, or in the early evening. If you have young children, think about what you know of their routine. Or you could pick a time when others are around to help look after them. Also, can you plan to read it every day – or do you need a gap to let things sink in?

Write the day and time:

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3. *Is it well planned?*

Next, do a quick reality check on your plan. Is it realistic and well planned? Are you trying to bite off more than you can chew? Is the task small enough for you to succeed, but not so small it makes no difference?

Is my planned task one that:

- Will be useful for helping me move forward? Yes ☐ No ☐
- Is clear, so that I will know when I have done it? Yes ☐ No ☐
- Is something that I value, or need to do? Yes ☐ No ☐
- Is realistic, practical and achievable? Yes ☐ No ☐

4. *What problems/difficulties could arise, and how can I overcome this?*

What could get in the way?

- *Things within you* – low motivation, forgetfulness, talking yourself out of it?
- *Or things outside you* – other people, bad weather, the need for money to do an activity, having to travel or perhaps a task that depends on someone else for success? Unpredictable things may also happen from time to time and interrupt your plans.

Have a back-up time planned for if you can't start working on the course when you first planned. For example, what if a friend unexpectedly drops by for a coffee, or your baby cries, wakes up or needs a nappy change?

Write your possible blocks in here:

Finally, think again about the task and decide whether you need to rewrite your plan to tackle these possible blocks.

5. *Write your final plan here:*

What are you going to do?

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When are you going to do it (day and time)?

During the course, you can make all sorts of future plans using the *Planner sheet* at the end of this workbook. Many people find that by writing things down it helps them organize things and get things sorted.

PLAN, DO AND REVIEW

Whatever you choose to do, the first step is to make a plan and then do (or not do!) it. The *Planner sheet* will help you create a clear and realistic plan. The next step is to use the *Review sheet* to consider how things have gone, and whether good or bad to learn from it. Copies of both sheets are found at the end of the workbook, and as with all the worksheets can be downloaded from www.lltff.com.

That last part is important. This isn't about having a time for self-congratulation or criticism. It's all about planning effective progress and learning about what works and doesn't in your own life right now.

Look at the *Review sheet*. It asks you to consider whether you completed the planned task or not. If you have managed to do it as planned, then well done. If not, then what stopped you? Was it something inside you like forgetting, or talking yourself out of it? Or was it something outside you like a child being ill, an unexpected accident or friend dropping by, or perhaps it rained? It's all about learning how to plan more effectively. How could you plan things differently the next time to tackle this?

Whichever tasks and workbooks you use on this course, keep coming back to this *Plan, Do, Review* approach. You can even use it to plan the weekly shop!

Other sources of support

www.lltff.com

This popular resource is designed to support readers of this course and is free to use. There's also a forum where you can make comments, or ask questions of other people who are using the same course.

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<http://shop.lltff.com>

Access a wide range of Five Areas resources, including free hand-outs and relaxation MP3 files.

Worksheets to help you practice

Practice is important to help you master this approach. You can download worksheets of all the key skills used in this workbook from www.lltff.com/worksheets/odlm

OTHER RESOURCES

Here are details of some other Five Areas resources that can be helpful at times of low or anxious mood.

- *24 hours to get a job that really fires you up.* (Highly Commended BMA book awards) (Kindle)
- *Overcoming anxiety: A Five Areas approach.*
- *Are you strong enough to keep your temper?* (Kindle)
- *I'm not good enough* (low confidence). (Kindle)
- *Stop smoking in 5 minutes.* (Kindle)
- *I feel so bad I can't go on.* (Winner, BMA book awards) (Kindle)
- *Fix your drinking problem in 2 days.* (Kindle) (Linked website: www.lltff4drink.com)
- *Enjoy your baby* (postnatal depression). (Kindle)
- *Reclaim your life from illness, disability, pain or fatigue.* (Kindle)
- *I'm not supposed to feel like this: A Christian self-help approach to depression and anxiety,* C Williams, P Richards and I Whitton (linked website: www.lltffwg.com).

A REQUEST FOR FEEDBACK

If there are areas in this course that you find hard to understand, don't work well for you or seemed poorly written, please let me know. I'm sorry I can't answer specific questions or provide advice on treatment.

Email: feedback@fiveareas.com

Address: Five Areas, PO Box 9, Glasgow G63 0WL, UK

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In your feedback, please state which workbook, book or website you are commenting on.

ACKNOWLEDGMENTS

The cartoon illustrations were produced by Keith Chan, kchan75@hotmail.com. The terms LLTTF and Five Areas are registered trademarks of Five Areas Resources Ltd.

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PLANNER SHEET

1. *What am I going to do?*

2. *When am I going to do it?*

Write in the day and time:

3. Is my planned task one that:

- Will be useful for helping me move forward? Yes ☐ No ☐
- Is clear, so that I will know when I have done it? Yes ☐ No ☐
- Is something that I value, or need to do? Yes ☐ No ☐
- Is realistic, practical and achievable? Yes ☐ No ☐

4. *What problems/difficulties could arise, and how can I overcome this?*

What could get in the way? Write your possible blocks in here:

Do you need to rewrite your plan to tackle these possible blocks?

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5. *Write down your final plan here*

What are you going to do?

When are you going to do it? (day and time)

Your back-up plan: Think of another back-up solution you could turn to if for whatever reason there are problems with your plan.

KEY POINT

If you feel worse with symptoms you can still choose to do the planned activity anyway – because it's important.

REVIEW SHEET

What did you plan to do?

Write it here.

What happened? Did you attempt the task?

Yes ☐ No ☐

If yes:

- What went well?
- What didn't go so well?
- What have you learned about from what happened?
- How are you going to apply what you've learned?

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If not:

What stopped you?

- *Internal factors* (e.g. forgot, not enough time, put it off, concerns I couldn't do it, I couldn't see the point of it, etc.).
- *External factors* (events that happened, work/home issues, etc.).
- How could you have planned to tackle these blocks?

Use the *Plan, Do, Review* approach to help you move forward.

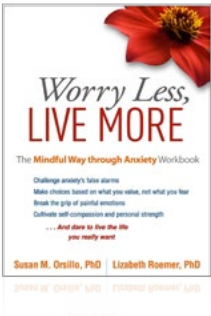
MY NOTES

3

UNDERSTANDING FEAR AND ANXIETY

*From Worry Less, Live More: The Mindful
Way through Anxiety Workbook*

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The following is excerpted from *Worry Less, Live More* by Susan M. Orsillo and Lizbeth Roemer. © The Guilford Press, 2016. All rights reserved.

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Daring to live the life we want requires us to respond skillfully and effectively to our emotions. And we are most able to do that when we have a clear understanding of why they arise and how they work. Everyone talks about feeling anxious, worried, tense, or scared from time to time, but we may not be fully aware of what each of these words means, how to notice them in ourselves, and why we have these experiences. It may seem like these things are obvious. But we've found that one very helpful step in changing the ways we respond to anxiety in our lives is to have a better, more scientific understanding of what these natural reactions are and how to detect them.

In this chapter, we will ...

1. Describe the difference between fear, anxiety, stress, and worry
2. Describe how to recognize clear and subtle signs of these states
3. Deepen our understanding of the fear response

GETTING THESE TERMS STRAIGHT

Fear—refers to the thoughts, emotions, and physical sensations that humans (and animals) naturally have *in the face* of threats. Threats can be physical dangers or the possibility of rejection. They can also be real and in the moment or imagined and in our minds.

You're crossing the street and you see a bus hurtling toward you; your heart races, your palms sweat, your thoughts scream out danger, and you run out of the way.

You hear someone yell something derogatory at you and move toward you in a threatening way. You feel your face flush and your blood pump through your body, as you are unsure whether to run or stand and defend yourself.

You are in a social situation where you might be judged, like when you give a presentation in front of a group, and you have the same physical response—racing heart, palms sweating—as well as thoughts that people do not like what you're saying or are not paying attention and urges to run from the room.

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You vividly imagine taking a risk—like trying out for a sports team or auditioning for a play— and you have the same thoughts and sensations you would have if it were actually happening.

Anxiety—is closely connected to fear, but occurs in *anticipation* of a feared situation. Often this can be accompanied by sensations of muscle tension or uneasiness or feeling on edge and easily startled, as well as thoughts about what might go wrong. Anxiety usually leads us to want to avoid a situation or reminders of a situation.

Before a job interview, a first date, or doing something else new, you might notice that your mind is racing and your body is tense in anticipation of something potentially going wrong. This might lead you to put off preparing for the interview or to consider canceling the date.

Understanding the nature of fear, anxiety, stress, and worry is an important foundation for making changes that can help you dare to live the life you want.

Stress—is the response we have—thoughts, feelings, physical sensations—to any demand or stressor. Stressors can be happy occasions (new relationships, a promotion) or extremely unpleasant events (death of a partner, witnessing an attack). They can be events that already happened (illness) or something that could happen (possibly being fired from a job). Stressors can be sudden events (like a test or an argument) or ongoing experiences (a long commute, discrimination). When we have a stress response, hormones and nerve chemicals are released to help us cope. But if we are chronically stressed, these physical changes can be harmful to our health. Both fearful events and worries can elicit stress responses. We can experience chronic levels of stress that we aren't even aware of but that affect our emotions and our actions in problematic ways, such as leading us to act impulsively or feel irritable or reactive.

After the birth of a child, there are multiple demands on one's time, energy, and financial resources. You may feel love and excitement, but may also feel many new pressures.

Worry—is the cognitive component of anxiety and can often occur even well in advance of a potential threat. When we engage in worry, we often ask ourselves “what if” questions. It can seem like this is solving a problem, but instead we often just go from one potential problem to another and have difficulty focusing on the task at hand.

We revisit worry in more detail in Chapter 2

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As I (L. R.) think about writing this book I ask myself, "What if we don't explain these concepts clearly enough for people?" and "What if there is a better way to start things off?"

As you drive to work, you find yourself thinking "What if something happens to my mother on the long car ride she has planned for the weekend?"

As you try to fall asleep, you keep thinking "What if I sleep through my alarm? What if traffic is bad tomorrow and I'm late for my appointment?"

WHAT ARE FEAR AND ANXIETY MADE UP OF?

TRY THIS

Take a moment to think about how you know when you're anxious or afraid. It might help to think of a recent example of a time you felt this way. What do you notice? Are there sensations in your body? What kinds of thoughts do you have, or what do you tell yourself? What kinds of feelings come/came up for you in this situation? What do/did you do? After you have thought about this for a few moments, look at the lists on pages 16–17 to see if you recognize any responses listed there.

Each of us varies in the ways we commonly experience fear or anxiety.

- We may notice a lot of sensations in our bodies.
- We may notice our minds get very busy with a lot of rapidly occurring negative, fearful thoughts.
- We may notice that our minds go completely blank.
- Our behavioral signs of anxiety may be very clear to us (e.g., we procrastinate, avoid situations where we might experience anxiety, do not pursue things that matter to us).
- Our behavioral signs of anxiety might be subtle and hard to notice (e.g., we don't realize how many social opportunities we passed up until we bring more attention to the choices we are making).
- We may just feel fear when we are afraid.
- We may have a lot of different emotions—fear, anger, disgust, hopelessness—that arise all at the same time.

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- We might be clear about exactly what emotions we are feeling.
- We might just notice we feel “bad” or “distressed” or even “numb” or “shut down.”

Physical Sensation		
Rapid heart rate	Blushing	Headaches
Sweating	Dry mouth	Restlessness
Dizziness or lightheadedness	Stomach distress	Fatigue
Shortness of breath	Tension or soreness in the neck, shoulders, or any other muscles	Irritability
Trembling or shaky feelings		

Thoughts / Cognitive Symptoms
Worries about what might occur in the future (e.g., “No one will talk to me at the party,” “I will fail this test,” “My parents will become ill,” “My children will not be happy,” “I will end up alone,” “I will have a panic attack at the supermarket,” “I am going to get sick from the germs in this bathroom,” “People won’t take me seriously at school”) Ruminations about the past (e.g., “I can’t believe I said that,” “My boss thought I did a terrible job,” “I wish I hadn’t snapped at my partner that way,” “Having nothing to say in that conversation was so humiliating”) Thoughts about being in danger (e.g., “I can’t do this,” “I am having a heart attack,” “I am losing my mind”) Narrowed attention toward threat or danger, inattention to evidence of safety

Other Emotions		
Sadness	Disgust	“Overwhelmed”
Anger	Shame	“Numb”
Surprise	Hopelessness	

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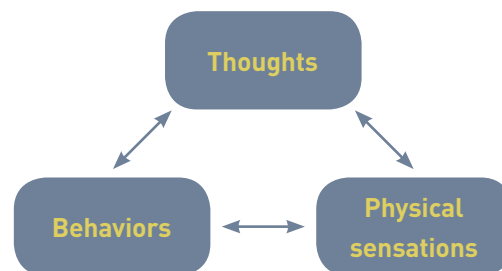
Behaviors

Repetitive behaviors or habits (e.g., biting fingernails, picking skin, playing with hair, tapping feet)

Avoidance or escape (e.g., turning down a social invitation; passing up a promotion; calling in sick to work; making an excuse to cancel a social engagement; leaving an event early; asking someone else to make a phone call for you; taking an alternative route to avoid a bridge or tunnel; using a ritual, security object, or lucky charm to get through an anxious experience)

Distraction techniques (e.g., overeating, smoking, watching television, having a few glasses of wine or a couple of beers, sleeping, shopping, putting excessive energy into work, exercising vigorously to try to “tire out” your body, coming up with a busy schedule to keep your mind off worries)

Yet, for all of us, no matter what sign of fear and anxiety we notice first, a reaction in one part of the system sets off reactions in the other. This is often called the cycle of fear and anxiety.



Tonia feels anxious in classes that require participation. When she wakes up in the morning before class, she feels her heart racing and a knot in the pit of her stomach, and she has thoughts like “I’m not going to be able to remember what I want to say” and “People are going to think I’m stupid.” The sensations she feels in her body produce more anxious thoughts like “Oh no, everyone is going to see how nervous I am” and “It seems like I am really starting to freak out,” and her anxious thoughts fuel more uncomfortable physical sensations (like sweaty palms). These thoughts and sensations also cue an urge to engage in certain behaviors. As she struggles with her thoughts, feelings, and urges, Tonia begins picking at her nails and reading and rereading her notes and almost misses her bus. Now she has thoughts like “I’m going to have to walk in late, and everyone

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will look at me and think I don't take school seriously, even though I really care about learning and this subject. Maybe I just shouldn't go at all." These thoughts lead her face to feel flushed, and she thinks, "Everyone will see how anxious I am." By the time she gets to class, she is experiencing so much physiological anxiety and her thoughts are racing so fast that she can't remember what she read the night before. She doesn't speak in class because she is overwhelmed by anxiety and afraid of the consequences. After class, she is frustrated with herself for not contributing when she had important points to make. She thinks, "This is only going to get worse."

Mario has an important interview for a promotion tomorrow. He plans a busy day to try to keep his mind off the interview: going to the gym in the morning, spending the day at the science museum with his family, and then watching a basketball game with his friends in the evening. He finds that as he does each of these activities his mind keeps going to the interview the following day. He thinks, "What if I don't give them the answers they're looking for? What if I can't show them how capable I am of being a leader? What if they already have someone else in mind for the job? What if my alarm doesn't go off and I'm late, making a bad impression?" With each thought he feels his body getting more and more tense. He finds himself so absorbed in planning out answers to imaginary questions that his children have to call out to him multiple times to show him an interesting exhibit at the museum. He notices his partner looks annoyed with him when this happens for the fifth time. He starts thinking, "Why can't I enjoy being here with my family? What's wrong with me? I keep thinking about the future and I'm not paying attention to the people I love. What if my partner gets fed up and leaves me?" He briefly notices feelings of sadness and fear, yet pushes these feelings away and continues running through anxious thoughts in his head. While his mind continues spinning through these worries, his partner has to call his name several times to indicate that it is time to go get lunch. Mario feels embarrassed and angry with himself that he keeps being distracted and vows to do better. Still, he finds himself thinking about the job interview tomorrow, how much he would like the position, and how worried he is that he won't be selected. His shoulders ache from all the tension he is holding.

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Suzanne describes herself as feeling “stressed out” and “overwhelmed” but has trouble pinpointing exactly what she is thinking and feeling. She describes her physical sensations as “agitation” or feeling “on edge” or sometimes feeling very slowed down and unmotivated. She feels like every moment is filled with tasks to do, and these tasks run through her head continually. She often finds herself in the middle of a room, uncertain why she entered the room in the first place. She thinks that other people manage their lives better than she does and that she is unreasonably “emotional” and that she really needs to “get a grip.” The more she thinks this, the more she finds herself feeling overwhelmed and the harder it is for her to accomplish her goals for the day. She rarely reflects on how she’s feeling about her life, but when she does, she feels generally dissatisfied. She hasn’t intentionally avoided pursuing more meaningful interpersonal relationships or advancement at work; she just always feels too overwhelmed to take on any new challenges and finds herself feeling stuck and hopeless.

In Chapter 3, we will more fully explore how to notice when habits of responding to early signs of fear and anxiety intensify and prolong distress. And Part II will explain how to develop new ways of responding that can move you forward.

Tonia, Mario, and Suzanne (and all the rest of us) each have very distinct patterns of physical sensations, thoughts, and actions. Yet in each case we can see how sensations lead to thoughts and vice versa and how both can influence behaviors, with behaviors feeding back to influence thoughts and sensations. And habits of responding (with worry, self-critical thoughts, avoidance, or emotionally shutting down) intensify the cycle, making distress more intense and interfering with how fully we can lead our lives. One important step to making changes

in these naturally occurring cycles is beginning to recognize them so that we can interrupt them and develop new patterns of responding.

TRY THIS

Consider the following questions: Tonia felt anxious about speaking in class and ended up staying silent. What do you think were some of the benefits of making that choice? What were some of the costs? In that moment of anxiety, Tonia acted as if the benefits

Recognizing the early signs of anxiety can help you choose to react in ways that aren’t as likely to prolong or increase your distress.

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outweighed the costs. If she were able to step out of the fear and anxiety of that moment, do you think she would come to a different conclusion?

How to “Monitor” or Become More Aware of Experiences throughout the Day

One way to begin recognizing your responses is to start to “monitor” or notice when you’re having certain thoughts, sensations, or emotions during the day. Throughout the book we will include forms to fill out to reinforce the habit of noticing your experience as it occurs and start to relate differently to it, as one important step in daring to live the life you want. You can use the forms right in the book, download the forms from www.guilford.com/orsillo2-forms so you can have extra copies, make your own forms, use a notebook to jot down observations, make notes on a smartphone, or take whatever approach is most manageable for you.

People find different strategies helpful for remembering to check in and pay attention to their experience:

- Checking in at certain times of day, like when you wake up, eat lunch, eat dinner, and go to bed
- Checking in as you switch tasks during the day
- Checking in when you notice you are feeling distressed (this can be more challenging, so be gentle with yourself while you try it)

TRY THIS

Now that you understand all the different signs of anxiety that may occur in your body, your mind, or through your behaviors, see if you can recognize these signs **as they occur** during the day. It might be helpful to use the Monitoring Your Fear and Anxiety form on page 20 to record them or download and use the monitoring form at www.guilford.com/orsillo2-forms.

More information on cultivating awareness and monitoring is in Chapter 6. If you find you really struggle with exercises involving noticing, you can skip ahead to that chapter for some more guidance.

Just note anything you notice in your body, mind, or behavior. Focus on simply observing what comes along with fear, anxiety, or stress for you. You may notice judgmental thoughts arising as you tune in to your responses. If that happens, see if you can just let those judgments be—no need to struggle with them or push them away—and as best you can, bring your attention back to simply observing. Observing and taking notes on your anxiety response is one way of learning to relate

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MONITORING YOUR FEAR AND ANXIETY				
Date/Time	Current Situation	Physical Sensations	Thoughts	Behaviors

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to it differently, so try it out and see what you notice. If you notice anything new in your experience, you can write this down too. Remember, the purpose of this exercise is to better learn to recognize the subtleties of your unique anxiety response, so it is important to record your observations in the moment as they are unfolding.

UNDERSTANDING THE FEAR RESPONSE

In this section we focus primarily on fear. See Chapter 2 for a deeper look at the experience of worry.

Most people who struggle with fear and anxiety wish that these feelings would just go away. It can seem like feeling these feelings means there is something wrong with us. And we often feel alone in our struggle. We don't often

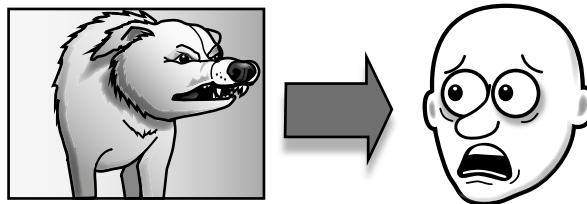
know when friends, neighbors, or coworkers are struggling with anxiety unless they choose to tell us, because we can't see others' racing thoughts, knots in their stomach, or muscle tension. So we judge our own "insides" against other people's "outsides" and often conclude we need to get rid of these feelings to lead a good life. While wishing anxiety and fear away is very understandable, particularly when our struggle with these feelings can be so exhausting and time-consuming, we are actually very lucky that we can so easily and readily experience fear. Fear serves an important function in our lives.

When we judge our own "insides" against other people's "outsides," we feel alone in our struggle with fear and anxiety.

Fear and the Fight-or-Flight Response

We are biologically prepared to detect threat and respond with fear (blood rushing to our limbs, a perception of danger, an urge to fight or flee) when we encounter any potential threat. This immediate response has helped us survive.

Imagine we saw this dog:



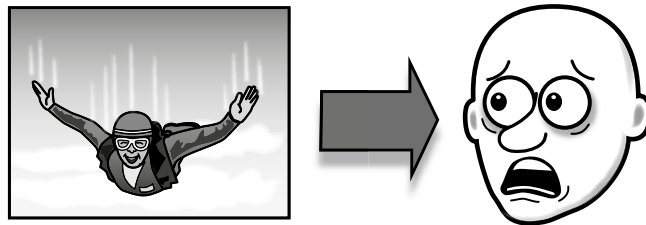
The best thing that can happen in this situation is that we experience an immediate spike in fear that leads us to run away from the dog, reducing the chances we get

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attacked. Our ancestors survived because of this instinctual response. They didn't question it at all, and followed the actions associated with the feeling.

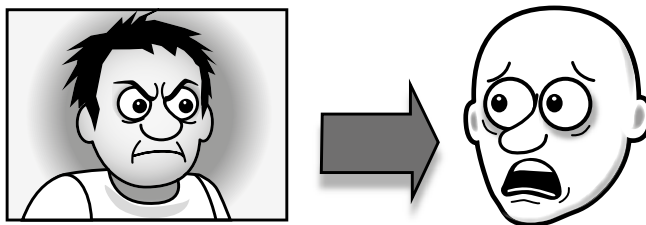
Fear is tied to an urge to act in very specific ways—when we experience fear we feel compelled to escape a situation or avoid entering it at all. We may also feel pressured to fight or attack the potential source of danger. This is referred to as a *fight-or-flight response*. Sometimes fear and anxiety in a situation where escape or attack does not seem possible lead to an urge instead to *freeze*, the way a deer does in headlights. These *action tendencies* can be experienced very strongly, so we often take the action suggested by our emotions. However, we do not have to act the way our emotions tell us to.

Imagine seeing this opportunity:



It is definitely true that we could get hurt skydiving. So noticing one's fear and avoiding this activity is one perfectly reasonable response. On the other hand, some people may really value "risky" activities like skydiving, skiing, or even riding a roller coaster. Some activities people think are "fun" require us to notice we are afraid, notice the urge to avoid or escape, and make the intentional choice to try the activity anyway.

A similar kind of learning happens with social cues. It makes sense that we learn to fear this angry person:



Someone who is angry could physically attack us or say something to us that could make us feel uncomfortable or guilty. So the safest response would be to avoid or escape from this person. Yet if this is a relationship with someone we value—like a

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boss or our partner—and the “threat” to us is uncomfortable (we are going to hear that we made a mistake on a report or acted inconsiderately) but not dangerous, we may actually want to approach the angry person. Again, rather than responding instinctively, we may need to notice our fear and also consider what matters to us personally when choosing a response. **Many of the activities that make life rewarding, enriching, and satisfying also elicit some fear. It is impossible to take risks or face challenges without feeling these feelings.** Using the awareness skills presented throughout this book, we can learn to pause when strong emotions arise and make a choice about how to respond given what is most important to us in that moment.

TRY THIS

Certain activities that make life rewarding, enriching, and satisfying—such as doing something adventurous in our free time, taking on a challenge at school or work, or asking someone out on a date—are pretty much guaranteed to elicit some fear. *Can you think of some action you have thought about taking or some activity you considered trying that has these two characteristics? Something that would likely add to your quality of life and would probably cause you to feel some fear?*

Fear and Generalization

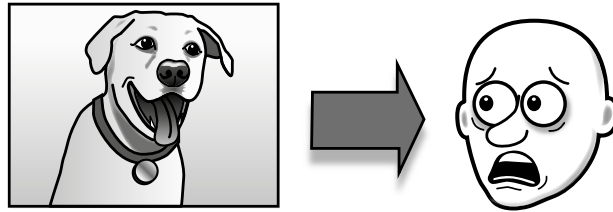
Another thing that can cause us to struggle with fear is that our learning can generalize from one situation to another. That is, we can learn fear to a specific cue so strongly that we also respond fearfully to other cues that were present at the time or to cues that are related to the initially threatening situation.

This ability to generalize can be extremely adaptive. For example, if you learned to fear and avoid black widow spiders after being bitten by a brown recluse spider, because your fear of spiders has generalized to all spiders, you might be less likely to get bitten by a dangerous, venomous spider.

Through our capacity for learning, generalizing protects us from real threats but also makes us more vulnerable to new anxiety triggers.

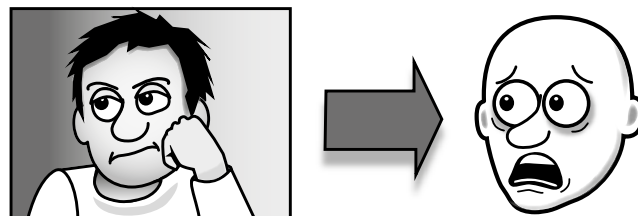
Generalizing can, however, also lead us to fear and avoid situations and objects and situations that are not at all dangerous. For example, while it might be helpful to feel fear and avoid the attack dog illustrated earlier, through our tendency to generalize our learning we can come to have the same kind of response when we see this much less threatening-seeming dog.

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Although this second dog may not seem as threatening to someone who has never encountered an attack dog, someone with that history may home in on the physical features this dog shares with the other dog that could be seen as threatening (e.g., sharp teeth and claws). This generalization might lead us to stay away from the less threatening dog instinctively, without thought, and keep us from learning that this dog is not dangerous.

Also, while it may be adaptive to feel fear in response to a threatening facial expression, through generalization learning we may find ourselves having a similar response to someone who looks more like this:



Expressions of boredom or disinterest can be painful if we care about the person making them, but they are not necessarily threatening and may not warrant the same kind of response we have to an angry expression. What causes us to make this kind of generalization? On the one hand, select *physical similarities* between the two expressions could promote generalization. For example, a slight frown can mean someone is starting to get angry or is focused intensely on thinking about something else. During the expression of both anger and boredom, one's eyebrows tend to be low on the eyes (whereas they tend to be rounded and high when one is happy). On the other hand, humans can also generalize their fear of one object or situation to another based on conceptual similarities between the two. Objects and situations develop *conceptual similarities* based on our learning. For example, milk, coffee, and lemonade are all liquids, and based on physical attributes milk and coffee are no more similar than lemonade and coffee. But through learning we have come to closely associate milk and coffee. So based on our unique life experiences, we can

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generalize the fear of one object or situation to another, even if the two are not linked to each other by physical characteristics.

This generalization of fear can be particularly strong when we have had traumatic, extremely distressing experiences. That kind of learning (being exposed to danger or humiliation) can lead to fear being triggered easily by a whole range of internal and external cues. This can make us feel like our fear is unreasonable or “crazy.” Yet the evolutionary aspect of this fear learning remains—at times of significant danger, it makes sense that we learn fear very strongly and in a very broad way, so as to maximize protection and our own safety. Unfortunately, this very intense, generalized fear response can severely restrict our lives because we fear and avoid such a wide range of cues. Understanding how and why we have such strong emotional responses across so many situations when we have experienced trauma is a first step toward learning to respond differently when our fear response is triggered.

“Unlearning” Fear

Recently scientists discovered something new about how fear is learned and how people can come to approach things they used to fear and avoid. Surprisingly, once we learn to fear something, we actually can never “unlearn” this association. So if Anh watches her friend Bree get bitten by the class pet Harry the hamster, Anh will always have some connection in her brain between hamsters and danger.

The good news is that we can learn new associations. For example, if Anh works in a pet store throughout high school, handles hamsters all the time, and never gets bitten, she will also have some connections in her brain between hamsters and harmlessness.

This means that to learn to be more comfortable around things, people, and activities we fear we have to do the *exact opposite* of what our emotions are telling us to do. The only way to become less fearful of something we value is to approach it and have some new experience with it that teaches us it is not threatening. We truly do have to dare to live the life we want—as developing the courage to approach the things we fear is the best way for us to change how we respond to them. And when we approach what we fear, we actually have to keep our minds open and present to the situation. If we are too frightened to pay attention and we distract ourselves, the new learning cannot occur.

Approaching what we fear is a daring act, but it's the best way to change how we respond to what scares us.

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Spiders and Snakes

Interestingly, we are much more likely to learn to fear things that used to be a threat to human survival—even though those threats are no longer as dangerous as they used to be. For example, people are seven times more likely to have a fear of spiders than they are a fear of driving. Yet approximately 37,000 U.S. citizens die in car

Although facing our fears may sound simple, in practice it can be extremely challenging. Many people live their whole lives avoiding people, situations, or activities that could elicit fear. Throughout the book we will introduce you to skills and strategies we hope will help you tap into your courage to live life fully and reduce your struggle with painful emotions.

accidents every year, whereas about eight people a year living in the United States die as a result of a venomous spider bite. We are biologically hardwired to instinctively fear those things that were most dangerous to our ancestors, even though they may no longer pose a significant threat. That may explain why it is difficult to rationalize away a fear even when we logically know we are not in danger.

Fear and anxiety are adaptive, natural responses that serve an important function in our lives. One reason we struggle with fear is that we think it can take control of our lives. Because we feel the

urge to fight, flee, or freeze in the face of fear, it can seem like these emotions cause us to behave in ways that may interfere with what matters most to us. Throughout the book, we will help you develop the skills of noticing when these urges arise and then choosing whether to follow the urge or to engage in a different action. This is an important step toward daring to live the life you want, because often the things that matter to us involve moving toward things we fear, rather than away from them. For instance, if we want to develop an intimate relationship, we have to open up and be vulnerable, even though anxiety will naturally arise due to fear of rejection or hurt.

TRY THIS

Fear and anxiety are strong habits for all of us, meaning that we respond to them without really noticing what we're doing. To make meaningful changes in our lives, the first step is to start noticing what is unfolding in the present moment. This will help us pause and consider new ways of responding. One way to learn to do this is to practice noticing even the most simple and automatic actions we take. Try Mindful Walking and see what you notice:

Take 5 minutes each day and walk with awareness. Inhale as you lift one foot, exhale as you place it on the ground, inhale as you lift the other foot, and exhale as you place it on

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the ground. Notice what it feels like to lift a foot and what it feels like as you place it on the ground. Notice your posture and the sensations of your breath. As your mind wanders, bring it back to your steps. Be gentle with your mind—it will naturally wander, and you may suddenly find that you sped up or started to do something else. Just return each time you notice to paying attention to what it feels like to walk. Some people do this in a circle, which you can try if you have room. If the weather permits, you can do it outside and possibly notice other sounds and smells while you're walking. Or you can do it in a very small space in your home or simply walk up and down a hallway or a room. There is no right or wrong way to practice. Just set aside some time to pay attention on purpose while you walk and see what it's like to be aware of this very habitual behavior. You might write down any observations and return to them later in the book when we talk more about building this awareness muscle.

Part II will visit awareness exercises in more depth, and Chapter 8 will describe mindfulness to you. But you can do practices without reading any of that first and just see what you notice.

Fear Is Learned

- Fear helps us avoid real physical dangers.
 - These are natural, human responses and are helpful to us.
- Fear is easy to learn.
 - Our nervous system has evolved so that we can readily detect and learn danger, to keep us safe.
- Fear and anxiety can easily spread to other things.
 - We easily learn to fear things that are similar to, or associated with, objects or situations that we perceive as threatening.
- Fear cannot be unlearned.
 - The only way we come to be less afraid of an object or an activity is to have lots of experience with it that teaches us we are safe.
- Some fears are biologically inherited.
 - We are more likely to fear things that threatened our ancestors' survival. We are "hardwired" to very quickly learn to fear and avoid snakes and spiders.

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QUESTIONS YOU MAY HAVE AT THIS POINT

Q: I feel like I am more anxious than other people I know—is this just how my personality is and it can never change?

A: Some people certainly experience more intense anxiety or respond more quickly to situations with anxiety. This can happen for a number of reasons. Some people are genetically predisposed to be more anxious. Others have experiences in their lives, like trauma, stressful life events, difficult family relationships, the absence of a strong social support system, discrimination, or limited financial resources, that lead them to learn to feel unsafe more easily in a wide variety of situations. And then, telling yourself that you are an “anxious person” can maintain this style of responding, as well as limit your life, all of which further feeds the cycle of anxiety. However, none of this means that it cannot change. It may take more practice and more patience to make changes in very well-worn patterns of responding. And some of us may continue to have an anxious response more easily. Yet we can still make substantial changes in the ways we respond to our anxious reactions by understanding them more fully, learning how to respond to them differently, and choosing actions that bring meaning to our lives. These changes can all change the intensity and duration of our cycles of anxiety.

See Chapter 3 for a more in-depth discussion of understandable responses to anxiety and distress and an introduction to strategies to help us change them; Chapter 12 presents a more expanded discussion of strategies.

Q: I've had very painful, traumatic experiences in my life. Is there really a way to address the anxiety that comes from those kinds of experiences?

A: Unfortunately, many of us have real-life experiences of threat, danger, humiliation, injustice, and/or violence that naturally lead to feeling unsafe in the world. Often recovery from these kinds of experiences requires processing them directly in some way, such as through therapy or a supportive network or affinity group. If you've done these things and still feel the very natural sense of anxiety and it is triggered easily by cues in the environment, you may find the strategies in this book helpful. A very important part of changing our relationship to these kinds of responses from traumatic or other harmful experiences is understanding that our reactions are natural given what we've been through. Having compassion for ourselves and these reactions can help us choose how we respond so that we can broaden our lives again and have new, rewarding experiences despite past or ongoing stressors.

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Q: You keep talking about needing to be more aware of the process of anxiety—I am painfully aware of my anxiety. Maybe this isn't the approach for me.

Developing understanding and compassion is challenging—we address this in more depth in Chapters 7 and 12.

A: This is the paradox of anxiety—on the one hand, anxiety leads us to be extremely aware of every anxious thought and sensation we have. So it makes sense to think the solution is to pay less attention or get it out of our minds. Unfortunately, as we discuss later, we can't actually completely avoid these sensations. Another option, therefore, is to broaden our awareness. Doing so can help us gain a broader perspective on the situation. We may also notice habits that increase our fear and find opportunities to respond differently. Making these changes can diminish our anxiety and help us lead a more fulfilling life.

In Chapter 4 we will explore the complexities and costs involved in trying to completely avoid painful emotions.

We examine the subtle but critical differences between accepting and tolerating painful emotions in Chapters 7 and 12.

Q: Are you saying that I have to tolerate anxiety because it won't go away?

A: Although we are saying that fear and anxiety are a natural part of life, we are not suggesting that you “grin and bear it.” We have found that by understanding anxiety, noticing its many components as it evolves, learning to relate to anxiety differently (with curiosity and compassion rather than self-criticism), and clarifying what is important, people can make significant changes in their lives, which are accompanied by less intense, less long-lasting experiences of anxiety, as well as increased feelings of joy and satisfaction.

Q: I'm not sure that I really experience anxiety. My heart doesn't race, and I don't feel scared very often. I just feel that my mind is very busy. I am always preparing for what comes next and what might go wrong, and I can't easily turn my attention to other things. Is that anxiety?

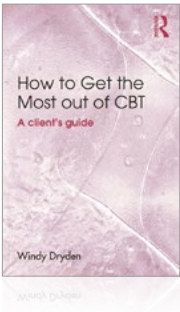
A: You are describing worry, a common cognitive component of anxiety. We address worry in more depth in Chapter 2.

4

DECIDE IF CBT IS FOR YOU

*From How to Get the Most Out of CBT:
A Client's Guide*

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The following is excerpted from *How to Get the Most Out of CBT* by Windy Dryden. © Taylor & Francis Group, 2015. All rights reserved.

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How is it that you have ended up either consulting or thinking of consulting a cognitive behaviour therapist? How much do you know about CBT? Have you actively sought out a CBT therapist or were you recommended to do so? These are some of the questions that come to my mind as I invite you to decide whether or not CBT is for you. Indeed, these are some of the questions that I do ask people who come to see me, either to consult me as a CBT practitioner or to seek my help to determine which approach to therapy is best suited to them. Since I don't know your answers to these questions, let me deal with the issue of how you can best decide if CBT is for you in a more general way.

COMMON FACTORS THAT SPAN DIFFERENT PSYCHOTHERAPEUTIC APPROACHES

In the field of psychotherapy and counselling, it is recognised that different approaches have both common factors (i.e. common to all therapeutic approaches) and specific factors (i.e. specific to the particular approach under consideration). The main common factors include:

- the development and maintenance of an effective working alliance between you and your therapist;
- the provision of a safe space in which you can discuss whatever is important to you;
- the mobilisation of hope whereby you come to see that you can effectively address your concerns; and
- experiencing your therapist as someone who is genuine with you, understands you and accepts you.

As I have said, these factors are common to all approaches to therapy and are not specific to CBT.

While I have entitled this chapter “Decide If CBT Is for You,” when it comes to the presence or absence of these common factors, I suggest that your focus be more on deciding whether or not the *therapist* whom you have come to see is the right *person* for you to consult than on whether or not CBT is right for you. Thus, your therapist may be technically proficient in CBT, but if you don't feel safe talking to her¹ about what really matters to you, you are right to have doubts about your therapist. Based on the above, here are some questions to ask yourself to help you make your decision about whether or not to work with your particular therapist:

- To what extent does my therapist understand my problems from my perspective?
- To what extent does my therapist accept me the way I am?

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- To what extent is my therapist genuine in the way she interacts with me?
- To what extent do I feel safe to discuss what really matters to me with my therapist?
- To what extent does my therapist inspire hope in me that I can effectively deal with my problems?
- To what extent does my therapist foster a working relationship with me focused on dealing with my problems?

While it is unrealistic for your therapist to score top marks on all these points, she should score highly enough for you to consider working with her over time. If she scores poorly on all these points, then, in all probability, she will not be able to help you much despite her proficiency in CBT. If the therapist scores highly on all but one or two points, then you should consider discussing your feelings with her on the points where she does not score highly. I will address the importance of discussing with your therapist matters to do with your therapy later in this book. For now, let me make two points. First, if you don't feel able to discuss your concerns with your therapist, then this may, in itself, indicate that your therapist is not right for you. Second, if you do decide to discuss your concerns with your therapist on these matters, the way your therapist responds is important. If she takes your concerns seriously and responds without defensiveness, then these are good signs that you can work with the therapist and that you can deal with any rifts in your relationship that may occur over the course of therapy. However, if your therapist appears to dismiss your concerns and/or responds defensively, then this does not augur well for the future and you should consider finding yourself a different therapist.

Having made this point, don't forget that therapists are human too and may have their off days. However, if a therapist responds dismissively and or defensively more than once, then I do urge you to think very carefully about continuing to work with that person.

CBT'S MAIN SPECIFIC FACTORS

In making a decision concerning CBT's suitability for you, it is important for you to understand some of the therapy's main features. I made the point in the introduction that CBT is, in fact, a psychotherapy tradition and that there are a number of different approaches that come under the umbrella of the CBT tradition. Having said that, let me outline some of CBT's main specific factors.

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CBT Focuses on the Way You Think and Act in the Context of Your Emotions and in the Situations in Which You Experience These Emotions

CBT stands for “cognitive behaviour therapy,” and thus you would expect that the therapy would focus on cognition and behaviour.

FOCUS ON BEHAVIOUR

Let's start with behaviour, as this is the easiest of the two terms to grasp. Your CBT therapist will focus a lot on the ways in which you behave, particularly in situations in which you experience your problem(s). However, your CBT therapist may also be interested to understand what may be termed “action tendencies.” These describe situations in which you feel an urge to act in a certain way but don't actually do so. Such action tendencies are particularly valuable in helping your therapist discover your hard-to-identify emotions (such as envy and hurt). Understanding the difference between an action tendency and an overt behaviour may also help you see that you don't have to act on your action tendency, which is particularly important with problems of anger and self-discipline.

The focus on behaviour in CBT is particularly linked to an understanding of your goals and values. Thus, expect your CBT therapist to enquire about the extent to which your problem-related behaviour helps you to meet your goals and to what extent it is consistent with your personally held values. Consequently, during therapy you can expect your therapist to encourage you to act in ways that help you to achieve your goals and are consistent with your values as well as to help you to identify, reflect on and deal with obstacles to the execution of such behaviour.

A particular focus on behaviour that your therapist may well take, particularly if you have problems with anxiety, is to consider your use of safety behaviours (i.e. behaviours which you use to keep yourself safe from threat, but in ways that *may* serve to maintain your anxiety problems). CBT practice is strongly underpinned by research, and while studies in the past showed the negative effects of such safety behaviours, more recent studies have shown that they may be useful in encouraging you to face your fears. Effective CBT therapists keep abreast of the research literature and modify their practice accordingly.

FOCUS ON THINKING

While the word “cognitive” is derived from the Latin verb *cognoscere*, meaning “to know,” in CBT it is used to refer to your thinking and particularly thinking that has a

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bearing on how you feel and act. Thinking can occur at different levels of your mind. There is surface-level thinking, which occurs in the form of automatic thoughts (i.e. thoughts that pass through your mind and are easily identifiable) and there is deeper-level thinking, which is less easily identifiable and takes the form of underlying beliefs. CBT therapists may vary in how much emphasis they place on these different levels of thought. If they deal with both levels, they may differ concerning the order in which they focus on them. What is important is that your therapist helps you to understand the role that your thinking (at whatever level) plays in your problems and elicits your agreement concerning how best to deal with them.

Another type of thinking that CBT therapists are interested in concerns where you place your attention when you are experiencing your problems. This is known as your attentional focus and is a useful area for you to explore with your therapist, whatever problems you have, for what you pay attention to tends to be closely related to your behaviour and emotions.

With respect to thinking, a recent development in CBT research and practice has been on ruminations, thoughts that go round and round in your head which you just can't seem to stop. It has been known for many years that such thinking is a key component of worry or generalised anxiety disorder (GAD), but more recently its important role in the perpetuation of depression and unhealthy anger, to take but two examples, has come to be realised.

You should be aware that CBT therapists from different approaches tend to vary concerning how they address your thinking. For example, practitioners of a CBT approach known as *rational emotive behaviour therapy* (REBT), pioneered by Dr. Albert Ellis, will help you early on in therapy to identify and challenge one or more of four deeper-level unhealthy beliefs (known as irrational beliefs) using arguments designed to encourage you to consider how consistent with reality these beliefs are, how logical they are and how useful they are. These therapists will also help you to construct healthy (or rational) alternatives to these beliefs which are more consistent with reality, more logical and more useful to you.

Other CBT therapists will focus initially on your more accessible surface-level thinking and help you to examine such thinking for its practical utility and to construct more helpful alternative ways of thinking. Later on these therapists, who have been trained in *cognitive therapy*, pioneered by Dr. Aaron T. Beck, will help you to identify and again examine for their practical utility a deeper set of underlying beliefs known as schemas and to construct more useful schemas. Unlike their

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REBT colleagues, cognitive therapists consider that the form of such schemas can vary greatly from individual to individual rather than be grouped into one or more of REBT's quartet of unhealthy (irrational) and healthy (rational) beliefs. Also, cognitive therapists are primarily concerned with the practical utility of thinking, and while they are also concerned with how consistent with reality such thinking is, they are less concerned with the logical status of such thinking.

While I have touched on some differences between REBT therapists and cognitive therapists, both are concerned to help you to focus on thinking that underpins your emotional problems with a view to help you to stand back and respond to it in some way. Other CBT therapists believe that encouraging you to respond to such thinking only results in you getting increasingly caught up in such thinking and may unwittingly help you to ruminate. For these therapists, who practise a form of CBT known as *acceptance and commitment therapy* (ACT), problem-related thinking is best dealt with by your accepting its existence without engaging with such thoughts (e.g. by challenging them) and by committing yourself to value-based behaviour despite the presence of these thoughts in your mind. These therapists do not speak of thoughts being distorted or realistic or beliefs being irrational or rational because they consider thoughts to be very much influenced by the context in which you find yourself rather than by the realistic or rational status of the thoughts.

From a therapeutic point of view rather than a scientific perspective, what really matters is whether the explanation provided to you by your CBT therapist concerning the role your thinking plays in your problems and what to do about it makes sense to you and whether you are prepared to proceed with a therapy that is based on these ideas. That is why it is so important in CBT for the therapist to be transparent in explaining her position on these issues to you.

CBT Focuses on How You Unwittingly Maintain Your Problems Rather Than on How They Originally Began. Consequently, CBT Focuses on What You Can Do Now to Address Your Problems

It is often thought that CBT therapists are not interested in your past. This is not correct, and in CBT you may talk about whatever it is you are bothered about, be it your past, your present or your future. Having said this, CBT therapists tend not to believe that helping you to understand the past roots of your present problems will be curative in the long term without you doing something about these problems in the present. CBT therapists generally hold to the view that relevant past experiences may have contributed to your current problems but do not account fully for

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these problems. CBT therapists explain this by pointing out that if 100 people all experienced exactly the same past experiences as you, not all of them would have developed the same problems as you. Some may have developed other problems and others would not have developed problems at all. Rather, it is the views you took from these experiences and still hold currently that largely account for your problems together with the behaviour that stems from these views.

For example, take the problem of jealousy. If you have such a problem, it may well be the case that you felt jealous of your sibling as a child. However, this insight will not help you if you continue to act in jealous ways in the present (e.g. by preventing your partner from doing things, checking on his or her whereabouts). Such behaviour will reinforce and strengthen the beliefs that underpin your jealous feelings and will nullify any effect that insight into the possible roots of your problem might have. As a result, unless your CBT therapist helps you to deal with the ways in which you currently, but unwittingly, maintain your problem, then it is unlikely that you will gain much long-term benefit from therapy.

CBT Focuses on Helping You to Put Into Practice Between Sessions What You Learn in Sessions

When you consult a CBT therapist, it is unlikely that you will derive any benefit unless you learn something in the therapy sessions. However, such learning is likely to be academic and thus of limited value to you unless you put it into practice between therapy sessions. Consequently, in CBT, expect your therapist to negotiate with you on ways of implementing your session-derived insights into relevant situations in your everyday life. The extent to which you do so will determine how much you get from CBT. Thus, I am often asked whether CBT is helpful. What is my answer? Yes, if you use it; no, if you don't! I will discuss the issue of applying what you learn in Chapter 6.

CBT Focuses on Helping You to Become Your Own CBT Therapist

While all approaches to counselling and psychotherapy have as an aim you learning how to help yourself in the future after therapy has ended, CBT therapists, more than other practitioners, implement this aim in specific ways. They do this by teaching their clients CBT self-help skills throughout the therapy process. Thus, your therapist may well use a CBT-related framework to teach you how to assess your own problematic thinking, feeling and behaviour in problem-related episodes and how to respond productively to these situations. You will then be encouraged to use this framework for yourself between sessions and helped to refine your developing

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skills in subsequent sessions by your therapist when you report back on how you implemented your skills. Given this emphasis on helping you to become your own CBT therapist, it is likely that your therapist will give you increasing responsibility to help yourself as therapy progresses. She will do this by gradually fading her own active contribution to the process, becoming more of a consultant and giving you feedback on your developing self-helping skills than by actively taking the lead as she did at the beginning of therapy.

Because CBT emphasises teaching clients self-help skills, there are a number of CBT-oriented workbooks available that can be used as an adjunct to therapy. Your therapist may suggest incorporating such a workbook into your therapy. While some clients value using such workbooks, others find them too formulaic and would prefer not to use them.

Flexible CBT therapists will be mindful of the fact that while CBT does emphasise the teaching of self-help skills as an integral part of the therapy, some clients do not want to learn these skills in such a deliberate manner. These flexible therapists adjust CBT accordingly. I will discuss the issue of becoming your own CBT therapist more fully in Chapter 8.

In this chapter, I have set out to give you a flavour of some of CBT's distinctive features while acknowledging that different CBT approaches emphasise some and de-emphasise others. I have also stressed that as CBT values explicitness, it is very likely that your therapist will make clear to you how she will use CBT to understand and deal with your problems. Thus, it should be easier for you to judge whether or not CBT is for you than it would be if you were consulting a therapist who practises a non-CBT approach. If you are still in doubt, most CBT therapists will suggest a brief "trial period" of therapy where you can experience CBT for yourself as a way of judging whether or not you wish to make a firm commitment to becoming a CBT client.

If you have decided that CBT is for you and you have found a properly trained therapist² to work with, you will need to make a number of practical agreements with her to ensure that therapy gets off on the right foot. This will be the subject of the next chapter.

Notes

1. In this book, when I refer to the therapist, I will refer to the person as "she." This was determined by the toss of a coin.
2. I suggest that you conduct an Internet search to locate the appropriate professional bodies in the country where you are that accredit CBT therapists in order to find a properly trained CBT practitioner if you have not already been referred to one.

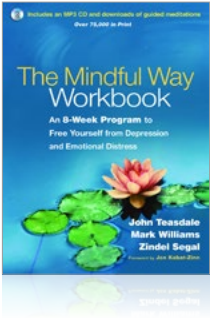
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DEPRESSION, UNHAPPINESS AND EMOTIONAL DISTRESS

*From Mindful Way Workbook: An 8-Week
Program to Free Yourself from Depression
and Emotional Distress*

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WHY DO WE GET STUCK?



The following is excerpted from *The Mindful Way Workbook* by John Teasdale, J. Mark G. Williams, and Zindel Segal. © The Guilford Press, 2014. All rights reserved.

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Jani would often wake very early in the morning, unable to sleep, with a heavy feeling in her body and thoughts going round and round, impossible to switch off. She'd sometimes get up to make a cup of tea, sitting in the kitchen with a blanket around her shoulders, reading bits of any magazine that she or her roommate had left lying around, or opening her laptop and trying to answer e-mails that had come in overnight. At last, exhausted, she'd go back to bed, only to find that the thoughts carried on, going round and round, but now with a new voice: "This is terrible. You'll be too tired to think straight today. Why is this happening again? Why can't you ever pull yourself together? What's wrong with you?"

For any of us it would be bad enough to wake up too early in this way. But Jani's mind just made things worse.

Reading through the story again, can you now see any similarities between the ways in which the "new voice" added its own twist to Jani's misery and your own past experience?

Put a ✓ next to any of these that you recognize:

- ☐ The voice added its own catastrophic interpretation ("This is terrible") to the situation.
- ☐ The voice was certain there would be awful consequences ("You'll be too tired to think straight").

The voice asked unanswerable questions that had the effect of:

- ☐ bringing to mind times in the past when things had gone wrong ("Why is this happening again? Why can't you ever pull yourself together?")
- ☐ focusing attention on weaknesses and failings ("What's wrong with you?")

Jani's experience illustrates a crucial and unexpected truth:

UNHAPPINESS ITSELF IS NOT THE PROBLEM

Unhappiness is part of the normal human condition. It is a natural response to certain situations. Left to itself, it will pass in its own good time, often surprisingly quickly.

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But, somehow, most of us don't feel able to let things take their natural course—when we feel sad or unhappy, we feel **we have to do something**, even if it's only trying to understand what's going on.

Paola: “I find I can't leave it alone once I get these moods. I sort of know it doesn't do me any good to worry and brood, but I can't help it.”

Paradoxically, it is those very attempts to get rid of unwanted unhappy feelings that get us stuck in ever-deepening unhappiness.

Our reactions to unhappiness can transform what might otherwise be a brief, passing sadness into persistent dissatisfaction and unhappiness.

Let's look more closely at what's going on here. We can distinguish three crucial stages:

Stage 1: Unhappiness arises.

Stage 2: The unhappy mood brings up negative thinking patterns, feelings, and memories from the past—this makes us more unhappy.

Stage 3: We try to get rid of the unhappiness in ways that actually keep it going and just make things worse.

THE ECHOES OF THE PAST

A few years ago Jani had been totally stressed by the amount of work she was expected to do in the job she held at the time. She'd become very down and constantly tried to “pull herself together” before eventually going to her physician, who prescribed antidepressant medication, which helped a bit.

She'd eventually left that job, but somehow she still blamed herself for giving in. Now, 7 years later in the early hours, as she struggled with not being able to sleep but not being really awake either, thinking of the day ahead, this echo of the past was making her feel worse.

Think back to a time when your mood began to go down. Make a ✓ next to any of the following adjectives that describe how you felt at that time—put a ✓ even if you felt that way only slightly.

- | | | | |
|-------------------------------------|------------------------------------|-------------------------------------|------------------------------------|
| ☐ dejected | ☐ depressed | ☐ despondent | ☐ a failure |
| ☐ inadequate | ☐ low | ☐ a loser | ☐ pathetic |
| ☐ sad | ☐ unlovable | ☐ unhappy | ☐ useless |

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There are actually two different kinds of words in this list. Some are simply descriptions of moods or feelings (*dejected, depressed, despondent, low, sad, unhappy*).

If you have been depressed in the past, thoughts of self-criticism or of being a failure are much more likely to get switched on by sad moods.

The others describe feelings that also seem to say something about the kind of person you are (*a failure, inadequate, a loser, pathetic, unlovable, useless*).

Research using this list of words has revealed something very important.

If you have been seriously depressed in the past, when you start to feel low now—*whatever the reason*—you are much more likely to begin to feel bad about yourself (and so check off words like *a failure*) than someone who has never been so depressed.

This is because, whenever we are very down, the mind is taken over by patterns of extremely negative thinking—thoughts that we are worthless, thoughts that we have let people down, thoughts that life is full of insurmountable difficulties, thoughts that the future is hopeless.

Bill: “Basically, I feel that I am just not good enough—and sooner or later people will find out.”

Links get forged between these thinking patterns and depressed, unhappy mood. The result? Sad mood arises now and old negative thinking patterns are right behind.

Tragically, these are exactly the feelings and thought patterns that would make anyone even more depressed.

And so the cycle continues: if you have been deeply depressed, it is much easier to slide back into depression again.

It’s not only thinking patterns that can get reawakened. Spells of depression will have often been triggered by experiences of major loss, rejection, or failure.

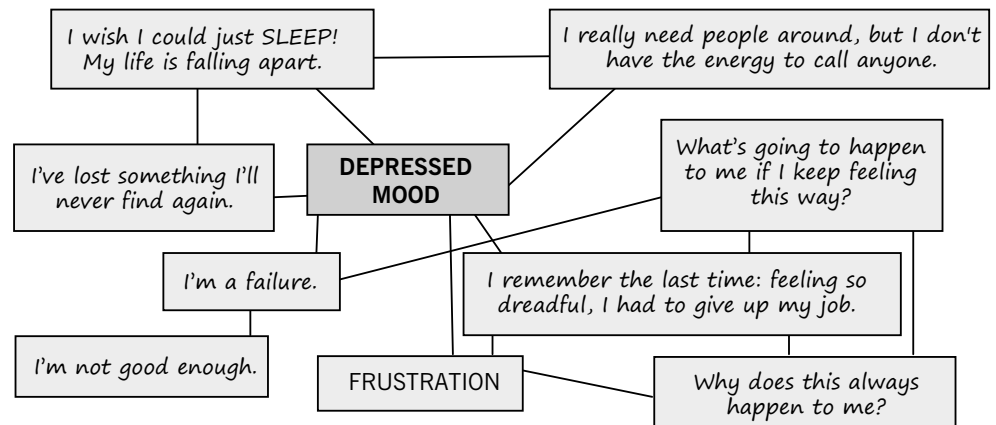
Anna: “Here I go again. I just feel like I have no future; that nothing has worked for me in the past and noth-ing will change. I can’t stand much more of this.”

When you feel sad or depressed again, memories of these losses and rejections—and all the weight of their tragedy—can break over you like a tidal wave. In that way, these thoughts and memories will make you even sadder, adding their own twist to a spiral of worsening mood.

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For Jani, her frustration with not getting to sleep and her fears about not being able to cope with her job evoked memories that made her feel even worse:



Just like Jani's depression, other emotions can color our experience in subtle (and not-so-subtle) ways, with consequences we cannot always see clearly.

For example:

- Feelings of anxiety can reawaken worrisome patterns of thinking— creating more anxiety, worries, and fears.

Olga: "What will happen if Bob gets ill again? Will I be able to cope? I don't want to be alone."

- Feelings of irritation and frustration can make us blame and criticize others, making us even more angry and frustrated.

Scott: "G. had no right to do this. If he does this again, it will be the last straw. I'm supposed to be responsible for running this project, not him."

- If we are overstressed by excessive demands, feelings of pressure may reawaken fears of being overwhelmed and force us to greater busyness and stress.

Pearl: "No one else can do this. It's all up to me. This deadline is really important."

The good news is that, with the right understanding and skills, we can break out of these mood–thought vicious cycles.

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Over and over, we have seen people learn to recognize these thought patterns for what they really are—just thought patterns—and then gracefully disengage from them by refocusing their attention.

The problem is that, through no fault of our own, most of us don't have the appropriate understanding and skills. In fact, you may have found that your bestintentioned efforts often have exactly the opposite effect to what you intended.

Let's see how.

HOW TRYING TO DIG YOUR WAY OUT OF TROUBLE CAN END UP DIGGING YOU IN DEEPER

Carmen: *"If I don't snap out of this soon, I'll be over the edge again. I've got to do something."*

If you have experienced low mood spiraling down to deeper depression in the past, you will know just how horrible that can be. It's completely understandable to feel the urge to get rid of the mood and stop the slide into something deeper.

Tony: *"I had such plans when I was young. Where did it all go? How did I end up like this? What is it about me? What's wrong with me?"*

Equally, if feeling constantly exhausted and unable to enjoy life reawakens a deep doubt about your worth as a person, what could feel more important than doing something about that?

If we look carefully, we can see what is happening here:

The mind is trying to get rid of unhappiness by thinking its way out of the problem.

Think back once more to a time when your mood started to go down. Did any thoughts like these go through your mind?

"What's wrong with me that I get so unhappy when other people seem to have friends and be happy?"

"What did I do wrong to end up feeling like this?" "What's going to happen to me if I keep feeling this way?"

These kinds of questions have no clear answer. Nonetheless, we feel compelled to keep chewing away at them—a process that psychologists call rumination.

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Psychologist Susan Nolen-Hoeksema spent many years investigating rumination and its effects. Her conclusions are stark:

Ruminating just makes us feel even worse:

We suffer the frustration of not being able to come up with answers.

We dredge up memories of failures and difficulties from the past to try to understand how we get things wrong. But focusing on our weaknesses and deficiencies in this way only drags us down further.

We anticipate the problems that will arise in the future if things don't change and dread the prospect of having to face the days, weeks, and months ahead.

We may even begin to wonder whether life is worth living at all.

Asha: *"All this thinking is getting me nowhere."*

Phil: *"It's always been this way. I'm sure I've lost friends by being so negative. What makes me do this? I remember the time when ..."*

Diana: *"Something in my life has been damaged forever."*

Far from freeing us from the downward spiral, **our attempts to get rid of unhappiness by thinking our way out of it are the very things that can deepen and prolong our sad moods.** These moods then bring up more unhappy memories and thoughts, and we now have new material to ruminate on.

If you have been clinically depressed in the past, rumination can create a slide in mood that will tip you into another episode of depression.

The problem with persistent and recurrent depression, unhappiness, and exhaustion is not "feeling low" in the first place. The problem is what happens next.

The core problem is how our minds react to feeling low, afraid, angry, or tired.

WHY DO WE GET LOCKED IN A STRUGGLE WE CAN'T WIN?

Rumination can cause us enormous difficulties. It can transform the simple experience of a fleeting emotion of sadness into serious depression; a fleeting emotion of irritation into a prolonged sense of indignation and anger; a fleeting moment of concern into a deep sense of anxiety.

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WHY DO WE GET STUCK?

So why do we do it? Why do we ruminate, brood, and worry, when, far from rescuing us from destructive emotions, this actually makes things worse?

To answer these questions, and to help us understand how we might begin to respond differently, let's step back and look briefly at the way the mind works more generally.

THE DOING MODE OF MIND

To solve a problem or to get things done, the mind usually works in certain predictable ways.

Take, for example, a day when you have to take a detour from your usual drive home to drop off a package at a friend's house, but you find yourself driving past the point where you meant to turn off into his or her street.

After a few moments, you notice the package is still there and realize that this is not what you had intended. You think **back**: "Ah, I should have taken the other turn." You think **forward**: "What shall I do next?"

You work out that the simplest way to sort things out is to turn around and make a deliberate effort, this time, to (1) turn off at the exit to your friend's house and (2) not just drive straight past it, as you normally would.

You act on these plans, take the right route, and deliver the package to your friend—mission accomplished!

The package eventually arrived at its destination through your mind's use of a very well-rehearsed and familiar mental routine.

This routine helps us get things done—achieve goals, solve problems, change things to make them more like we want them to be.

We call this the "**doing**" mode of mind. You'll find its core features listed in the box below.

To work effectively, at each point the doing mode has to *hold in mind*, and then *compare*, three ideas:

1. where you are at each moment (the **current state**)
2. where you want to be (your destination, **goal**, or **desired outcome**)
3. where you don't want to be (your nondestination or **outcome you want to avoid**)

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By holding and comparing these three ideas the mind can see how well the current state of affairs matches up with the goal you want to achieve and is different from the outcome you want to avoid.

Knowing whether these gaps are increasing or decreasing allows the doing mode to “steer” the mind and body in the right direction and reach the desired goal and/or avoid the undesired destination.

We are not necessarily conscious of all these processes. Many of them are carried out automatically in the background of awareness.

By using the same “doing” strategy the human mind can achieve some remarkable goals—from developing computers to building cities and putting a man on the moon.

Seven Core Features of the Doing Mode

1. it often comes on line **automatically**.
2. it uses **thoughts and ideas**, holding them in mind as you work.
3. it **dwells in the past and future** to help get where you want to be.
4. it keeps in mind what to **avoid**—where you **don’t** want to end up.
5. it **needs things to be different**, forever focusing on the gap between where you are and where you want to be.
6. it takes **thoughts/ideas as real** (it would not be useful to keep doubting your destination).
7. left to itself, it **continues to focus on the goal** until the task is complete, or until you are too tired and depleted to continue. The demands of the doing mode can be quite **harsh** and **unkind**.

IF DOING IS SO USEFUL, WHAT GOES WRONG?

If we want to achieve goals by making changes in the world around us—like building a house—the doing mode of mind is brilliantly effective.

So, it makes sense that the mind turns to the same basic strategy when we want to achieve goals in our inner, personal worlds—goals like feeling happy, not feeling anxious, being a spontaneous person, or not being someone who gets depressed.

It is here that things can go horribly wrong.

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For now there is a major difference. To solve problems, the doing mode has to hold in mind the ideas of where we are, where we want to be, and where we don't want to be. To work at all, these have to be held at the back of the mind all the time until the problem is solved or let go of.

For *external* problems like driving to a destination, holding these ideas in mind does not itself affect the distance left to travel.

But what about those times when the goal is *internal*—to be happy, or *not* to have certain unwanted feelings, or *not* to be a certain kind of person?

Remember how the doing mode works. Now we have to hold in mind “I’m unhappy”; “I wish I were happier”; “I don’t want these horrible feelings to come back.” What happens now?

Try saying these sentences to yourself two or three times:

“I’m unhappy.”

“I wish I were happier.”

“I don’t want these horrible feelings to come back.”

The doing mode needs to keep in mind the gap between the kind of person we are and the kind of person we want to be. But this just reminds us how much we are falling short of where we feel we need to be, creating more unhappiness.

What was your experience? You probably felt worse. Most people do.

The gap between where you are now and where you want to be just got bigger.

It is not just the ideas held in the mind that cause problems; it is comparing them. Sometimes the mind can see what’s going on and just let go of the project to get rid of unwanted feelings altogether.

At other times the mind feels compelled to continue: if we’ve known many times when sadness has led to depression, there will be an understandable *fear* of unhappiness—we’ll feel a need to avoid experiencing unhappiness at all costs.

That way, we believe, we’ll prevent ourselves from slipping back yet again into the depths of emotional turmoil. Here the mind just cannot let go—it will feel like we need *at all costs* to get rid of the negative feelings.

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WHY DO WE GET STUCK?

This is where the doing mode becomes the “**driven-doing**” mode:

Driven-doing is the mode of mind where we feel we just cannot let go of trying to get what we want or get rid of what we don't want.

Ruminative worry is just one form of driven-doing—the mind *redoubling its efforts* to apply the power of the doing mode to a problem to which it is tragically unsuited.

Rumination turns to the doing mode to “fix” sadness and unhappiness because this pattern of mind really does work very well when we have to get things done in the external world.

But when it comes to fixing what is going on in our internal worlds, in what we see as “me,” rumination and doing backfire disastrously.

So what can we do instead?

There are two key steps to responding more skillfully:

1. Learning to recognize ruminative worry and driven-doing as they arise in our moment-by-moment experience and to see them for what they are.
2. Cultivating an alternative mode of mind that allows us to respond more skillfully to sadness, unhappiness, and other unpleasant emotions and unwanted inner experiences.

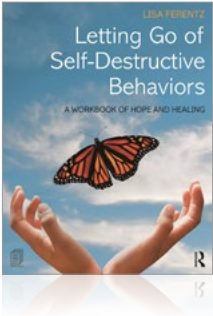
In Chapter 3 we will introduce that alternative mode and describe where and how mindfulness fits into the overall scheme of things.

6

CONNECTING TO OTHERS AND MANAGING YOUR EMOTIONS

*From Letting Go of Self-Destructive
Behaviors: A Workbook of Hope
and Healing*

6 :: CONNECTING TO OTHERS AND MANAGING YOUR EMOTIONS



The following is excerpted from *Letting Go of Self-Destructive Behaviors* by Lisa Ferentz. © Taylor & Francis Group, 2015. All rights reserved.

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One of the things that will really help you work through your self-destructive behaviors is to understand the role of attachment and its impact on your ability to handle your emotional states. Seeking out and maintaining safe, healthy attachment is a universal need, and a part of our biology as human beings. Attaching to people you love lets you feel protected, safe, physically comforted and soothed, and connected to others.

When you come into the world your first task is to successfully attach to the primary people who are taking care of you. You have to attach in order to survive because you are completely helpless and dependent! If your family is loving, predictable, safe, and emotionally available, then attaching is easy to do. As an infant you were born with a very limited number of resources for self-soothing. You had a sucking reflex, you could look away when something startled or upset you, and you could “zone out” or dissociate to tune out experiences that distressed your fragile system. Everything else that you learned to do was supposed to come from the soothing you got from your caretakers.

When you were upset as an infant you cried as a way to reach out to others, and you were appropriately communicating your desire for physical and emotional comfort and connection. If you were spoken to in a soothing singsong voice, rocked, stroked, and gazed at lovingly, your body relaxed and you felt comforted and calm. If your caretakers responded in these loving and soothing ways they were supporting your legitimate needs and teaching you to trust in the fact that others would be available to you when you needed them.

Therefore, the attachment pattern that got established for you was crucial, since it directly connected to your future ability to self-soothe and to manage your emotional states. If there was secure attachment, you looked for soothing and learned to trust that you would get what you needed. If you consistently received comfort, in time, you mastered the ability to use your own internal resources for self-soothing when

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your caretakers were temporarily unavailable. Over time, the positive “learned” experiences of comforting got imprinted on your body and you began to regulate yourself in ways that mirrored what your loving caretakers were doing. If you were allowed to develop healthy ways to self-soothe, it means that as you faced the future stressors of adolescence and adulthood you would be able to cope effectively and not feel the need to turn to unhealthy strategies such as cutting, bingeing, or drinking.

But if you *do* rely on these behaviors to cope and self-soothe, there is the possibility that you didn’t get the consistent, secure attachment and comforting responses you needed and deserved. This was not your fault. You had no control over emotionally unavailable caretakers who may have entered into parenthood with their own unresolved trauma, addictions, significant mental illness, a chronic medical condition, intense family stressors, undiagnosed and untreated anxiety or depression. Sadly, many primary caretakers are emotionally unavailable or easily triggered by a child’s need for close attachment. In these situations, when the child cries out for comfort, the caretakers will go into a fight/flight or freeze response because they are threatened or overwhelmed by the sound of crying. This means the infant’s need for soothing will instead be met with parental aggression, avoidance, withdrawal, or spacing out.

If your caretakers were unavailable, inconsistent, abusive, easily triggered, dysfunctional, or violent, then your necessary task of attaching was complicated and compromised. You may have attempted to use crying, charming smiles, reaching out gestures, cooing and making other sounds to engage your disinterested or unresponsive caretakers. *Their lack of response was not because you were unlovable or undeserving.*

You may have gotten the message that you were “emotionally needy,” “high maintenance,” “overly sensitive,” “demanding,” or “selfish.” The reality is parents often give children these labels when *they* are unable to meet a child’s legitimate emotional needs. It’s a way for parents to take the focus off of their own shortcomings. They put the blame on the child instead. But it is the *adult’s* obligation and responsibility to create a secure attachment, not the child’s. If your caretaker was neglectful, abusive, or non-protective, the impact was the same: you felt unworthy of love and attention. Over time, this repeated “childhood propaganda,” or untrue message, becomes accepted by you as a core truth. It may be one of the reasons why you stopped showing emotions and lost the ability to effectively communicate your needs to others. It also deprived you of the ability to learn how to do self-protection and self-care.

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As you explore the possible connection between using self-destructive behaviors to manage overwhelming feelings and not having secure attachments in childhood, take a moment to revisit memories of how your caretakers responded when you were emotionally upset. In the journal exercise below, answer the questions from whatever memories you have, along with whatever information you've been given from siblings and other relatives about the caretaking you received growing up. You can focus on your caretaker's words, expressed feelings, body language, and behavioral responses. The purpose of this is not to keep blaming your parents. Rather, it is to help you make sense out of why you still find it hard to manage your different emotions. When you answer these questions, know that whatever thoughts or feelings come up for you are normal and okay. Take the time you need to either breathe and re-group after finishing the statements, or to stop mid-way through if you start to feel too uncomfortable. Remember to refer back to the earlier suggestions about self-soothing and re-grounding.

JOURNAL EXERCISE: EXPLORING CARETAKER RESPONSES TO YOUR EMOTIONAL NEEDS

1) When I felt angry as a child, the typical response from my caretakers was:

2) If I became sad and started to cry, the typical response from my caretakers was:

3) If I needed encouragement or support for something that was challenging in my life, the typical response from my caretakers was:

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4) When I was frightened and needed reassurance about my safety or wellbeing, the typical response from my caretakers was:

5) If I physically got hurt, the typical response from my caretakers was:

6) If I made a mistake of any kind, the typical response from my caretakers was:

Take a few moments and let your answers sink in. This may be the first time you've considered a possible connection between your caretakers' responses to your physical, emotional, and psychological needs and the extent to which you do or don't know how to engage in consistent self-care. If you grew up in a dysfunctional family, it is likely that legitimate emotional and physical needs continued to be downplayed or ignored, and you learned to feel a sense of shame for wanting anything. Even if your caretakers were not directly abusive, you might have been traumatized if they acted like non-protective bystanders. This means that although they did not directly harm you, their inability or unwillingness to shield you or get you away from someone harmful became a form of abuse and caretaker betrayal in and of itself.

Even as you continued to grow up, you needed to find ways to attach to unavailable caretakers. It was probably impossible for you to think of your parents as toxic, unloving, or bad: that reality would make it too difficult to attach and bond with them. Instead, you could unconsciously "make sense" out of your parents' neglectful or abusive actions by taking ownership of the poor parenting, telling yourself you were not loved because there was something wrong with *you*. When you tell yourself "my parent isn't bad, I am bad," you can still successfully attach to that parent. Although

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this may have been necessary to believe at the time, self-blame may be at the core of the negative thoughts that haunt you and, in your mind, make it okay to use self-punishing behaviors.

Attachment patterns can be complicated in families. Obviously, the ideal kind of attachment is a “secure” one: caretakers are consistent, reliable, willing and able to bond, protect, and nurture. Other styles of attachment can be confusing. Sometimes a caretaker is available and responds appropriately and at other times their response to the same situation is totally inappropriate. Oftentimes, the people who are supposed to be the most supportive are the ones who are the scariest or they, themselves, are frightened and become helpless. Keep in mind that you can only learn what you witness and experience. So a lack of secure attachment and comfort will profoundly affect whether or not you develop your own tools for soothing later in life.

In addition, if you were in survival mode, you took on the behaviors, body postures, and movements that were most adaptive in your family. This means you acted, moved, and spoke in ways that kept you safe and were considered “acceptable” by caretakers. Often this manifests in collapsed or closed off posture, avoiding direct eye contact with others, and using a voice that is passive, not assertive. Although these responses kept you safe, you couldn’t gain mastery over healthier ways to communicate or advocate for yourself. Your body language communicated a shamed self, and gave others information about how you expected to be treated in relationships. In the long term, this can create a tragic self-fulfilling prophecy, as predators seem to have radar for sensing more vulnerable children and adults, and are able to exploit and revictimize them without consequences.

As you read this, keep in mind that most people do not come into the therapy process or approach this work with an awareness of how early childhood attachment and relationship dynamics profoundly impact current behaviors and a core sense of self. These are realizations that unfold, in time, with the help of a trained professional.

Fifty-year-old Brian spent much of his childhood in foster care and was repeatedly abused and neglected. Early in treatment he said:

Ok, I get that I had a really horrible childhood, and I even understand that I have the right to be angry about it. But that doesn’t explain my addiction to porn and why I keep winding up with women who hurt me and cheat on me. I feel like despite my childhood, I am responsible for the mess I am making of my adult life. Now there’s no one to blame but myself.

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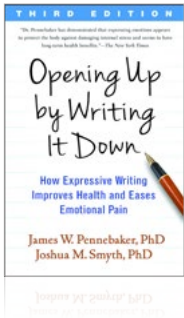
Despite the sad reality of the impact of insecure and disorganized attachments, the exciting news is that when you are able to reach out to other people and accept the safe attempts they make to connect with you (including with a healthy partner, close friend, or therapist) it can help to repair those earlier attachment losses. As you continue to find the courage to reach out and the curiosity to pursue safe and trusting relationships, you will also be able to begin experimenting with healthier self-soothing strategies. Learning how to trust other people allows you to tap into your own inner wisdom and that's when true healing can begin!

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THE INVENTION OF THE EXPRESSIVE WRITING APPROACH

*From Opening Up by Writing It Down: How
Expressive Writing Improves Health and
Eases Emotional Pain*

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The following is excerpted from *Opening Up By Writing It Down* by James W. Pennebaker and Joshua M. Smyth. © The Guilford Press, 2016. All rights reserved.

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It would be so compelling to tell the story of how the first studies on expressive writing grew out of our traumatic experiences in our childhoods and how we independently discovered the healing power of writing on our own. It would be gripping, initially heartbreaking, and ultimately redemptive. But also false. The expressive writing method was actually the result of a series of serendipitous research findings. Okay, maybe not gripping, but still an interesting story.

THE CASE OF TRAUMATIC SEXUAL EXPERIENCES

Early in his career, Jamie and his students were putting together a questionnaire on health issues. The idea was to break out of the traditional way of thinking and simply ask a large group of students a broad range of questions about their lives. In putting together the questions, the group decided to ask about people's childhoods, their favorite foods, maybe even color preferences. One member of Jamie's research team suggested that they include an item on traumatic sexual experiences in childhood. There was no specific reason for including the question—but it was a question no one appeared to have asked before, and it made intuitive sense that such experiences might be important. So, toward the end of the 12-page questionnaire, they added a question that very few researchers ever ask:

"Prior to the age of 17, did you have a traumatic sexual experience (e.g., rape, being molested)? Yes_____ No_____"

Of the 800 college women who later completed the survey, about 10 percent answered in the affirmative. Overall, the women who reported traumatic sexual experiences in childhood did not differ from others in terms of age, social class, race, or even number of close friends. Most striking, however, was that those who reported a sexual trauma evidenced more health problems than any other group we had ever seen.

Soon afterward, a writer for the magazine *Psychology Today*—one of the most popular magazines of the early 1980s—was able to get 24,000 adults to complete a health survey that included the traumatic sexual experience question. Overall, 22 percent of the women and 10 percent of the men reported having a childhood traumatic sexual experience. These rates roughly corresponded with those found in numerous national polls on the topic.

Even though the reported sexual trauma had occurred almost 20 years earlier, it was associated with large increases in ulcers, the flu, heart problems, cancer diagnoses, and virtually every other category of health problem. In fact, those who reported a

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traumatic sexual experience as a child had been hospitalized *nearly twice as often* as those who did not report such traumas.

On the *Psychology Today* questionnaire, respondents were asked to include their name and telephone number for possible future telephone interviews. Fifteen people who claimed to have experienced a sexual trauma were called by Carin Rubenstein, the author of the magazine piece. In her article, she writes:

One woman was raped at 16; another was a victim of incest at 8; yet another had been fondled at the age of 5 by a man selling ponies. A 51-year-old woman from Los Angeles told me that she had been raped, at 5, by her neighbor, who was a friend of the family . . . "I never told anyone about it. You're the first," she said. Later on, not making the connection, she remarked, "I've always had health problems with organs in that area . . . since I was 5." (p. 34)

Every person with whom Rubenstein talked reported an experience that all of us would agree was traumatic. In addition, the majority had not discussed this traumatic event with anyone when it had occurred. If they eventually did discuss their trauma, it was not until many months or years later.

What makes sexual traumas so devastating?

It is clear that childhood sexual traumas influence long-term health. However, changes in health following the traumas may not reflect sexuality per se. Rather, traumas may be insidious because people often cannot talk about them. They must actively inhibit their wanting to discuss these intensely important personal experiences with others.

Later surveys from thousands of people—both students and nonstudents—supported this. Having nearly any kind of traumatic experience is bad for your health. However, if you keep the trauma secret, it increases the odds that you will have health problems. Not surprising, of all the traumas we have studied (death of a family member, victim of violence, moving, failure, personal losses, etc.), people are typically least likely to talk about a sexual trauma.

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EXPRESSIVE WRITING AND ILLNESS PREVENTION

If secrets are so bad for us, would talking to others bring about benefits to our health? In the mid-1980s, psychotherapists began providing the first solid research evidence that therapy was good for both mental and physical health. In fact, there had been a couple of largely overlooked insurance studies showing that when insurance companies started coverage for psychotherapy, the extent of and costs associated with physical health care dropped.

What if we set up an experiment where we had people come into the lab and talk to someone about their secret traumatic experiences? You can immediately see the problems with such a study. Where would we find people who would be willing to come in and talk to some stranger about their darkest secrets? Even if we found them, would they really be willing to come to a lab for this? And who should they talk to? How should the people listening to the traumas react? This was too complicated.

And it was about this time that Jamie recalled an experience of his own that had happened eight years earlier. About three years after their wedding, he and his wife were dealing with some formidable issues in their marriage. For the first time in his life, he was despondent, even depressed. Even though he was a graduate student in psychology, he never considered going to a therapist. Instead, after a couple of weeks, he started writing. He wrote about their relationship, his career, his childhood, basically everything that was important to him. In almost no time, the clouds parted. He realized how central his wife was to his very existence.

Recalling this experience, Jamie realized that he could have people write about upheavals in their lives rather than talk to others. Plus, writing would be much simpler to do in an experimental setting.

And so the expressive writing paradigm was born.

THE ORIGIN OF EXPRESSIVE WRITING

Together with a new graduate student, Sandra Beall, Jamie outlined the following study: The plan was to recruit a group of college students to write about either traumatic experiences or superficial topics. With the students' permission, the student health center would release the number of illness visits each student made in the months before versus after the experiment.

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On how many occasions should people write? How long should each writing session last? There was no blueprint for this. Because only a certain number of rooms were available between 5:00 and 10:00 p.m. for four consecutive days, the arithmetic was easy. Jamie and Sandy could run the required number of students if each person wrote for 15 minutes on each of four days. (There is an irony here. People often ask why expressive writing is typically designed to be done for 15 minutes on four days. The answer is that the first study arbitrarily used this approach and it worked, and this approach has been routinely copied since that time.)

On the day of the experiment, students came into a small office. After the study was described and students gave their consent, those assigned to write about their thoughts and feelings about a trauma were told the following:

“Once you are escorted into the writing cubicle and the door is closed, I want you to write continuously about the most upsetting or traumatic experience of your entire life. Don’t worry about grammar, spelling, or sentence structure. In your writing, I want you to discuss your deepest thoughts and feelings about the experience. You can write about anything you want. But whatever you choose, it should be something that has affected you very deeply. Ideally, it should be something you have not talked about with others in detail. It is critical, however, that you let yourself go and touch those deepest emotions and thoughts that you have. In other words, write about what happened and how you felt about it, and how you feel about it now. Finally, you can write on different traumas during each session or the same one over the entire study. Your choice of trauma for each session is entirely up to you.”

Those in the comparison or control group were asked to write about superficial or irrelevant topics during each session. For example, on different days they were asked to describe in detail such things as their dorm room or the shoes they were wearing. The two groups were in the same location, interacting with the same experimenters, and engaging in the activity of writing for the same amount of time; what differed was the content of writing—one group wrote about their deepest thoughts and feelings, and the comparison group wrote about emotionally neutral (and likely quite uninteresting) topics. Thus, the purpose of the control group was to evaluate what effect writing in an experiment per se had on health changes, independent of what was believed to be the important contribution of the content of the writing. Any

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differences between the two groups should, therefore, be due to the *content* of the writing, not any aspects of their participation in the study.

For the students, the immediate impact of the study was far more powerful than we had ever imagined. Several of the students cried while writing about traumas. Many reported dreaming or continually thinking about their writing topics over the four days of the study. Most telling, however, were the actual writing samples. Essay after essay revealed people's deepest feelings and most intimate sides. Many of the stories depicted profound human tragedies.

One student recounted how his father took him into the backyard on a hot summer night and coolly announced his plans to divorce and move to another town. Although the student was only nine years old at the time, he vividly remembers his father's voice: "Son, the problem with me and your mother was having kids in the first place. Things haven't been the same since you and your sister's birth."

On all four days of the experiment, one woman detailed how, at age 10, her mother asked her to pick up her toys because her grandmother was visiting that evening. She didn't pick up her toys. That night, her grandmother arrived, slipped on one of the toys, and broke her hip. The grandmother died a week later during hip surgery. Now, eight years later, the woman still blamed herself every day.

Another woman described being seduced by her grandfather when she was 13. She depicted the terrible conflict she experienced. On one hand she admitted the physical pleasure of his touching her and the love she felt for her grandfather. On the other, she suffered with the knowledge that this was wrong, that he was betraying her trust.

Other essays disclosed the torture of a woman not able to tell her parents about her being a lesbian, a young man's feelings of loss about the death of his dog, or the anger about parents' divorces. Family abuse, alcoholism, suicide attempts, and public humiliation were also frequent topics.

That a group of college students had experienced so many horrors and, at the same time, had so readily revealed them was remarkable. The grim irony is that, by and large, these were 18-year-old kids attending an upper-middle-class college with above-average high school grades and good College Board scores. These were the people who were portrayed as growing up in the bubble of financial security and suburban tranquility. What must it portend for those brought up in more hostile environments?

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The results of the study were fascinating, but also a bit unexpected. Compared to people in the control group, we found that people who wrote about traumatic experience evidenced:

- **Immediate increases in feelings of sadness and anxiety after writing.** Students likened it to the feelings that they had after watching a sad movie. Writing about emotional topics does not produce some kind of immediate release or euphoria.
- **Long-term drops in visits to the student health center for illness.** Those who wrote about emotional upheavals had half the number of illness-related visits to the health center in the six months after the study than people in the control condition.
- **Greater sense of value and meaning as a result of writing.** Not only did people express this in questionnaires afterward, but students would sometimes stop Jamie on campus and thank him for letting them be in the experiment.

The overall pattern of results was exciting. But for every question that the experiment had answered, a dozen more questions appeared.

Perhaps the most basic issue concerned the trustworthiness of these findings. Were the effects real? Does writing about traumas really affect physical health? Perhaps the experiment had just affected people's decisions to visit the student health center. Or even worse, maybe the findings were simply due to chance. Every now and then, for example, you can toss a coin ten times and come up with heads every time.

Additional studies needed to be conducted.

FREEWITING

As a useful practice exercise, and one that can enhance creativity and foster your capacity for expression, find a quiet time and place to practice writing. For this exercise, write whatever comes into your mind for 10 to 20 minutes. Try to write the entire time without stopping. Don't worry about style or grammar; the important thing is to keep writing continuously for the entire session. Just let yourself write, a sort of limbering-up exercise. We will return to more structured expressive writing later in the book.

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EXPLORING THE IMMUNE SYSTEM: WRITING ABOUT TRAUMAS IS BETTER THAN WE THOUGHT

Soon after the first expressive writing study was submitted, Jamie teamed up with Janice Kiecolt-Glaser, a clinical psychologist, and her husband, Ronald Glaser, an immunologist, both with the Ohio State University College of Medicine. In the mid-1980s, they were leaders of a new field called psychoneuroimmunology—the mind–body exploration of how mental states and strong emotions might influence the immune system. Together they were blazing a trail by showing that overwhelming experiences such as divorce, major exams in college, and even strong feelings of loneliness adversely affected immune function. They had recently published an article showing that relaxation therapy among the elderly could improve the action of the immune system.

The work by Jan and Ron was groundbreaking because it relied on techniques that directly measured the action of T-lymphocytes, natural killer cells, and other immune markers in the blood. It made good sense for Jan, Ron, and Jamie to work together—so they set out to see if expressive writing could directly influence these direct measures of how the immune system was functioning.

The experiment that they designed together was similar to the first confession study. Fifty students wrote for 20 minutes a day for four consecutive days about one of two topics. Half wrote about their deepest thoughts and feelings concerning a trauma. The remaining 25 students were expected to write about superficial topics. The major difference was that all the students consented to have their blood drawn the day before writing, after the last writing session, and again six weeks later.

As before, the experimental volunteers poured out their hearts in their writing. The tragedies they disclosed were comparable to those in the first experiment. Instances of rape, child abuse, suicide attempts, death, and intense family conflict were common. Again, those who wrote about traumas initially reported feeling sadder and more upset each day of writing, relative to those who wrote about superficial topics.

Collecting the blood and measuring immune function was a novel experience that added to the frenzy. As soon as the blood was drawn, it was driven to the airport to make the last flight to Jan and Ron's lab in Columbus, Ohio. Once the blood samples arrived, the people in the immunology lab worked around the clock in an assemblyline manner. The procedure involved separating the blood cells and placing a predetermined number of white cells in small petri dishes. Each dish contained differing amounts of various foreign substances, called mitogens. The dishes

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were then incubated for two days to allow the white blood cells time to divide and proliferate in the presence of the mitogens.

In the body, there are a number of different kinds of white cells, or lymphocytes, that serve a regulatory function in the immune system. The cells help govern and coordinate aspects of our immune responses. T-lymphocytes, for example, can stimulate other lymphocytes to make antibodies. Antibodies, along with parts of the body's defense system, can identify and kill bacteria and viruses foreign to the body. These aspects of the immune system help keep us healthy. The immune measures that were used simulated this bodily process in the dishes. Just as viruses and bacteria can stimulate the growth of T-lymphocytes in the body, the mitogens did the same in the laboratory dishes. If the lymphocytes divide at a fast rate in response to the mitogens, we can infer that at least part of the immune system is working quickly and efficiently.

What were the findings? People who wrote about their deepest thoughts and feelings surrounding traumatic experiences evidenced enhanced immune function compared with those who wrote about superficial topics. Although this effect was most pronounced after the last day of writing, it tended to persist six weeks after the study. In addition, it was again observed that health center visits for illness dropped for the people who wrote about traumas compared to those who wrote on the trivial topics.

There were now two experiments that showed similar patterns. Taken together, the studies indicated that writing about traumatic experiences could be beneficial. The effects were not due to simple catharsis or the venting of pent-up emotions. In fact, the people who just blew off steam by venting their feelings without any thoughtful analysis tended to fare worse. Further, both experiments indicated that writing about feelings associated with traumatic experiences was painful in the short term. Virtually no one felt excited, on top of the world, or cheerful immediately after writing about the worst experiences of their lives.

In the surveys sent out several months after the experiments, people were asked to describe what long-term effects, if any, the writing experiment had on them. In sharp contrast to the reports immediately after writing, nearly everyone who wrote about traumas now described the study in positive terms. More important, approximately 80 percent explained the value of the study in terms of insight. Rather than explaining that it felt good to get negative emotions off their chests, the respondents noted how they understood themselves better. Some examples:

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It helped me think about what I felt during those times. I never realized how it affected me before.

I had to think and resolve past experiences. . . . One result of the experiment is peace of mind, and a method to relieve emotional experiences. To have to write emotions and feelings helped me understand how I felt and why.

Although I have not talked with anyone about what I wrote, I was finally able to deal with it, work through the pain instead of trying to block it out. Now it doesn't hurt to think about it.

The observations of these people and most others who participated in these early studies are almost breathtaking. They tell us that our own thought and emotional processes can help us heal.

BEYOND HEALTH: WRITING AND OCCUPATIONAL SURVIVAL

The early studies were just the beginning of a research odyssey that has taken the expressive writing literature in several directions. Soon after the results of the immune study were published, Stefanie Spera called. Stefanie was a psychologist with an outplacement company in Dallas. An outplacement company typically works with large corporations in the midst of “downsizing”—a polite way of saying the company was laying off a significant number of employees. The outplacement company offers a variety of services to those who have been laid off, including providing office space, secretarial support, and jobhunting skills.

Stefanie called because a large computer company had laid off about 100 senior engineers four months earlier and not one of them had found a new job. She was curious to know if expressive writing could help speed up these engineers getting new jobs.

Over the next few weeks, a sense of how the layoff had occurred started to emerge. The corporation had never had to lay people off in its history. On a Wednesday morning in January, about 100 people, averaging 52 years of age, were individually called into their supervisor's office and informed that they were being terminated with no possibility of being rehired. The employees, most of whom had been with the company since graduating from college almost 30 years earlier, were then escorted to their workspace by a security guard who watched them clean out their desks. They

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were then taken to the front door, relieved of their keys and security badges, and bid farewell. No forewarning, no retirement watches.

Six months later, an expressive writing study was under way with almost 50 people. Even though they were a rather embittered and hostile group, they were desperate to try anything that might increase their odds of finding another job.

The basic study was quite simple. Half were asked to write about their deepest thoughts and feelings about getting laid off for 30 minutes a day for five consecutive days. The other half wrote for the same period about how they used their time—a strategy based on “time management” (time management was all the rage at that time in the corporate world, despite little if any actual support for such a technique being helpful). A third group of 22 former employees did not write at all and served as another comparison group.

As with our other studies, those who were asked to write about their thoughts and feelings were extremely open and honest in their writing. Their essays described the humiliation and outrage of losing their jobs as well as more intimate themes—marital problems, illness and death, money concerns, and fears about the future.

The potency of the study was surprising. Within three months, 27 percent of the experimental participants landed jobs compared with less than 5 percent of those in the time management and no-writing comparison groups. By seven months after writing, 53 percent of those who wrote about their thoughts and feelings had jobs compared with only 18 percent of the people in the other conditions. Particularly striking about the study was that the participants in all three conditions had all gone on exactly the same number of job interviews. The only difference was that those who had written about their feelings were offered jobs.

Why did writing about getting laid off help these people find jobs more quickly? The key probably has something to do with the nature of anger. Those who had explored their thoughts and feelings were more likely to have come to terms with their extreme hostility toward their previous employer. Recall that these former employees felt betrayed by their company. Even during the initial interviews, it was difficult to stop them from venting their anger. In all likelihood when most of them went on interviews for new jobs, many would let down their guard and talk about how they were treated unfairly and lash out at their former employer—perhaps quite inappropriately so. Those who had written about their thoughts and feelings, on the other hand, were perhaps more likely to have come to terms with getting laid off and, in the interview, came across as less hostile, more promising job candidates.

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DOES WRITING WORK?: THE FIRST ROUND OF META-ANALYSES

The first expressive writing study was published in 1986, and the layoff study came out in 1993. Other labs were now starting to conduct and publish writing studies. Most of the studies worked, but some didn't.

At Stony Brook University, a lab headed by Arthur Stone was beginning to run some interesting writing studies. Arthur was a scientist known for having a critical mind and was keenly capable of finding the flaws and limitations of psychology projects. Unfortunately (or, as it turned out, fortunately), his skeptical eye soon was locked on the expressive writing research. Several of his students were interested in expressive writing work, and one of these was Josh.

At this point, over a dozen studies had been conducted and published in the scientific literature. Josh reasoned that this would be an opportunity to apply a statistical method known as meta-analysis to the expressive writing studies. Put simply, a meta-analysis allows us to examine multiple studies in a cumulative fashion, attempting to find out what the overall message (or finding) is from all the studies collectively. By doing this, we can begin to get a more precise estimate of an effect—in this case, to determine if there was strong evidence that expressive writing was helpful.

In other words, this method could tell us whether or not expressive writing was leading to health improvements relative to writing about emotionally neutral topics. Such an approach can address other important questions as well. Are there particular outcomes that appear to show greater or lesser benefit from writing? For example, does writing work better for physical health outcomes or for depression?

In many ways, Josh was well suited to this task. He had adopted his adviser's skepticism but, at the same time, was not wedded to any particular outcome. Josh had another interest—ways to measure hard, objective outcomes. By way of background, social science has a reputation for relying on people's self-reports, which are considered soft (or not related to anything important) in scientific parlance. Perhaps disclosing deep thoughts and feelings through writing was leading people to overestimate their health in their reports—maybe they felt some emotional connection to the researchers after this powerful disclosure process and were trying to help the researchers out. By examining different types of outcomes, Josh could look at a wider array of objectively measured outcomes such as those measuring immune function. After combing through the scientific articles and selecting the dozen or so best studies, Josh applied the meta-analytic methods. Several promising

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findings emerged. Most important, people who wrote about their deepest thoughts and feelings related to stressful or traumatic experiences had reliable improvements in health in the two to three months after writing. Although there were also improvements in people's self-reports of their health, there were equally large effects on people's physiological functioning.

There were some unexpected findings as well. The results of questionnaires that asked about health behaviors—such as healthy eating, exercising, taking medication, and the like—were not influenced by writing. Although some had suggested that writing may be beneficial as a result of better self-care activities, this explanation was not supported by Josh's analysis. Finally, he found that writing reliably but temporarily increased people's feelings of distress. Interestingly, the degree to which people felt distressed was unrelated to subsequent long-term mental or physical health changes. If you were thinking that a "no pain, no gain" explanation could account for the value of writing, it is not that simple. Even though most people felt somewhat distressed by writing, it turns out that suffering more in your writing doesn't lead to more improvements later.

One other observation was critically important: All of the early writing studies relied on people who were physically healthy. If this method is good for people's health, he asked, why haven't any researchers looked at people suffering from chronic disease?

When Josh's meta-analysis was published in the *Journal of Consulting and Clinical Psychology*, in 1998, it had an immediate impact. Researchers around the world realized that there might be something to this expressive writing and began conducting an array of innovative and interesting studies. Within the next several years, a surge of study findings were published that included wildly broad and diverse samples—people with a variety of acute and chronic disease, with major and minor mental health problems. Other studies employed people who were quite healthy but who were trying to master new skills, do better in college, or exhibit greater creativity.

The net effect of Josh's meta-analysis is that it demonstrated the potential value of expressive writing. His paper, however, challenged researchers at the time to explain why it worked. Clearly, when people wrote about emotional upheavals, something important was happening. But what? What precisely happens when people are given the opportunity to disclose their secrets and emotions to others?

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TRY EXPRESSIVE WRITING

Find a quiet time and place for this next writing exercise. Write for 20 to 30 minutes, focusing on your deepest emotions and thoughts about a stressful or upsetting experience in your life. Whatever you choose to write about, it is critical that you really let go and explore your very deepest emotions and thoughts. Write continuously, and don't worry about spelling, grammar, or style.

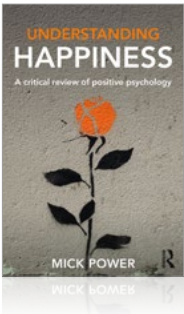
Warning: Many people report that after writing, they sometimes feel somewhat sad, although this typically goes away in a couple of hours. If you find that you are getting extremely upset about a writing topic, simply stop writing or change topics.

8

HAPPINESS: AN OVERVIEW

From Understanding Happiness: A critical review of positive psychology

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The word 'happiness' should be retired because it's so ambiguous.

(Daniel Kahneman, 2012)

INTRODUCTION

The world's greatest living psychologist (at least according to his friend Steven Pinker) and only living psychologist to have won the Nobel Prize for economics, Daniel Kahneman, is on record to have declared himself a pessimist (Kahneman, 2011). However, many people might still be puzzled at Kahneman's declaration that 'happiness', at least as a word, should be abandoned and replaced with something more useful given how much time, effort and money people spend on the pursuit of happiness. Perhaps, you might think, that is taking pessimism just a bit too far. But let us consider a few facts and figures.

In relation to tourism and happiness, in 2013 the World Tourism Organization (see www.unwto.org) estimated that there were over 1 billion international travellers worldwide who spent over US\$1.4 trillion in order to travel. In relation to alcohol and happiness, in the UK alone the expenditure on alcohol has been estimated for 2012 at approximately £38 billion (Institute of Alcohol Studies, www.ias.org.uk). Similarly, in the week leading up to Christmas 2011 UK shoppers had spent £8 billion in shops by Christmas Eve at a rate of an estimated £2.5 million every minute, with the total expenditure for Christmas in the UK coming in at a staggering £69.1 billion (*Daily Mirror*, 19 December 2011, accessed at www.mirror.co.uk). There are bucketloads of such figures that could be rolled out, but they all lead to the one question: What is it that we are trying to buy? Haven't we been told that money can't buy love? That money can't buy happiness? So why do we seem to behave as if the opposite were true?

To return to the world's greatest living psychologist, what is even more puzzling about Daniel Kahneman is that as a self-confessed pessimist he moved from Israel to work in the United States. If there is one country in the world that has elevated the pursuit of happiness to a major cultural preoccupation, then it has to be the US. Enshrined in the American Declaration of Independence are the immortal words:

We hold these truths to be self-evident – that all men are created equal; that they are endowed by their Creator with certain inalienable rights, that among these are life, liberty, and the pursuit of happiness.

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Those of us who are a little sceptical of such high ideals might suggest that Americans have been more preoccupied with the pursuit of wealth than the pursuit of happiness, given that more than half of the world's billionaires live in the US (Wilkinson and Pickett, 2010). However, a more constructive response might be to point to the development of the positive psychology movement in the US and to argue that surely this movement follows in the great tradition begun in the Declaration of Independence. The founder of this movement, the psychologist Martin Seligman, describes its origins to have taken place in his back garden when his 5-year-old daughter Nikki asked him why he was always so grouchy. As Seligman writes:

Nikki ... was throwing weeds into the air and dancing and singing. Since she was distracting me, I yelled at her, and she walked away. Within a few minutes she was back, saying, 'Daddy, I want to talk to you.'

'Yes, Nikki?'

'Daddy, do you remember before my fifth birthday? From when I was three until when I was five, I was a whiner. I whined every day. On my fifth birthday I decided I wasn't going to whine anymore. That was the hardest thing I've ever done. And if I can stop whining, you can stop being such a grouch.'

This was an epiphany for me. In terms of my own life, Nikki hit the nail right on the head. I was a grouch.

(Seligman, 2002, p. 28)

This insightful question from a wise 5-year-old seems to have led to a mid-life crisis in which the inventor of Learned Helplessness, a state that would surely make anyone a grouchy old man, rediscovered his inner positive self and then wrote, as it states on the front cover, *The New York Times* Bestseller *Authentic Happiness* (2002), and a whole truckload of similar books besides.

One crucial point that we must make about the positive psychology movement (we will return to it frequently in later chapters) is that surely it must be annoying to older generations to see some of their ideas repackaged and recycled? Wasn't that a positive psychology movement back in the 1950s when the great (and, of course, subsequently very rich) Norman Vincent Peale wrote classics such as *The Power of Positive Thinking*, *The Power of Positive Living*, *The Amazing Results of Positive Thinking*, *The Power of Positive Thinking for Young People*, and, my favourite title of all, *Stay Alive*

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All Your Life? I guess you begin to get the idea. Anyway, Norman's conquest of positive thinking includes examples that sound just like Seligman in his back garden:

Altogether too many people are defeated by the everyday problems of life. They go struggling, perhaps even whining, through their days with a sense of dull resentment at what they consider the 'bad breaks' life has given them. ... By learning how to cast them from the mind, by refusing to become mentally subservient to them, and by channelling spiritual power through your thoughts, you can rise above obstacles which ordinarily might defeat you.

(Peale, 1953, pp. vii–viii)

On the basis of this continual pursuit of happiness and all things positive, you might naively assume that Americans should come top of the happiness league tables that are now generated from large-scale surveys of how people feel. Absolutely not! An extremely insightful and highly recommended book by Richard Wilkinson and Kate Pickett, *The Spirit Level: Why Equality is Better For Everyone* (2010), includes a wealth of charts and figures demonstrating why the US, of all the developed countries, typically comes bottom on almost all indicators that are relevant to health, well-being and quality of life. Figure 1.1 presents the case, showing that of all the developed nations, the US has the greatest income inequality, which in turn is linked to a variety of negative indicators such as the Index of Health and Social Problems.

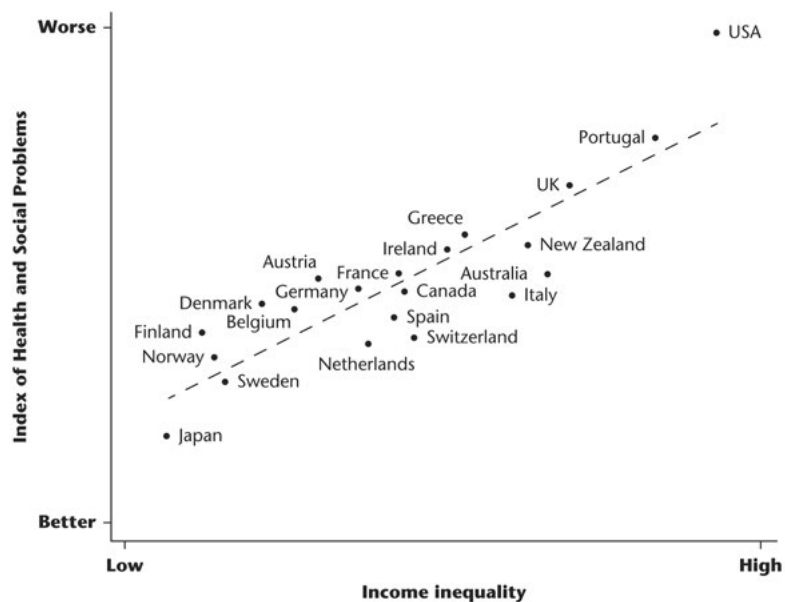


FIGURE 1.1 Income inequality and the Index of Health and Social Problems.
Source: Wilkinson and Pickett (2010). Reproduced with permission.

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In order to understand the consequences of such an unequal society whilst being bombarded by positive thinking and the pursuit of happiness, Wilkinson and Pickett point to the work of the US psychologist Jean Twenge, who, in publications such as *Generation Me* (2006), has identified some worrying longitudinal trends in the US. In a summary of studies carried out between the 1950s and the 1990s, Twenge found that there has been a continuous upward trend in the levels of anxiety over those 40 years for both men and women in the US. Over a similar time period, studies also seem to show that people in the US report increasingly positive levels of self-esteem on standardised measures of esteem. The paradox therefore seems to be that Americans are becoming both more anxious and more positive about themselves at the same time, which seems to present a puzzling scenario. In explanation, Twenge has argued that high self-esteem can come in two varieties: the first is a genuine healthy style that is open to experience and to feedback from others; in contrast, the second is a type of defensive egotism or narcissism that is not open to experience or to feedback from others, but which provides a defence against social-evaluative threats. We will examine this defensive self-esteem and a number of other similar problems in detail in Chapter 2. However, it is important to point out the possible links to the 'Have a nice day – Have a nice life' think positive movement with which the US is currently preoccupied.

The problem with all such simplistic philosophies is that they come with a psychological blindness that can put people at risk, whilst leading to the apparent paradoxes such as why we are wealthier but not happier, or why we think more positively but act more negatively. In Chapter 3 we will consider more sophisticated psychological models such as our own SPAARS model (Power and Dalgleish, 2008) and Kahneman's (2011) arguments for two major systems: System 1, which operates largely automatically and outside of awareness, and System 2, which is largely conscious and controlled. However, at this point we will just note the entertaining book by Barbara Ehrenreich *Smile or Die: How Positive Thinking Fooled America & The World* (2009) (published in the US under the title *Bright-Sided*). In a nutshell, Ehrenreich argues that the think positive movement is just one of the many tricks by which the rich enjoy being rich but try to keep the poor, the infirm, the unemployed and the disabled quiet about their situations: 'Just think positive and you too could be President!' However, and just to set the balance straight, we are certainly not arguing that positive psychology is all bad, but that in the popular press unfortunately it has become synonymous with a simplistic happiology industry; in later chapters we will examine many of the plusses, including issues about strengths and virtues, the importance of forgiveness, gratitude and acts of kindness in our interpersonal

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relations, and the assessment and improvement of our quality of life. The issue for us is that the positive psychology movement in its popular presentation appears to throw the baby out with the bathwater. For example, so-called 'negative' emotions, which got such a bad press in Seligman's writings at the beginning of the movement, are essential parts of us and, when used in the appropriate way, also add to our strengths, virtues, and improve the quality of our interpersonal relationships (see Chapter 3).

POPULAR CONCEPTIONS AND MISCONCEPTIONS OF 'HAPPINESS'

One of the puzzles of modern economics is that, despite the genuine increase in the wealth of the developed nations, there has not been an equivalent increase in the happiness of people populating those nations (e.g. Diener and Biswas-Diener, 2008). One interpretation has been that although physical capital has increased, there has been a concomitant decline in social capital, that is, in the quality of the social support and social networks with which we all enrich our lives. Although the reasons for the decline in social capital are likely to be complex (Layard, 2011), we can take one simple example: the impact of television over the past 50 years has been considerable. A study of television's impact in the Kingdom of Bhutan, located to the east of Tibet in the Himalayas, has come up with some dramatic results. The Kingdom of Bhutan has taken a unique approach to the state of its population in that it has introduced an economic population measure known as Gross National Happiness (GNH), which sits alongside other economies' preoccupation with Gross National Product (GNP). In the time since television was introduced into Bhutan in 1999, there has been a dramatic decline in social capital, or GNH, to the extent that the Bhutanese Government is likely to cut down the number of TV channels and the amount of TV coverage that will be available in the future (Layard, 2011; MacDonald, 2003).

One of the problems that economists struggle with is the relationship between objective and subjective indicators of states such as happiness; our emphasis throughout this book is very much that it is not the objective event or situation but the subjective appraisal of an event or situation that is more important in determining the consequent emotional state (see Power and Dalglish, 2008). Although there may be thresholds below which material deprivation and poverty do impact on happiness (Diener, 2003), above these thresholds the impact on happiness and on quality of life is likely to be more subjective or appraisal-based (e.g. Power, 2003), which is why the relationship between health and well-being indicators with income inequality emerges in the wealthy developed nations.

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A second issue that the economic approach to happiness raises is that the type of happiness referred to by economists, that is, the type that might relate to how many zeros there are in your income and how many cars are parked outside your house, is not the same as the brief momentary states of happiness that are the equivalent of states of anxiety, anger or sadness. The term 'happiness' is an umbrella term that needs to be divided into at least two different meanings (e.g. Argyle, 2001): one refers to brief transitory emotions such as joy, amusement or ecstasy, and the other refers to concepts such as meaning in life, self-actualisation, life satisfaction and a mood-like state of continuing contentment (e.g. Layard, 2011). The second type should not be labelled 'happiness', as we will argue throughout this book, because of its confusion with the first type of 'happiness'. Authors' opinions vary as to the extent aspects of these two general categories should be included under each heading, but, allowing for these differences, these two types of happiness have also been referred to as the hedonic and eudaimonic approaches respectively (e.g. Ryan and Deci, 2001), or a related distinction between 'experienced well-being' and 'life satisfaction' by Daniel Kahneman (2011).

Samuel Franklin, in his elegant *The Psychology of Happiness* (2010), also argues that it is the eudaimonic rather than the hedonic by which we should lead our lives, and he suggests, with others, that the 'pursuit of happiness' enshrined in the American Declaration of Independence refers to this virtuous form of happiness. Although the primary focus in this book will be on the more permanent states variously referred to as life satisfaction, contentment, actualisation and so on, nevertheless, some of the discussion will inevitably spill over into considerations of the transitory emotions of happiness when we consider states such as romantic love (see Chapter 2). We are also mindful of the fact that the distinction between hedonic and eudaimonic may be more a theoretical than a practical one. Paul Dolan, for example, has argued in *Happiness by Design: Finding Pleasure and Purpose in Everyday Life* (2014) that we should not see pleasure and purpose as separate but rather as two sides of the activities we engage in and that the best activities may give us a sense of both.

HAPPINESS – TEMPORARY EMOTION VERSUS PERMANENT STATE

So, what is 'happiness'? Attempts to define happiness within the psychology literature generally reflect the breadth of scope that, according to Averill and Moore (2000), is so mocking of analysis. For example, Wessman and Ricks (1966) proposed that happiness:

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Appears as an overall evaluation of a quality of the individuals own experience in the conduct of his vital affairs. As such, happiness represents a conception abstracted from the flux of affective life indicating a decided balance of positive affectivity over long periods of time. (*pp.* 240–241)

Similarly, Veenhoven (1984) suggests that happiness is ‘the degree to which an individual judges the overall quality of his life-as-a-whole favourably’ and is ‘not a simple sum of pleasures, but rather a cognitive construction which the individual puts together from his various experiences’.

It seems clear that conceptualisations of happiness such as these are referring, as we noted above, to a different type of construction that are not the same as brief states of anger, sadness, fear or disgust. Clearly, there are positive emotions that are circumscribed in the same way as the ones we have discussed: the emotions of joy, exhilaration, ecstasy and so on are most usually about the achievement of a particular valued goal and, indeed, the term happiness is also frequently used in this way. However, we must also distinguish happiness as an emotion from pleasure that results from drive satisfaction, consistent with the distinction that we have made previously between emotions, drives and sensations (Power and Dalgleish, 2008; Power, 2014); thus, the satiation of food, thirst, sex or other drives may be accompanied by feelings of pleasure, which may in turn lead to the appraisal of happiness, but could lead under many circumstances to appraisals other than happiness (e.g. Rozin, 1999). In this chapter, therefore, we will begin by briefly examining circumscribed positive emotions such as joy before proceeding to a more detailed discussion of happiness as defined above by authors such as Veenhoven and Wessman and Ricks.

Joy may be conceptualised as the emotional state related to an appraisal that a valued goal has been achieved, or that movement towards such an achievement has occurred. So, for example, somebody might feel joy when she is able to go and book her summer holiday. Such an analysis of joy clearly distinguishes it from what is called life satisfaction. Joy is very much an emotional reaction to a specific goal in a specific domain, whereas life satisfaction, it seems, casts its appraisal net much wider. It is perfectly feasible for an individual to experience joy with respect to a specific goal whilst not being generally happy when all goals in all domains are considered together (cf. Fredrickson, 2005); similarly, it is possible for an individual to be happy in general, that is, high in life satisfaction, whilst also feeling some fear, anger or sadness as a result of appraisals concerning specific goals in specific domains.

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The circumscribed positive emotion of joy is very much the antithesis of the negative emotions, though it can contribute to the experience of emotional conflict. Descartes, in his *The Passions of the Soul* (1649/1989), recounts the tale of a man who, whilst being sad at his wife's death, was also unable to contain his joy at his new-found freedom because she was no longer alive to trouble him. Such conflict between feelings of joy at the achievement of goals that we may feel uncomfortable with and negative emotions towards those goals is frequently the subject of therapeutic work (see Power, 2010). Much of this conflict not only incorporates our own goals but also those of others, such as in the experience of the wonderful German emotion of *Schadenfreude*, which is very much the opposite of envy, and which involves joy at another's misfortune, and can often prove distressing. Whether it involves being secretly pleased that one of our classmates did not succeed in getting a distinction in his exams, or feeling a surge of exhilaration when someone's perfect relationship breaks down, feelings of *Schadenfreude* can disturb us because they reveal wants, needs and goals that we perhaps did not realise we had and that feel uncomfortable and incongruent with our idealised models of our selves.

In the Introduction we noted several definitions of the broad emotional state of happiness. Although these attempts at definition capture some of the breadth of the concept, research into the nature of happiness has, for the most part, been conducted outside of such definitional guidelines or theoretical frameworks. Such research has tended either to ask people what they feel makes them happy, or has examined the correlates of happiness in people who claim to be happy (e.g. Veenhoven, 2000). The findings from these approaches have revealed a number of issues and paradoxes that, in our view, underline the need for a carefully thought-out, theoretical framework before we can achieve any measure of understanding of the concept of happiness. In the sections that follow we consider some of this research and some of the issues that it has generated. We must note, though, that in order to assess whether or not individuals are happy, researchers have devised a number of fairly straightforward self-report measures, but because of the wide-ranging nature of the concept, it is not always happiness that these inventories actually measure. There now exist questionnaires that look at 'positive affect', 'subjective well-being', 'satisfaction with life', 'quality of life' and a number of other related constructs. Although there are clearly debates about the relationships between these different concepts, it is our broad assumption in this chapter that they are all more or less intended as synonyms for the concept of happiness when taken in its broadest sense, but which we take to mean life satisfaction and quality of life (e.g. Layard, 2011).

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On the prototypical happiness questionnaire, the respondent is asked, on a single- or multiple-item scale, how happy he or she is. Examples of such questionnaires include the Satisfaction with Life Scale (Diener *et al.*, 1985), the Depression-Happiness Scale (McGreal and Joseph, 1993), the Memorial University of Newfoundland Scale of Happiness (Kozma and Stones, 1980) and the Oxford Happiness Inventory (Hills and Argyle, 1998) (see Argyle, 2001, and Larsen and Fredrickson, 1999, for reviews). Convergent validity for such measures of happiness is surprisingly good. For example, Sandvik *et al.* (1993) found a strong relationship between self-reports of emotional well-being and interview ratings, peer ratings, reports of the average ratio of pleasant to unpleasant moods, and an index of a memory for pleasant and unpleasant events. Furthermore, such measures are claimed to be uncontaminated by social desirability (Diener *et al.*, 1991), and show structural invariance across time and cultural group (e.g. Vitterso *et al.*, 2002).

However, when looked at more closely, there may be fundamental limitations with the approach and with the conclusions that are reached. For example, the most famous of the happiness researchers is Ruut Veenhoven in the Netherlands, who has established the World Database of Happiness, and who is founder of the *Journal of Happiness Studies* (see www.worlddatabaseofhappiness.eur.nl). Veenhoven's surveys are typically based on a single item along the lines of:

All things considered, how happy would you say you are these days?

This question is typically rated on a 4-point scale that runs from Unhappy to Very Happy, though his website lists about a thousand variants of this and related questions that range from 3-point to 10-point rating scales. We will consider later in the section on quality of life why such single-item questions are weaker in terms of reliability and validity in comparison to multiple-item scales, especially when they are translated across different languages and cultures in which the term 'happiness' has very different interpretations. For example, on the basis of these single questions, Veenhoven has argued that individualistic cultures are happier than collectivist cultures and that income inequality does not affect

happiness. In direct contradiction to these conclusions, we know from the work of Wilkinson and Pickett (2010) (see Figure 1.1 above and Figures 1.2 and 1.3 later) that income inequality really does matter for health, well-being and happiness, and that many collectivist cultures (e.g. Japan) score very highly on the relevant indicators. The moral, as we argue throughout this book, is that because of its multiple ambiguities, surveys with single items that ask you to rate 'happiness' are

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fraught with conceptual and empirical problems and should be avoided except where 'happiness' refers to the brief emotion state and, in English, is better referred to as 'joy' or 'elation' or something similar.

PLEASURE VERSUS PAIN

The possibility that happiness might be defined as the absence of pain and the presence of pleasure has engaged philosophers from the Greeks onwards (see Darrin McMahon's *The Pursuit of Happiness*, 2006, for an entertaining account) and, more recently, has provided the backbone that underlies both psychoanalytic and behavioural approaches within psychology. But to begin with the philosophers, if there is one philosopher who still lends his name to indulgent pleasures, then it has to be Epicurus. Epicurus says 'The pleasure of the stomach is the beginning and root of all good, and it is to this that wis-dom actually refer' (Long and Sedley, 1987, p. 117). Even during his life, Epicurus was misunderstood because of such remarks, and misrepresented as an orgiastic hedonist, but to quote further from him:

But when we say that pleasure is the end, we do not mean the pleasures of the dissipated and those that consist in having a good time ... but freedom from pain in the body and from disturbance in the soul. (p. 114)

If we are in any doubt that pleasure does not equate with hedonistic indulgence for Epicurus, then the following quote surely seals it:

For what produces the pleasant life is not continuous drinking and parties or pederasty or womanizing or the enjoyment of fish and other dishes of an expensive table, but sober reasoning which tracks down the causes of every choice and avoidance, and which banishes the opinions that beset souls with the greatest confusion. (p. 114)

In fact, Epicurus comes close to the philosophers of the virtuous life, to whom we will return in the next section. This was just an opportunity to put the record straight for poor old misunderstood Epicurus.

In his *Nicomachean Ethics*, Aristotle makes the point that, just as slaves cannot be happy if they are denied the opportunity to pursue virtuous activities (higher order goal fulfilment), neither can the victim of torture be happy merely by virtue of being a good person (that is, because of the lack of fulfilment of lower order biological

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goals). These are contentious points; the first seems to be an argument against any form of hedonism, whilst the latter seems to provide objections to certain forms of spiritualism or religious happiness.

The principal objection to taking an entirely bottom-up approach to happiness (that is, that happiness can derive from the fulfilment of low-level goals in the various goal domains) is that individuals seem to habituate fairly rapidly to such events (e.g. McIntosh and Martin, 1991). So, events that at one time seem very positive come to be perceived as less positive when people get used to experiencing those events. For example, the brand new Alfa Romeo sitting in the driveway may be a source of great joy; however, after a few months the owner will become used to seeing the car parked outside and it will no longer be a source of such positive affirmation. Or the first time you try the Anglo-Indian dish chicken tikka masala you might be overwhelmed with pleasure, but try and eat it every day and you are soon likely to grow tired of it and perhaps even nauseous at the thought of it. This habituation process suggests that the path to happiness does not lie with increasingly indulgent satisfaction of such low-level goals, but requires also the satisfaction of higher level, less materially dependent, more psychologically important goals and needs. The circle returns to Epicurus.

What about the possibility of happiness when there is no such fulfilment of low-level basic goals and needs? It seems unlikely that the biological and psychological systems, with their evolutionary imperatives to satisfy basic goals and needs such as hunger, thirst and physical comfort, could be continually short-circuited such that the absence of satisfaction of these needs is not an impediment to the individual's overall happiness. As Aristotle retorts: 'Those who say that the victim on the rack or the man who falls into great misfortune is happy if he is good are . . . talking nonsense' (*Nicomachean Ethics*, 1153b, 19). This notion of the fulfilment of basic needs prior to the achievement of higher order aims is central to a number of theories in humanistic psychology (e.g. Maslow, 1968) and is central to the more recent development of the measurement of quality of life (e.g. Power, 2003), which we will return to later in this chapter. However, we will step briefly back into the recent history of psychology where the ideas of pleasure and pain have played a much more significant role than perhaps they do in modern academic psychology.

Behaviourism was the dominant force in psychology throughout much of the twentieth century, especially in the US, although the development of the computer and cognitive science has led to its substantial decline. At the core of behaviourism are the principles of pleasure and pain, which provide the conditions for learning

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through either classical conditioning (identified by Pavlov) or operant conditioning (identified by Skinner and others) (see Power and Champion, 2000). For example, in the famous Pavlov studies of the conditioning of salivation in dogs, the dogs learned that the sound of the dinner-bell led to the pleasure of food, so that eventually they salivated just at the sound of the dinner-bell even when no food was presented. Equally, in conditioned emotional responses, if a painful stimulus such as an electric shock occurred after the dinner-bell instead of the pleasurable lunch, then the dogs would show signs of distress at the sound of the bell because they learned to associate it with pain.

Skinner extended this analysis to show that we can also learn through operant conditioning that if behaviour leads to the termination of a painful stimulus such behaviour is quickly learned through so-called negative reinforcement. However, one of the main turning points in psychology came when Skinner applied the behavioural pleasure–pain analysis to ‘verbal behaviour’. Skinner’s book *Verbal Behavior* (1957) was shredded by a young linguist, Noam Chomsky, in a couple of pages of review in 1959, in which Chomsky argued that it would be mathematically impossible to learn a language with only pleasure–pain-based learning mechanisms. As Chomsky demonstrated, most of what we say or write consists of unique utterances and sentences that we could not have learned previously (though, in support of Skinner, I have one or two academic colleagues who seemed to have learned their lectures Skinnerian style). Chomsky’s proposed solution was that we have an innate universal grammar- or language-acquisition device, from which our spoken language develops with the minimal language input that it receives in social interaction (Chomsky, 1965). However, Chomsky’s solution has in turn been heavily criticised (e.g. see Eysenck and Keane, 2015). The important point to make here is that, although pleasure and pain mechanisms are of importance in much of what we do, the developments in language and other areas of cognitive science have highlighted that they cannot provide the whole story and that much of what we do is not motivated simply by seeking out pleasure or avoiding pain, as any sado-masochist would tell you. We will consider more general frameworks later in this chapter when drive-related pleasure and pain are considered as one level in complex hierarchical structures that may provide a more adequate approach to well-being and quality of life.

Finally in this section, we note the well-known thought experiment that the political philosopher Robert Nozick presented in his book *Anarchy, State and Utopia* (1974), which is often cited against simple models of hedonism and, in its more recent Jeremy Bentham form, utilitarianism. Nozick imagined a science fiction scenario of an Experience Machine that would offer its user the possibility of non-stop pleasure

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for the rest of his or her life. Would you choose to spend the rest of your life in such a state of perfect pleasure? No, answered Nozick, an answer that demonstrates that simple hedonistic models must be wrong. However, although Nozick's conclusion perfectly suits our argument in this book, and in a perfectly biased world we would simply rest our case at this point, in fact the situation is not quite as straightforward as Nozick would have us believe. In a series of classic studies in psychology, Olds and Milner (e.g. 1954) carried out brain stimulation studies in rats, in which they found an area of the brain, the lateral hypothalamus that, if directly stimulated with a small electrical current or chemical substance, led the rats to seek out constant stimulation in preference to food, water or the opportunity to copulate, to the point where the rats died from a lack of food and water. Olds and Milner, therefore, had previously carried out the experiment that Nozick simply imagined. Of course, you might argue, ok, just because it works like that for rats, it is not like that for us humans. Although this may be true for the majority of humans, unfortunately there are a minority who become addicted to drugs and alcohol in a way that leads to deterioration in their health, just as the rats continued the brain stimulation until they died. Perhaps Aristotle had already answered this question.

BACK TO PHILOSOPHY AND THE VIRTUOUS LIFE

Aristotle's concept of *eudaimonia* provides a key starting point in any discussion of happiness. In his *Nicomachean Ethics* (1095a, 15–22), Aristotle says that *eudaimonia* means 'doing and living well'. One important move in Greek philosophy to answer the question of how to achieve *eudaimonia* is to bring in another important concept in ancient philosophy, *arete* (that is, 'virtue'). Aristotle says that the *eudaimonic* life is one of 'virtuous activity in accordance with reason', to which, as we noted earlier, even Epicurus subscribed. Socrates presented a more extreme version of the virtuous life with his disagreement with those who thought that the *eudaimonic* life is the life of honour or pleasure, and he criticised the Athenians for caring more for riches and honour than the state of their souls. The Stoic philosophers developed this Socratic viewpoint with Stoic ethics presenting a strong form of *eudaimonism*. According to the Stoic philosophers such as Zeno, virtue is necessary for *eudaimonia*. To quote from Stobaeus, the finest compiler of writings from Greek philosophy:

The bastion of Stoic ethics is the thesis that virtue and vice respectively are the sole constituents of happiness and unhappiness. These states do not in the least depend, they insisted, on the possession or absence of things conventionally

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regarded as good or bad – health, reputation, wealth etc: It is possible to be happy even without these.

(Long and Sedley, 1987, p. 357)

The Stoic philosophers thereby denied the importance to eudaimonia of external goods and circumstances, in contrast to what Aristotle had proposed. He thought that severe misfortune (such as the death of one's family and friends) could rob even the most virtuous person of eudaimonia.

The proposals of Aristotle and the more extreme versions of eudaimonia from Socrates and the Stoic philosophers have had a considerable influence on modern proposals, with writers such as Maslow on self-actualisation and Erikson on adult development, which we will examine in more detail in the section on quality of life later in the chapter. However, to give away the punch-line first we have to side with Aristotle because we believe that it is only for some people under special or extreme circumstances that the denial or excessive control of basic needs and drives can provide a sense of 'virtue' and goodness (as we will examine in Chapter 6 with some extreme religious practices). However, for the majority of us ordinary folk, who are neither saints nor martyrs, we achieve the virtuous life through continuing satisfaction of the basic needs in addition to whatever higher ideals we might possess.

THE HEDONIC TREADMILL

There exists a plethora of research findings concerning the relationship between measures of happiness on self-report measures, such as those described above, and a variety of demographic and resource variables (see Argyle, 2001, and Layard, 2011, for comprehensive summaries). To overview briefly we will start with the very complex and changing views on the possible relationship between income and happiness. It has been found that income is related to well-being, though for some time it was assumed that this relationship was only significant below a minimal level of income and was therefore much stronger in poorer countries (Vitterso *et al.*, 2002).

The effect seems possibly to be to do with relative income compared to others in a community or culture rather than the absolute level of income (Wilkinson and Pickett, 2010). However, more careful analyses by Deaton (2008) of the extensive Gallup World Poll data from 132 countries showed that when the logarithm of income or GDP is used rather than the raw value, no such threshold occurs in the data when country

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log income per person is plotted against life satisfaction. Kahneman and Deaton (2010), in further detailed analyses of 450,000 US respondents in a subsequent Gallup survey, replicated the positive relationship between life satisfaction and log income, but in contrast showed that a measure of current emotional well-being did show a threshold effect at about US\$75,000. That is, with an income below \$75,000 in the US, respondents' emotional well-being increased with income, but then plateaued at around this value with no further increases in current well-being. As Kahneman and Deaton demonstrate, the question 'What is the relationship between income and happiness?' needs to be deconstructed into which type of 'happiness' (current affect or life satisfaction) and into which type of income variable (log income, relative income, income inequality). A more recent analysis of longitudinal Gallup World Poll data across 135 countries by Diener *et al.* (2013) showed similar findings for log GDP data, with the strongest effect being for life evaluation. When they used household income, effects were also obtained for current well-being, but they found that the effects were primarily mediated by material possessions, optimism and financial satisfaction, and that they did not adapt across time.

Other variables have been less well studied than the income question. Age and education show only small correlations with subjective reports of happiness (Diener, 1984). Okun and George (1984) found a surprisingly small correlation between health and reports of happiness when objective measures of health are employed, though there is a small but clear effect (Power *et al.*, 2005). Finally, unemployment has been a predictor of unhappiness in some studies (e.g. Clark, 2003), and marriage has been a consistent but weak positive predictor of subjective reports of happiness, with married individuals being reportedly happier than single individuals (e.g. Helliwell and Putnam, 2005). Even putting aside basic methodological issues, for example, that perhaps it is happy people who get married, rather than married people who are happy (though recent evidence suggests that both statements may be true to some extent: see Lucas *et al.*, 2003, for summary findings from a longitudinal German study), the general lack of positive correlational findings between what objectively might be thought of as desirable resources or qualities and subjective reports of happiness requires explanation.

It seems that objective life situations and resources are, at best, only weak predictors of happiness. Indeed, this has been starkly illustrated in some comparative studies; for example, in a classic study, Brickman *et al.* (1978) found that recent lottery winners are often no happier than control participants, only slightly happier than recently paralysed accident victims, and no happier than they were before they won the lottery. Overall, Andrews and Withey (1976) found that age, sex, race, education,

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income, religion, occupation, employment status and size of city only accounted for approximately 11 per cent of the total variance in subjective judgements of happiness. Similarly Kammann (1982) concluded that objective life circumstances routinely account for less than 5 per cent of the variance in subjective judgements of happiness. One of the attempts to account for how both highly negative events and highly positive events affect levels of happiness in the medium to long term was suggested by Philip Brickman and his colleagues in their classic study of lottery winners versus accident victims. They suggested that, as with many physiological processes, there are important adaptational psychological processes that impact on us. Brickman and colleagues argued that, equivalent to these physiological adaptation processes, hedonic adaptation occurs through similar homeostasis, such that we return to our original level of happiness as we 'recover' from both highly positive events such as winning the lottery, and highly negative events such as serious traffic accidents.

In his book *Happiness: Facts and Myths* (1990), Michael Eysenck referred to this adaptation process as the 'hedonic treadmill', and this is the term that has come to be widely used. The proposal is that we all have a 'happiness set-point', such that like on a treadmill, we always return to the point at which we started whatever happens in the intervening period. Eysenck presented numerous examples that, included increasing wealth, increasing achievement and increasing sporting ability, in all of which the expectations change such that an athlete could be happier after winning his school 100 metre championships than after coming second in the Olympic Games. The failed hope or expectation in the Olympics from not winning gold can be viewed even in a negative manner with second place being seen as a failure. Or, perhaps, to give a current example from the world of banking, the banker who receives million dollar bonuses every year will simply come to expect these as normal, with 'happiness' only occurring if the bonus is several times more than expected. Of course, the same bonus dropped into an African Shanty Town and shared between 10,000 would give such a level of utilitarian happiness that would make even Jeremy Bentham cry with joy (preserved as he is in his glass case in University College London). The important point that Brickman, Eysenck and others made is that happiness is simply a temporary state from which we all recover relatively quickly, perhaps even returning to a set-point to which we are genetically shackled.

The conclusions from the 'hedonic treadmill' theory would be very tempting for the approach that we take in this book, in which we argue that chasing happiness is as likely to be successful as chasing the pot of gold at the end of the rainbow or the

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mirage in the desert. As conducive to our own argument as this theory is, we must, however, point to some weaknesses in the theory that mean it has to be modified (cf. Diener *et al.*, 2006). In the next section, we will consider some of Seligman's and others' work on the contribution of circumstances and activity to levels of happiness, which runs counter to hedonic treadmill theory, but at this point we just want to note the work of Laura Carstensen (e.g. 2006) on so-called 'socio-emotional selectivity theory'. The theory argues that, as time horizons reduce with age, people become increasingly selective, investing greater resources in emotionally meaningful goals and activities. Ageing becomes associated with a preference for positive over negative information in both attention and memory (the 'positivity effect'). Older adults often spend more time with familiar individuals with whom they have had rewarding relationships, which increases positive emotional experiences and reduces emotional risks as individuals become older. At the same time, older adults report that they are better able to regulate their negative emotions compared to when they were younger, so typically report fewer and less extreme 'downs' in comparison to younger people. The net effect is that people are often surprised at the wisdom and emotional capacity that they develop with age, as we have shown in our studies on attitudes to ageing (Laidlaw *et al.*, 2007), to which we will return in Chapter 3. The important point to make here is that there may actually be a slow drift upwards in the 'set-point' across the lifespan, even if in the short to medium term the Brickman treadmill model has some validity for some types of life events for some of the people some of the time. Overall, however, the hedonic treadmill theory does not account for why some life events do change our happiness levels upwards or downwards, nor why people on average become more positive across their lifespan.

MARTIN SELIGMAN

Another major area of criticism of the hedonic treadmill approach to happiness comes from the work of Martin Seligman and colleagues. For those who have reached the age of consent, you can log onto Seligman's website at www.authentichappiness.sas.upenn.edu, where you can join a community claimed to be 2 million users worldwide, who regularly complete positive psychology questionnaires, presumably because if you know the right answers you will get positive feedback and feel good about yourself!

Seligman and colleagues, such as Sonja Lyubomirsky, Mihaly Csikszentmihalyi and others, have produced what they have called the 'happiness formula' (e.g. Seligman, 2002):

$$H = S + C + V$$

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In this formula, *H* refers to the enduring level of happiness that you experience, *S* refers to the genetic or biological set-point, *C* refers to the conditions or circumstances of your life, and *V* refers to the factors under your voluntary control. Seligman has summarised a range of studies of twins and adoptees to show that behavioural traits have a relatively high amount of genetic determination, such that 50 per cent of our personality comes from our genetics. However, in contrast to the hedonic treadmill approach, which basically assumes that all personality is predetermined, Seligman and colleagues have included two factors that contribute the remaining 50 per cent to levels of happiness and that can be changed; namely, the circumstances and voluntary control of activities factors.

The circumstances in the formula include factors such as income and wealth, marriage, fulfilling social lives and religious belief. These factors all have extremely complex relationships to enduring happiness, which any examination of the relationship between money and happiness indicates. For example, in his book *Authentic Happiness* (2002), Seligman summarised analyses of surveys of income and national happiness such as those from the World Values Survey in the 1980s and 1990s, which demonstrated that there is a threshold in income below which you do find a positive correlation between income and happiness, the threshold being at about US\$8,000 in the 1990s, though the more recent figures provided by Wilkinson and Pickett (2010) suggested that this threshold had increased to about US\$25,000. Above these thresholds, Seligman and others have summarised the income data to show that there is no correlation between income and happiness, which leads Seligman to conclude erroneously that income redistribution is unnecessary because it does not make the poor any happier once they are at this minimum threshold. However, we now know from the more recent work of Deaton (2008), Diener et al (2013) and Kahneman and Deaton (2010) summarised earlier that when log income data is used there is no threshold effect and that, contrary to Seligman, there is a continuing increase in life satisfaction and experiential well-being with increases in income. A second problem, as we showed earlier in Figure 1.1, is in relation to health and social problems; it may not be absolute income that is crucial in relation to indicators of health and well-being in a society, but the inequality in income.

Figures 1.2 and 1.3, taken from Wilkinson and Pickett's analyses, dramatically demonstrate the differences between absolute income and income inequality. These figures show the impact in developed nations of income (Figure 1.2) versus income inequality (Figure 1.3) on the UNICEF index of child well-being in the world's 22 richest nations. Figure 1.2 clearly supports Seligman's claim that there is no relationship between absolute income and child well-being in the world's rich

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nations, despite the wide income range covered, from Portugal at the lowest to the US at the highest.

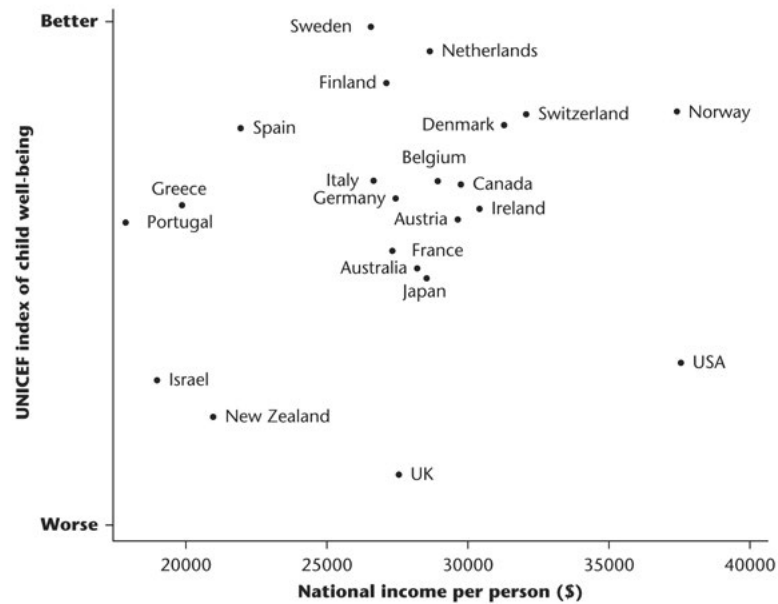


FIGURE 1.2 Absolute income and the UNICEF index of child well-being.
Source: Wilkinson and Pickett (2010). Reproduced with permission.



FIGURE 1.3 Income inequality and the UNICEF index of child well-being.
Source: Wilkinson and Pickett (2010). Reproduced with permission.

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Figure 1.3 highlights what happens when absolute income is replaced by income inequality. Now there is a strong relationship with child well-being revealed, with the highest level of income inequality in the US being associated with low child well-being, and the low income inequality, especially in the Scandinavian countries, being associated with high levels of child well-being. Seligman and his colleagues need to rethink and re-analyse their data, as well as their politics.

The final element in the Seligman equation is the 'V' or voluntary control factor. Again, this is a wide-ranging concept, and it originally included directives from Seligman such as 'Do not dwell on the past', 'Get rid of your negative emotions', 'Forgive and forget' and 'Increase your optimism'. This earlier work of Seligman could be seen to contribute to the split in the American psyche that Jean Twenge, as we noted earlier, has identified with both the increase in self-esteem and the concurrent increase in anxiety and depression. Contrary to Seligman, you do not need to walk away from your negative emotions and from your past as if you were merely a piece of computer hardware that could be reformatted and the disk wiped clean (cf. Kashdan and Biswas-Diener, 2014). Fortunately, more recently Seligman himself and many within the positive psychology movement have shifted away from this one-coloured coat and now wear a coat of many colours; thus, the excellent *Oxford Handbook of Positive Psychology* edited by Lopez and Snyder (2009) presents a much more diverse approach, with many thoughtful analyses of topics such as wisdom, resilience, emotion expression, courage and humility that we will return to throughout this book (especially in Chapter 7) and with which we are in agreement.

WELL-BEING, QUALITY OF LIFE AND OTHER ABSTRACT STATES

Some years ago in Britain the newspapers carried a story of a man who committed suicide when he realised that he had not won the National Lottery. The situation was this: the man in question had selected a set of numbers that he entered into the lottery each week, subscribing six or seven weeks in advance. Eventually, he hit lucky and his numbers matched those drawn out of the hat for the maximum jackpot. However, on checking again the following day after a night of celebration, he found that his lottery entry was not valid because his advance subscriptions had lapsed the week before. Unable to cope with having thought that he had won millions of pounds when he had actually won nothing, the man committed suicide. What is perhaps most puzzling about this case is the fact that, prior to his false lottery win, the man concerned was reportedly very happy. The objective circumstances of his life that had previously made him happy were no different and were exactly the same after his

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false win; however, it seems that the man's goals and dreams in various domains of his life had shifted in line with his supposedly new-found fortune and the inability to attain these new goals and needs seems to have led to desperate unhappiness and his suicide.

The proposal that life satisfaction and well-being may be a function of goal fulfilment across various domains and levels, even allowing for the fact that individuals can invest more heavily in one domain or another, entails that such states are necessarily dynamic, with the possibility of change, even after long periods of apparent stability. A sense of satisfaction and well-being lasting minutes, hours, days, weeks or months will arise out of a process of psychological negotiation in which the goals and needs in one domain are pursued and realised. As Diener and Biswas-Diener have emphasised in their book *Happiness: Unlocking The Mysteries of Psychological Wealth* (2008), our sense of satisfaction is often more about the process than it is about the end-point of that process. As anyone who has children will recognise, even if earlier in life having children might have seemed like the end-point of a goal, having children is in fact just the beginning of a process of being a parent that never actually comes to an end. As with many goals, one apparent end-point merely opens up a new set of goals and processes. To take a different example, imagine that your goal had been to be a professor in a university. Once you have achieved that goal, there is then a whole new set of goals and processes that open up, such as walking around absent-mindedly, wearing different coloured socks and having a hairstyle that looks like you dress in front of the Van de Graaff generator every morning before you go to work. Joking, of course.

In order to understand how people make judgements about their position in life and their sense of satisfaction or dissatisfaction with their life, we have been working for many years to understand what is known as 'quality of life'. The phrase 'quality of life' is used in many different ways, and a major issue that faces this area of work is how the term should be defined and conceptualised. One of the key distinctions that has been made is that between health-related and non-health-related quality of life (e.g. Spilker, 1996). The starting point for a number of the health-related definitions has been the well-known World Health Organization (WHO) (e.g. 1958) definition of health as: 'a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity'. The inclusion of the phrase 'well-being' in the WHO definition has led some researchers to focus too narrowly on self-reported psychological well-being, or the even more restricted notion of 'happiness', as being the only aspect of quality of life of importance (e.g. Dupuy, 1984). However, 'well-being' and 'happiness' have to be seen as the narrower terms that may be important

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aspects of quality of life, but which are not the only aspects to be considered. The challenge has been to specify the range of health-related and non-health-related aspects of quality of life that should also be included, such that 'quality of life' is not simply another term for 'well-being'.

The original WHO definition of health provides us with an excellent starting point for defining quality of life (WHOQOL Group, 1995), but it leaves open two key questions. First, what other areas should be included in addition to the physical, mental and social? And, second, should the conceptualisation include, for example, objective characteristics of the individual in addition to the individual's subjective evaluation? Other definitions and measures of quality of life take many varied approaches to these two questions. Nevertheless, there may now be an emerging consensus for both of these key issues. In addition to the physical, mental and social aspects, there is now a recognition that spiritual and religious aspects need to be included in health-related quality of life (e.g. Power *et al.*, 1999; see Chapter 6), and a range of aspects of the individual's physical environment needs to be included in non-health-related quality of life.

In relation to the second issue of the objective and the subjective, although many of the earlier measures of quality of life included both objective characteristics (e.g. being able to run for a bus or walk up a flight of stairs) and subjective characteristics (e.g. rating satisfaction/dissatisfaction with level of physical mobility), the more recent measures have focused solely on the subjective (WHOQOL Group, 1998a). It seems to make sense now that subjective and objective indicators should be kept separate. To give an extreme example, how can an individual living in poverty in a village in India report a higher level of happiness and quality of life than a multi-millionaire in Manhattan? This problem has led economists such as the Nobel-prizewinning Amartya Sen (Sen, 2001) to suggest that subjective indicators should be rejected because of their discordance with objective economic indicators. However, as psychologists within the positive psychology movement would agree, the discordance between the objective and the subjective is crucial and often provides a testament to how the human spirit can overcome and even flourish under adversity.

Our starting point, therefore, was to agree a definition of quality of life as follows:

individuals' perception of their position in life in the context of the culture and value systems in which they live and in relation to their goals, expectations, standards and concerns. It is a broad ranging concept affected in a complex way by the persons' physical health,

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psychological state, level of independence, social relationships
and their relationship to salient features of their environment.

(WHOQOL Group, 1995, p. 1404)

A series of focus groups were then carried out in 15 different centres worldwide in order to generate different items and facets for the different domains of health, relationships and personal environment that were included in the definition. A pilot measure was developed after analysis of the focus group material, which was then tested and further refined in a series of field tests (WHOQOL Group, 1998a, 1998b).

TABLE 1.1 WHOQOL DOMAINS AND FACETS

Physical domain	Pain and discomfort
	Energy and fatigue
	Sleep and rest
Psychological domain	Positive feelings
	Thinking, learning, memory and concentration
	Self-esteem
	Bodily image and appearance
	Negative feelings
Level of independence	Mobility
	Activities of daily living
	Dependence on medical substances and treatments
	Work capacity
Social relationships	Personal relationships
	Social support
	Sexual Activity
Environment	Physical safety and security
	Home environment
	Financial resources
	Health and social care: accessibility and quality
	Opportunity for new information and skills
	Recreation and leisure activities
	Physical environment (pollution / noise / climate)
	Transport
Spirituality / religion / personal beliefs	

Source: WHOQOL Group (1998a, 1998b)

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The resulting domains and facets are shown in Table 1.1 and these have been found in subsequent studies to be the important domains and facets that apply across the adult lifespan in over 40 cultures that have now been tested (e.g. Power, 2003).

One of the opportunities that data collected in this way offered was the possibility of examining the actual structure and content of quality of life. Partly inspired by Maslow's (e.g. 1968) original hierarchy of needs and subsequent work in personality theory, a number of influential approaches have conceptualised quality of life as a hierarchical structure or pyramid with overall well-being at the top, broad domains (such as physical, psychological and social) at the intermediate level, and then specific facets or components of each domain at the bottom (e.g. Spilker, 1990). This overall hierarchical approach was adopted by the WHOQOL Group. As a preliminary test of this predicted hierarchy, a table of facet and domain inter-correlations was produced. The most notable finding was that, whereas the experts had relegated sexual activity to the physical domain (facet-to-corrected-domain $r = 0.16$), the data showed that respondents considered sex to be part of the social relationships domain ($r = 0.41$), to which it was moved. The difference may of course tell us something about experts versus real people! On a variety of psychometric criteria (see WHOQOL Group, 1998a), five of the facets (sensory functions, dependence on non-medicinal substances, communication capacity, work satisfaction, and activities as a provider/supporter) were dropped from the generic measure. (It was noted, however, that some of these might need to be included in subsequent illness-specific or group-specific modules.) With these deletions, there were now 24 specific facets and several items measuring overall quality of life. In deciding on the number of items to choose for each retained facet, the decision was taken to select four items per facet, because four is the minimum number required for the scale reliability analyses that were carried out in subsequent psychometric testing of the instrument. These decisions led to the selection of $25 \times 4 = 100$ items (including the four general items); thus, the revised field trial WHOQOL became known as the WHOQOL-100.

The hierarchical model of quality of life that we have developed since the early 1990s in the form of the WHOQOL provides a much more complex multidimensional view of Aristotle's eudaimonia than do one-dimensional approaches that simply measure 'well-being' or ephemeral emotion states of 'happiness'. However, one of the important early questions that happiness proponents directed at us was whether or not our complex account of quality of life could, at the end of the day, simply be reduced to happiness and that a Veenhoven-type single-item measure of happiness (such as the 'All things considered, how happy would you say you are these days?' noted earlier) would do just as well as our 100-item WHOQOL. There is the

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technique of multiple regression that can tackle such a question at least from the statistical point of view. To this day, I can remember sitting in my office at WHO in Geneva in September 1995 when John Orley, the then coordinator of the WHOQOL, asked me to run the analysis based on the data from our first 15 centres to see if there really was more to quality of life than mere 'happiness'. It was with some trepidation that I loaded up the statistics package, SPSS, and carried out the multiple regression analysis.

The outcome of the statistical analysis is summarised in Table 1.2, a revised version of the equation which uses data from our field trial study (WHOQOL Group, 1998a) from 7,701 respondents in 19 different centres worldwide. The basic assumption of the analysis in Table 1.2 was that if 'happiness' (a facet covered in the psychological domain as 'positive feelings') were to explain all of quality of life, then only this facet would be significant when all the remaining 23 WHOQOL-100 facets were subsequently included in the analysis. However, the actual analysis showed that a further 19 of the facets continued to offer a significant contribution (as shown in the column labelled 'Sign.' standing for 'significance level' in Table 1.2) to the statistical account of overall quality of life, even when the positive feelings facet was included first in the multiple regression. That is, all of the different domains, including physical, psychological, relationships, independence, environment and personal beliefs, make a separate but important contribution to our overall assessment of our own quality of life. The statistics provided powerful support for both our own conceptualisation and the information that came back from focus groups worldwide that 'quality of life' is a far broader concept than the notions of happiness or well-being and that it is not reducible to these one-dimensional concepts. As mentioned above, the concept of quality of life was construed initially as a useful adjunct to traditional concepts of health and functional status. An overall health assessment, therefore, would have included a single measure of the person's physical health, a measure of functioning and a measure of quality of life. Early attempts at assessments that went beyond physical health status sometimes took the form of a rating on a single scale, but, as we have stated, these scales unfortunately condensed a complex multidimensional concept into a single dimension. To devise a measure of quality of life that is both reliable and valid, a broad range of potentially independent domains covering all important aspects of quality of life is necessary. Furthermore, to devise a measure that is reliable and valid cross-culturally requires a different approach to instrument development (e.g. Bullinger *et al.*, 1996). Our collection of data from a large number of different cultures has allowed the question to be asked, therefore, of whether or not there is something universal about the aspects of our

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lives that contribute to our overall sense of well-being and quality of life. Although the term 'quality of life' itself does not translate well into all languages, our analyses across a wide variety of cultures suggest that there are universal aspects of this concept that may well be linked in to other universals in areas such as language, emotion and social relationships (e.g. Power and Dalgleish, 2015). As the saying goes, not only should we add years to life, but we should also add life to years.

TABLE 1.2 REGRESSION EQUATION FOR HAPPINESS AND QUALITY OF LIFE

<i>Model</i>	<i>Beta</i>	<i>t</i>	<i>Sign</i>
Positive feelings	0.136	12.119	.000
Energy	0.157	13.565	.000
Sleep	0.034	3.514	.000
Think	0.029	2.613	.009
Esteem	0.145	12.165	.000
Body Image	-0.091	-8.708	.000
Neg feel	-0.062	-5.978	.000
Mobility	0.04	3.838	.000
Daily activities	0.039	2.89	.004
Work	0.166	15.466	.000
Relationships	0.122	9.932	.000
Support	0.025	2.533	.011
Sex satisfaction	0.055	5.992	.000
Home	0.089	8.135	.000
Finances	0.07	6.978	.000
Social services	0.096	10.346	.000
Leisure	0.14	13.024	.000
Environment	0.024	2.49	.013
Transport	-0.093	-9.036	.000
Meaning	-0.038	-4.401	.000

Finally, one very straightforward question to ask cross-culturally is whether or not the confusion that occurs in English between 'happiness' – the brief emotion state – versus 'happiness' – the long-term state of satisfaction with life – also occurs in other languages. For example, in German, the word *Gluck* shares the same problem as the English word happiness in that it can refer either to the momentary state or to long-term satisfaction. In Spanish, there is a similar problem with the word *felicidad*, which also refers to both, but which is then distinguished by which verb (*ser* or *estar*) accompanies the noun. In Portuguese, *alegria* is more likely to be used for the momentary state, and *felicidade* for long-term satisfaction, but again it is acceptable

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to use either word to refer to both. But our favourite is in Russian in which счастье ('little happiness') refers to the momentary state of happiness and счастливый ('big happiness') refers to the permanent state of well-being. It is both reassuring and worrying at the same time, therefore, that languages other than English share some of the confusion between the two types of 'happiness', even if they do not always have a word or phrase for 'quality of life'.

THE HAS AND THE HAS-NOTS

In order to conclude this chapter and line up some of the key issues for the forthcoming chapters, we can attempt to summarise some of the main points from what we have said so far together with their implications for the understanding of well-being and quality of life. One simple formula for what counts and does not count in life may be to categorise people into the HAS and the HAS-nots:

1. H = Healthy lifestyle
2. A = Adaptability
3. S = Sociability.

The final point is that having a healthy lifestyle in general contributes to well-being and quality of life. We will consider in later chapters how factors such as physical exercise, good diet, good sleep habits, avoidance of smoking and low to moderate alcohol use can all help towards a positive state of physical and mental health. And we emphasise healthy lifestyle as opposed to simply a state of health because the increasing numbers of chronic diseases that occur in our ageing affluent societies mean that many people are living longer but with accompanying chronic conditions such as diabetes, arthritis, high blood pressure and so on. The point is that it is possible to manage these chronic conditions well or badly such that many people with chronic conditions still report good quality of life, as we have found in our international studies of older adults (Power *et al.*, 2005).

The second feature that we have noted is adaptability, which refers to the capacity of an individual, a family or a larger group to cope well with significant changes in the conditions of their lives, whether these changes affect just those individuals or the larger group as a whole. The story of how *Homo sapiens* colonised Europe and survived through an ice age is indicative of just how adaptable our species has proven to be, and we will return to this issue in Chapter 6. However, there is still considerable variation between individuals and groups in how well they adapt. In Chapter 3, when we consider the range of negative emotions, we will be particularly

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interested in individual differences that people have in dealing with their negative emotions and how problems in emotion and emotion regulation can cause difficulties both physically and psychologically for individuals, and also impact on the quality of relationships, as we will explore in Chapter 2. The third feature for the HAS is sociability. We do not imply a one-size-fits-all model of sociability, but note that for healthy sociability there are considerable variations. In Chapter 2, we will examine the importance of an intimate confiding relationship for everyone, whether or not that is a heterosexual physical relationship or some other intimate relationship. Equally, good sociability includes a range of other types of relationships that serve different social support functions. Now we are not denying that it may be possible for an ascetic monk to live alone quite contentedly in a cave on a remote island rock such as Skellig Michael, miles off the coast of Kerry in south-west Ireland. In fact, such a monk would be motivated to live such a life because of his relationship to an imaginary being, and, as we will see in Chapter 6 on religion, there can be many benefits from a deep commitment to such belief systems. The problem is how you cope if your god appears to abandon you and does not provide the support that a real person should.

Finally, we know that the world is full of HAS-nots, and maybe even one or two of you have got as far as to start reading this book. Although we have not set out to write a self-help guide on how to be a HAS rather than a HAS-not, and in contrast to the shelf-loads of such books that fill all popular psychology bookshops, there may nevertheless be plenty of suggestions buried beneath all the evidence that we will examine. Undoubtedly, there will be pointers provided on how to avoid the excesses of too much or too little and how, instead, to take the 'good enough' route through life that will maximise your opportunities without being sunk by the chains of perfectionism, nor defeated by the sibyls of unrealistic pessimism. If we convert even one HAS-not into a HAS along the way, there would be much proverbial rejoicing in the positive psychology heaven. However, throughout this book we also want to present the other side of the argument with the case for the HAS-nots. Our species has become the most successful to date because of its diversity and its capacity to be creative. If we all wandered around smiling and making lemonade out of lemons, we would probably still be living in caves and telling ourselves how lucky we were just to be there. Oh what a beautiful cave!

Dissatisfaction with what we have has been a great motivator for invention – for tools that are more powerful and sharper than the previous ones, for modes of transport that are more efficient and faster, for social organisations that freed some people up from subsistence existence to be creators and inventors. Progress in technology and

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progress in social organisation have resulted from complex and multiple different types of motivation, not simply the pursuit of happiness. We must acknowledge, therefore, that our cultures and societies have benefited as much from the HAS-nots as from the HAS. You need only to read the biographies of the great contributors to science, such as Copernicus, Darwin and Newton to say that by positive psychology standards they might well have been discarded into the bucket of the HAS-nots. This proposal is not to deny that the extremes at the negative end of H, A and S are problematic, but equally problematic are the extremes at the positive end, the people whom the French writer Gustav Flaubert referred to as the 'stupid happy'. The stupid happy score 10 out of 10 on all the authentic happiness scales, but, as we will show in subsequent chapters, they are just as likely to develop illnesses and die younger than the people who score low on these scales. Ultimately, however, our argument is in favour of diversity, for societies in which Prozac does not have to be put into the water for all. There are situations under which it is better to anticipate the worst outcome rather than the best, because terrible things do sometimes happen, when the illusion that we control our universe is readily shattered, as we will show in subsequent chapters.

ACT at Work

Frank W. Bond

*Department of Psychology, Goldsmiths College,
University of London, New Cross, London, UK
and*

Steven C. Hayes

Department of Psychology, University of Nevada, Reno, NV, USA

Redesigning work and management processes in order to reduce workers' exposure to sources of stress has been advocated by occupational health psychologists, in both the UK (e.g., Cox et al., 2000) and USA (e.g., National Institute of Occupational Safety and Health [NIOSH], 1996). Furthermore, outcome research has provided empirical support for work redesign interventions that target such variables as lack of worker control, role ambiguity, poor workplace communication, career concerns, poor organisational climate, heavy workload, or shift work (e.g., Bond & Bunce, 2001; Landsbergis & Vivona-Vaughan, 1995). Despite these successes, it also seems to be important to address the psychological styles or approaches that workers bring to stressful work situations. First, some sources of stress may not be completely avoidable (e.g., underpromotion). Second, workers' efforts to modify stressful situations may themselves be inhibited by poor psychological coping strategies (e.g., use of avoidance strategies). Finally, work-related stress does not occur in a vacuum, and psychological styles that increase stress reactions at home (e.g., being overcontrolling) may also result in feeling stress at work. Consistent with this line of thought, researchers investigating stress at work have found that psychological styles, such as type-A behaviour pattern (TABP) and locus of control, are reliable predictors of occupational stress (e.g., Jex, 1998).

Research by Bond & Bunce (submitted) confirms the significance of these two individual characteristics in predicting stress, but it also highlights the even greater importance of another psychological style that has yet to receive much attention in the work stress literature: acceptance. Psychological acceptance refers to people's willingness to experience difficult private events (e.g., thoughts, emotions, sensations, and memories), without avoiding, struggling, or suppressing them. Bond & Bunce found that poor psychological acceptance predicts, in one year's time, mental ill-health and poor productivity to a significantly greater extent than does TABP, locus of control, and work-related sources of stress. They also found that low levels of acceptance predict these stress outcomes across several types of industries (e.g., nursing, advertising, and central government civil

service). Such findings clearly suggest a need to consider this particular individual characteristic when assessing and promoting occupational health.

Acceptance and commitment therapy (ACT, pronounced as a whole word, not by its constituent letters, A-C-T) (Hayes et al., 1999b; see also Zettle & Hayes, Chapter 3 in this volume) is one cognitive-behaviour-based treatment that is particularly focused on the improvement of psychological acceptance. In this chapter, we describe how ACT can be used in a brief, group-based, worksite stress-management intervention (SMI). It is the protocol for this SMI that forms the core of this chapter, but before presenting it, we will briefly discuss ACT theory and outcome studies (although see Zettle & Hayes, Chapter 3 in this volume, for a more thorough discussion of both). Importantly, we also discuss how Bond & Bunce (2000) validated our worksite ACT protocol in a UK media organisation, and how they demonstrated that this intervention worked for the reasons actually specified by ACT theory.

ACT: THEORY AND RESEARCH

ACT is based on a functional contextual approach to psychological dysfunction (e.g., Hayes, 1993) that maintains that, due to a person's previous learning history, certain situations (e.g., being criticised by the supervisor) evoke unwanted thoughts, emotions, and sensations. Nevertheless, these negative, internal events lead to strain (defined here as impairment of psychological or physiological well-being [Bunce, 1997]), only in a certain context: one in which they are (a) taken literally (e.g., If I *feel* fearful, then there is *really* something to fear) and (b) are construed as events that need to be changed or avoided. In such a context, these unwanted, internal events function as the problems that they are construed to be and, thus, result in strain; however, these same negative thoughts, emotions, and sensations will not lead to impaired functioning if they are placed in a different context: one in which they are accepted (Hayes, 1994). (See Hayes et al. [1999a] for a more in-depth discussion of functional contextualism and its relationship to ACT.)

In a context of acceptance, people are mindful that they are engaged in processes called thinking and feeling, but they are, in a sense, "mindfully inactive". They merely observe and describe their internal events without participating in them, that is, without taking them literally, and without needlessly trying to change, avoid, or otherwise control them, no matter how unwanted or unpleasant they may be (Hayes, 1987). Observing and describing unwanted internal events increases the degree to which people can *act* to obtain the goals that they have in a given situation, without first waiting for their own historically produced *reactions* (e.g., fear and sadness) to change (Hayes, 1987). Such action will increase the probability that people can attain more positive reinforcement, and avoid more punishment, than they could by participating in, and thus becoming entangled in, any unwanted internal events (e.g., worry, unhappiness, or fear). Contact with these improved contingencies will, in turn, help prevent or reduce strain. To clarify this diagrammatically, ACT maintains the following:

Unwanted thoughts, emotions, and sensations + (the context of) literality,
change or avoidance → strain

—and—

Unwanted thoughts, emotions, and sensations + (the context of) acceptance
→ unwanted thoughts, emotions, and sensations *that do not result in strain*

It should be emphasised that adopting a context of acceptance may not, at least at first, prevent a given stressor from evoking negative thoughts, emotions, and sensations; however, the more people accept these negative internal events, the more likely it is that the given stressor will cease to trigger them, due to extinction processes (Linehan, 1993). Nevertheless, it is not helpful for ACT clients to think that acceptance is a way to avoid (even eventually) experiencing unwanted internal events. Most people have learning histories that will result in certain situations evoking unwanted internal events; and in these inevitable circumstances, it is in the best interests of clients (or SMI trainees) to react willingly (i.e., with acceptance) to these internal events.

Unlike ACT, many other cognitive behaviour approaches to emotional and behavioural change have traditionally maintained that people with emotional disorders become less distressed, because the “*content*” of their thoughts becomes more functional (e.g., Beck, 1976; Beck et al., 1979); that is, because their thoughts change, for example, from “I am a *failure*” to “I am an *adequate person*”. The context in which people hold their “dysfunctional” thoughts (e.g., willingly) has not been emphasised very strongly (with the exception, perhaps, of rational emotive behaviour therapy (REBT) [see Ellis & Robb, 1994]). Recently, however, acceptance strategies have been integrated, to varying degrees, into CBT treatments, and with good reason as well. As Hayes et al. (1996) note, a large literature now documents an association between emotional willingness and positive outcome in psychotherapy. Specifically, acceptance-based psychotherapy methods have been developed for problems such as borderline personality disorder (Linehan, 1993), substance abuse (Marlatt, 1994), relationship problems (Koerner et al., 1994), and generalised anxiety disorder (Wells, 1995; Chapter 6, this volume). Nevertheless, ACT was the first CBT to promote psychological acceptance without also targeting psychological content for change, and it seems to be the one that focuses on acceptance processes to the greatest extent. Other CBTs that employ acceptance techniques (e.g., REBT) still emphasise the content of cognition over the context in which it is held. As a result, the “content CBTs” and ACT advocate different primary goals for dealing with undesirable cognitive content; that is, change vs. acceptance, respectively.

Although the outcome literature on ACT is increasing rapidly (see Hayes et al. [1999a] for a recent review), it is not yet known whether ACT will generally produce better outcomes than will content-oriented CBTs. Research thus far, however, demonstrates (a) fairly equivalent, beneficial effects of these two kinds of treatment, and (b) that each works through different processes. Regarding this second point, ACT quickly reduces the believability, or literality, of people’s negative, internal events (e.g., because I am fearful, it does not mean that there is *really* something to fear), but the frequency of these experiences is reduced more slowly (Zettle & Hayes, 1986). Conversely, content-oriented cognitive therapy (CT) (Beck et al., 1979) reduces the frequency of people’s negative, private events fairly quickly, but the believability of these events, when they inevitably occur, is reduced more slowly (Zettle & Hayes, 1986).

ACT AS A WORKPLACE STRESS MANAGEMENT INTERVENTION

Given the ability of ACT to improve acceptance and psychological well-being (see Hayes et al., 1999a), Bond & Bunce (2000) hypothesised that ACT might be effective as a worksite SMI, and so tested this hypothesis in a randomised, controlled trial in a large UK media

organisation. ACT was compared with a waiting-list control group and a risk management approach designed to increase worker innovation (the latter is not discussed further in this chapter, due to its irrelevance to CBT, but see Bond & Bunce, 2000).

Bond & Bunce (2000) wished to test the efficacy of ACT using a brief, group-based implementation method. To this end, they adapted the “2 + 1” method of psychotherapy delivery (e.g., Barkham & Shapiro, 1990) in which participants receive three, three-hour sessions, two on consecutive weeks, and a third three months later. This particular brief format has the advantage of allowing participants to carry out homework assignments that encourage the application of learned acceptance strategies in their work environments. They then receive feedback on their efforts during the following sessions. Such a delivery strategy, unique in SMI research, may be useful in helping people to learn more effective behaviours in less therapeutic time (e.g., Bond & Dryden, 1996), thereby making an SMI more cost-effective and thus more attractive to organisations.

As Bond & Bunce (2000) predicted, the results of their study indicated that ACT significantly improved people’s mental well-being and tendency to innovate, in comparison to the control group. (Mental well-being was measured by the General Health Questionnaire [GHQ] [Goldberg, 1978] and the Beck Depression Inventory [BDI] [Beck et al., 1961].) Furthermore, research using the “four tests for mediation” recommended by Baron & Kenny (1986) (Kenny, 1998) showed that the ACT intervention improved people’s mental health (GHQ) and propensity to innovate, because it increased their psychological acceptance (as measured by the Acceptance and Action Questionnaire [AAQ] [Hayes et al., submitted]). Unlike content CBTs, ACT did not lead to these improvements as a result of making participants’ cognitive content more functional. Indeed, there was no significant pre-test/post-test change for the ACT group on the measure of cognitive content used in this experiment (i.e., the Dysfunctional Attitude Survey [DAS] [Weissman, 1979]). Thus, it appeared that psychological acceptance, and not cognitive content, was the mechanism, or mediator, by which ACT improved the stress outcomes of participants in the UK media organisation.

These findings support the ACT hypothesis of the mechanism by which this intervention effects change. They are also consistent with several other studies on ACT in such areas as depression (Zettle & Hayes, 1987; Zettle & Rains, 1989), coping with psychotic symptoms (Bach & Hayes, 2002; García & Pérez, 2001), anxiety (Luciano & Gutierrez, 2001; Roemer & Orsillo, 2002), and pain (Geiser, 1992; Hayes et al., 1999a; Luciano et al., 2001; see also McCracken, 1998), among other problems.

It appears, then, that our ACT protocol, presented in this chapter, can decrease strain and improve innovation propensity in a work environment, and it does so because it increases psychological acceptance (Bond & Bunce, 2000). Understanding the mechanism by which ACT works is, of course, helpful for theoretical reasons, but it is also essential for therapeutic ones. For, unless we understand why an intervention works, it is impossible to maximise its effectiveness. To elaborate, by knowing that acceptance mediates the effects that ACT has on stress-related outcomes, psychologists can propose and test techniques that may manipulate this variable more effectively, and thus better help workers to improve their mental health and effectiveness at work.

Other than that of Bond & Bunce (2000), we are not aware of another outcome study that has (a) evaluated rigorously the mechanisms by which an SMI works, or (b) used such a detailed SMI protocol in a successful, randomised controlled trial. For these reasons, we believe that our programme may be helpful to practitioners who wish to implement a

brief, group-based SMI, and particularly a brief, group-based, CBT SMI, in a workplace setting.

ACT MANUAL

Training Session I (Week 1)

At the beginning of this session, the participants are told that any personal information that is revealed or discussed in this or any session is confidential and should not be shared or discussed with anyone at all outside the room (including partners and family members). Anyone not prepared to make such a commitment to privacy is asked to leave. In addition, the participants are told that they are free to leave this or any subsequent training session at any time and for any reason, and the fact that they left will not be reported to anyone.

What is Stress?

Firstly, the participants are asked, “What are your symptoms or signs of stress?”, and their responses are noted on a whiteboard. After several people provide some examples, an overhead projector slide of various signs of stress is presented. This slide notes many physical (e.g., sleep loss), mood (e.g., irritability), thinking (e.g., poor memory), and action signs (e.g., crying) of stress. After these are read out and questions are dealt with, it is noted that these signs of stress are relevant to the individual and that they centre on mental and physical well-being, but it is also noted that the organisation, as a whole, may also show signs of stress, and that these signs centre on organisational ineffectiveness, employee turnover, and sick absence rates. An overhead projector slide of organisational signs of stress is then displayed, and then there is a brief discussion and question time on these organisational signs of stress.

What Causes Work Stress?

The participants are told that there are two sources of work stress: work/organisational-related factors and individual characteristics. The latter are discussed first, and they are displayed on an overhead projector. They include personality (e.g., optimistic vs. pessimistic), pre-existing psychological well-being (e.g., history of anxiety or depression that predates the current job), social support (e.g., good friends and/or a helpful partner), and home-based factors (e.g., a bad partnership or noisy neighbours). These factors are discussed briefly and then an overhead projector slide of work-related sources of stress (taken from Cooper & Marshall [1976]) is displayed, comprising factors intrinsic to the job (e.g., long hours and work overload), role in the organisation (e.g., conflicting job demands, or lack of clear objectives or duties), relationships at work (e.g., those with superiors, subordinates, and colleagues), career development (e.g., job security, retirement, and stagnation), and organisational structure and climate (e.g., lack of freedom and autonomy).

The participants are told that these workshops will not focus on changing the individual or work-related sources of stress; rather, they will focus on changing how individuals react to these types of stressful events. The participants are told the following:

Consider for a moment that you are a bathroom or kitchen sink. The sources of stress that we have just discussed are like taps that can pour water (or stress) into a sink, and the more taps that flow, the more water there is that is poured into the sink. Now, under most circumstances, a sink will not overflow with water, if it is unplugged, but if it is plugged, the sink will become overwhelmed with water and overflow, causing damage. The goal of these stress-management sessions is not to stop the water from flowing, but, rather, to help you to unplug your sinks so that the stresses that you encounter will not overwhelm you.

Beginning ACT

Since most people believe that undesirable thoughts and emotions (i.e., negative psychological content) interfere with effective and enjoyable living, the participants have probably spent a great deal of effort trying to eradicate this content, and, first of all, examples of this eradication effort are elicited from the participants. For example, they may have tried to avoid, change, justify, rationalise, deny, ignore, or tolerate their psychological content. In order to elicit these examples, the trainer asks a question, such as, “How have you tried to deal with your (anxiety, unhappiness, worries [i.e., unwanted psychological content])?” It is likely that the participants will respond to this question by listing a number of popular and culturally reinforced methods. These might include alcohol, relaxation training, religion, meditation, avoidance, social reassurance, distraction, positive thinking, “analysing” the situation, and maybe even psychotherapeutic techniques that involve changing one’s beliefs.

And How Has That Worked?

ACT facilitates a contest between two main players. On the one hand, there is the mind of the participants. By “mind” it is meant a set of rules and constructions that order the world (e.g., “Men should be hard as nuts”). On the other hand, there is the direct experience of each participant; for, the participants have directly contacted certain contingent relationships in the world; that is, in a given situation, if I do X, then Y will occur. These contingent relationships and one’s “mind” are in fundamental conflict. The goal of ACT is to “blow” the trainees’ minds, so that their experiences can play a greater role in their lives (i.e., so that they can more often make direct contact with contingencies of reinforcement). To accomplish this goal, the concept of “workability” is stressed; that is, the participants are told, “Ask yourself: have your rules delivered? If you do what your mind tells you (if you follow your verbal constructions), are the consequences you actually experience those that the rules specify?” If the answer is “yes”, then it is unlikely that the participants display cognitive, emotional, physiological, and behavioural stress symptoms. Since, for many, the answer to this question is not “yes”, it is likely that their “mind” has not been very

beneficial in helping them to alter the impact of their unwanted emotions, cognitions, and physiological sensations (i.e., their negative experiences).

To determine how helpful their mind has been, the trainer asks how the participants' favoured, experiential avoidance strategies have assisted them in altering the form of, frequency of, or susceptibility to their negative experiences. For example, consider this situation in which a participant wanted to stop worrying needlessly about work:

TRAINER: What have you tried to do to stop worrying?

PARTICIPANT: I've tried to talk myself out of the worry by "thinking things through", relaxing, doing something not related to work—like watch TV or clean my house.

T: Right. Good. Let me ask you this. Your mind says, "Don't worry: relax, watch TV, think things through, and then you won't worry." Right?

P: Right.

T: OK, and how has that worked? For example, as you've done what your mind has told you to do, have you been able to stop worrying or even worry less, and have you then been able to enjoy or be involved fully in what you need to do?

P: Sometimes, but not always, and even if I can stop worrying, it's only very temporarily.

T: Right, so, overall, would you say that you have been able to stop worrying unnecessarily?

P: No, and it's a continuous battle: I spend so much time trying to stop worrying that I become exhausted. Really, it's not unusual for me to get headaches because of all of the worrying that I do.

T: Isn't that interesting? It seems like a paradox, doesn't it? I mean, you do what your mind says: do something to stop worrying: watch TV, think it through, clean your house, but it doesn't work: you still worry.

P: Yes, but what can I do?

T: What does your mind tell you to do?

P: A lot. Relax! You're crazy. Do less work. It's not the end of the world if things go wrong. Take a break, etc.

T: Have you done these things?

P: Of course—I've certainly tried.

T: And how have they worked? Have they paid off in a fundamental way, so that by doing them you have transformed the situation and you are no longer bothered by worrying? Or are you, unbelievably, sinking in deeper, worrying more, feeling worse?

P: I feel like I'm sinking deeper: I can't relax!

T: Incredible, isn't it? I mean if we had an investment adviser with this track record we would have sacked him long ago, but here your mind keeps leading you into efforts that don't really, fundamentally pay off, but it keeps following you around, nattering on, and

it is hard not to give it one more try. I mean what else can you do but go along with what your mind tells you to do? But maybe we are coming to a point in which the question will be “which will you go with? Your mind or your experience?” Up to now the answer has been “your mind”, but just notice what your experience tells you about how well that has worked.

Conversations such as these are conducted with several members of the group, and it is always emphasised that listening to one’s mind is not always effective in relieving the effects of stress, worry, unhappiness, etc. (and it’s not even always effective in helping people to achieve the goals that they wish to accomplish). After three of these conversations, the trainer speaks as follows:

Isn’t this interesting? We do what our mind tells us to do about our thoughts and feelings, but it doesn’t seem to help. In fact, it can even make us feel worse. Why is this the case? Well, human language has given us a tremendous advantage as a species, because it allows us to break things down into parts, to formulate plans, to construct futures we have never experienced before, and to plan action. And it works pretty well. If we look just at the 95% of our existence that involves what goes on outside the skin, it works terrifically. Look at all the things the rest of creation is dealing with and you’ll see we do pretty well. Just look around this room; almost everything we see in here wouldn’t be here without human language and human rationality: the plastic chair, the lights, the heaters, our clothes, the television. So, we are warm, it won’t rain on us, we have light and TV: with regard to the stuff with which non-humans struggle, we are in good shape, really. You give a dog or a cat all these things: warmth, shelter, food, social stimulation, and they are about as happy as they know how to be.

So, the really important things in our life, important to us as a species competing with other life forms on this planet, have been done with human language. In a very significant way, we human beings realise this, and this realisation is encapsulated in a pervasive rule that most all of us buy into: if you don’t like something, figure out how to get rid of it, and get rid of it. Now, this rule works extremely well in 95% of our life. But, not in the world inside the skin: that last 5%. It’s a pretty important 5%, as well, because that is where satisfaction, contentment, and happiness lie. I don’t want you to take my word for this; look at your own life, and in your own experience, not in your logical mind, is it not the case that, in the world inside the skin, the rule actually is, if you aren’t willing to have it, you’ve got it?

For example, this woman struggles with worry; she’s not willing to have it. But, if it is really, really important not to worry or be anxious, and if you then start to worry or be anxious, that is something to worry and become anxious about. Let’s put a name to this phenomenon. Let’s say: in the world outside our body, conscious, deliberate, purposeful control works great. So, figure out how to get rid of what you want to get rid of and do it. But in the areas of consciousness, your life history, self, emotions, thoughts, feelings, behavioural predispositions, memories, attitudes, bodily sensations, and so on, such control often isn’t helpful. In this world under our skin, the solution isn’t deliberate control; the problem is control. If you try to avoid your own history, your own worries, your own anxiety, depression, and their triggers, you are in a no-win struggle. See, what I am saying here is that we can’t control our thoughts and feelings, or anything that happens under our skin, in our mind, in our body.

Consider this. Suppose I had all of you hooked up to the best polygraph that's ever been built. This is a perfect machine, the most sensitive ever made. When you are all wired up to it there is no way you can be stressed or anxious without the machine knowing it. So, I tell you that you have a very simple task here: all you have to do is stay relaxed. If you get the least bit anxious, however, I will know it. Now, I know you want to try hard and I want to give you an incentive to do so, so I also happen to have a .44 magnum pistol which I'll hold to your head. If you just stay relaxed, I won't blow your brains out, but if you get nervous (and I'll know it because you're wired up to this perfect machine), I'm going to have to kill you. Your brains will be all over the walls. So, just relax!—What do you think would happen? Guess what you'd get? Bam! How could it be otherwise? The tiniest bit of anxiety would be terrifying. You'd all be thinking, "Oh my God! I'm getting anxious! Here it comes!" Bam! You're dead meat. How could it work otherwise?

You, of course, have the perfect polygraph already hooked up to you: it's called your nervous system. It is better than any machine humans have ever made. You can't really feel something and not have your nervous system in contact with it, almost by definition. And you've got something more powerful and more threatening than any gun—your own self-esteem, self-worth, the workability of your life. So, you actually are in a situation that is very much like this. You're holding the gun to your head and saying, "relax!" or "don't worry", or "snap out of this unhappiness". So what you then get is—bam!: more worry, more anxiety, more unhappiness, and you throw in a little humiliation, self-downing and feelings of inadequacy to boot. So, see if this isn't true for you: people's thoughts and emotions cannot be effectively, nor always, controlled by their own thoughts and other people's instructions. Sadly, even pleasant thoughts and emotions can't be verbally controlled.

Consider this. What if I pointed out a person to each of you and said that if you fall in love with that person within two days, I'll give you £10 million. Could you do it? Could you fall in love on command, in order to earn lots of money? What if you came back in two days and said, "I did it, pay me the money!", and then I said, "Sorry, it was a trick: I don't have that kind of money." What are you going to do? If you really can fall in love that way, it wouldn't matter if I tricked you, but I suspect you would be awfully mad. In other words, it's not just getting rid of emotions that is difficult, but it is equally difficult to create them, even ones you like, in any kind of controllable, highly predictive, deliberate way.

Willingness as an Alternative Strategy

We've seen how trying to control our thoughts, feelings, and other products of our mind is not a reliable, or even helpful, strategy for reducing our problems. But what is the alternative? It's willingness. If one is willing to have an emotion or thought, and not get rid of it or alter it, then one can escape the inevitable consequences of control; that is, if you refuse to have it, you've got it. This consequence is inescapably the case both because the control strategy doesn't work, and because the strategy itself results in an increase in the events it is designed to prevent. That is, the more you don't want to be anxious, the more anxious you will probably become.

One way to conceive of the different outcomes of willingness and control is through clean versus dirty discomfort. Clean discomfort is the discomfort that we all experience in our lives as a function of living. Clean discomfort varies in level; it might be relatively low at times, as when we feel irritated at someone for putting us down, or it may be high, as when we have a major argument with our partner or we lose a job. Life serves up painful events, and our painful reactions to them are natural and entirely acceptable. It is when we are unwilling to accept these natural reactions—the clean discomfort—that we wind up with what we term dirty discomfort. Dirty discomfort is emotional pain created by our efforts to control the normal, natural clean discomfort that we experience. That is, when we are trying to avoid, control, or get rid of the clean discomfort, a whole new set of painful feelings, emotions, and thoughts appear. This dirty discomfort is the unnecessary addition of pain on top of pain: fear of fear, guilt over guilt, shame over guilt, blame over fear, or blame over unhappiness, as in “you’ve only yourself to blame for being unhappy in this situation”. This simple additive process results inevitably in an increased likelihood that people will use control/avoidance strategies and, thus, carry on in a vicious circle of ever increasing control and pain.

Acceptance, or willingness, involves moving in the opposite direction: towards the pain, rather than away from it; towards the emotions, thoughts, and feelings that we dislike. Consider this. Suppose you were caught in quicksand. Naturally, you’d try to get out. But almost everything you’ve learnt about how to get out will create problems for you: if you try to walk, jump, climb, or run, you just sink in deeper, because you end up trying to push down on the sand. If you struggle, wiggle, push with your hands, or crawl, you sink in deeper. Often, as people sink, they panic and start flailing about, and down they go. In quicksand, the only thing to do is to create as much surface area as possible: to stretch out on the quicksand, getting in full contact with what you’ve been struggling with, but without more struggle. That will be hard: not hard in the sense of effortful, because to apply effort is to struggle; no, I mean hard as in tricky. It is tricky, because your mind is telling you to struggle in a situation where using this strategy is counterproductive; but let’s not be too hard on your mind, as it has been taught that this control strategy works, and it can’t see anything else to do. Furthermore, it has learnt this ineffective control strategy so well that you can’t just tell it to stop using it and expect that it will.

Now, we are going to practise some acceptance strategies; strategies that encourage you to get into full contact with your bodily sensations, thoughts, and emotions without struggling with them; without trying to control them: that is, without trying to make them go away or avoid them.

Willingness Exercise I: “Just Noticing”

I’d like each of you to sit comfortably and close your eyes while we do an exercise. I am going to ask you to “just notice” various things that happen inside your skin: in your body. Your goal in this exercise is to act as if you were watching a film or TV; that is, your goal is to “just notice” what is occurring in your body; it is not to change it, avoid it, or struggle with it in any way: it is just to notice it. Remember, you are watching a film or TV; you are an audience member; you are not the director who controls what will be on the screen; your role is not that of the editor, who takes away scenes that he or she thinks

should not be seen or experienced; and your role is not that of the producer, who finances the film and decides whether or not it will appear at all. Rather, your job, again, is just “to notice” what is actually shown on screen: what your body and mind provide you with.

Now, I’d like you to notice your breathing—see how your breath comes into your body, streams down into your lungs and goes back out of your body again. Remember, do not change how you are breathing, but just notice how you do it. [This breathing observation continues for about two minutes. Meanwhile, the trainer says things like, “if you find your mind drifting away to other things, just gently bring it back to just noticing your breathing.”]

Now I would like you to notice a bodily sensation that you may have right now. Maybe it’s a cramp, a tingling sensation, or a pleasant “warmth” in a muscle. Perhaps it may be in your legs, your arms, your neck, or your back. What I would like you to do is to focus on that bodily sensation and, without trying to stop it or alter it in any way, see whether the sensation stays the same or changes in any way. If it does change, just notice how it changes; if it doesn’t change, notice that as well.” [During the next two minutes or so, the trainer says things like, “if you find your mind drifting away to other things, just gently bring it back to just noticing your bodily sensation.”]

Now, I’d like you to imagine yourself walking through a quiet, comfortable valley that is green and lush. The sun is just how you like it and so is the temperature. As you are walking through this valley, you see and hear a gentle stream and decide to walk over to it. You find a perfect spot and sit down by the stream. While sitting there, I’d like you to look into the stream and notice how the water flows gently and clearly along. Also, I’d like you to notice how a trail of leaves flows down the stream, gently passing you. On these leaves, I’d like you to place any thoughts that you have and let the leaves carry your thoughts down the stream, away from your sight.

[During the five minutes that this part of the exercise is done, the trainer says things like, If you find this difficult to do, that’s all right, just put that thought, “This is difficult”, on a leaf and let it float downstream, as well. If your mind wanders from the leaves and the stream, just bring it gently back to the leaves and place another thought on a leaf. If you are wondering whether or not you are doing the exercise “correctly”, place that thought on a leaf and watch it go down the stream.]

Now, I would like you to picture this room in your mind, see where in the room you are sitting, and imagine what you will see when you open your eyes; and, when you are ready, open your eyes.

[The trainer then asks, “How did you find this exercise?” and if it has not come up during the discussion, the trainer next asks, “How does this exercise relate to what we have been discussing?” As should be evident, this “just noticing” exercise begins to show the participants how they can view and watch their thoughts and bodily sensations, without having to alter them or stop them. The trainer also notes that this exercise is useful to do when people start to feel stress.]

Identifying “Stress Buttons”

The participants are now asked to list on a piece of paper the various “stress buttons” that they have. They are told that these are situations, thoughts, emotions, or sensations that cause them stress. For example, it may be that office confrontations, thoughts of failure, unhappiness, concern, or shallow breathing may trigger a stress reaction. Before they start

writing down these “stress buttons”, the participants are asked to share, with the group, examples of their own triggers, in order to model what is expected. After several people have identified some of their own stressors, questions are elicited, and then participants write down their own “stress buttons”.

Homework

For homework, participants are asked to do the following:

1. Notice, in the week between sessions, how cognitive avoidance, cognitive struggle, and a lack of awareness of what they are thinking (or cognitive fusion) promotes stress, when their “stress buttons” have been pressed.
2. Spend at least 10 minutes each day doing the “just noticing” exercise.

These two homework assignments are written down on paper and handed out to the participants at this time. Any questions regarding the homework are taken and answered.

Willingness Exercise II: Face-to-Face

This exercise consists of simply looking at another person for about two minutes. It may seem longer when you actually do it, but that’s all the time it takes. What the exercise will consist of, if you are willing to do it, is getting a couple of chairs and pulling them close together. Your job is simply to be with the other person. You don’t have to say anything, or do anything, or communicate anything, just be with the other person, experience the other person. Now, your mind will tell you all sorts of reasons why you can’t do that: it will make you feel strange, or maybe you will have a desire to laugh, or maybe you’ll be worrying about how your breath smells, or you’ll be bored or distracted. But the purpose of the exercise is simply to notice these things, to experience everything that you think and feel, and to notice how you sort of come-and-go from being not at all present to experiencing the other person. [As the participants do this, the trainer says things like, “See if you can stay with the reality that there’s another person over here, looking at you. See if you can let go of the sense of wanting to do this ‘right’. If you find yourself thinking about this, or evaluating it, just notice that you’re doing that and then come back out into the room and get in touch with the exercise. See if you can connect with the experience of discomfort in simply being present to another person.” After the exercise, the participants’ reactions are processed, and, if it has not already come up, the participants are asked, “How does this exercise relate to what we have been discussing?”]

At the end of the session, the participants are asked not to discuss with any colleagues the content of this or any other session, or the ideas discussed in it, for six months. This request was made in order to help preserve the internal validity of the Bond & Bunce (2000) experiment in which this manual was first used.

Training Session II (Week 2)

At the beginning of this session, the participants are reminded that any personal information that is revealed or discussed in this or any session is confidential. Anyone not prepared to make this commitment to privacy is asked to leave. In addition, the participants are told that they are free to leave this or any subsequent training session at any time and for any reason, and the fact that they left will not be reported to anyone.

The session begins with a discussion about what the participants noticed, during the previous week, concerning the relationship between avoidance, struggle, and stress, particularly in the areas related to their stress buttons. After this discussion, any problems related to the “just noticing” assignment are considered.

Right Versus Wrong

The Participants are addressed as follows:

If a miracle were to happen, and suddenly all of the stuff that you’ve been struggling with, everything that makes you stressed, would just cease to be a problem, without any change in the circumstances or people or things that are around you, who would be made wrong by that? For example, if your “unreasonable” deadlines and “horrible” colleagues did not change, but now they did not cause you stress, who would be wrong about why you had been stressed? Would it be you, to any extent? That is, you said that you were stressed as a result of your dreadful colleagues, and they are still as bad as ever, yet now you are not stressed. What happened? Were *you wrong* in thinking that your colleagues made you feel so poorly?

I’m not saying that you have to believe me, but consider for a moment the possibility that you yourself, independent of anyone or anything else, have the ability to make your life bigger, richer, less stressful—that, in fact, nothing has to change before that can start. Now, I wish to be clear that I am not saying that you are to *blame* for the stress or painful emotions that you feel, but I am saying that you are responsible; that is, you are *able* to *respond* to stressful situations, such as horrible colleagues, in a manner that can lead you to have a richer, more stress-free life. [The point here is to help the participants see that they can turn away from the justifications that they have been giving for their stress: the deadlines that they have, or their terrible, dictatorial bosses.]

You may think that what I am saying is rubbish, and if you do, there we are, fine: tell me, and we can discuss it. Often, a reason for being unwilling, that is, of wanting control over your thoughts and emotions, is a desire to be right about something (e.g., the reasons why you are stressed: your horrible colleagues). But no one can make you be unwilling (or willing), so you and everyone else have a choice between, on the one hand, being willing and thus enhancing your lives, doing what works for you, and, on the other hand, being right. If you strive to be right, you make decisions, act, and behave in ways that may help you to be right (e.g., I’ll prove that my boss makes our lives hell); but, in so doing, you become a puppet to being right, and your ability to live a full, meaningful life loses out to your greater “need” to be right. So, the choice is this: are you going to be right, or are you going to be alive? If you’re striving to be right, your life gradually

deadens; if you're striving to live your life fully, your desire to be right will only get in the way of living.

What I am talking about can be compared with a traditional psychology experiment that uses a rat in a maze. If an experimenter puts some food at the right side of the maze, the rat will run round and eventually learn to find the food there every time. Now, if the experimenter moves the food to the left side of the maze, at first the rat will run, over and over again, down to the right where the food no longer is; but, eventually, the rat will begin to run in a different direction and will learn to find the food on the left. Now, people are not like rats, because, sometimes, people will keep running to the right *for the rest of their lives*. What they tell themselves may be things like, "It used to be there!" and "It's not fair! It should still be there!" and "It never used to work that way!" Also, people can tell themselves, "Things shouldn't have changed!" and "They never said that I would be expected to do this!" In the interest of being right, these people never learn to do something different, something that would work better for them in their lives. Does anyone have an example of what I mean by how being right can rob people of a more effective and stress-free life?

But

I'd like to talk about the word "but" for a moment. It's a funny little word that can draw us into a struggle with our thoughts and feelings when we use it to explain our behaviour in terms of private events (i.e., thoughts and feelings). In so doing, we end up pitting one set of private events (e.g., a *desire* to succeed) against another (e.g., I feel so *angry* with my boss). "But" literally means that what follows the word contradicts what went before the word. It originally came from the words "be out". When we use it we often say, "This private event be out that private event." That is, "I should be nice to her as she is my boss, but she is a nightmare; so, to hell with her, I'll say what I like!" As you can see, "but" is literally a call to fight, so it is no wonder it pulls us up into the war zone with our own thoughts and feelings, which makes us more stressed, as we saw from last week's assignment.

Let's consider some examples of this "call to arms": "I love my partner, but I get so angry at him." How about this one?: "I want to do a good job at work, but I am anxious or upset right now." Notice that although both say, "One internal event (e.g., love) should be contradicted by another internal event (e.g., anger)", what the person actually experienced in both cases was two things: for example, love *and* anger. The bit of the sentence after the *but*, or "be out", part isn't a *description* of what happened—it is a *proscription* about how private events, thoughts, and feelings should go together. This proscription, however, is exactly what we are trying to back out of. No one *experienced* that two private events have to be resolved; rather, two private events were experienced: love and anger, for example. If the word "but" is replaced by the word "and", the statement is almost always much more reflective of reality. So, in our examples, it is much more accurate to say, "I love my partner *and* I get angry at him", or to say, "I want to do a good job at work, *and* I am anxious or upset right now." Both things are true: the wanting to do a good job

at work and the being anxious and upset. So, I'd like everyone to be aware of this when they use the word "but", and then substitute it for the word, "and", because this switch may make you more sensitive to one of the ways that we get pulled into the struggle with ourselves that makes us more stressed. What are people's thoughts on all of this?

The Observer Exercise

If you would, I'd like you all to close your eyes and follow my voice. Just relax. For a moment now, turn your attention to yourself in this room. Picture the room; picture yourself in this room. Now, begin to go inside your skin, and get in touch with your body. Notice any feelings that are there. Notice any emotions that you are having. Notice any thoughts that you are having. Now, get in touch with the observer: the part of you that noticed the bodily sensations, the feelings, the thoughts. As the observer, hear and follow this.

My body is constantly changing. My body may find itself in different conditions of health or sickness. It may be rested or tired; strong or weak. It started out as a baby and grew continuously, and will continue to change and gradually grow old. I may lose part of my body. It may become fat or thin. Yet, through all of this, the part of me that is observing my body has been constant. I have been me my whole life. Thus, I have a body, yet I don't experience that I *am* my body. When my body changes, I am still me. Focus your attention on this central concept. Allow yourself to realise this as an experienced fact, not just as a thought, or belief, or point of view. Think of all the ways that your body has changed, all of the situations it has been in, while *you* have remained constant. [A period of silence follows.]

Now, consider this: I have roles to play, and yet I am not my roles. My roles are many and constantly changing. Sometimes I'm in the role of a worker, a family member, a friend, or a citizen of a country. I play some role all the time. If I were to try not to, then I'd be playing the role of not playing a role. Even now, part of me is playing a role—the role of a trainee in a work-effectiveness class. Yet, all the while, the observer, the part of me I call "I", is watching. I can play my constantly changing roles, yet all the while I can be there, as a constant, steady observer of it all. So, I have roles, and yet I am not my roles. Allow yourself to realise this as a fact that you have experienced. You know that it is true, and you've known it all along, although sometimes you may forget it; nevertheless, it is how you have *experienced* your life roles, and you are simply allowing yourself to realise that you are observing your own roles.

Now, still as the observer, consider this. I have many emotions. My emotions are countless, contradictory, changing. They may swing from love to hatred, from calm to anger, from stressed to relaxed, and yet *I* have been here through all of these changes and contradictions. Even now, *I* am experiencing emotions—interest, boredom, embarrassment, relaxation. And, throughout, I am capable of observing it all. Though a wave of emotion may come over me, it will pass in time. The observer part of me knows that I am having this emotion, and yet I *am not* this emotion. The emotions that I experience are constantly changing, but the observer remains constant through it all, noticing the changes; thus, I have emotions, but I am not my emotions. Focus your mind on this central concept: I have emotions, but I am not my emotions. Allow yourself to believe this,

because *you* have experienced it, not because *I* have told you so. So, think of things you have liked, and don't like any longer; of fears that you once had that are now resolved. Yet, despite these emotional changes, you experience yourself as a constant. You are there through it all. [A period of silence follows.]

Finally, let's turn to what may be the most difficult area: your own thoughts. Consider this: I have thoughts, but I am not my thoughts. My thoughts are constantly changing. In my life, I have gained new ideas, new knowledge, and new experience. I can think something falsely and then find out the truth and think something entirely different. Sometimes my thoughts are foolish and make little sense. Sometimes thoughts come up automatically, from out of nowhere. Yet, all the while, the observer part of me is seeing these thoughts. The observer part of me knows that I have thoughts, and yet I am not my thoughts.

Allow yourself to realise this as an experienced fact. This is the way that it is, though often we forget it. And notice, even as you realise this, your stream of thoughts continues to flow. Indeed, you may get caught up with them, yet in the instant that you realise that, you also realise that a part of you is standing back, watching it all. So now, watch your thoughts for a few moments, and see that watching happens, and then observe that as well. [A period of silence follows].

So, you are *not* just your body, your thoughts, your feelings, your roles. These things are the content of your life, while *you* are the arena, the context, the space in which they unfold. As you see that, notice how you can distance yourself from the things you've been struggling with, and putting up with. You've been trying to change your roles, to get rid of your "bad" feelings, to control your mind; and the more you do that, the worse it gets, the more entangled you become: the less *you* are even "there". You've been trying to change the content of your life, but you don't have to change your roles, thoughts, feelings, and memories before your life can work, as these things are not *you* anyway; so, why struggle with them? Instead, why not focus your efforts on doing things, right now, that will help you to achieve the goals and dreams that you have for yourself. [A few minutes of silence follows.]

Now, again, picture yourself in this room, and now picture the room [the room is described]. When you are ready to come back into the room, open your eyes.—What were people's experiences of this exercise?" [Analysis of the experience is avoided; rather, the focus is on the experience itself.] This exercise [the trainer continues] demonstrates, among other things, that you, the observer you, can take a direction in your work, in your life, regardless of what your thoughts and feelings are saying to you. Your observer you, the true you, can see what is there, feel what is there, and still say, "This is what I need to do to get where I want to go!"

The "Milk, Milk, Milk" Exercise

TRAINER: Let's do another exercise. It's an eyes-open one for a change. I'm going to ask you to say a word. Then you tell me what comes to mind. I want you to say the word, "milk". Say it once.

PARTICIPANT: Milk.

T: Good. Now, what came to mind when you said that?

P: I have milk at home in the refrigerator.

T: OK, who else? What happens when we say “milk”?

P: I picture it.

T: Good. What else?

P: I can taste it, sort of.

T: Right, and can you imagine what it might feel like to drink a glass? Cold, creamy, coats your mouth, goes *glug, glug* when you drink it? Fine, so let’s see if this fits: what shot through your mind were things about actual milk, and your experience with it; and all of that happened when we just thought of and said the word, “milk”. Incredible, isn’t it? We weren’t looking at or drinking any milk: none at all. But yet we were experiencing milk psychologically. You and I were seeing it, tasting it, feeling it, yet we were only handling the word, not the real thing. Now, here is the little exercise, if you’re willing to try it. It’s a little silly, and so you might feel a little embarrassed doing it, but I am going to do it with you so we can all be silly together. What I am going to ask you to do is to say the word “milk” out loud, rapidly, over and over again and then notice what happens. Are you willing to give it a try?—Right, Let’s go. Say “milk” over and over again. [As the participants do this, so does the trainer, but he periodically interjects comments like: “As fast as you can until I tell you to stop. Faster! Keep going faster!” The exercise lasts for three minutes.]

T: OK, now stop. Did you notice what happened to the psychological aspects of milk that were here a few minutes ago?

P: They disappeared after a while.

P: All I could hear was the sound: it sounded very strange. In fact, I had a funny feeling that I didn’t even know what I was saying for a few minutes. It felt unreal.

P: We all sounded like a flock of birds after a while.

T: Right, the creamy, cold, gluggly stuff just went away. The *first* time you said it, it was as if “milk” were really “meaning-full”; it was almost solid. But, when you said it again and again and again, you began to lose that meaning, and the words began to be just a sound. So, you see, when you say things to yourself, in addition to any meaning behind those words, isn’t it *also* true that these words are just words? The words are just smoke: there isn’t anything solid to them. For example, consider the words, “I’m feeling so stressed”; milk, milk, milk, milk. What’s the difference? When you have a thought or feeling, it looks as though it’s more than what you’ve experienced it to be. It creates an illusion that it is *what it says it is*. Milk, milk, milk, milk, milk. But no matter what the words are, they are just that: words. They are just symbols we experience. Of course, they’re related to things; it’s not that the words are meaningless or will ever be meaningless, I don’t mean that. What I mean is that when the hologram shows up, looking so solid (“I’m feeling so stressed”; “I can’t

cope”; “turn left in the maze”), you are not actually experiencing the *real thing*; that is, thinking “stress” is not the same thing as experiencing stress; saying “milk” doesn’t leave you with milk on your hands. In both cases, you are having a thought, and that’s perfectly acceptable, but saying the word “milk” or “stress” doesn’t mean that you are going to end up drowning in it either; it doesn’t even mean that milk or stress are near you. So, the problem isn’t the word “milk”; the problem is that you think the word is real, that it is not just a hologram; and so, like a drunk person, you begin fighting or running from the polka-dotted elephant. But that’s because you are losing yourself in the meaning of these scary thoughts, these words in your head; you are believing that what they say is true: that there really is milk right there, that a stressful event must be imminent; but, remember, while words can trigger fearful thoughts and feelings, they are, at the end of the day, just words; just symbols of the fear, and not the fear itself; and, since they are just symbols, just holograms, why must they be resisted and fought? [At this point, questions and comments are invited.]

The Tin Can Monster Exercise

[Each participant is given an audio cassette with this exercise on it.] Over the last two sessions, we’ve talked about being willing to experience our thoughts and feelings as what they really are, that is, thoughts and feelings, and not what they say they are (e.g., danger and loss). Based upon this concept of willingness, I would like you to consider a very important question: *out of the place from which there is a distinction between you and “that” (“that” meaning the stuff you struggle with, the thoughts, feelings, or evaluations that come forward when you’re doing something), are you willing to feel that, think that, experience that as it is, not as it says it is, and do what works for you in the situation?* This question is the core of what we have been doing. It is a question that we can never stop asking ourselves, because willingness is not an outcome; it is a process. Willingness is a choice to do something and, in that context, to have happen whatever it is that is going to happen.

To be willing can often be like facing a giant, 30-foot monster who is made up of tin cans and string. It is almost impossible to face it willingly. However, if we disassemble him into all the cans, string, and wire that he’s made of, each of those individual pieces is easier to face up to. The exercise that we are now going to do, if you are willing to do it, is intended to help you give up the struggle with emotional discomfort and disturbing thoughts by disassembling them. It is also designed to give you the experience of the natural ebb and flow of willingness, to realise that it is not something that you will “get” and “have” forever.

Now, if you are willing, get comfortable and relaxed, shut your eyes, and think of a specific feeling or situation. The situation should be something that you are currently struggling with, something that is important to you. I want you to try to get in touch with the feeling or situation that you’ve chosen. Notice first if there are any bodily sensations that are associated with what you’ve chosen. As you think about it, just gently notice what your body is feeling, what your body is doing, and if you notice some specific

reaction, just pull it up and look at it. Now, I want you to spend some time letting go of the struggle with that specific bodily feeling. The goal here is not that you like the feeling, but that you are willing to have it and not struggle with it. Good. Now I want you to look for another bodily sensation attached to this situation. [The trainer asks the participants to repeat this procedure for bodily sensations three times.]

Right, I want you to notice whether there are any emotions associated with the feeling or situation that you selected. Just gently put yourself in the situation, and notice whether there are any emotions that appear, and, if so, just gently turn your attention to the first emotion that you noticed. I want you to spend some time just trying to let go of the struggle with that emotion. [This formula is repeated three times for emotions, thoughts, and behavioural predispositions (e.g., wanting to run away). After each repetition, the participants are told “good” in response to their efforts to give up the struggle with each reaction.]

Right, now for the last part. I want you to imagine that you have all the memories of your life on little snapshots, like the index cards filed away in a filing cabinet, all the events of your life from your birth until the present. I want you to get back in touch with the situation that you have selected, and now I want you to open the filing cabinet and start gently flipping through your cards of memories. You can either flip back from the present time to your past, or start as far back as the memories go, and start flipping towards the present, but if you find yourself pausing at any picture, stop flipping and spend a few minutes with the memory. It doesn't matter whether or not it seems to make sense as to why you stopped at that picture; if you find yourself stopping at any point, just stay there for a moment. As you stop, just notice who else is in the picture. What were you feeling and thinking at the moment? What were you doing? Now, see if you can let go of the struggle that may be associated with the memory; it might be a pain associated with the picture, or an unwillingness to leave it because of the happiness associated with it; whatever your reactions to the memory, just see if you can gently let go of your struggle with it, if you can let yourself be willing to have that memory exactly as it is, have exactly what happened to you as it happened. That doesn't mean you'll like it, but that you are willing to have it [Three more memories are explored.] Right, when you're ready, I want you to close the filing cabinet, and picture the room as it was when you shut your eyes and began the exercise. When you can picture it, just open your eyes and come back to the present. Does anyone want to share any of their experiences?

During the next week, I should like you to play this tape and do this exercise once daily. When you do it, please use as your material the “stress buttons” that you wrote down last week. After you have done it every day for one week, I would like you to do it one time each week until we next meet in three month's time.

Letting Go of the Struggle, in Order to Achieve your Values and Goals

If you are willing, I would like to go around the room and have each person tell us what they value, what they wish to do and achieve, that has been interfered with by the struggles that they have had with their stress. Then, I would like each person to commit publicly to let go of these needless struggles that are interfering with their values; but I

only want you to make such a commitment, if you are really prepared to choose to give up this struggle. If there are values that you have, goals that you wish to achieve, that you feel unprepared to state here, try stating them to your partner, or a close friend or family member; and if you find that you are not willing to do that, at least state them to yourself in the mirror. The goal here is to make as public a commitment as you can to letting go of the struggle that is getting in your way of achieving the values and goals that you have.

After this, the trainer asks for comments and answers any final questions. The participants are also reminded again about the homework.

At the end of the session, the participants are asked not to discuss with any colleagues the content of this or any other session, or the ideas discussed in it, for six months. Again, this request was made in order to help preserve the internal validity of the Bond & Bunce (2000) experiment in which this manual was first used.

Session III: The Three-Month Follow-up (Week 14)

This follow-up session is designed as a “booster session”, in that its aim is to practise the willingness techniques that were covered in the first two sessions and address any questions and comments that the participants have.

At the beginning of this session, the participants are reminded that any personal information that is revealed or discussed in this or any session is confidential. Anyone not prepared to make this commitment to privacy is asked to leave. In addition, the participants are told that they are free to leave this training session at any time and for any reason, and the fact that they left will not be reported to anyone.

Firstly, the **“just noticing”** exercise done in session I is repeated. After this, the participants are asked to reveal the biggest hooks, or internal events, that they had a hard time letting go of. The **tin can monster** exercise is then done, using these hooks as the material. After comments and questions are elicited about this exercise, the **face-to-face** exercise is done again. After the trainer deals with comments and questions about it, a **group-sharing** exercise is done. Specifically, all participants, if willing, are asked to stand up in front of the group, one by one, and state which of their values or goals have been interfered with by stress struggles. Then, with a view towards achieving, or living, these values and goals, they are asked to declare what their intentions are in the area of (a) letting go of needless struggle, and (b) letting go of “right and wrong” in work and home.

After this exercise, the **observer** exercise from session II is repeated, and any final questions and comments are addressed. Lastly, the participants are encouraged to continue to employ willingness exercises at least three times a week; and if they choose to do these exercises, hopefully they will do so, not because the trainer is recommending it, but because their experience tells them that acceptance of negative thoughts and feelings is less stressful than avoidance of them. In addition, the participants are reminded that acceptance is a process; it is not an end state: no one will reach it permanently. It is necessary, therefore, constantly to be mindful and aware that thoughts and feelings are just that: thoughts and feelings, and not the events that they represent (i.e., internal events are not to be taken literally).

THE FUTURE AND ROLE OF ACT AT WORK

Due to the initial success of this brief, group-based, ACT protocol, we and our colleagues are currently testing its efficacy in several other (i.e., non-media-related) industries (e.g., education, advertising, and banking). In so doing, we are examining, in particular, ACT's ability to reduce stress-related outcomes that are of vital concern to organisations; these include absenteeism, turnover, and impaired performance. We believe that it is crucial to demonstrate that this intervention can improve such work-related outcomes, *in addition* to mental health ones. Failure to show such work-related improvements is likely to limit severely the number of organisations (and industry sectors) that would use ACT, or any other individual-focused intervention, as a means of improving their employee's mental health.

While we are confident of ACT's ability to reduce work-related stress, we do not believe that organisations can responsibly use this, or any other type of individual-focused intervention, as their sole strategy for dealing with occupational stress. We agree with the UK health and safety legislation (see Cox et al., 2000) that holds organisations responsible for identifying and reducing stress-related risk factors that centre on the design and management of work (e.g., low levels of control, role conflict, and repetitive work). Thus, we consider individual-focused interventions, such as ACT, important adjuncts to a more comprehensive risk management approach to occupational stress. Indeed, we hypothesise that the combination of work redesign and ACT may serve as the most comprehensive and effective strategy for improving mental health and work-related outcomes. We expect that future research will investigate this possibility.

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The Cognitive Behaviour Framework

A Whistle-Stop Tour

This chapter will briefly outline the basic principles and practices of cognitive behaviour therapy. A central principle of cognitive behaviour therapy is that thoughts, emotions, behaviours and physiology are part of a unified system. A change to any one part will be accompanied by changes to the other parts. For example, if a person's television exploded as she watched it, she may experience immediate physiological changes (a surge of adrenalin); a rapid shift of behaviour (from calmly watching the screen to rushing for a fire blanket or to telephone for help); a feeling of anxiety and thoughts or cognitions such as '!!*!!*: the house is burning I'm going to die'. Cognitive behaviour therapy recognizes the unified nature of this process. The scenario presented here does not represent a psychological problem, but where psychological problems exist, change is sought through focus on unhelpful thoughts. It is important to stress that components such as emotions and behaviours are not ignored or considered unimportant and we will demonstrate in later chapters how these are given full consideration in a process which brings about therapeutic change.

In reading about the exploding television, you may have thought that you would not have experienced a rush of adrenalin or felt or acted in the same way. Cognitive behaviour therapy accounts for this by suggesting that each person brings to a situation a different range of feelings, physiological responses and behaviours, and that thoughts guide these. Feelings and responses are not caused intrinsically by the situation itself, but largely by the way in which each of us views it. This was realized at least as long ago as the first century AD when the stoic philosopher Epictetus noted that people 'are disturbed not by things but by the views which they take of them'. Well known theorists and creators of various therapeutic approaches have taken a similar view, including Arnold (1960); Beck (1964); Ellis (1962, 1995); Kelly (1955); and Lazarus (1997).

Beck developed cognitive therapy and his work initially centred on depression (1963). The central themes of cognitive behaviour therapy are:

- that thoughts can trigger emotions and behaviour;
- that emotional disorders arise from negatively biased thinking (which leads to unhelpful emotions and behaviour);
- that emotional disorders can be helped by changing such thinking (which is assumed to be learned).

Two components of thinking are focused upon in cognitive behaviour therapy: automatic thoughts and underlying beliefs.

Automatic thoughts

'Automatic thoughts' was a term given by Beck to the thoughts and images which occur involuntarily in a person's stream of consciousness (Beck and Greenberg, 1974). Similar terms are 'internalized statements', 'self-statements' or 'things you tell yourself' (Ellis, 1962), and 'self-talk' (Maultsby, 1968).

Underlying beliefs

These are the beliefs and assumptions which generate the thoughts and images forming the content of automatic thoughts. The relationship between automatic thoughts and underlying beliefs can be better understood by briefly considering the idea of schemas. Schemas are abstract mental plans that serve as guides to action, as structures for remembering and interpreting information and as organized frameworks for solving problems. Each of us adopts a vast range of schemas which allow us to make sense of the world and to place any new information or experiences into context (e.g. see Rosen, 1988; Rumelhart, 1981; Rumelhart and Ortony, 1977; Young et al., 2003). They are like vast filing systems which each of us uses to order the world and which begin to be formed from early childhood. Once formed, they guide information processing and behaviour (Bartlett, 1932) and shape how a person thinks, feels and behaves concerning self, others and the world. Importantly, they are hierarchical. For example, matter can be perceived as animate or inanimate; animals as vertebrates or invertebrates and so on (see Figure 2.1).

Similarly, in the case of the exploding television, a person who holds an underlying belief (i.e. a belief further up the hierarchy) such as 'fire always causes death' is likely to feel panic stricken and beset with automatic thoughts such as 'I am going to die', accompanied by negative images such as a charred body engulfed by fire. The person who holds the belief that 'fire is dangerous if not controlled' may feel sufficient concern to act appropriately and may experience automatic thoughts such as 'cut the electrics and get help now', accompanied by helpful and constructive coping imagery. A person who holds a belief at the top of the hierarchy, such as 'life is dangerous', will be predisposed to anxiety in a wide range of situations. A belief such as this, at the top of a hierarchy and affecting a large portion of a person's life is known as a core belief.

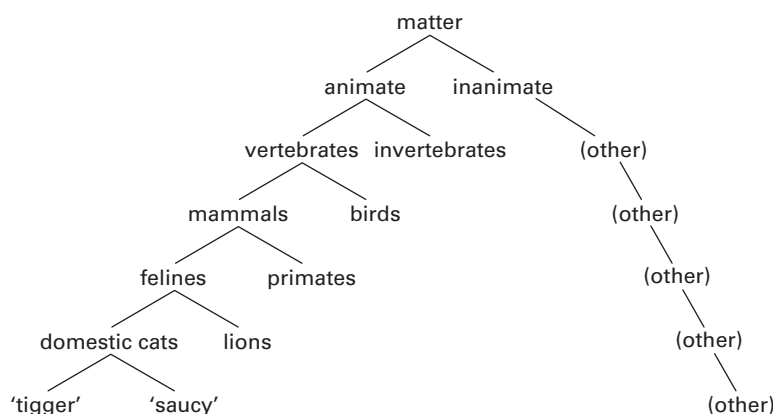


Figure 2.1 Schema representation

We follow Beck's view that the schema is the cognitive structure (or 'filing system') within the mind, whereas the core belief is the specific content (Beck, 1964), but a number of practitioners use the two terms interchangeably (Padesky, 1994). Core beliefs operate at a rudimentary level and are global, rigid and over-generalized.

Intermediate beliefs

Between core beliefs and automatic thoughts lie a number of intermediate beliefs (Figure 2.2). Intermediate beliefs are composed of attitudes, rules and assumptions:

Attitude: It is dreadful to be in danger.

Rule (expectation): I must always be safe and act with caution.

Underlying assumptions: If I always act with caution then life will be less dangerous.

If I don't act with caution then life will be dangerous. (The corollary assumption.)

We have briefly outlined the cognitive structures that hold the key to psychological difficulties which individuals experience. How does a cognitive behaviour therapist help a client to overcome these difficulties through brief therapy? Obviously, basic counselling skills are a prerequisite and are identified by Truax and Carkhuff (1967) as follows:

- 1 Be accurately empathic.
- 2 Be 'with' the client.
- 3 Be understanding, or grasp the client's meaning.
- 4 Communicate empathic understanding.
- 5 Communicate unconditional positive regard.

<i>Core beliefs</i>	<i>Intermediate beliefs</i>	<i>Automatic thoughts</i>
Global	Attitudes	Stream of thought
Rigid	Rules/expectations	Images
Over-generalized	Assumptions	Words/phrases

Figure 2.2 Schemas

While these are basic counselling skills, there are a number of characteristics fundamental to cognitive behaviour therapy which enable the therapist to help the client by explaining, exploring and building upon the principle of thoughts leading to feelings and behaviours and some bodily reactions, as described above. Some therapists find it useful to symbolize this principle as ABC (Ellis, 1962, 1977, 2003), as indicated below:

- A = *Activating event*: e.g. television exploding.
- B = *Belief*: e.g. 'I'm in grave danger'.
- C = *Consequence*: e.g. anxiety.

The A (activating event or adversity) may be an actual event (such as the exploding television), an experience or even a thought, daydream, image or emotion about which

a person might have thoughts or beliefs. The B (belief) may be a core belief, an underlying assumption, an intermediate belief or an automatic thought. ‘Many times, maladaptive ideation occurs in a pictorial form instead of, or in addition to, the verbal form’ (Beck, 1970). The C (consequence) may be anything the thought or belief (B) leads to, such as an emotion (fear), a behaviour (running away) or a physiological response (palpitations).

Cognitive behaviour therapy progressively encourages the client to recognize and accept their emotions, to detect first their automatic thoughts and then the related underlying beliefs (both intermediate and core). Only if the connection between thoughts (beliefs) and emotions (consequences) is recognized can further progress be made through cognitive behaviour therapy. Once this process is started the client is encouraged to look for evidence in support of unreasonable and unhelpful beliefs. The inability to find such evidence poses a challenge to such unhelpful beliefs, which are in turn transformed into more adaptive, helpful beliefs. These new and more realistic beliefs no longer support the distressing emotions and behaviours. To aid this process, an *automatic thought form* is commonly used (see item 7, p. 18, and Figure 2.7, p. 19). Beliefs and emotions are rated to aid belief change and evaluation over time (see scaling beliefs, p. 69).

Thinking Errors

Thinking errors are also known as cognitive distortions or twisted thinking (Burns, 1980, 1989). Identifying automatic thoughts and recognizing the thinking errors contained within them play major roles within cognitive behaviour therapy. Individual clients tend to make consistent errors in their thinking. The therapist helps the client to identify these errors or cognitive distortions. Clients are introduced to the notion that when people feel emotionally distressed they often have accompanying thoughts which seem believable at the time but which on closer scrutiny are not always consistent with objective reality and are unhelpful to the client. Most of the negative thoughts which disturb a client are distorted and unrealistic. A client will be taught how to identify the thinking errors which lead to negative moods. It is important to convey to the client that they are not alone in having these distortions. They are common to humankind but proliferate with emotional distress. We identify the most common thinking errors in Figure 2.3.

1 All or nothing thinking (also known as dichotomous thinking)

This is where a client evaluates herself, other people, situations and the world in extreme categories. This type of thinking tends to be absolutist and does not allow for shades of grey. An example is the young mother who views herself as all bad because she becomes frustrated with her young child and views other mothers as always being patient with their children. So she sees other mothers as all good and herself as all bad, which is unrealistic.

2 Personalization and blame

Personalization is a thinking error in which a person totally blames himself for all that goes wrong and relates this to some deficiency or inadequacy in himself. He holds himself personally responsible for an event which is not entirely under his control.

- 1 All or nothing thinking
- 2 Personalization and blame
- 3 Catastrophizing
- 4 Emotional reasoning
- 5 Should or must statements
- 6 Mental filter (selective abstraction)
- 7 Disqualifying or discounting the positive
- 8 Over-generalization
- 9 Magnification and minimization
- 10 Labelling
- 11 Jumping to conclusions (arbitrary inference)
 - Mind reading
 - Fortune telling

Figure 2.3 *Thinking errors*

An example is the young trainee who believes her trainer is brusque with her because she made a mistake. She overlooks the part others may play and confuses the possibility that she may have contributed to what happened but was not totally responsible. The opposite is to blame others. She blames others for her problems or circumstances and does not believe she has contributed to the problem. An example is the man who totally blames his wife for the break-up of their relationship.

3 Catastrophizing (fortune telling)

This is where the person predicts the future negatively and believes things will turn out badly. This thinking error is common with anxiety problems (Ellis, 1962) where clients tend to dwell on the worst possible outcome of any situation. An example is the successful manager about to make a presentation for the company. She becomes preoccupied with thoughts that she will make a mess of her presentation, let down her company, lose her job and become destitute.

4 Emotional reasoning

This is where a person draws conclusions about an event based entirely upon their feelings and ignores any evidence to the contrary. For example, the young man who has been waiting 30 minutes for his new partner to arrive feels sad and rejected. He says to himself, 'I've been ditched' and fails to consider that his partner may have been delayed at work, missed the bus, got a flat tyre, etc.

5 Should or must statements

This is where the person has a fixed idea of how she, others or the world 'should' or 'must' be. Preferences or expectations are elevated to rigid demands. When these demands are not met the person feels emotionally distressed and over-estimates how bad it is that her expectations have not been met. For example, the gymnast performing a difficult manoeuvre on the parallel bars said to herself, 'I really shouldn't have made so many mistakes'. This led to her feeling so angry and frustrated with herself that she did not practice for many days.

6 Mental filter (selective abstraction)

The person pays particular attention to one negative detail and dwells on it endlessly, regardless of any other positive aspects. He does not view the picture as a whole and only concentrates on one negative aspect. For example, an architect receives many positive comments about how he tackled a difficult work project. However, one colleague was critical about one aspect of the project. This comment played on his mind for a couple of days.

7 Disqualifying or discounting the positive

This is where the client ignores the positive of any situation and tells herself that these positive experiences do not count. For example, a father cooks good meals on most occasions for his family, but does not give himself any praise. He produced a nourishing but unappetising meal on one occasion and thought of himself as being an awful and unimaginative cook.

8 Over-generalization

The person thinks that because an unpleasant experience happened to her once, it will always happen. This is where she makes sweeping generalized conclusions on the basis of one situation. For example, a woman who attended a job interview but did not get the position believed she would be rejected for every job.

9 Magnification and minimization

The person who makes this thinking error when evaluating himself, other people or situations will tend to exaggerate or magnify the negative components and minimize or play down the positive. When being appraised at work he over-estimates the importance of some areas where change is needed and pays little attention to a considerable range of positive aspects highlighted by the appraisal. He concludes that this shows he is inadequate.

10 Labelling

This is where the person views herself or others in all or nothing terms but goes beyond this by applying a label which is usually derogatory. For example, the mother we referred to in item 1 might label herself 'a heartless bitch'. When this thinking error is applied to others a client dislikes or disagrees with, he may say to himself, 'He's an arse'. The client will see the person as globally bad and may then feel angry and hostile. The client who makes an error at work may label herself as 'totally stupid'.

11 Jumping to conclusions (arbitrary inference)

A person whose thinking is distorted in this way infers that a particular outcome will be negative, without having any evidence, or even if the evidence points to a positive outcome. There are two main types of this thinking error:

(a) Mind reading

The client thinks she knows what others are thinking and does not consider other more plausible or likely possibilities. An example would be the client with social anxiety who thinks her work colleagues see her as inadequate in a wide range of situations.

(b) Fortune telling

A person predicts that events in the future will turn out badly. For example, a person attending a routine chest X-ray assumes he has cancer (see ‘Catastrophizing’ above, where fortune telling is greatly exaggerated).

Clusters of thinking errors

It is important to note that a particular automatic thought, belief or inference may contain not one, but several of these thinking errors and we will exemplify this later. In introducing these thinking errors, we noted that an individual may tend to make consistent errors in their thinking. For example, a particular person may frequently use a cluster such as jumping to conclusions, ‘should’ statements, emotional reasoning and magnification/minimization. Similarly, emotional problems such as anxiety, depression or guilt each tend to have a prominent cluster of cognitive distortions or thinking errors surrounding them. Figure 2.4 gives the clusters of thinking errors common to particular emotional problems. This can be used as an aid to help the therapist to identify the most significant thinking errors for a particular problem. However, it is important to note that the clusters which a particular individual brings to a specific emotional problem will frequently have an idiosyncratic element.

Anxiety panic, nervousness	Depression	Anger	Guilt	Hurt	Morbid jealousy	Shame/ embarrassment	
	✓	✓		✓		✓	All or nothing thinking
	✓	✓	✓	✓		✓	Personalization/blame
✓	✓				✓	✓	Catastrophizing
✓	✓			✓			Emotional reasoning
✓	✓	✓	✓	✓	✓		‘Should’ or ‘must’ statements
	✓		✓		✓	✓	Mental filter
✓	✓						Discounting the positive
✓	✓	✓					Over-generalization
✓	✓			✓			Magnification/ minimization
		✓	✓			✓	Labelling
							Jumping to conclusions
✓	✓	✓		✓	✓	✓	(a) Mind reading
✓	✓					✓	(b) Fortune telling

Figure 2.4 Thinking errors for common problems

Fundamental Characteristics of Cognitive Behaviour Therapy

So far in this chapter, we have outlined the mechanisms by which a person experiences emotional difficulties or distress. In the next section we will guide you through the

fundamental characteristics of cognitive behaviour therapy by which this model is applied to the client's problems. Figure 2.5 is a list of the fundamental characteristics of cognitive behaviour therapy.

- 1 Therapeutic style
- 2 Formulation of problem
- 3 Collaborative relationship
- 4 Structure to sessions and to therapy
- 5 Goal-directed therapy
- 6 Examines and questions unhelpful thinking
- 7 Uses range of aids and techniques
- 8 Teaches client to become own therapist
- 9 Homework setting
- 10 Time limited
- 11 Audio-recording sessions

Figure 2.5 *Fundamental characteristics of cognitive behaviour therapy*

1 Therapeutic style

We start with therapeutic style because this is often quite different in cognitive behaviour therapy to some other forms of counselling. In cognitive behaviour therapy, it is assumed that the therapist brings to the therapeutic encounter a range of skills and knowledge. From this it follows that the therapist, particularly in the early stages of therapy, directs the course of therapy. The therapist's style is therefore active and directive. The therapist is sensitive to balance this with communicating empathic understanding and being 'with' the client, together with grasping the client's meaning (see items 2, 3, 4, pp. 10–11). Part of this style is about the ability to elicit information appropriate to the client's problems that will include the client's cognitions. The therapist also seeks out client strengths that may aid the change process.

2 Formulation of problem

The therapist gathers a wide range of information about the client and her problems. This is based not only on verbal information from the client but also on the therapist's own observations. A full assessment is carried out early on in therapy (discussed more fully in Chapter 3) and a *cognitive formulation or conceptualization* of the client's problems is made. This is constantly revised in the light of new information. In particular, the therapist identifies current thinking associated with and maintaining problematic behaviours and emotions, precipitating factors connected to the current problem and reinforcing unhelpful beliefs, and the client's unhelpful interpretation of major developmental events. The cognitive conceptualization is more fully discussed in Chapter 3.

3 Collaborative relationship

The therapist is open about his formulation and usually shares it with the client at some point for discussion together and for further development. Unlike some forms of therapy, cognitive behaviour therapy is transparent and does not present therapy as magical.

It is psychoeducational and informs the client (in easily understood language) about the process of therapy. To this end, reading material (i.e. bibliotherapy) is often used (Lazarus, 1971; Macaskill and Macaskill, 1991). The therapist socializes the client into such aspects as recognizing automatic thought, or in particular, negative automatic thoughts (NATs) and other unhelpful beliefs, and recommends certain courses of action such as homework, but encourages the client to participate fully and to take over responsibility for many aspects of therapy from an early stage.

The openness of therapy is further aided by client and therapist setting an agenda together for each session, as well as giving and receiving feedback. Through such collaboration and by using other therapeutic skills not specific to cognitive behaviour therapy (outlined above), the therapist quickly establishes a therapeutic relationship with the client. With some clients who have deep and far reaching psychological problems (as may be the case when working with people with personality disorders) it may be necessary to take more time and care in ensuring this is firmly established before entering too far into therapy, as outlined elsewhere in this chapter. Where this is so, brief therapy may not be appropriate unless a discrete and limited problem area is being worked with (see Beck et al., 1990a).

4 Structure to sessions and to therapy

As discussed in the last section, sessions are structured by an agenda, the contents of which is collaboratively agreed upon by therapist and client. The agenda has a common format applied across all sessions except the first. Although it need not be slavishly adhered to, the novice therapist is best advised to follow it during their course of training. The use of the agenda has several benefits which become more pronounced in brief therapy. It allows the client and therapist to use their limited time most efficiently. It helps to ensure that important material which client or therapist wish to bring to the session is not left to the end or entirely forgotten. It also aids collaboration by encouraging a problem-solving and business-like attitude (rather than encouraging a 'patient' or 'sick' role). The agenda also provides a convenient framework for monitoring progress throughout therapy. It gives a structure to help the client to understand the central themes of cognitive behaviour therapy and to apply this structure to herself once therapy is terminated. The structure of a typical session is as follows:

- 1 Check client's mood.
- 2 Brief review of week.
- 3 Set agenda for current session.
- 4 Feedback and link to previous session.
- 5 Review homework.
- 6 Discuss agenda items.
- 7 Negotiate homework.
- 8 Seek feedback at end of session.

The structure of therapy has a number of elements which are concerned with effectiveness and efficiency. These elements of cognitive behaviour therapy are relatively constant over its course, but some aspects change as therapy progresses. Cognitive behaviour therapy recognizes that different stages are involved and optimizes this

development by incorporating relevant tasks and responsibilities (for client and therapist) at the pertinent stage. For example, identifying negative automatic thoughts will be central to the start of therapy, while modifying core beliefs will usually not be attempted until later in the process. In brief therapy the therapist may also alert the client to the positive automatic thoughts (PATs) and helpful intermediate and core beliefs he may already be using in many situations. The structuring of therapy in this way allows for the termination of therapy, relapse prevention and booster sessions to be planned in advance, and obstacles to be confronted and overcome. The structure of therapy will also be determined by the particular problem with which the client presents. For example, if a client is diagnosed in accordance with the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5) of the American Psychiatric Association (APA, 2013) as having a particular disorder, the structure indicated in Chapter 10 (Treatment Protocols) may be followed.

5 Goal-directed therapy

We have already described the way in which the therapist and client collaboratively work together. When the client brings material to the session, the therapist encourages her to put this into behavioural terms. For example, if the client says that she feels depressed, the therapist will check out with her how this depression manifests itself in her. If she says, among other aspects, that she now seldom socializes, that she no longer keeps a healthy diet or that she has abandoned activities that she previously found enjoyable, the re-establishment of these tasks may become specific goals for therapy. Similarly, anyone who seeks therapy seeks change in some way: the therapist will work collaboratively with the client to decide on the changes which are germane to therapy. Goals will be decided upon collaboratively, together with the tasks necessary to bring about such goals. There may be occasions when the client initially wants to achieve a goal which is counter-productive to her progress; for example, where she seeks goals consistent with the unrealistic belief that she must be perfect in every way. In such cases, the therapist will discuss this frankly with her. It is only through appropriate goal setting that brief therapy is possible. Realistically, brief cognitive behaviour therapy is 'limited goal therapy' (see Lazarus and Fay, 1990) and if a client has many therapeutic goals then he or she may be unsuited to a brief intervention.

6 Examines and questions unhelpful thinking

The cornerstone of cognitive behaviour therapy is that thoughts guide behaviour, emotions and (in some cases) physiological responses. A person will need to alter their thinking if they are to overcome various psychological problems. Depending on the nature of the problem, this may entail working on automatic thoughts, intermediate thoughts or core beliefs. Unlike some forms of cognitive therapy such as rational emotive behaviour therapy – REBT (Ellis, 1994), cognitive behaviour therapy does not confront unhelpful thinking head on, but uses a process of socratic questioning whereby the client is encouraged to look at the evidence for holding his unhelpful (unrealistic or negative) beliefs (Beck et al., 1979: 66–71). It is important to check out the meaning of a client's beliefs before attempting to work on them. This becomes particularly important when working across cultures (Ruddell, 1997: 22–4). Another major difference between rational emotive behaviour therapy and cognitive behaviour therapy is that the

latter works mainly at an inferential level during the beginning stage of therapy, while REBT works mostly at an evaluative level. For example, consider the person mentioned when we viewed catastrophizing (p. 11). The successful manager drew the inference that she would ‘make a mess’ of her presentation. The cognitive behaviour therapist may invite her to look at the evidence for this inference; if the evidence were lacking, this would remove the foundations on which her other beliefs (about letting her company down, losing her job and becoming destitute) are founded. However, the rational emotive behaviour therapist would prefer to focus on the evaluations which may ensue from such an inference such as ‘I must not make a mess of the presentation, and if I did, I couldn’t stand it’.

A list of questions about thoughts and beliefs from which the therapist may draw has been developed by Palmer and Dryden (1995) and Palmer and Strickland (1996). They are reproduced in Figure 2.6 and are best used in the context of the process of cognitive behaviour therapy outlined throughout this book. These questions are also given in Appendix 1 as a *self-help aid*, which the client may use themselves.

- Is it logical?
- Would a scientist agree with your logic?
- Where is the evidence for your belief?
- Where is the belief written (apart from inside your own head!)?
- Is your belief realistic?
- Would your friends and colleagues agree with your idea?
- Does everybody share your attitude? If not, why not?
- Are you expecting yourself or others to be perfect as opposed to fallible human beings?
- What makes the situation so terrible, awful or horrible?
- Are you making a mountain out of a molehill?
- Will it seem this bad in one, three, six or 12 months’ time?
- Will it be important in two years’ time?
- Are you exaggerating the importance of this problem?
- Are you fortune telling with little evidence that the worst case scenario will actually happen?
- If you ‘can’t stand it’ or ‘can’t bear it’ what will really happen?
- If you ‘can’t stand it’ will you really fall apart?
- Are you concentrating on your own (or others’) weaknesses and neglecting strengths?
- Are you agonizing about how you think things should be instead of dealing with them as they are?
- Where is this thought or attitude getting you?
- Is your belief helping you to attain your goals?
- Is your belief goal-focused and problem-solving?
- If a friend made a similar mistake, would you be so critical?
- Are you thinking in all or nothing terms: is there any middle ground?
- Are you labelling yourself, somebody or something else? Is this logical and a fair thing to do?
- Just because a problem has occurred does it mean that you/they/it are ‘stupid’, ‘a failure’, ‘useless’ or ‘hopeless’?
- Are you placing rules on yourself or others (e.g. shoulds or musts, etc.)? If so, are they proving helpful and constructive?
- Are you taking things too personally?
- Are you blaming others unfairly just to make yourself (temporarily) feel better?

Figure 2.6 Questions to help examine unhelpful thinking

7 Uses range of aids and techniques

We have already noted that a central task of the cognitive behaviour therapist is to arrive at a cognitive conceptualization of the client's problems. The conceptualization, which may be developed over time, guides the course of therapy. This leaves the therapist reasonably free to import a range of techniques and aids to help bring about a change in the client's unhelpful thinking, which will in turn lead to the client overcoming his emotional and behavioural problems. However, any such techniques and aids are firmly embedded within the framework and principles of cognitive behaviour therapy rather than being 'add ons'. An example of a technique is the use of behavioural experiments with panic disordered clients, such as voluntary hyperventilation (Clark, 1986). Aids include a wide range of instruments – questionnaires – (see Ruddell, 1997) to measure aspects of a client's problem, such as Rapid Assessment Instruments (RAIs; see Corcoran and Fischer, 1987). Probably the most commonly used aid in cognitive behaviour therapy is the Beck Depression Inventory (BDI; Beck et al., 1961; Beck and Steer, 1987). Another aid is the *automatic thought form*, of which there are many variations. Our own form (completed) is shown in Figure 2.7 and discussed previously (p. 10). A blank version is shown as Appendix 2.

8 Teaches client to become own therapist

One of the main reasons cognitive behaviour therapy is open, structured, collaborative, relatively brief, goal directed and uses homework is to hand over the process of bringing about change to the client before therapy is terminated. The scope and fluency of the skills the client learns will depend in part on the nature and severity of the client's problem, which in turn will influence the length of therapy. A crucial factor in the client becoming his own therapist is his active participation in the process of therapy within formal sessions as well as in assignments outside them.

9 Homework setting

The average length of a therapeutic session is about one hour. If a client only focuses on his problems during the session, this means that within a week 167 hours will not be put to good therapeutic use. In turn, this leads to a lack of continuity in working with a particular problem because the therapeutic process of recording, monitoring and assessing thoughts will not be easily carried into daily living. A useful analogy here is that of maintaining physical health: people do not usually object to cleaning their teeth, bathing, washing their hair regularly and so on – in just the same way it is important to allot time to maintaining our mental health. A problem sometimes encountered with the setting of 'homework' is the word itself. Some people connect it with schoolwork, which in turn may have negative associations. It is important to check this out with a client and if this is the only obstacle, to use a term, preferably chosen by him, which does not have such negative connotations. 'Assignment' is a commonly used alternative. In passing, such thoughts can be used as therapeutic material. A further reason for clients to continue their work outside sessions is that it is sometimes not until they are in the midst of a particular situation, or type of situation, that they are able to capture their most salient automatic thoughts, sometimes known as 'hot cognitions'. Another important reason for homework is that it more ably equips the client with the means to continue therapy alone once the sessions come to an end, since he will already have developed a degree of proficiency from

What happened? This might be an event, thought, image or memory you had
Watching TV - Programme about marriage

What thought or thoughts went through your mind? Rate how much you believe each thought.

(a) I'll screw this relationship up like the others 90 %
(b) I feel I'm useless 85 %
(c) %
(d) %

What emotions did you feel? How strong was each?

(a) Anxiety 65 %
(b) Depression 70 %
(c) %
(d) %

Tick the thinking errors for each thought

	a	b	c	d
All or nothing thinking		✓		
Personalization/blame		✓		
Catastrophizing	✓			
Emotional reasoning		✓		
'Should' or 'must' statements				
Mental filter				
Discounting the positive	✓	✓		
Over-generalization	✓			
Magnification/minimization				
Labelling		✓		
Jumping to conclusions - mind reading				
Fortune telling		✓		

Alternative view for each thought

	Rate		Rate
(a) I'm doing something about it by having counselling. Some parts of relationship are okay and Kas really likes me	80 %	(b) I know in my heart of hearts I'm not really useless	85 %
(c)	%	(d)	%

Rate emotions now

(a) Anxious	20 %	(b) Depressed	30 %
(c)	%	(d)	%

Figure 2.7 Automatic thought form

carrying out homework assignments. Assignments vary enormously, depending on the client, the problem and the stage of therapy. A common assignment is to complete an automatic thought form (see Figure 2.7).

10 Time limited

Time constraints have been more fully discussed in Chapter 1 in relation to brief cognitive behaviour therapy. We wish only to add here that cognitive behaviour therapy is limited in time, compared to many other approaches, due to its directive, collaborative, goal directed and structured approach.

11 Audio recording sessions

The practice of audio recording sessions is not central to cognitive behaviour therapy but it is often carried out and can be useful for the following reasons. First, it can be helpful for supervision and may provide the supervisor with different and more detailed primary information about the session than the secondary description given by the (trainee) therapist possibly after some time has elapsed. As with other aspects of cognitive behaviour therapy, audio recording is carried out with the collaboration of the client, alongside a discussion of the reasons for use of a digital recorder. Second, the client may benefit from recording the session himself as it enables him to revisit aspects of the session at home. This reinforces the practice of working therapeutically both inside and outside the session (see ‘Homework setting’ above), contributes to the openness of therapy (see ‘Collaborative relationship’ above) and helps enable clients to develop therapeutic skills which they can apply to their own problems (see ‘Teaches client to become own therapist’ above). If the client decides to make an audio recording of the session on their cell/mobile phone or smart phone, it is important that they are made aware of the lack of security of digital recordings and files using phones. With young people, in particular, the intentional sharing of digital audio files with friends or family may have unintended consequences now or in the future especially if the files are uploaded to the internet or shared.

The following extract is taken from a typical session. The dialogue will be interrupted with commentary indicating the points at which the factors above, *the fundamental characteristics of cognitive behaviour therapy*, are demonstrated. The therapist and client have already greeted each other and both have set their audio tapes in motion for the session.

Therapist: How have you been feeling this week, Tom?

Client: It’s not been so bad the last couple of days, but at the weekend, I went right down again – back to square one.

Therapist: So it seems like your mood has been quite up and down . . .

Client: Hmm.

Therapist: Can we put that on the agenda to look at in more detail?

Client: Yeah.

Even from this brief extract, taken from the start of a session, some of the characteristics of cognitive behaviour therapy are in evidence. The therapist commences by checking the client’s mood and will shortly move on briefly to review the client’s week. These are the first components in the structure of a typical session as discussed in the fourth fundamental characteristic, ‘Structure to sessions and to therapy’ (see above). It is also apparent that the therapist is interacting with the client and demonstrates the directive

quality of his therapeutic style by focusing on agenda setting (see ‘Therapeutic style’ above). However, the therapist tempers this directive aspect by eliciting feedback from the client, which helps in establishing a ‘Collaborative relationship’ (see above).

Therapist: Could I see your depression inventory? [*Client hands therapist a Beck Depression Inventory (BDI) completed prior to the session.*]

Therapist: Thanks... Is there anything else that happened during the week that you would like to talk about today?

Client: The weekend was awful. Kas started on at me and things went from bad to worse...

Therapist: Okay. We’ve put what happened at the weekend on the agenda. Before we set the rest of the agenda, I’d briefly like to ask you about last week’s session – what stands out in your mind about it?

Client: Well, it was a relief to find out that I can do something about my problems after all this time and that they’re all connected. In a way, that made it seem more manageable. Also, the forms you gave me helped me to see that the way I think about things can make my problems worse, so if I start to think about things in a different way I could start to feel better. It was helpful to have this jotted in my notebook as a reminder.

The therapist regularly monitors the client’s mood throughout the course of therapy as well as within individual sessions. As noted under item 7 above, ‘Uses range of aids and techniques’, the therapist commonly uses the Beck Depression Inventory (BDI) to monitor the client’s mood. Many therapists arrange for the client to complete appropriate inventories in the waiting area immediately prior to commencing a session, as the therapist in the extract above demonstrates. The client also alludes to two aids: a therapy notebook and an automatic thought form (see Figure 2.7). In brief cognitive behaviour therapy, clients are encouraged to keep a therapy notebook in which salient details of the session are noted as well as points outside the session which the client believes are important. An important aspect of the therapy notebook is to link the sessions together and to link therapy with other parts of the client’s life. The automatic thought form was used by Tom as part of an assignment following the last session (see item 9 ‘Homework setting’). Although the client in this extract says that the forms helped him to link his thoughts and feelings together, this was a theme already discussed in some detail when the client’s problems were initially formulated (see item 2 ‘Formulation of problem’) and initial goals for therapy set (see item 5 ‘Goal-directed therapy’). The client’s ‘insight’ of linking thoughts and feelings as a result of working with automatic thought forms outside the session is one of the benefits of the psychoeducational approach adopted by cognitive behaviour therapy. This is also an early stage in the development of the client learning to work independently of the therapist (see item 8 ‘Teaches client to become own therapist’). Throughout therapy, the therapist focuses not only on what the client wishes to change, or their difficulties, but recognizes and focuses on the client’s strengths and how these can be integrated into the process of therapeutic change (see Kuyken et al., 2009; Mooney and Padesky, 2002). The extract also sees the therapist following a typical session structure by moving on to set an agenda (see item 4 ‘Structure to sessions and to therapy’).

Therapist: I'm glad you've been so quick to make the link between thoughts and feelings. That's a really important part of this work... Did you find anything in the last session that troubled you?

Client: No, not once I got here!

Therapist: [laughs] Is there anything else that happened during the week that you would like to include on today's agenda?

Client: No, I don't think so.

Therapist: Right. And is there anything else you would like to put on today's agenda?

Client: No... Only what we said I'd try out during the week. I didn't manage to do it all.

Therapist: Okay, shall we put last week's homework on the agenda too, and spend some time looking at the difficulties you had in completing it?

Client: Yeah, okay.

Therapist: You might remember me saying that we will usually look at the work you've done outside of the sessions, with me during the session?

Client: That's right.

Therapist: That's most of our agenda set for today. Do you recall what else we usually have on the agenda too?

Client: I suppose we need some time to look at things I can do ... like the forms I did last week...

Therapist: Your assignment?

Client: Yeah, that's it.

Therapist: Okay, and we can allocate some time to set the coming week's assignments and can finish by looking at your thoughts and feelings about this session. Can we move straight on to look at your assignments? You said that you hadn't been able to complete it all.

Client: That's right. I did three of the sheets before the weekend, but after our big row I felt really down and didn't feel up to doing any more.

Therapist: You said earlier that the forms you took away with you were helpful in letting you make a link between the feelings you had and the thoughts that triggered them and I'd like to carry that forward to how your thoughts and feelings about your row with Kas led to you being unable to carry on with the work you had started and had found helpful up to that point... Do you want to look at the work you have done first, or would you prefer to focus on the difficulties you had that prevented you from doing the rest?

Client: Well, after we had the row, it was like all the stuff I'd done up to that point got wiped out ... and it made the work with the forms seem meaningless. So I guess I'd like to look at that part first.

Therapist: Okay. You said the work you had done up to that point had got 'wiped out' and you also said earlier that you had completed three of the automatic

thought forms. Can I check this out with you? Would you think of a time shortly before you had the row with Kas... Can you let me know whether or not the work with the forms was useful up to that point?

Client: Yeah, as I said earlier, I did find it helpful linking the thoughts and feelings and was feeling quite optimistic but when we had the row it just counted for nothing.

Therapist: When you were using the forms, part of your assignment was to identify disturbing feelings and link these with particular thoughts. Do you recall if the form asked you to identify anything else?

Client: [*looking at homework*] What, the alternative thoughts?

Therapist: It's one of the sections on the form between your automatic thoughts and your alternative thoughts!

Client: Oh, you mean the thinking errors.

Therapist: Yes! That's it! Now you said that the work had been useful up to the row, but then its usefulness had been 'wiped out' by the row. Can you identify any thinking errors here?

Client: Well, I suppose I was over-generalizing.

Therapist: Can you expand on that for me?

Client: Well, because we had a big row, and that was really dreadful, it doesn't mean that everything else going on for me has to be affected and it doesn't turn something that seemed helpful at the time into uselessness.

Therapist: How much do you believe that, Tom?

Client: Well, as I'm saying it to you now, I believe it a lot and it makes sense, but at the time, I just felt on a 'downer' and couldn't think of anything else except how dreadful it all was.

Therapist: Okay. Let me just feed that back to you. You said that when you had an argument with Kas, everything was 'wiped out'. Then looking back at the incident, you recognized that the row – although very disturbing to you at the time – didn't alter the usefulness of your new way of looking at your problems... If you had another similar row this week, do you think you would be able to view it in this new way?

Client: [*long pause while client looks away*] It's like I have two heads. There's the one I have as I sit here with you, and also as I'm filling out the forms, where I can look at things in a new light. But there's the one that's always been there, and can't bring in these new ways of thinking, when I'm right in the middle of a situation.

Therapist: That's a very good point you've made. I think it is important for me to emphasize that we have only had a couple of sessions and that it will take time and practice to use the new skills you are developing in the heat of the moment. It is partly for this reason that I ask you to write out your difficulties and alternative responses on the forms so that you can gradually incorporate your new ways of thinking and feeling into your everyday life.

In this extract, the therapist and client continue to work through a typically structured session and proceed to discuss agenda items. The therapist assists the client to *examine and question unhelpful thinking* (item 6), which includes identifying *thinking errors*. The therapist decided not to question the client's thinking errors of magnification or catastrophizing when he used the word 'dreadful' to describe an event. He made a mental note of this for later discussion and also led the client towards identifying and challenging a number of other thinking errors surrounding this one event. In this way, the therapist continues to socialize the client into cognitive behaviour therapy. This is aided by helping the client to recognize the importance of work outside of the sessions (see item 8 'Teaches client to become own therapist') and also by helping the client to appreciate that it is unhelpful to view therapeutic change in all or nothing terms (see 'Thinking errors', p. 10). The client's 'two heads' will only become one as he practises and assimilates new skills enabling him to bring new thinking patterns to current difficulties.

It is an accreditation and good practice requirement that every practitioner receive regular, high-quality professional supervision from an experienced CBT therapist to maintain quality standards and to protect the client.

The present chapter has focused on the fundamental aspects of cognitive behaviour therapy and has concluded with annotated extracts which help to exemplify them while giving a flavour of cognitive behaviour therapy in practice. In the following chapters we consider the process in greater detail.

Practice points

- 1 Learn the cognitive model and use it daily in your own life to help you to understand it. Beliefs are learned and can be changed.
- 2 Use the ABC to help you to remember:
 - A = Activating event (event, image, thought, memory, etc.).
 - B = Belief (automatic thoughts and underlying beliefs).
 - C = Consequences (emotions, behaviour, physiology and further thoughts).
- 3 Recognize that thinking errors are central to the cognitive model: know them thoroughly.
- 4 Use basic counselling skills as a foundation for your practice of cognitive behaviour therapy.
- 5 Structure the therapy session by negotiating an agenda.
- 6 Familiarize yourself with the fundamental characteristics of cognitive behaviour therapy.
- 7 Remind yourself and your client of the importance of the other 167 hours outside of the therapy session every week to put cognitive behaviour therapy into practice in the real world.
- 8 Arrange for appropriate supervision.
- 9 With the client's permission make a recording of the session for the purposes of supervision and encourage the client to record the session too and listen to it as an in-between therapy session exercise.

Original Paper

Resource-Based Internet Intervention (Med-Stress) to Improve Well-Being Among Medical Professionals: Randomized Controlled Trial

Ewelina Smoktunowicz^{1,2}, PhD; Magdalena Lesnierowska¹, PhD; Per Carlbring², PhD; Gerhard Andersson^{3,4}, PhD; Roman Cieslak^{1,5}, PhD

¹Department of Psychology, SWPS University of Social Sciences and Humanities, Warsaw, Poland

²Department of Psychology, Stockholm University, Stockholm, Sweden

³Department of Behavioural Sciences and Learning, Linköping University, Linköping, Sweden

⁴Department of Clinical Neuroscience, Karolinska Institutet, Stockholm, Sweden

⁵Trauma, Health, and Hazards Center, University of Colorado Colorado Springs, Colorado Springs, CO, United States

Corresponding Author:

Ewelina Smoktunowicz, PhD

Department of Psychology

SWPS University of Social Sciences and Humanities

Warsaw

Poland

Phone: 48 506354565

Email: esmoktunowicz@swps.edu.pl

Abstract

Background: Medical professionals are exposed to multiple and often excessive demands in their work environment. Low-intensity internet interventions allow them to benefit from psychological support even when institutional help is not available. Focusing on enhancing psychological resources—self-efficacy and perceived social support—makes an intervention relevant for various occupations within the medical profession. Previously, these resources were found to operate both individually or sequentially with self-efficacy either preceding social support (cultivation process) or following it (enabling process).

Objective: The objective of this randomized controlled trial is to compare the efficacy of 4 variants of Med-Stress, a self-guided internet intervention that aims to improve the multifaceted well-being of medical professionals.

Methods: This study was conducted before the COVID-19 pandemic. Participants (N=1240) were recruited mainly via media campaigns and social media targeted ads. They were assigned to 1 of the following 4 groups: experimental condition reflecting the cultivation process, experimental condition reflecting the enabling process, active comparator enhancing only self-efficacy, and active comparator enhancing only perceived social support. Outcomes included 5 facets of well-being: job stress, job burnout, work engagement, depression, and job-related traumatic stress. Measurements were taken on the web at baseline (time 1), immediately after intervention (time 2), and at a 6-month follow-up (time 3). To analyze the data, linear mixed effects models were used on the intention-to-treat sample. The trial was partially blinded as the information about the duration of the trial, which was different for experimental and control conditions, was public.

Results: At time 2, job stress was lower in the condition reflecting the cultivation process than in the one enhancing social support only ($d=-0.21$), and at time 3, participants in that experimental condition reported the lowest job stress when compared with all 3 remaining study groups (ds between -0.24 and -0.41). For job-related traumatic stress, we found a significant difference between study groups only at time 3: stress was lower in the experimental condition in which self-efficacy was enhanced first than in the active comparator enhancing solely social support ($d=-0.24$). The same result was found for work engagement ($d=-0.20$), which means that it was lower in exactly the same condition that was found beneficial for stress relief. There were no differences between study conditions for burnout and depression neither at time 2 nor at time 3. There was a high dropout in the study (1023/1240, 82.50% at posttest), reflecting the pragmatic nature of this trial.

Conclusions: The Med-Stress internet intervention improves some components of well-being—most notably job stress—when activities are completed in a specific sequence. The decrease in work engagement could support the notion of dark side of this phenomenon, but further research is needed.

Trial Registration: ClinicalTrials.gov NCT03475290; <https://clinicaltrials.gov/ct2/show/NCT03475290>

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KEYWORDS

well-being; job stress; internet intervention; self-efficacy; perceived social support; medical professionals

Introduction

Background

Well-being at work has been the focus of numerous workplace interventions, as it is associated with important outcomes both for the organization and the individual, including performance [1], absenteeism [2], and turnover intentions [3]. One occupational group that is particularly at risk of losses in terms of well-being is health care professionals [4-6]. In this study, we tested the efficacy of an internet intervention that aims to enhance psychological resources—perceived social support and self-efficacy—to improve the well-being of medical professionals. As this study was conducted before the outbreak of the COVID-19 pandemic in 2020, we managed to capture the everyday functioning of medical professionals.

Regardless of occupation, workplace interventions represent various approaches. Some of them are contextual and focus on factors (job demands and resources) that are specific to a given organization [7,8], whereas others are context-free because they target personal characteristics, such as self-efficacy, hope, and optimism [8]. Finally, there are psychotherapies [9,10] that focus on specific mental health disorders, such as depression and anxiety, albeit in the context of work.

Although many of these interventions have been found to be effective, their accessibility is usually constrained, and they can be difficult and costly to scale-up. Technology overcomes some of these limitations. Over the past three decades, internet interventions have been shown to deliver effective treatments for a variety of mental health problems [11-13]. More recently, their efficacy has also been tested in work environments with promising results [14,15]. Some of these interventions have been conceptualized within the cognitive behavioral therapy (CBT) framework [16], whereas others employed mindfulness [17] or stress management techniques [18]. However, although some programs [19] were based on theoretical models such as the stress and cognitive appraisal model by Lazarus and Folkman [20], they rarely test frameworks that have been specifically developed in the context of occupational health (eg, job demands-resources model [21]) or adapted to it (eg, conservation of resources theory [COR] [22]).

COR [22] originated as a conceptual framework of general stress but has since repeatedly served as a backbone for research on occupational well-being [23,24]. One of its fundamental premises is the resource investment principle, also known as a gain spiral. It refers to a process in which people who possess certain resources (eg, time) invest them to generate new ones (eg, learn new skills) and ultimately lower their risk of future losses. This process has already been empirically demonstrated [25-27]; however, studies have tended to focus on contextual

job resources and have rarely been experimental (with notable exceptions [28]). Meanwhile, personal resources are by definition context-free. A risk of focusing on them in an intervention is that they are difficult to customize to the unique combination of demands and resources in a given workplace. However, at the same time, such an intervention has the advantage of not being limited to only a few workplaces but is applicable across organizations.

Resources that have repeatedly been shown to be associated with well-being at work are social support and self-efficacy. Social support is usually defined in relation to its source (eg, from supervisor, friends, or family), and within those relationships, it can be either received or provided [29]. Social support that does not refer to the outside source is a perceived one and is defined as one's belief that help will be provided when needed [30], making it similar to optimism and hope [31]. Moreover, the relationship between received and perceived support is only moderate [31], showing that they are distinct concepts. Therefore, it is justified to enhance perceived social support independently from a specific social context. Finally, social support is a process that generates new resources [32], which makes it uniquely suitable as a chain in the gain spiral. Meanwhile, self-efficacy, another key resource, is defined as "beliefs in one's capabilities to organize and execute the courses of action required to produce given attainments" [33]. Its 4 major sources have been identified as mastery and vicarious experiences, persuasion, and reinterpretation of emotional and physiological states [33], with the first considered the most powerful [34]. As self-efficacy is more predictive of a given outcome when it is task- or domain-specific [35], it is more beneficial to enhance contextual self-efficacy in a workplace intervention (eg, related to coping with job stress) than a general one.

Hobfoll et al [32] suggested that people abundant in personal resources such as self-efficacy are in a better position to elicit support from others but also that this relationship works in reverse, and social support can prevent the depletion of personal resources. Called the cultivation and enabling hypotheses, this two-way relationship was later conceptualized in the form of 2 arguments [36]. According to the former, people with higher self-efficacy are more likely to reach out and solicit social support when needed, whereas the latter suggests that using members of a social network as models—in particular those that are similar and/or face similar situations—or benefiting from their verbal assurance can increase the sense of efficacy. Not only theoretical but also empirical support was found for both propositions, however mostly in the context of health [37] and traumatic stress [38,39] but not for outcomes specific to work. In addition, although many of the studies were longitudinal, few were of experimental design. Therefore, in

this study, we aim to verify the efficacy of enhancing social support and self-efficacy to improve well-being in a randomized controlled trial (RCT) and test whether the sequence in which they are strengthened is important.

The results of the sixth European Working Conditions Survey show that people working in the health sector experience the highest level of work intensity [40]. Their well-being impacts the services they provide as well as the functioning of health care systems [41]. Thus, improving it is important not only for themselves but also for the patients and their entire workplace communities.

The term well-being covers numerous occupational phenomena. In this intervention, we focus on 5 that are prevalent in the health care setting: job stress, job burnout, work engagement, job-related traumatic stress, and depression. Our primary focus is on job stress and burnout. Stress is a response to a situation when resources are not available or are insufficient to offset demands that are placed upon employees [42], whereas job burnout, which has recently been added to the 11th Revision of the International Classification of Diseases [43], is defined as “a prolonged response to chronic emotional and interpersonal stressors on the job” [44]. Depending on the conceptualization, it comprises 1, 2, or 3 dimensions [45]. Although it is difficult to assess the prevalence of occupational issues such as job stress and job burnout among medical professionals, there is evidence to show that these problems are encountered globally, including in the cultural West [5,46], sub-Saharan Africa [47], and Japan [48]. A recent systematic review that extracted data from populations of 45 countries reported that up to 80.5% of physicians experienced burnout [49], making this a problem that needs to be addressed urgently.

Work engagement is defined as “a positive, fulfilling, work-related state of mind that is characterized by vigor, dedication, and absorption” [50]. Although it is negatively related to job burnout, they are not opposite poles of one dimension. Recent findings show that employees exhibit different burnout-engagement profiles: from those representing clear discrepancies (ie, high engagement with low burnout and vice versa) to profiles showing that people can be low, moderate, or high on both these dimensions at the same time [51]. Thus, interventions that target job burnout should also measure their impact on work engagement.

Job-related traumatic stress is another outcome that belongs to the occupational context, albeit not to all professions. Defined as stress resulting particularly from indirect exposure to aversive details of traumatic events via face-to-face contact with traumatized individuals or exposure to drastic materials [52], it has been widely recognized as a significant occupational burden among health professionals that needs addressing [53,54].

Contrary to previous outcomes, depression is not a typical occupational phenomenon. However, it is an important component of work-related well-being: it is prevalent across professions and has been found to predict poor job performance [55] and sickness absences [56]. Moreover, although criteria distinguishing depression from job burnout are defined, it remains problematic to differentiate them [57].

Objectives

The main goal of this study was to test whether a Med-Stress internet intervention dedicated to medical professionals would be effective in reducing job stress, burnout, depression, and job-related secondary stress and in increasing work engagement through the enhancement of 2 psychological resources: perceived social support and self-efficacy. We compared 2 experimental conditions, in which self-efficacy and social support were enhanced sequentially, with 2 active controls, in which only self-efficacy or only social support was targeted. We expected that experimental conditions, which are designed to build 2 resources, would prove to be more beneficial than active control ones—that target only a single resource—in terms of reducing negative outcomes (ie, job stress, burnout, depression, and job-related secondary stress) and improving a positive one (ie, work engagement) both at posttest and at a 6-month follow-up. Furthermore, we aim to test whether one sequence of resources—in line with either the enabling or cultivation hypothesis—would turn out to be more beneficial than the other.

Methods

Study Design

The Med-Stress study was designed as a 4-arm parallel RCT comparing the posttest and follow-up effects of 2 experimental and 2 active control conditions. The trial was conducted on web. This study was approved by the Ethical Review Board at SWPS University of Social Sciences and Humanities (opinion 4/2018 issued on February 13, 2018) and registered on ClinicalTrials.gov (NCT03475290). The study protocol has been previously published [58].

Recruitment and Study Participants

Participants were recruited between October 2018 and April 2019 via targeted social media campaigns, advertisements via radio and press, professional forums, and through medical organizations. The following inclusion criteria were applied: (1) being at least 18 years old and (2) representing the health-related profession that involved direct patient care. A total of 1575 people were registered for the study; however, 335 did not complete the required baseline assessment. The final sample size was $N=1240$.

Power Calculation

To ensure a statistical power of 0.90 to detect the posttest effect of comparisons between study conditions, we conducted an a priori sample size estimation using G*Power 3.1 3.1 [59]. Taking into account the previous studies on the effectiveness of internet interventions for general and job-related stress [15,60], we aimed to detect the minimum effect sizes of $d=0.40$ for the comparisons between conditions at 2 measurement points while controlling for baseline scores at an alpha error level of .002, reduced to correct for multiple comparisons (ie, 5 between-group comparisons and 5 outcomes). A power analysis showed that a sample of 607 was needed. As we expected a high dropout rate, we aimed to recruit a sample of 1200. As the analyses were ultimately conducted using linear mixed effects models (see the *Statistical Analysis* section for more details),

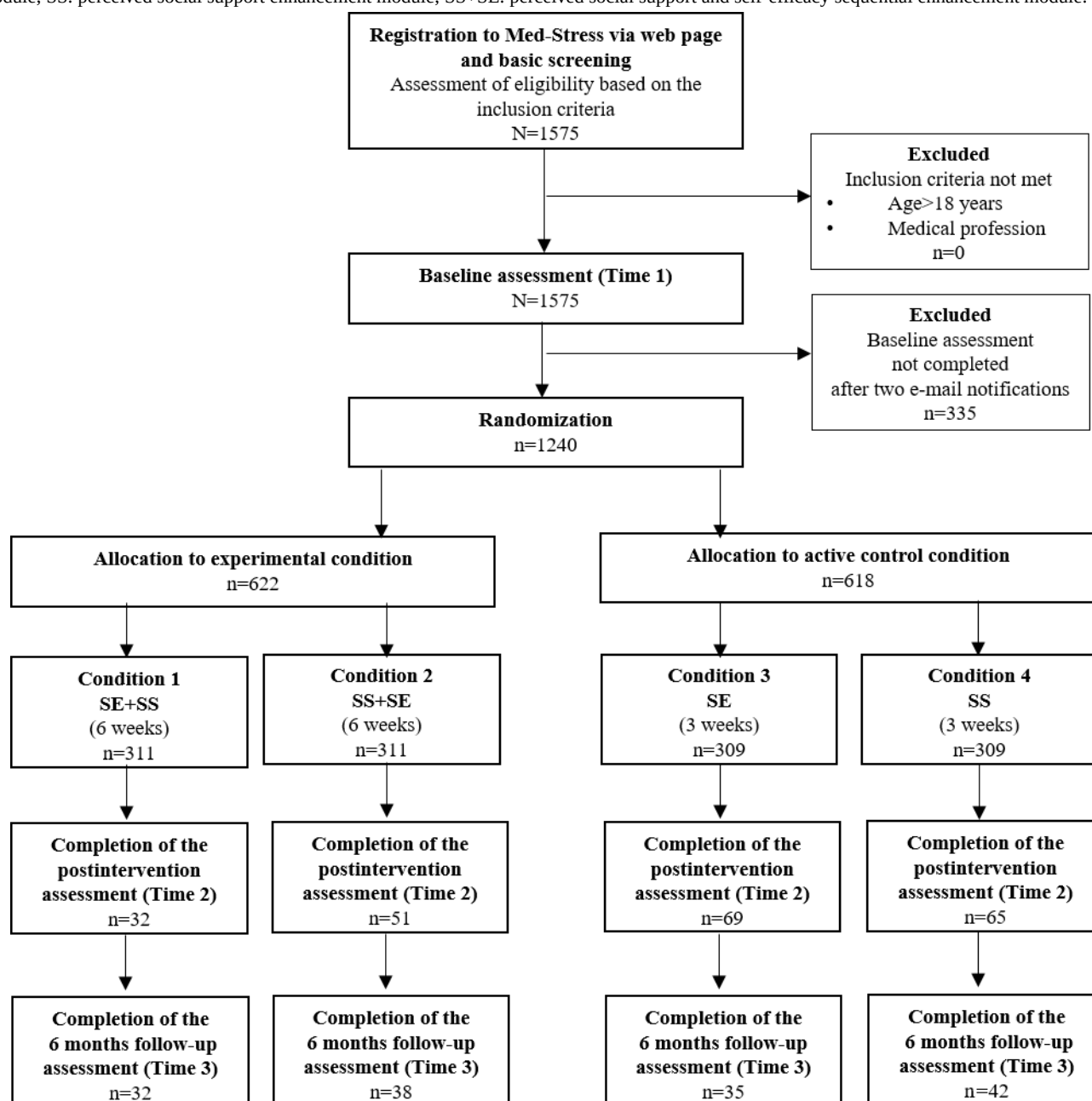
we conducted an additional analysis using the powerlmm package in R [61] to find that with the sample of 1240 participants, we had the power to detect an effect size of approximately $d=0.21$ in pairwise comparisons. The dropout rate was higher than expected (1023/1240, 82.50%), and thus, we did not have enough power to test for robustness of effects in the per protocol analysis.

Procedure

The study flow is presented in Figure 1. Participants were directed to the Med-Stress website, where they filled out a screening to ensure they met the inclusion criteria. The registration process was finalized when they signed a web-based

informed consent form. Subsequently, participants were asked to fill out a baseline (time 1) assessment. Only those who completed this step were randomized to 1 of the 4 conditions. The intervention lasted either 6 weeks (experimental conditions) or 3 weeks (active controls). Activities in the intervention were released once a week for a participant. Although we encouraged participants to complete all tasks, this was not a prerequisite to access subsequent exercises. Immediately after completing the intervention (at the posttest, time 2) and 6 months after the baseline assessment (follow-up, time 3), participants were invited to fill out the questionnaires on web. Each time, participants received 3 email reminders (2 automatic and 1 personalized).

Figure 1. Flow of participants. SE: self-efficacy enhancement module; SE+SS: self-efficacy and perceived social support sequential enhancement module; SS: perceived social support enhancement module; SS+SE: perceived social support and self-efficacy sequential enhancement module.



Intervention

Med-Stress is a self-guided internet intervention. It contains 2 main modules that were made available to participants in different variants depending on randomization to study conditions: (1) self-efficacy and perceived social support sequential enhancement modules (SE+SS; experimental condition), (2) perceived social support and self-efficacy sequential enhancement modules (SS+SE; experimental condition), (3) self-efficacy enhancement module (SE; active control condition), and (4) perceived social support enhancement module (SS; active control condition). There were 3 exercises per module. Each consisted of psychoeducational animated clips and interactive tasks requiring both web-based and offline activities. To complete all tasks within each exercise,

participants needed up to 1.5 hours. In addition, everyone had access to 4 modules: relaxation, mindfulness, cognitive restructuring, and lifestyle, which were optional and available throughout the intervention. Such a design allowed the participants to partially self-tailor the intervention. Exercises in each module were evidence based and developed in a participatory manner, as a result of the preimplementation study among 744 medical professionals conducted at the Med-Stress developmental stage (Lesnierowska et al, unpublished data, 2018). Exercise descriptions are presented in Table 1, and the example screenshots are shown in Figure 2. The program was run on the Iterapi platform developed at the Department of Behavioral Sciences and Learning at Linköping University, Sweden [62].

Table 1. The content of the Med-Stress intervention: obligatory and optional modules.

Modules	Exercises	Aims
Obligatory		
Self-efficacy enhancement	1. Mastery experience 2. Vicarious experience 3. Action planning	To enhance self-efficacy via recollecting past successes in coping with job stress, taking inspiration from other people (models), and creating a coping plan, based on the “if-then” approach to address potential coping barriers. The exercises are released weekly.
Perceived social support enhancement	1. Received support and cognitive distortions 2. Social skills and peer support 3. Action planning	To enhance perceived social support through verifying distorted assumptions about obtaining support, recalling past situations when the actual support was received, engaging into real receive-provide support interactions with other participants (chat-rooms), and creating a coping plan, based on the “if-then” approach aimed at obtaining social support. The exercises are released weekly.
Optional		
Relaxation	1. Breathing 2. Progressive relaxation 3. Visualizing the body’s warmth and weight 4. Visualizing a calm place	To support the coping process by stress relief techniques and actions. The exercises in this module aim at active relaxation. The exercises are available during the whole intervention.
Mindfulness	1. How do body and mind react to thoughts? 2. Body scanning 3. Breathing and sounds 4. Being mindful of emotions	To support the coping process by stress relief techniques and actions. The exercises in this module aim at deliberate attention on a present experience. The exercises are available during the whole intervention
Cognitive restructuring	1. Opinion or fact 2. Identifying thinking traps 3. How important will this be in the future?	To support the coping process by stress relief techniques and actions. The exercises in this module aim at skills such as distinguishing opinions from facts or identifying cognitive traps. The exercises are available during the whole intervention
Lifestyle	1. Physical activity 2. Pleasant activities	To support the coping process by stress relief techniques and actions. The exercises in this module aim at implementing physical activity or pleasant activities that foster stress reduction. The exercises are available during the whole intervention

Figure 2. Med-Stress internet intervention: (a) mastery experience and (b) mindfulness exercises.

Randomization Procedure

Participants were automatically allotted to 1 of the 4 study conditions in line with the predefined randomization protocol generated via randomizing software (randomizer.org, a free service offered by Social Psychology Network). We applied block randomization with a block size of 4. The allocation ratio was 1:1:1:1 to ensure an equal number of participants in each study condition. Masking was partially achieved: allocation was

not revealed to the participants; however, they had previously been informed that the intervention would last 3 or 6 weeks depending on the intervention variant.

Measures

Primary Outcomes

Job Stress

Job stress was assessed using the 14-item Perceived Stress Scale [42]. Instructions were adapted for the occupational context. The questionnaire includes 14 items describing the range of stress symptoms rated from 0 (never) to 4 (very often). Cronbach alphas were .85 at time 1, .86 at time 2, and .89 at time 3.

Job Burnout

Job burnout was measured using the Oldenburg Burnout Inventory [63]. The scale consists of 16 items. Participants were asked to respond on a scale ranging from 1 (strongly agree) to 4 (strongly disagree). The reliability of the scale was $\alpha=.87$ at time 1, $\alpha=.89$ at time 2, and $\alpha=.91$ at time 3.

Secondary Outcomes

Depression

Depression was assessed using the Patient Health Questionnaire-9 [64]. The scale has 9 items with responses ranging from 0 (not at all) to 3 (nearly every day). Cronbach alphas were $\alpha=.87$ at time 1, $\alpha=.88$ at time 2, and $\alpha=.89$ at time 3.

Work Engagement

Work engagement was measured using the Utrecht Work Engagement Scale-3 [65]. This shortened tool includes 3 items with a response scale ranging from 0 (never) to 6 (always). The Cronbach alphas were .73 at time 1, .78 at time 2, and .83 at time 3.

Job-Related Traumatic Stress

Job-related traumatic stress was assessed using the Posttraumatic Stress Disorder Checklist 5 [66] with modified instructions. It consists of 20 items. Participants assessed the severity of the symptoms on a scale ranging from 0 (not at all) to 4 (extremely). The Cronbach alphas were $\alpha=.95$ at time 1, $\alpha=.96$ at time 2, and $\alpha=.96$ at time 3.

Other Measures

Self-Efficacy to Manage Job Burnout and Work Stress

Self-efficacy to manage job burnout and work stress was measured using the Work Stress and Job Burnout Self-Efficacy Scale [67]. The scale has 28 items rated from 1 (I am definitely not capable) to 7 (I am definitely capable). The Cronbach alphas were .93 at time 1 and .96 at both time 2 and time 3.

Social Support Self-Efficacy

Social support self-efficacy was measured using the modified version of the Berlin Social Support Scales, Subscale 3 [68]. The tool was adapted to the context of social support self-efficacy. It consists of 5 items with a response scale ranging from 1 (I am definitely not capable) to 7 (I am definitely capable). The reliability was $\alpha=.88$ at time 1, $\alpha=.92$ at time 2, and $\alpha=.83$ at time 3.

Perceived Social Support

Perceived social support was assessed using the Who Can You Count On Scale [69]. The questionnaire consists of 32 items with a response scale ranging from 1 (to very little extent) to 5 (to a great extent). The Cronbach alphas were .95 at time 1 and .96 at time 2 and time 3.

Secondary Trauma Exposure

Secondary trauma exposure was measured using The Secondary Trauma Exposure Scale [70]. Participants indicated their answers on a yes or no scale in response to whether they indirectly experienced each of the 10 listed traumatic events. The volume, frequency, and ratio of exposure were assessed using a single item with response scales ranging from 1 (none or never) to 7 (a few thousands or every day), and from 0% to 100% for the exposure ratio. Due to scale construction, the Cronbach coefficient was not computed.

Expectancy of Improvement and Perceived Credibility of the Intervention

Expectancy of improvement and perceived credibility of the intervention were assessed using the Credibility and Expectancy Questionnaire Version II [71]. The scale consists of 6 items representing 2 subscales: perceived intervention credibility and expectancy of improvement. Responses ranged from 1 (not at all) to 9 (very). The reliability for the perceived intervention credibility subscale was $\alpha=.81$ and for the expectancy of improvement was $\alpha=.89$.

Intervention Usage

We assessed the use of obligatory and optional modules. For the former, engaging in at least one weekly assigned task was coded as activity. Optional modules contained a number of similar exercises that participants could choose from, thus engaging in at least one of them was coded as completion.

Statistical Analysis

Analysis of variance (ANOVA) for continuous and chi-square tests for categorical data were used to conduct a randomization check and dropout analysis. We did not formulate predictions regarding differences in resources (ie, perceived social support and self-efficacy) between study conditions, but we did expect them to increase between time 1 and time 2 and to remain stable by time 3. To test the change over time, we applied linear mixed effects models with a restricted maximum likelihood estimation, an unstructured covariance matrix, and time as a fixed factor compared between 3 measurement points.

To verify the effect of intervention conditions on the 5 outcome variables over the course of 3 time points, we applied linear mixed effects models with restricted maximum likelihood estimation, unstructured covariance matrices of random effects, and random slopes and intercepts [72], separately for each of the 5 outcomes. Time variable was coded as time 1=0 (baseline), time 2=1 (posttest), and time 3=2 (follow-up). To make comparisons between each active control group and experimental group as well as between 2 experimental conditions, we created 3 dummy variables in which 2 controls and 1 experimental condition subsequently served as a reference category. It is important to note that our research question was

not about the differences in changes over time between groups but about the differences in the outcomes between study conditions at 2 time points: the posttest and at the follow-up. This was reflected in the study protocol [58] where we planned to conduct separate analyses at 2 measurement points. Ultimately, to take into account the dependency between multiple measurements, we used linear mixed effects models. For each outcome, we applied a top-down approach by first fitting a loaded model with fixed and random effects, and we subsequently compared it with 2 nested models: one without random slope and one without random intercept. In the case of job-related secondary traumatic stress, we additionally controlled for exposure to traumatic events, which is a critical factor for the development of posttraumatic symptoms [52]. Finally, we tested whether there were predictors of the 2 primary outcomes (job stress and burnout). To do so, we first calculated the residual gains by subtracting the standardized initial score (Z1) multiplied by the correlation between the initial and posttreatment scores from the standardized posttreatment score (Z2) [73]. Next, we inverted the gains' values so that positive scores reflected reduction. These residual gains were entered separately as outcome variables in the regression analyses performed with 10,000 bootstrapped replications. All analyses in this study were conducted on the intention-to-treat sample. The multiple imputation (MI) procedure was applied using 10 iterations. All baseline scores, available posttest and follow-up scores, and all variables that were found to differentiate completers and dropouts (see below in *Preliminary Results*) were introduced as predictors of imputation [74]. Ultimately, the total sample of 1240 participants was included in the analyses (311 participants in each experimental condition and 309 participants in each control condition; Figure 1). Analyses were conducted using IBM SPSS Statistics 25.0.

Results

Sample

The final sample (N=1240) consisted of physicians (448/1240, 36.13%), nurses (308/1240, 24.84%), dentists (93/1240, 7.50%), physiotherapists (75/1240, 6.05%), midwives (65/1240, 5.24%), paramedics (62/1240, 5.00%), and other medical professionals (191/1240, 15.40%). Overall, job tenure ranged from less than 1 year to 40 years (mean 11.46, SD 10.26), whereas tenure at participants' current position was mean 6.69 years (SD 7.31). On average, participants worked 47.38 hours per week (SD 18.53). The majority of responders were employed in public health care institutions (924/1240, 74.52%). Women constituted 86.61% (1074/1240) of the sample. Participants' ages ranged between 20 and 66 years (mean 36.21, SD 10.18). Nearly 17% of the sample had received specialist support (psychotherapy or pharmacotherapy) to cope with job stress at some point in their professional career.

Preliminary Results

Dropout Analysis and Randomization Check

Out of the 1240 participants, 217 completed the posttest assessment and 147 completed the 6-month follow-up. Intervention dropout, defined as a loss to posttest, was 82.50%

(1023/1240). The Little [75] test revealed that data were not missing completely at random ($\chi^2_{172}=254.9$ $P<.001$). The subsequent ANOVA and chi-square tests confirmed the missing at random (MAR) pattern, depending on the level of primary outcomes at the baseline, condition assignment, and selected job and demographic characteristics. We found that completers were assigned to shorter intervention modules ($\chi^2_1=14.9$; $P<.001$), were older ($F_{1,1238}=52.10$; $P<.001$; $\eta^2=0.04$), with longer total job tenure ($F_{1,1238}=43.58$; $P<.001$; $\eta^2=0.03$) and tenure in the current position ($F_{1,1238}=28.20$; $P<.001$; $\eta^2=0.02$), and had been in their respective relationships for longer ($F_{1,1238}=17.11$; $P<.001$; $\eta^2=0.01$). They also showed a higher initial expectancy ($F_{1,1238}=12.33$; $P<.001$; $\eta^2=0.01$) and perceived credibility of the intervention ($F_{1,1238}=10.57$; $P=.001$; $\eta^2=0.01$). Those who completed the intervention presented lower baseline job stress ($F_{1,1238}=9.05$; $P=.003$; $\eta^2=0.01$), job burnout ($F_{1,1238}=5.44$; $P=.02$; $\eta^2=0.01$), depression ($F_{1,1238}=18.09$; $P<.001$; $\eta^2=0.01$), and job-related posttraumatic stress ($F_{1,1238}=7.91$; $P=.005$; $\eta^2=0.01$). No other differences were detected. The MAR pattern of missingness justified conducting MI [74].

Participants randomized into 4 study conditions did not differ with regard to either the study variables at baseline or on descriptive characteristics such as age, tenure, and sex. Finally, participants across the 4 conditions did not differ with regard to their expectations of the treatment or its perceived credibility.

Intervention usage was similar between the experimental and control conditions. In the experimental condition reflecting the cultivation process (self-efficacy precedes perceived social support), 209 participants (out of 311 randomized) engaged in at least one exercise in the first module, and 33 of them engaged in at least one exercise in the second module. For the experimental condition reflecting the enabling process (n=311), these numbers were 206 and 43, respectively. In each control condition (n=309), 205 participants engaged in at least one exercise in the self-efficacy or social support enhancement module. Usage of optional modules was similar across the conditions and varied between 98 participants benefiting from at least one optional module in the social support control condition to 124 participants in the self-efficacy control condition.

Descriptive Statistics

Means, standard deviations, and Pearson correlations for all measures are found in [Multimedia Appendix 1](#).

Effects on Self-Efficacy and Perceived Social Support

We found that there was a significant difference in job stress and burnout-related self-efficacy between time 1 and time 2 ($B=-0.08$, $SE=0.01$; 95% CI -0.11 to -0.06) but not between time 2 and time 3 ($B=0.00$, $SE=0.01$; 95% CI -0.02 to 0.03). Pairwise comparisons showed that self-efficacy plateaued after posttest but remained significantly higher than at baseline. We observed exactly the same pattern for social support

self-efficacy, with a significant difference between time 1 and time 2 ($B=-0.18$, $SE=0.03$; 95% CI -0.23 to -0.13) and a nonsignificant difference between time 2 and time 3 ($B=0.00$, $SE=0.02$; 95% CI -0.03 to 0.04). In the case of social support, there were significant differences both in the time 1 and time 2 lag ($B=-0.19$, $SE=0.01$; 95% CI -0.21 to -0.17) and time 2 and time 3 lag ($B=-0.06$, $SE=0.01$; 95% CI -0.08 to -0.03). Pairwise comparisons showed that after social support increased between baseline and posttest, it significantly dropped at the follow-up; however, it remained higher than at baseline.

Hypotheses Testing

First, for each outcome, we compared a reference model with fixed effects of condition assignment, time, and time by

condition interaction as well as 2 random effects associated with each participant—intercept and time—with nested models in which either random effect associated with time or random intercept was not included. For all the outcomes, the differences in -2 Restricted Maximum Likelihood log-likelihood between the reference and nested models were statistically significant, and therefore, we retained both random effects. Moreover, for each of the outcomes, we found them to be statistically significant, indicating that within each study condition, there were considerable variations between participants before they started the intervention and in how they responded over time (Table 2).

Table 2. Results of linear mixed effects models.

Outcome	Fixed effects		Random effects	
	Estimate (SE)	95% CI	Estimate (SE)	95% CI
Job stress				
Intercept ^a	2.17-2.23 (0.03)	2.11 to 2.17, 2.22 to 2.45	0.21 (0.01)	0.19-0.23
SE ^b to SE+SS ^c	0.02 (0.04)	-0.06 to 0.10	— ^d	—
SE to SS+SE ^e	-0.03 (0.04)	-0.1 to 0.05	—	—
SS ^f to SE+SS	-0.02 (0.04)	-0.10 to 0.06	—	—
SS to SS+SE	-0.07 (0.04)	-0.15 to 0.01	—	—
SE+SS to SS+SE	0.05 (0.04)	-0.03 to 0.12	—	—
Time ^a	-0.08 to -0.09 (0.01)	-0.11 to -0.10, -0.07 to -0.06	0.01 (0.00)	0.01-0.01
Time×SE to SE+SS	-0.05 (0.01)	-0.08 to -0.03	—	—
Time×SE to SS+SE	0.01 (0.01)	-0.01 to 0.04	—	—
Time×SS to SE+SS	-0.07 (0.01)	-0.09 to -0.04	—	—
Time×SS to SS+SE	0.00 (0.01)	-0.03 to 0.03	—	—
Time×SE+SS to SS+SE	-0.07 (0.01)	-0.09 to -0.04	—	—
Job burnout				
Intercept ^a	2.61-2.70 (0.03)	2.56 to 2.65, 2.67 to 2.76	0.21 (0.01)	0.19-0.24
SE to SE+SS	0.03 (0.04)	-0.05 to 0.11	—	—
SE to SS+SE	0.04 (0.04)	-0.03 to 0.12	—	—
SS to SE+SS	-0.06 (0.04)	-0.14 to 0.02	—	—
SS to SS+SE	-0.05 (0.04)	-0.12 to 0.03	—	—
SE+SS to SS+SE	-0.01 (0.04)	-0.09 to 0.06	—	—
Time ^a	-0.02 to -0.06 (0.01)	-0.08 to -0.03, -0.05 to -0.00	0.01 (0.00)	0.00-0.01
Time×SE to SE+SS	-0.03 (0.01)	-0.05 to -0.00	—	—
Time×SE to SS+SE	-0.04 (0.01)	-0.04 to -0.00	—	—
Time×SS to SE+SS	0.02 (0.01)	-0.00 to 0.05	—	—
Time×SS to SS+SE	0.01 (0.01)	-0.02 to 0.03	—	—
Time×SE+SS to SS+SE	0.02 (0.01)	-0.01 to 0.04	—	—
Work engagement				
Intercept ^a	4.06-4.21 (0.06)	3.95 to 4.09, 4.18 to 4.32	0.89 (0.04)	0.81-0.98
SE to SE+SS	-0.03 (0.08)	-0.19 to 0.13	—	—
SE to SS+SE	-0.04 (0.08)	-0.20 to 0.12	—	—
SS to SE+SS	0.12 (0.08)	-0.04 to 0.28	—	—
SS to SS+SE	0.10 (0.08)	-0.06 to 0.26	—	—
SE+SS to SS+SE	0.01 (0.08)	0.15 to 0.17	—	—
Time ^a	-0.09 to -0.19 (0.02)	-0.23 to -0.13, -0.16 to -0.05	0.03 (0.01)	0.02-0.04
Time×SE to SE+SS	-0.05 (0.03)	-0.10 to 0.01	—	—
Time×SE to SS+SE	-0.02 (0.03)	-0.07 to 0.04	—	—
Time×SS to SE+SS	-0.14 (0.03)	-0.19 to -0.08	—	—
Time×SS to SS+SE	-0.10 (0.03)	-0.16 to -0.05	—	—
Time×SE+SS to SS+SE	-0.03 (0.03)	-0.09 to 0.02	—	—

Outcome	Fixed effects		Random effects	
	Estimate (SE)	95% CI	Estimate (SE)	95% CI
Depression				
Intercept ^a	1.16-1.21 (0.04)	1.08 to 1.13, 1.23 to 1.28	0.36 (0.02)	0.33, 0.39
SE to SE+SS	0.05 (0.05)	-0.06 to 0.15	—	—
SE to SS+SE	-0.05 (0.05)	-0.15 to 0.05	—	—
SS to SE+SS	0.05 (0.05)	-0.05 to 0.15	—	—
SS to SS+SE	-0.05 (0.05)	-0.15 to 0.06	—	—
SE+SS to SS+SE	0.09 (0.05)	-0.01 to 0.20	—	—
Time	-0.04 to -0.07 (0.01)	-0.10 to -0.06, -0.05 to -0.01	0.02 (0.00)	0.02, 0.02
Time×SE to SE+SS	-0.01 (0.01)	-0.05 to 0.02	—	—
Time×SE to SS+SE	0.01 (0.02)	-0.02 to 0.05	—	—
Time×SS to SE+SS	-0.05 (0.02)	-0.08 to -0.01	—	—
Time×SS to SS+SE	-0.02 (0.02)	-0.06 to 0.01	—	—
Time×SE+SS to SS+SE	-0.03 (0.02)	-0.06 to 0.01	—	—
Job-related traumatic stress^g				
Intercept ^a	1.26-1.38 (0.05)	1.17 to 1.29, 1.36 to 1.48	0.60 (0.03)	0.54, 0.65
SE to SE+SS	-0.10 (0.07)	-0.23 to 0.03	—	—
SE to SS+SE	-0.09 (0.07)	-0.22 to 0.04	—	—
SS to SE+SS	0.02 (0.07)	-0.11 to 0.16	—	—
SS to SS+SE	0.03 (0.07)	-0.10 to 0.16	—	—
SE+SS to SS+SE	-0.01 (0.07)	-0.14 to 0.12	—	—
Time ^a	0.00-0.03 (0.02)	-0.06 to -0.00, 0.01 to 0.07	0.05 (0.00)	0.04, 0.06
Time×SE to SE+SS	-0.02 (0.03)	-0.07 to 0.03	—	—
Time×SE to SS+SE	0.03 (0.03)	-0.02 to 0.08	—	—
Time×SS to SE+SS	-0.07 (0.03)	-0.12 to -0.02	—	—
Time×SS to SS+SE	-0.03 (0.03)	-0.08 to 0.02	—	—
Time×SE+SS to SS+SE	-0.04 (0.03)	-0.09 to 0.01	—	—

^aAs the values differ slightly depending on which study condition is used as a reference category, a range of estimates and confidence intervals are provided.

^bSE: self-efficacy enhancement module.

^cSE+SS: self-efficacy and perceived social support sequential enhancement module.

^dThe lack of data in these cells follows statistical analysis: these values are only provided for intercept and time.

^eSS+SE: perceived social support and self-efficacy sequential enhancement module.

^fSS: perceived social support enhancement module.

^gControlling for exposure to traumatic events.

We expected that the experimental conditions (ie, the ones comprised 2 modules: SE+SS and SS+SE) would be more effective than each active control condition (ie, SE and SS) in reducing job stress and job burnout (primary outcomes) as well as depression and job-related posttraumatic stress (secondary outcomes) at posttest (time 2), and we expected these effects to remain at a 6-month follow-up (time 3). In the case of work engagement, we expected it to be higher at time 2 in the experimental conditions in comparison with active controls, and we expected the effects to remain at a 6-month follow-up

(time 3). We were also interested in whether the 2 experimental conditions would differ in their efficacy.

Job Stress

The mixed effects model for job stress showed no significant effect of condition assignment on job stress for any of the between-group comparisons, but it did show a significant effect of time for all comparisons (Table 2), indicating that stress decreased with time. Significant interaction effects were found for time and the comparison between the experimental condition SE+SS with active control SE ($B=-0.05$, $SE=0.01$; 95% CI

–0.08 to –0.03) and with active SS control ($B=-0.06$, $SE=0.01$; 95% CI –0.09 to –0.04) as well as for time and the comparison between the 2 experimental groups ($B=-0.07$, $SE=0.01$; 95% CI –0.09 to –0.04). Bonferroni post hoc tests revealed that at time 2, job stress was significantly lower in the experimental SE+SS condition when compared with the active control SS ($d=-0.21$; 95% CI –0.37 to –0.05). However, at time 3, it was lower in the SE+SS condition than in the remaining 3 conditions: for SS, $d=-0.41$, 95% CI –0.57 to –0.25; for SE, $d=-0.24$, 95% CI –0.40 to –0.09; and for SS+SE, $d=-0.24$, 95% CI –0.39 to –0.08.

Job Burnout

Assignment had no main effect on job burnout for either of the comparisons between study conditions, but there was a main effect of time for all the comparisons (Table 2), indicating that burnout decreased with time. Interaction effects of time and the comparisons between groups were either not significant or marginally significant at a level that did not warrant further investigation due to the risk of inflated type I error resulting from multiple comparisons.

Work Engagement

We found no main effect of condition assignment on work engagement, but we did find a main effect of time (Table 2) for the comparisons between all study conditions, showing that work engagement decreased with time. Significant interactions were found for time and the comparisons between SS active control and 2 experimental groups: SE+SS ($B=-0.13$, $SE=0.03$; 95% CI –0.19 to –0.08) and SS+SE ($B=-0.10$, $SE=0.03$; 95% CI –0.16 to –0.05). Bonferroni post hoc tests demonstrated that there were no significant differences between study conditions at time 2, but at time 3, work engagement was significantly lower in the experimental condition SE+SS than in the active control SS ($d=-0.20$; 95% CI –0.36 to –0.04).

Depression

Similar to the case of work engagement, for depression, we found an effect of time, indicating that depression decreased with time, but no effect of condition assignment for any of the between-group comparisons (Table 2). The only significant interaction was found for time and the comparison between active control SS and experimental condition SE+SS ($B=-0.05$, $SE=0.02$; 95% CI –0.08 to –0.01). However, Bonferroni post hoc tests revealed that there were no meaningful differences between these conditions at either posttest or at follow-up.

Job-Related Traumatic Stress

For job-related traumatic stress, we additionally controlled for the fixed effect of the exposure to traumatic events. Neither condition assignment nor time had an effect on the outcome (Table 2). A significant interaction was observed for time and the comparison between active control SS and experimental SE+SS condition ($B=-0.07$, $SE=0.03$; 95% CI –0.12 to –0.02). Bonferroni post hoc analysis showed that this difference was significant only at time 3, with lower stress in the experimental condition ($d=-0.24$; 95% CI –0.40 to –0.08).

Predictors of Outcomes

To identify potential predictors of the 2 primary outcomes, job stress and job burnout, we ran bootstrapped regression analyses with the outcomes reflecting change between pretest and posttest (residual change score) and 95% bias-corrected bootstrap CIs. Age, gender, marital status, average weekly work hours, type of organization (public vs private), receiving treatment in the past, and general tenure were not associated with changes in job stress and job burnout. However, tenure in the current workplace turned out to be a significant predictor of both change in job stress ($\beta=-.22$; 95% CI –0.02 to –0.01) and in burnout ($\beta=-.18$; 95% CI –0.02 to –0.01): greater reduction in both outcomes was observed among participants with shorter tenure. Changes in job stress and burnout were also associated with the level of expectations of the intervention. Participants who had higher expectations of the treatment and perceived the intervention as more credible experienced a greater decrease in stress (credibility: $\beta=.16$; 95% CI 0.04–0.09 and expectancy: $\beta=.17$; 95% CI 0.04–0.09) and in burnout (credibility: $\beta=.10$; 95% CI 0.01–0.06 and expectancy: $\beta=.25$; 95% CI 0.06–0.10).

In terms of using the intervention, change in job stress was predicted by engaging in exercises in the self-efficacy enhancing module ($\beta=.11$; 95% CI 0.06–0.22): at the posttest, stress was lower among the participants who engaged in the self-efficacy module. Using social support enhancing exercises and optional modules (relaxation, mindfulness, cognitive restructuring, and lifestyle) was not associated with change in job stress. A change in job burnout was predicted neither by engagement in self-efficacy nor by social support exercises. Only using optional modules was associated with a reduction in job burnout at the posttest ($\beta=.07$; 95% CI 0.01–0.16).

Discussion

Principal Findings

This study had 2 objectives. First, we wanted to test whether enhancing 2 psychological resources, perceived social support and self-efficacy, would be more effective for improving the well-being of medical professionals than strengthening only 1 of them. Second, we aimed to experimentally verify the enabling versus the cultivation hypothesis, that is, to test whether the sequence in which social support and self-efficacy were targeted would have an effect on well-being. To reflect the complexity of a natural occupational context, we tested the efficacy of the intervention for 5 outcomes. We found that job burnout did not depend on assignment to the study condition. However, allocation to study condition did have an effect on another primary outcome, job stress. Immediately after the intervention, we found that one experimental condition (ie, the one reflecting the cultivation hypothesis by first enhancing self-efficacy and later social support) was more effective in reducing stress than the control condition that solely targeted social support, but not the one dedicated to enhancing only self-efficacy. This result partially supported our expectations. However, 6 months later, the same experimental condition was found to be the most effective. This result supported the cultivation process of stress reduction over an enabling one. When we analyzed how participants used the Med-Stress intervention regardless of

condition assignment, we found that there was a reduction in job burnout, and it was predicted by using optional modules that comprised CBT-based exercises. Perhaps these exercises, and not those dedicated to building personal resources (ie, self-efficacy and perceived social support), were responsible for the decrease in burnout, albeit small, in participants across all conditions. Job stress, on the other hand, was not associated with completing optional modules; the only significant predictor of its decrease was using self-efficacy exercises.

No meaningful differences between study conditions were detected for depression. In the case of job-related traumatic stress, between-group differences at the posttest were negligible, whereas at the follow-up, we again found that participants in the condition reflecting the cultivation hypothesis reported greater reduction in stress than those in a control condition enhancing solely social support, although the effect was small. Work engagement also revealed different patterns at posttest and follow-up. Immediately after the intervention, there were no meaningful differences in work engagement between study conditions; however, 6 months later, work engagement was lower in the same experimental condition reflecting the cultivation hypothesis than in the control condition that aimed to build only social support. Moreover, contrary to what might be expected, work engagement decreased over time. In fact, at the follow-up, work engagement was significantly lower in the same condition that we found was beneficial for the reduction of both types of job-related stress. In other words, participants who first completed self-efficacy enhancing exercises and then the ones dedicated to building perception of social support reported lower stress and lower work engagement. This pattern of results suggests that a decrease in work engagement might be beneficial to medical professionals. In fact, although work engagement is overly considered a positive state, associated with numerous beneficial consequences for a person, such as better self-reported health, a so-called dark side of engagement has also been identified, suggesting that over-engaged workers might experience undesired outcomes [76]. Moreover, the results of a recent meta-analysis [77] showed that only half of the interventions aimed at improving work engagement were found to have a positive effect. The rest had no effect or had a negative effect, although the latter were in the minority. Interestingly, the intervention that resulted in decreased work engagement was dedicated to service workers in older care, a group that is similar to the one in Med-Stress [78]. This might indicate that, at least for these occupational groups, being highly engaged in work can be detrimental. At the same time, this interpretation needs to be treated with caution as one of the options that require further analysis.

Taken together, these results show that the experimental condition reflecting the cultivation hypothesis—self-efficacy preceding social support—is the most beneficial in the long run, although this finding needs to be treated with caution for the following reasons: first, it was not found for all components of well-being. Second, for those that it was found (job stress and traumatic stress), the effect sizes were small. Third, in the case of work engagement, more research is needed to establish whether the decrease over time was beneficial or detrimental to the participants. Small effect sizes are perhaps not that

surprising considering that gain cycles, contrary to loss spirals, are considered to be weaker and take longer to unfold [23]. As 6 months is a relatively short period, subsequent measurement points are needed, and therefore, we have scheduled another follow-up in the future. In addition, small effect sizes could be attributed to the comparisons being made between experimental conditions and active, not passive, controls [79].

Perhaps reaching a result that is not completely coherent for all outcomes is not that surprising when these outcomes are numerous and represent different aspects of well-being. By targeting multiple outcomes with one program, we risk losing precision, but we gain a more naturalistic intervention.

Strengths and Limitations

Med-Stress responds to a call “to conduct feasibility studies and randomized controlled studies on the effect of low-intensity interventions and technology supported (eg, web-based) interventions in low- and middle-income countries (LMICs), preferably using an active control condition as comparison, to ensure we disseminate effective treatments in LMICs” [80]. LMICs are reported to experience a disparity between the need for mental health services and their provision. We would argue that the need for such treatments is also urgent among certain groups, including health care providers. Content development of Med-Stress was preceded by a web-based focus study to better understand the working conditions of the target audience. We found that despite high levels of stress and burnout among medical professionals, no support programs were offered at the organizations where responders worked. Benefiting from traditional help often proves difficult for them due to irregular working schedules and an insufficient number of professionals. Thus, the availability of an internet intervention of low intensity and focused on personal resources that are context-free could be beneficial in alleviating some work-related mental health issues. This became particularly salient a few months later, when the COVID-19 pandemic broke and medical professionals found themselves in need of support under conditions where it could not be easily provided. In fact, to respond to this crisis, we developed a shorter, contextualized version of the Med-Stress intervention, called Med-Stress SOS [81].

Our study has several limitations. First, the dropout was high, which affects the interpretation of the results. We mitigated this risk by applying the MI method. There is an ongoing argument regarding the threshold over which the dropout rate is deemed unacceptable; however, the latest simulations indicate that it is not the proportion but the pattern of missingness that should be taken into account when considering data imputation. Specifically, in the case of the MAR pattern, the MI procedure, which includes all identified variables that differentiate dropouts from completers as imputation predictors, leads to the least biased results [74,82]. As the pattern of missingness in this study was indeed MAR, we followed these recommendations when conducting MI. Moreover, it should be said that internet interventions, in particular self-guided ones, do tend to have high dropout rates. Although such a loss to follow-up is rare, it is not unprecedented [83,84]. However, despite our attempts to mitigate the risks resulting from high dropout, the obtained results need to be treated with caution. This loss reflects the

pragmatic nature of this trial: it was conducted in Poland, where internet interventions are still rare, and we suspect that users did not have a framework into which this form of psychological help could be easily incorporated. A high recruitment rate compared with high dropout probably reflects participants' initial enthusiasm and curiosity that diminished over time. This was also reflected in the poor usage of the intervention, particularly in the experimental conditions in which the second assigned module was used much less frequently than the first one. In fact, comments that we received post intervention seem to support this notion: users referred to the content of the exercises (not user-friendly enough), pace of the intervention (new exercise released each week), and the fact that it was a web-based intervention and not an app-based intervention. High dropout was also a reason why we could not conduct a per protocol analysis: the sample of completers was too small to provide sufficient power. Ultimately, we had enough power to detect effects of minimum $d=0.21$, and therefore, all findings could not be generalized; however, the effects of job stress and job-related traumatic stress were large enough. Some of the

phenomena in this study, in particular job burnout and work engagement, are multidimensional, and analyzing them as such could provide a more in-depth understanding of the intervention's impact on them. However, this would further increase the already high number of comparisons that needed to be accounted for in the study design.

Conclusions

This study offers both theoretical and practical contributions. It is a pragmatic trial that offers an insight into how people really use this intervention [85]. It is an experimental verification of the cultivation versus enabling process of stress reduction, demonstrating cautious support for the former. Moreover, we empirically tested the concept of gain cycles for 5 components of well-being. In practical terms, Med-Stress is a support program that was found to be effective in enhancing the well-being of medical professionals. This intervention has the advantage of being broadly accessible to health care workers who currently receive no help in coping with their mental health problems.

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Authors' Contributions

ES, ML and RC conceptualized the study. ES and ML conducted statistical analyses and prepared original draft. GA, PC, and RC reviewed and edited the manuscript. All authors have read and approved the final manuscript.

Conflicts of Interest

None declared.

Multimedia Appendix 1

Means, SDs, and correlations for study variables.

[DOCX File, 25 KB-Multimedia Appendix 1]

Multimedia Appendix 2

CONSORT-EHEALTH checklist (V 1.6.1).

[PDF File (Adobe PDF File), 1585 KB-Multimedia Appendix 2]

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Abbreviations

ANOVA: analysis of variance

CBT: cognitive behavioral therapy

COR: conservation of resources theory

LMIC: low- and middle-income country

MAR: missing at random

MI: multiple imputation

RCT: randomized controlled trial

SE: self-efficacy enhancement module

SE+SS: self-efficacy and perceived social support sequential enhancement module

SS: perceived social support enhancement module

SS+SE: perceived social support and self-efficacy sequential enhancement module

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Terapia rozmową
Zeszyt ćwiczeń

**Zeszyt
ćwiczeń 1**

CBT, stany lękowe i depresja

Główny zeszyt
ćwiczeń

 **0300 365 2000**

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Przejdź przez powyższe etapy i zaznacz po ich ukończeniu

Wprowadzenie do CBT

W jaki sposób terapia poznawczo-behawioralna (CBT) może pomóc?

Terapia poznawczo-behawioralna (CBT) to jedna z form leczenia psychologicznego, która koncentruje się na „tu i teraz”.

Czasami gdy czujemy się przygnębieni lub zaniepokojeni, jesteśmy bardziej skłonni postrzegać rzeczy w negatywny sposób, a nasze myśli mogą nam przeszkadzać.

Podobnie może się też zmienić nasze zachowanie i rzeczy, które robimy lub których nie robimy, również stając się nieprzydatne lub negatywne.

Nie możemy bezpośrednio zmienić tego, jak się czujemy emocjonalnie lub fizycznie, ale znajdując przeszkadzające myśli i zachowania, rzucając im wyzwanie i zmieniając je, możemy dokonać zmian pośrednio.

Często to nasza reakcja na określony czynnik wyzwalający lub sytuację powoduje, że czujemy się w ten sposób.

To nie czynnik wyzwalający lub sama sytuacja powoduje, że czujemy się przygnębieni lub niespokojni, ale sposób, w jaki interpretujemy tę sytuację lub działamy z jej powodu.

Sposób, w jaki myślimy i działamy, często jest naszym sposobem radzenia sobie z czynnikiem wyzwalającym lub sytuacją.

Choć mogą być pomocne na krótką metę, te mechanizmy radzenia sobie mogą okazać się nieprzydatne w dłuższej perspektywie, zwłaszcza jeśli pozostaną niezmienione.



Wprowadzenie do CBT

Po co korzystać z CBT?

Terapia poznawczo-behawioralna pomaga nam nauczyć się technik zmiany nieprzydatnych mechanizmów radzenia sobie i daje nam narzędzia do wdrażania tych bardziej pomocnych.

Na przykład możemy nauczyć się zmieniać strukturę negatywnych myśli, skuteczniej radzić sobie ze zmartwieniami i rozwiązywać problemy, a także zmieniać nasze nieprzydatne zachowanie.

Udowodniono, że CBT, jako metoda samopomocy, jest jednym z najskuteczniejszych sposobów poprawy dobrostanu i została zatwierdzona przez Narodowy Instytut Doskonałości Klinicznej (NICE; www.nice.org.uk) jako zalecana metoda leczenia na stany lękowe i depresję.

Pomysł

- Terapia poznawczo-behawioralna zwykle nie skupiania się na czynniku wyzwalającym lub samej sytuacji, ponieważ chcemy skupić się na tym, co możemy kontrolować i zmieniać, czyli nasze myśli i zachowanie.
- To właśnie one podtrzymują nasze cykle złego nastroju i/lub niepokoju w odpowiedzi na czynnik wyzwalający lub sytuację.



Wprowadzenie do CBT

Jak wygląda leczenie?

Terapia poznawczo-behawioralna różni się od innych terapii tym, że wymaga od osoby, która jest nią leczona, wykonywania zadań poza sesjami w celu nauczenia się umiejętności i technik wymaganych do wprowadzenia zmian w sposobie myślenia, odczuwania i zachowania.

Te umiejętności i techniki dają nam zestaw narzędzi do wykorzystania teraz i w przyszłości, gdy życie następnym razem zrobi się trudne.

Jak zacząć?

Krok 1

Zidentyfikuj czynnik lub czynniki wyzwalające (przykłady na stronach 6 i 7).

Krok 2

Zidentyfikuj symptomy w następujących obszarach:

- Myśli
- Uczucia
- Odczucia fizyczne
- Zachowanie

Każdy z tych różnych obszarów często wpływa na inne.



Krok 3

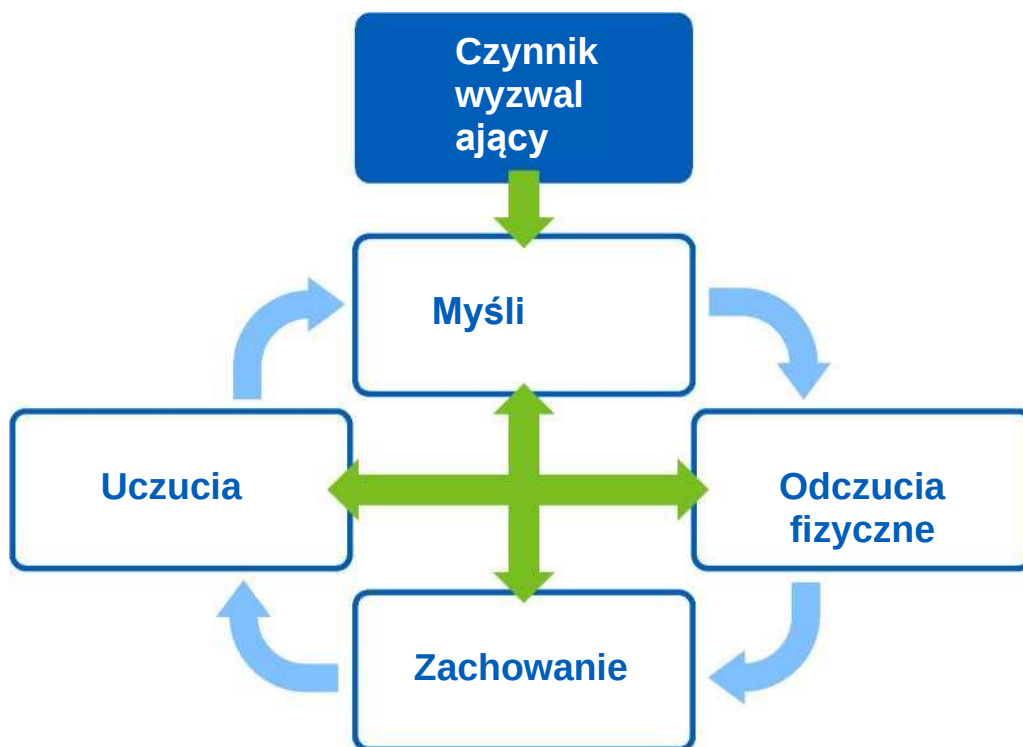
Wykorzystaj techniki, których nauczono Cię w trakcie sesji z terapeutą, aby przerwać te cykle.

Jeśli masz jakieś pytania na temat tego, jak działa CBT, porozmawiaj ze swoim terapeutą.

Drożdżowa bułeczka ze znakiem krzyża

Sposób, w jaki myślisz o problemie (**myśli**), może wpływać na reakcję Twojego ciała (**sfera fizyczna**) i emocjonalną (**uczucia**). Może również zmienić sposób, w jaki możesz zadziałać w odpowiedzi na ten czynnik wyzwalający (**zachowanie**).

Prostym sposobem pokazania tego procesu jest poniższy diagram „gorącej bułeczki ze znakiem krzyża”, którego nazwa wzięła się od krzyża pośrodku i okrągłego kształtu.



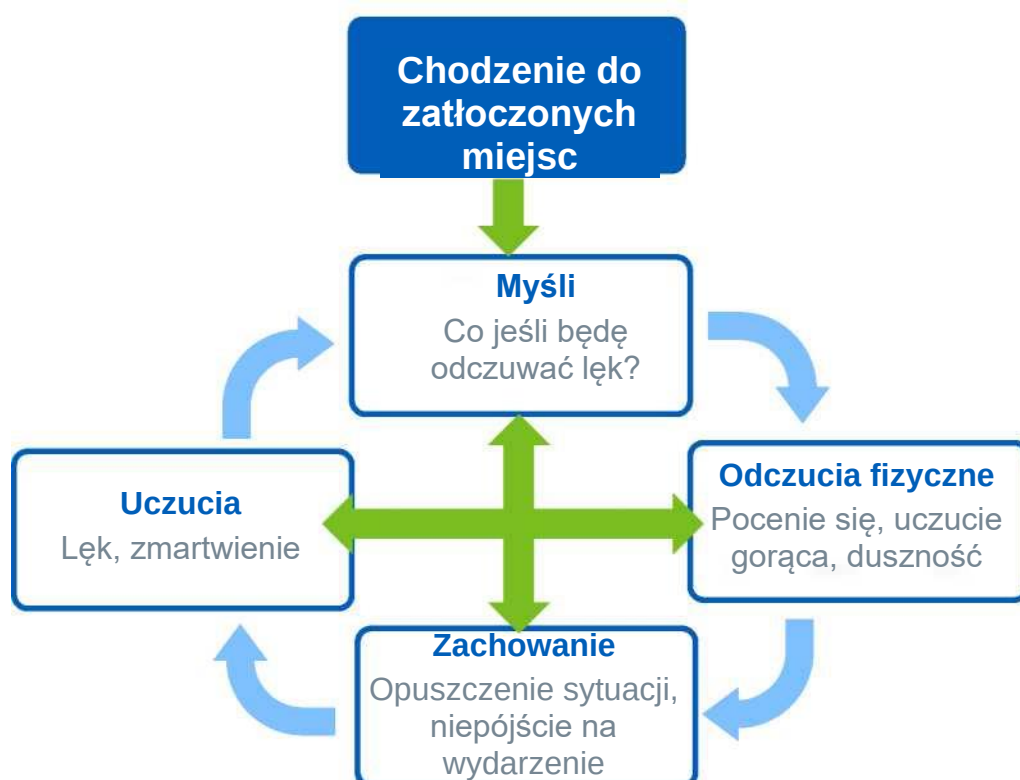
Spróbuj zrobić swoją własną bułeczkę na stronie 13.

Stany lękowe

Czym jest lęk?

Lęk jest normalną i naturalną reakcją na sytuacje, które postrzegamy jako zagrażające. Każdy odczuwa lęk, chociaż poziom lęku i sytuacje, w których się pojawia, mogą się różnić w zależności od osoby. Na przykład niektórzy ludzie boją się przemawiać przed dużą publicznością, podczas gdy inni chętnie to robią.

Przykład:



Co mogłoby utrzymywać cykl lękowy?

Unikanie

Daje nam krótkotrwałą ulgę, ale w dłuższej perspektywie wzmacnia nasze poczucie, że nie jesteśmy w stanie sobie poradzić, ponieważ nie dowiadujemy się, że w danej sytuacji dalibyśmy sobie radę (byłoby dobrze).

Błędna interpretacja

Niezrozumienie objawów fizycznych może prowadzić do myślenia, że dzieje się coś innego (np. zawał serca, choroba fizyczna itp.).

Walcz lub uciekaj

Lęk pełni bardzo ważną funkcję – chroni nas przed niebezpieczeństwem. Kiedy stajemy w obliczu zagrożenia, nasze ciała wytwarzają adrenalinę, aby przygotować nas do podjęcia działania. Stan ten powszechnie określany jest jako „walcz lub uciekaj”.

Kiedy stan „walcz lub uciekaj” staje się problemem?

Lęk staje się problemem, gdy zaczyna zakłócać naszą zdolność do funkcjonowania w życiu codziennym. Może się tak zdarzyć, gdy zaczniemy postrzegać neutralne, bezpieczne sytuacje jako zagrażające. Osoby cierpiące na stany lękowe mają tendencję do przeceniania zagrożenia w pewnych sytuacjach przy jednoczesnym niedocenianiu swoich możliwości poradzenia sobie z nim.

Prowadzi to do tego, że dana osoba czuje się przytłoczona lub pozbawiona gruntu pod nogami.



Dlaczego odczuwamy lęk?

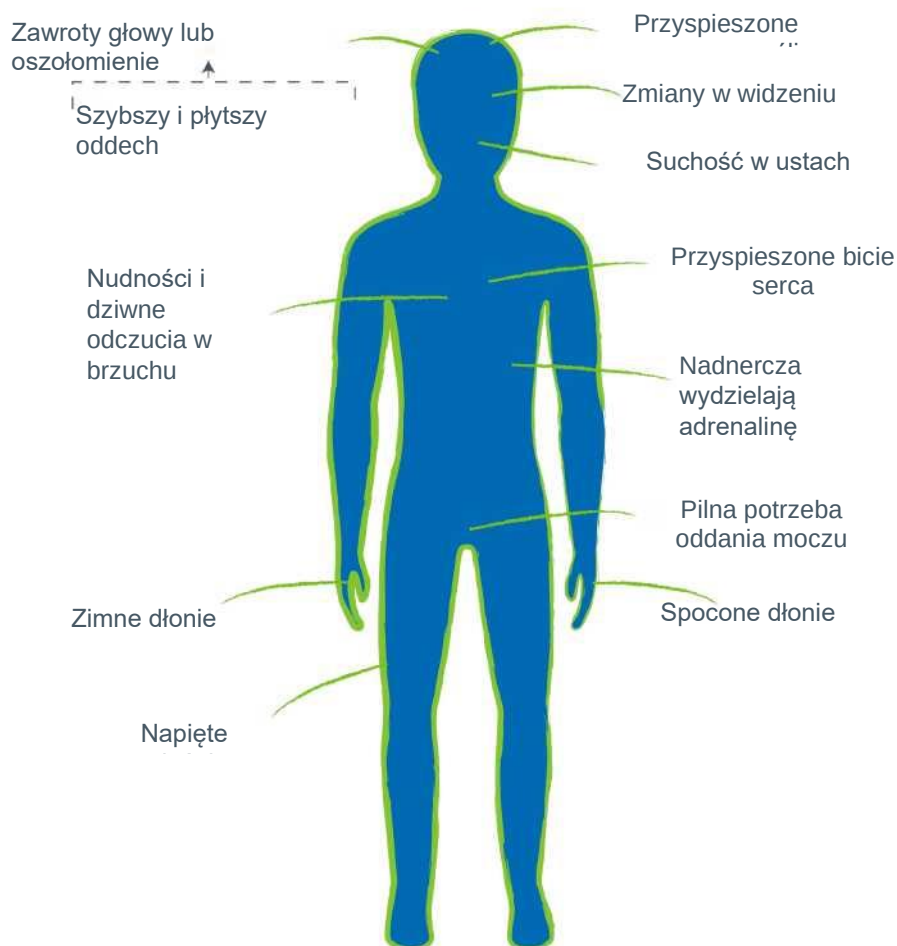
Wyobraź sobie taki scenariusz: Jaskiniowiec kontra Tygrys

Jaskiniowiec wyrusza na polowanie czegoś na obiad, gdy nagle spotyka głodnego tygrysa.

Co musi zrobić, żeby przetrwać? Musi albo walczyć z tygrysem, albo uciec.

Aby walczyć lub uciekać, organizm musi przygotować się na wiele sposobów:

- **Mózg** wysyła sygnały do ciała, aby pompowało adrenalinę do krwiobiegu i dużych mięśni szkieletowych ramion i nóg.
- **Serce** pompuje wtedy szybciej i potrzebuje więcej tlenu, przez co zwiększa się częstość oddechów.
- **Ciało** musi się ochłodzić, więc się poci, powodując wilgoć lub podjęście naczynek pod powierzchnię skóry, co objawia się rumieńcem.
- Aby być jak najlżejszym, może zaistnieć potrzeba opróżnienia jelit lub częstego oddawania moczu.



W naszym współczesnym, codziennym życiu jest bardzo mało prawdopodobne, że będziemy mieli do czynienia z głodnymi tygrysami – wciąż jednak jesteśmy narażeni na rzeczywiste lub wyimaginowane zagrożenia albo stresory, a nasze ciała reagują na nie w ten sam sposób.

Jeśli sytuacja nie stanowi fizycznego zagrożenia (np. dawanie prezentacji), taka reakcja nie pomoże w poradzeniu sobie z nią, a w rzeczywistości może być dość przerażająca.

Po lewej: Typowe objawy fizyczne reakcji „walcz lub uciekaj”.

Depresja

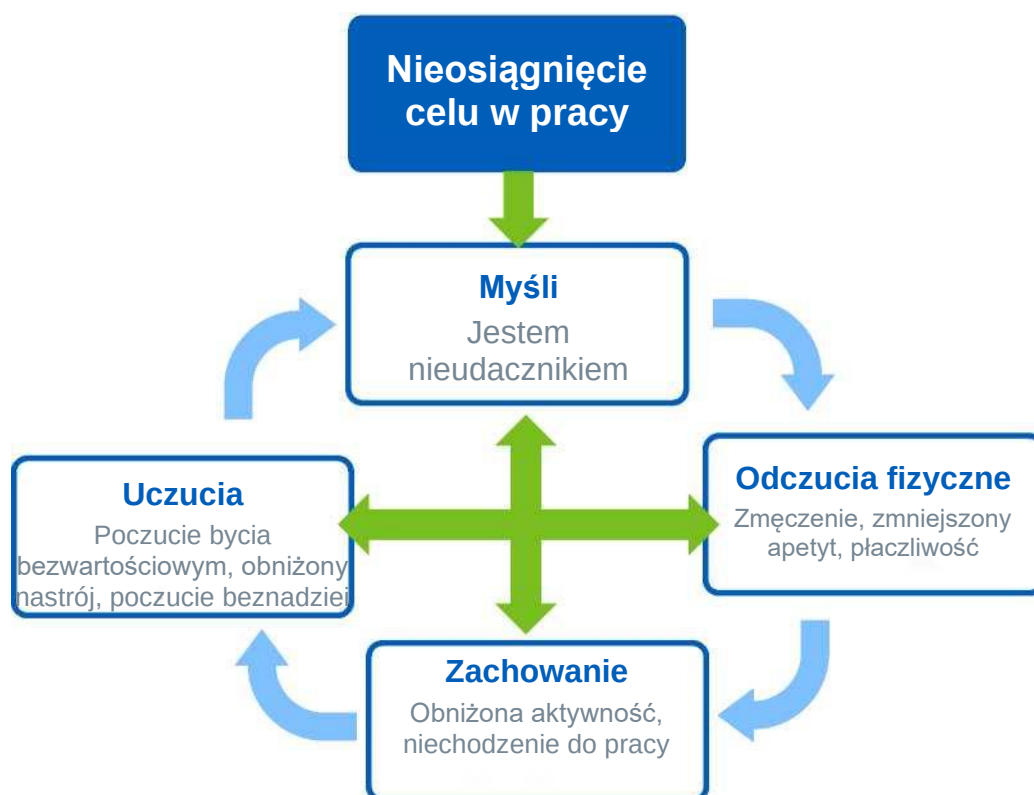
Czym jest depresja?

Uczucie zmęczenia i obniżonego nastroju jest normalną częścią życia. Jest to szczególnie prawdziwe, gdy jest reakcją na przygnębiające lub trudne wydarzenie lub sytuację.

Zwykle przyczyny złego nastroju są jasne (np. stresująca sytuacja, trudności w związku, jak np. poczucie bycia zawiedzionym przez kogoś, lub problem praktyczny), a spadek naszego nastroju trwa tylko przez krótki czas.

Czasami może być tak, że nastrój danej osoby spada z niewielkiej lub żadnej oczywistej przyczyny i początkowo może być trudno ustalić przyczynę. W niektórych przypadkach depresja może się pogorszyć i całkowicie zdominować życie danej osoby. Kiedy ktoś czuje się bardzo źle przez ponad dwa tygodnie, dzień po dniu, tydzień po tygodniu, nazywa się to chorobą depresyjną. Kiedy depresja pojawia się w ten sposób, wpływa na nastrój i sposób myślenia danej osoby. Może również prowadzić do zmiany zachowania i powodować szereg objawów fizycznych w organizmie.

Przykład:



Depresja

Co mogłoby utrzymywać cykl depresji?

Obniżona aktywność

Obniżona aktywność często występuje, gdy dana osoba przechodzi depresję. Poziom aktywności może się zmniejszyć z powodu braku energii, niskiej motywacji i faktu, że wykonywanie różnych czynności wymaga dużo większego wysiłku.

Ograniczona aktywność może oznaczać, że mamy trudności z wykonywaniem rutynowych czynności, takich jak obowiązki domowe, a także czynnościami niezbędnymi, takimi jak płacenie rachunków lub rozwiązywanie ważnych problemów, oraz czynnościami przyjemnymi, takimi jak spotkania z przyjaciółmi i hobby.

Nie angażując się już w te czynności, tracimy możliwość odczuwania przyjemności, radości i osiągnięć, które zwykle daje nam ich wykonywanie. Chociaż unikanie aktywności daje nam krótkotrwałe poczucie ulgi, w dłuższej perspektywie wpływa na nasz nastrój, wzmacniając go i utrzymując. Im mniej robimy, tym gorzej się czujemy, a im gorzej się czujemy, tym mniej robimy. To błędne koło.

Obsesyjne rozmyślanie

Obsesyjne rozmyślanie ma miejsce wtedy, gdy masz ciągle i powtarzające się myśli o czymś.

Czasami możemy przyłapać się na tym, że ciągle wracamy myślami do rzeczy, które wydarzyły się w przeszłości lub do decyzji, które podjęliśmy, a które doprowadziły do naszej obecnej sytuacji. Może to dokładać do naszego obniżonego nastroju, sprawiać, że będziemy mieć trudności z angażowaniem się w działania i byciem obecnymi w danej chwili, a także prowadzić do innych pułapek negatywnego myślenia.

Im więcej mamy negatywnych myśli, tym gorzej się czujemy, a im gorzej się czujemy, tym więcej mamy negatywnych myśli. To błędne koło.



Twoja pierwsza sesja

Jak przygotować się do pierwszej sesji?

Jeśli masz zarezerwowaną pierwszą wizytę ze swoim terapeutą, uzupełnij poniższe strony.

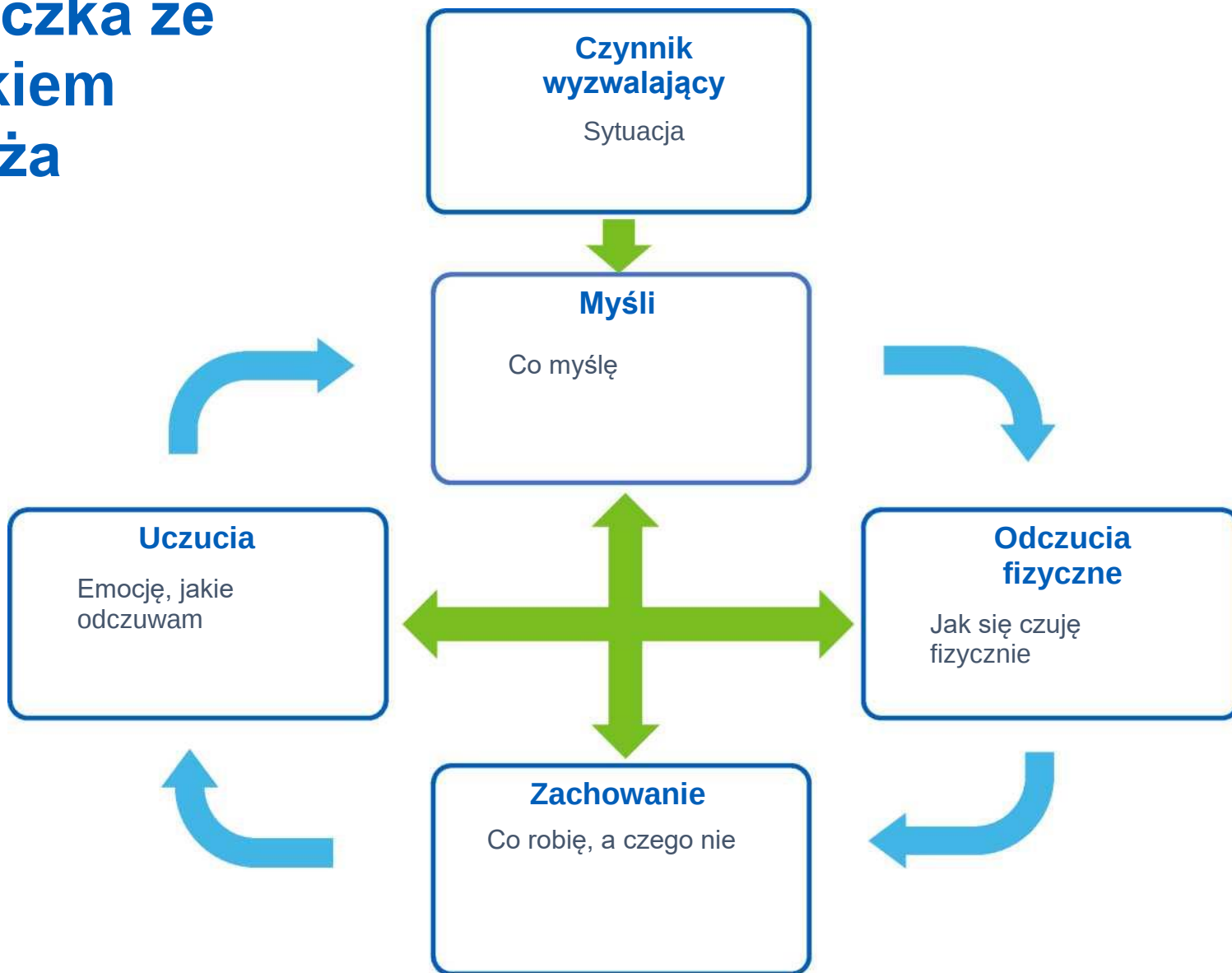
Jeśli wizyta odbywa się osobiście, ważne jest, aby zabrać je ze sobą i przekazać je terapeutcie. Jeśli wizyta odbywa się przez telefon lub za pośrednictwem łącza wideo, należy mieć je wtedy przed sobą.



Rzeczy do uzupełnienia

- ☐ **Gorąca bułeczka ze znakiem krzyża** – spróbuj stworzyć własną gorącą bułeczkę, biorąc pod uwagę czynniki wyzwalające, to, jak się czujesz fizycznie i emocjonalnie, co w rezultacie robisz, a czego nie robisz, oraz myśli, które przychodzą ci do głowy.
- ☐ **Opis problemu** – podsumuj swoje główne problemy
- ☐ **Wyznaczanie celów** – Co chcesz osiągnąć w trakcie leczenia?
- ☐ **Kwestionariusze**

Drożdżowa bułeczka ze znakiem krzyża



Zdefiniowanie problemu

Pomocne może być zastanowienie się nad swoimi obecnymi trudnościami w tak zwanym opisie problemu. Uzupełnij opis problemu poniżej:

Moje zdefiniowanie problemu	
Moim głównym problemem jest...	
Czynnikiem wyzwalającym jest...	
Myślę, że...	
Zacząłem(-am)/przestałem(-am)...	
Fizycznie odczuwam...	
To wszystko wpłynęło na...	

Stawianie sobie celu

Jakie są Twoje cele leczenia?

Planując cele leczenia, ważne jest, aby upewnić się, że są to cele **SMART**.

- **S**pecific (konkretne) – Co, gdzie, kiedy, z kim?
- **M**easurable (mieralne) – po czym poznasz, że dane działanie zostało wykonane?
- **A**chievable (osiągalne) – czy masz wszystko, czego potrzebujesz, aby to osiągnąć?
- **R**ealistic (realistyczne) – czy cele są realistyczne?
- **T**ime-focused (oparte na czasie) – o której godzinie, jakiego dnia i na jak długo?

Przykład:

„Ćwicz” zmienia się w „Idź pływać w każdy piątek o 9:00 przez 30 minut, samemu”

Moje cele SMART	
Cel SMART 1:	
Cel SMART 2:	
Cel SMART 3:	
Cel SMART 4:	

Kwestionariusze

Czym są kwestionariusze?

W ramach leczenia będziesz poproszona(-y) o wypełnienie kwestionariuszy dotyczących każdej sesji. Możesz je rozpoznać z pierwszej wizyty oceniającej. Omówiono w nich typowe objawy lęku, depresji, fobii i ich wpływ. Możesz także otrzymać dodatkowe kwestionariusze, jeśli będą one powiązane z Twoimi obecnymi trudnościami. Nie używamy ich do diagnozowania jakichkolwiek schorzeń psychicznych, ale do sprawdzenia, jak się czujesz i czy występują jakieś objawy, o których powinniśmy wiedzieć w trakcie leczenia, aby móc zapewnić Ci najlepsze możliwe wsparcie.

Jak wypełnić kwestionariusze?

Zazwyczaj są one wysyłane do Ciebie e-mailem i należy je wypełnić przed spotkaniem, jeśli udzieliłaś(-eś) zgodny na wysyłanie do Ciebie formularzy internetowych. Pomocne jest ich wypełnienie przed każdą sesją.



[illegible]

Notatki

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Dodatkowe zasoby



Zeszyty ćwiczeń

Centre for Clinical Interventions (CCI) – <https://www.cci.health.wa.gov.au/>

Wiele szczegółowych zasobów i zeszytów ćwiczeń w formacie PDF dostępnych do bezpłatnego pobrania, skupiających się na różnych problemach związanych z depresją i stanami lękowymi.

Northumberland, Tyne and Wear NHS – <https://web.ntw.nhs.uk/selfhelp/>

Poradniki samopomocy do bezpłatnego pobrania w formacie PDF i wydrukowania, dotyczące różnych problemów, w tym lęku i depresji.

Aplikacje na smartfony (do pobrania ze sklepu Apple i Google Play)

WRAP – aplikacja do odzyskiwania dobrego samopoczucia

Mood Tools – pomoc w depresji

Fear Tools – pomoc w stanach lękowych

Insight Timer – aplikacja do medytacji

Strony internetowe

Mind – <https://www.mind.org.uk/>

Mental Health Foundation - <https://www.mentalhealth.org.uk/>

NHS - <https://www.nhs.uk/mental-health/>

Użyteczne kontakty...



Terapia rozmową:

E-mail:

Inne kontakty:

Zespół kryzysowy w Berkshire:

Samarytanie:

0300 365 2000

(Otwarte od 8:00 do 20:00 od poniedziałku do czwartku,
a w piątek od 8:00 do 17:00)
talkingtherapies@berkshire.nhs.uk

0800 129 9999

(24 godziny na dobę, specjalistyczna pomoc w przypadku nagłego kryzysu psychicznego)

116 123

(24 godziny, usługa poufnego wysłuchania)

W nagłym wypadku zadzwoń na numer 999

NHS Direct / poza godzinami pracy:

111

(24 godziny na dobę, problemy związane ze zdrowiem)



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Temperament, poczucie własnej skuteczności i jakość życia kobiet regularnie uprawiających sport w porównaniu z kobietami nieaktywnymi fizycznie Temperament, self-efficacy and quality of life of women regularly doing sport in comparison with physically inactive women

Barbara Grześkowiak, Anna Siwy-Hudowska

SWPS Uniwersytet Humanistycznospołeczny w Warszawie, Wydział Psychologii
University of Social Sciences and Humanities in Warsaw, Psychology Department

Streszczenie

Wprowadzenie i cel pracy

Celem pracy było zbadanie różnic w subiektywnej ocenie jakości życia i jej sfer oraz w zakresie cech temperamentu i poczucia własnej skuteczności pomiędzy kobietami regularnie uprawiającymi sport a kobietami deklarującymi brak aktywności sportowej, a także sprawdzenie, jakie jest znaczenie temperamentu i poczucia własnej skuteczności dla jakości życia kobiet w porównywanych grupach.

Materiał i metoda

Badanie było przeprowadzane w dwóch grupach kobiet: 35 uprawiających sport i 35 nieaktywnych fizycznie (brak jakiegokolwiek zaangażowania w uprawianie sportu). Osoby wypełniły kwestionariusze samoopisowe: Kwestionariusz Jakości Życia, Kwestionariusz Temperamentu EAS, Skalę Uogólnionej Własnej Skuteczności (GSES) oraz ankietę dotyczącą danych demograficznych i sportowych.

Wyniki

Kobiety uprawiające sport miały wyższe poczucie własnej skuteczności oraz wyżej oceniły wszystkie sfery własnej jakości życia niż kobiety nieaktywne sportowo. Ponad 57% kobiet nie uczestniczących w regularnych zajęciach sportowych chciałyby poprawić swoją kondycję fizyczną. Wykazano także, że aktywność fizyczna poprzez podnoszenie poczucia własnej skuteczności może mieć pozytywny wpływ na subiektywną ocenę jakości życia. W zakresie cech temperamentu kobiety aktywne fizycznie charakteryzowały się istotnie niższym niezadowoleniem niż kobiety nieaktywne fizycznie. Wraz ze spadkiem niezadowolenia przy występowaniu aktywności fizycznej zwiększa się jakość życia badanych kobiet.

Wnioski

Uprawianie sportu sprzyja doświadczaniu pozytywnych zmian w sobie. Może to skutkować podniesieniem poczucia własnej skuteczności, co pozwala lepiej myśleć o sobie i efektywniej uzyskiwać cele, które są ważne dla jakości życia jednostki.

Słowa kluczowe: jakość życia, temperament, poczucie własnej skuteczności, aktywność fizyczna.

Abstract

Introduction and purpose

The aim of the work was to analyse the differences in subjective quality of life and its domains and, moreover, in the field of temperamental traits and self-efficacy between the women regularly doing sport in comparison to those who do not. The impact of temperament and self-efficacy on the quality of life was also verified.

Material and method

The research was carried out on two groups of females: 35 active and 35 physically inactive (doing no sports whatsoever). Participants completed self-description questionnaires: Quality of Life Questionnaire, EAS Temperament Questionnaire, SES Self-Efficacy Scale and the form of demographic and sport details.

Results

Active women had higher self-efficacy and better assessed each of domains of the quality of life than inactive women. More than 57% of inactive women wanted to improve their physical condition. It was shown also that physical activity through self-efficacy improvement can positively influence subjective quality of life. In the field of temperamental traits, active women were characterized by significantly lower emotionality-distress than inactive women. When emotionality-distress decreased with the presence of physical activity, the person's quality of life increased.

Conclusions

Doing sport is conducive to experiencing of positive changes in one's life. This may result in higher self-efficacy which allows women to think better about themselves and more effectively reach goals that are important for the quality of life.

Keywords: quality of life, temperament, self-efficacy, physical activity, sport.

Wprowadzenie

Nie będzie przesadą stwierdzenie, że wszyscy ludzie chcieliby mieć wysokie poczucie zadowolenia ze swojego życia, chociaż dróg jego osiągnięcia jest pewnie tyle samo ile ludzi. Jeśli podejmowane działania nie podnoszą satysfakcji z życia, to przez to mogą ją wręcz obniżać. Dlatego wydaje się zasadne prowadzenie badań nad tym, jakie zachowania i czynności sprzyjają poprawie jakości życia, a także jakie stałe predyspozycje osobowościowe są istotne i różnicują osoby o wyższej ocenie jakości życia, od tych, które oceniają je niżej. Uprawianie sportu to jeden z wielu, i co ważne, łatwo dostępnych i tanich sposobów podniesienia satysfakcji z życia. Przeprowadzone badanie opiera się na porównaniu właściwości kobiet regularnie uczestniczących w zajęciach sportowych z kobietami nieaktywnymi sportowo.

Aktywność fizyczna wpływa na funkcjonowanie fizyczne i psychiczne człowieka. Analizując aspekty psychiczne, czyli pozytywne i negatywne emocje związane z uprawianiem sportu, nie da się jednak pominąć czynników biologicznych, czyli wpływu hormonów, neuroprzekaźników zarówno na subiektywne odczucia (psychikę) jak i na funkcjonowanie organizmu (fizjologię). Negatywne odczucia, które mogą pojawić się przy okazji różnych form zajęć sportowych, to stres związany ze współzawodnictwem lub oczekiwaniem określonego wyniku. Także lęk przed porażką lub smutek w przypadku rozczarowania swoim wynikiem. Pozytywne emocje to najczęściej zadowolenie z zaangażowania w aktywność, satysfakcja i duma z realizacji założonych celów lub osiągnięcia określonego wyniku. Na pozytywne emocje mają wpływ wydzielane hormony (endorfiny), które dają poczucie spokoju i odprężenia a jednocześnie hamują wytwarzanie hormonów stresu, które mogą zdominować odczucia w sytuacji zbyt intensywnego treningu wytrzymałościowego [1]. Zadowolenie ze swojej sprawności fizycznej jest ważnym elementem oceny jakości życia [2]. Badania pokazały, że ludzie biorący udział w programie ćwiczeń po jego zakończeniu ogólnie wyżej oceniali poziom swojego życia i większy wigor w wykonywaniu codziennych czynności [3]. Kilka godzin po treningu fizycznym pojawiają się emocjonalne skutki w formie ostrej, nazywane szczytem góry lodowej. Przejawiają się wysokim poziomem wigoru oraz niskim poziomem negatywnych nastrojów, takich jak

depresja, lęk, wrogość, przygnębienie czy znużenie. Regularne, wielotygodniowe uprawianie sportu przynosi z kolei długotrwałą poprawę samopoczucia, także zwiększenie wigoru i pewności siebie. Pozytywne emocje pojawiają się przy okazji realizacji wspólnych ćwiczeń, zadowolenia z uprawiania danej dyscypliny sportowej, osiągania coraz lepszych efektów. Bardzo ważne są także relacje społeczne związane z uprawianiem sportu. Powtarzalne ćwiczenia aerobowe przy muzyce, w których pojawia się rytmiczne oddychanie, w znanym sobie środowisku, bez silnego współzawodnictwa sprzyjają poczuciu zadowolenia. Regularne ćwiczenia przynajmniej 2-3 razy w tygodniu po okresie 3-6 miesięcy dają już subiektywne poczucie większej sprawności i wydolności fizycznej. Po takim czasie pojawia się także długotrwała poprawa nastroju i zmniejszenie negatywnych emocji [2]. Z drugiej strony, zbyt intensywne ćwiczenia lub zbyt długotrwałe mogą powodować przeciążenie, przemęczenie i spadek motywacji. Wyższy dobrostan i satysfakcja z uprawiania sportu pojawią się wówczas, gdy sama aktywność sportowa jako czynność daje poczucie zadowolenia i jest zgodna z możliwościami osoby ćwiczącej. Odpowiednia długość i intensywność ćwiczeń oraz powtarzalność i rytmiczność, jak również brak silnej rywalizacji oraz poczucie wspólnoty z innymi osobami ćwiczącymi sprzyja zwiększeniu zadowolenia z samego uprawiania sportu, a to prowadzi do wyższej subiektywnej jakości życia [4].

Jakość życia od dawna była rozumiana na wiele różnych sposobów i łączona z wieloma obszarami ludzkiej aktywności. Początkowo była łączona z posiadaniem różnych dóbr materialnych, określonego, wysokiego dochodu pozwalającego na zapewnienie obiektywnie wysokiego standardu życia. W psychologii ujmuje się jakość życia z jednej strony jako obiektywnie istniejące zjawisko, obejmujące takie elementy, jak: stan zdrowia, wykształcenie, sytuacja rodzinna, poziom dochodów, posiadanie dóbr materialnych, zaś z drugiej strony – subiektywne czynniki psychologiczne, takie jak: bilans pozytywnych i negatywnych emocji, samorealizacja oraz zadowolenie z życia i poszczególnych jego sfer [5]. W modelu opracowanym przez badaczy z Centrum Promocji Zdrowia Uniwersytetu w Toronto jakość życia definiuje się najpełniej jako „stopień, w jakim jednostka korzysta z możliwości, jakie niesie ze sobą życie” [6]. W tym modelu jakość życia ocenia się przez pryzmat trzech komponentów: 1) istnienie – przez co należy rozumieć dosłowne istnienie fizyczne, ale także psychologiczne i duchowe, 2) przynależność – określana przez kontakty fizyczne, społeczne i socjalne, 3) stawianie się – odnosi się do realizacji celów, nadziei i aspiracji. Zaletą tego modelu jest szerokie spojrzenie oraz

wyróżnienie rozwoju jako ważnej składowej jakości życia. Światowa Organizacja Zdrowia [7] patrzy na jakość życia także z wielu perspektyw. Ważne są zarówno czynniki obiektywne, takie jak: zdrowie fizyczne i psychiczne oraz relacje społeczne i środowisko. Nie bez znaczenia pozostaje również subiektywna ocena sytuacji życiowej jednostki poprzez jej własny system wartości w powiązaniu z realizacją celów, oczekiwań i standardów. Przykładem subiektywnej perspektywy oceny własnego życia i jego sfer (pracy, zdrowia) jest transakcyjny model jakości życia Anny Zalewskiej, w którym oceny wyrażane są w dwóch ujęciach: afektywnym i poznawczym. System ocen emocjonalnych i poznawczych zależy od trzeciego czynnika, którym jest poziom reaktywności emocjonalnej jednostki (w innych ujęciach teoretycznych określany jako neurotyczność lub emocjonalność). Ta ogólna cecha temperamentu uwarunkowana biologicznie wyraża siłę i szybkość reakcji na pojawiający się bodziec. Osoby wysoko reaktywne charakteryzują się wysoką pobudliwością i aktywowalnością, lepiej się czują w środowisku o mniejszej ilości bodźców. Osoby nisko reaktywne dla optymalnego funkcjonowania potrzebują większej ilości bodźców. Zatem reaktywność modyfikuje zależności między emocjonalnymi i poznawczymi ocenami jakości życia i jego sfer [8]. Buss i Plomin [9] wyodrębnili trzy cechy temperamentu: emocjonalność (Emotionality), aktywność (Activity) oraz towarzyskość (Sociability), które od pierwszych liter angielskich słów stworzyły skrót EAS. Emocjonalność definiują jako skłonność do reagowania silnym pobudzeniem, „to tyle co niezadowolenie, skłonność do łatwego i silnego rozstrojenia” [10]. W obszarze tej cechy mieszczą się tylko emocje negatywne tj. niezadowolenie, strach, złość, które jako pierwsze pojawiają się w rozwoju dziecka. Nie uwzględnia się tutaj emocji pozytywnych jako niespełniających kryteriów definicyjnych oraz odznaczających się niskim poziomem aktywacji. Aktywność, czyli ruchliwość i poziom wydatkowanej energii, przejawia się w każdym zachowaniu człowieka. Można ją opisać pod kątem dwóch składników: wigoru i tempa, które są ze sobą dodatnio skorelowane. Wigor jest określany jako poziom energetyczny zachowania, zaś tempo jako szybkość reakcji. „Towarzyskość oznacza tendencję do przedkładania towarzystwa innych osób nad samotność” [10]. Cecha ta rozwija się poprzez pozytywne wzmocnienia z interakcji społecznych. Do tych wzmocnień należą: obecność innych ludzi, wspólna aktywność, zainteresowanie ze strony innych, wzajemne reagowanie na siebie oraz inicjowanie kontaktów społecznych. Im większe nasilenie cechy towarzyskości, tym silniejsze reagowanie na niedobór kontaktu z ludźmi. Wysoki poziom emocjonalności ujemnie koreluje z poczuciem szczęścia i oceną jakości życia. Natomiast

wysoki poziom ekstrawersji, dodatnio korelującej z aktywnością w teorii Bussa i Plomina, wykazuje dodatni związek z odczuwanym szczęściem i wysoką oceną jakości życia [11]. W przypadku satysfakcji z uprawiania sportu także ważne są: zdolność do przeżywania pozytywnych emocji i jednocześnie umiejętność radzenia sobie z nadmiernym podekscytowaniem i lękiem [12]. Ekstrawertycy odczuwają więcej pozytywnych emocji w kontaktach społecznych, niż w sytuacjach, gdy muszą działać w pojedynkę. Także ćwiczenie lub trenowanie wspólnie z innymi osobami może być dla ekstrawertyków jednym z powodów wzrostu ich satysfakcji z uprawiania sportu. Osoby z wysokimi wynikami na skali satysfakcji ze sportu charakteryzują się wyższą ekstrawersją i niższą neurotycznością, podczas gdy mniejsze znaczenie mają ugodowość i sumienność [12]. Okazuje się również, że wyniki satysfakcji z życia i sportu są ze sobą umiarkowanie powiązane. Najlepszymi predyktorami satysfakcji z życia i sportu okazały się ekstrawersja i neurotyzm, chociaż ich wartość predykcyjna była wyższa dla zadowolenia z życia niż ze sportu. Można więc powiedzieć, że osoby mniej neurotyczne, bardziej ekstrawertyczne, będą wyżej oceniały jakość swojego życia i będą szczęśliwsze [13].

Elavsky i współpracownicy [14] w 5-letnim badaniu podłużnym wykazali, że jakość życia może być oceniana wyżej nie tyle poprzez bezpośredni wpływ aktywności fizycznej, co poprzez poczucie własnej skuteczności, które z kolei jest wzmacniane przez fizyczną aktywność. Istotny związek pomiędzy regularną aktywnością sportową a poczuciem własnej skuteczności, które z kolei miało wpływ na jakość życia zauważono także w badaniu McAuley'a i współpracowników [15]. Zależności te były weryfikowane w badaniu Paxton, Motla, Aylward i Nigga [16] w modelu linearnym oraz modelu jednoczesnego wpływu aktywności sportowej na poczucie skuteczności i zdrowie psychiczne, a pośrednio na jakość życia. Oba te modele zostały potwierdzone w badaniu empirycznym, a wskaźniki dopasowania były zbliżone, co sugeruje, że obie zmienne (skuteczność i zdrowie psychiczne) są komplementarne w związku pomiędzy aktywnością fizyczną i jakością życia [4]. Badanie potwierdziło, że aktywność fizyczna wpływa na jakość życia za pośrednictwem własnej skuteczności i zdrowia psychicznego, nadal jednak zależności te nie są w pełni jasne i wymagają dalszych badań.

Poczucie własnej skuteczności to konstrukt psychologiczny odnoszący się do przekonań na temat własnych możliwości osiągnięcia celów. Nadają one kierunek postępowaniu i wybieraniu celów, zadań, co do których mamy wewnętrzne przekonanie, że są osiągalne. Wg Bandury, nasze przekonanie o skuteczności postępowania determinuje nasze oczekiwanie wyniku. Jeśli

oczekujemy powodzenia naszych przedsięwzięć, to dodatkowo motywujemy się do dalszego działania. Jeśli w stosunku do jakiegoś zadania mamy odwrotne założenia, to znaczy, jeśli przeczuwamy problemy, to takie nastawienie dodatkowo wpływa zniechęcająco i w konsekwencji może doprowadzić do porażki, która potwierdzi nasze pierwotne przypuszczenia [17]. Wg Bandury są przynajmniej cztery sposoby kształtowania poczucia własnej skuteczności: (1) poprzez wcześniejsze doświadczanie sukcesów i porażek, (2) poprzez obserwowanie innych ludzi, zwłaszcza ważnych osób czy autorytetów, którzy dzięki wytrwałości i zaangażowaniu osiągają swoje cele, (3) poprzez uzyskiwane informacje zwrotne na temat naszych działań i ich konsekwencji, (4) poprzez pozytywne odczucia emocjonalne i fizyczne związane z dążeniem do celów i ich osiągnięciem [4].

Podsumowując powyższe rozważania warto podkreślić, że ocena jakości życia jest zależna od cech osobowych i środowiska oraz ich interakcji. Zatem, na ocenę życia osoby podejmującej aktywność fizyczną, wpływa zarówno jej konstrukcja temperamentalna, jak i czynnik sytuacyjny. Przez czynnik sytuacyjny rozumie się istnienie dostępnych, konkurencyjnych czynności istotnych dla tej jednostki.

Cel pracy

Zasadniczym celem pracy było zbadanie różnic w subiektywnej ocenie jakości życia i jej sfer oraz w zakresie cech temperamentu i poczucia własnej skuteczności pomiędzy kobietami regularnie uprawiającymi sport a kobietami deklarującymi brak aktywności sportowej, a także sprawdzenie, jakie jest znaczenie temperamentu i poczucia własnej skuteczności dla jakości życia kobiet w porównywanych grupach.

Materiał i metoda

Badanie było przeprowadzane w dwóch grupach kobiet: nieaktywnych fizycznie, a więc tych, które deklarowały brak jakiegokolwiek zaangażowania w uprawianie sportu oraz aktywnych fizycznie, do których były kwalifikowane te, które deklarowały regularne uczestnictwo w zajęciach sportowych od przynajmniej 6 miesięcy z częstotliwością minimum 2-3 razy w tygodniu przez minimum 3 godziny tygodniowo. Większość kobiet aktywnych sportowo, które wzięły udział w badaniu, od wielu lat uczestniczyło w zajęciach typu fitness organizowanych przez Kluby FitnessPark Warszawianka i Agora w Warszawie. Szczegółowe rodzaje podejmowanych aktywności zawarto w tabeli 1.

Tabela 1. Rodzaje aktywności sportowej w grupie aktywnych fizycznie

Typ zajęć	<i>Częstość</i>	<i>Procent</i>	<i>Procent skumulowany</i>
Mieszane (Cardio&Muscle)	31	32,98%	32,98%
Body&mind	16	17,02%	50,00%
Bieganie	11	11,70%	61,70%
Choreograficzne	8	8,51%	70,21%
Pływanie/aqua	8	8,51%	78,72%
Wzmacniające (Muscle)	5	5,32%	84,04%
Rower	5	5,32%	89,36%
Siłownia	4	4,26%	93,62%
Inne	4	4,26%	97,87%
Hi-Low impact (Cardio)	2	2,13%	100,00%
Ogółem	94	100,00%	

Wszystkie osoby badane proszone były o wypełnienie pakietu kwestionariuszy samoopisowych: Kwestionariusza Jakości Życia [18], Kwestionariusza Temperamentu EAS Bussa i Plomina w adaptacji Włodzimierza Oniszczenko [19], Skali Uogólnionej Własnej Skuteczności (GSES) opracowanej Schwarzera i Jerusalema w polskiej adaptacji Zygryda Juczyńskiego [20] oraz ankiety socjodemograficznej, skonstruowanej na potrzeby tego badania. Kwestionariusz Jakości Życia jest narzędziem mierzącym zgeneralizowaną jakość życia oraz jej sfery: zdrowie fizyczne (ocena zdrowia, dolegliwości, przyjmowanie leków), samopoczucie (ocena samopoczucia psychicznego, zadowolenia i szczęścia), codzienne funkcjonowanie (ocena motywacji i energii do codziennego działania oraz tego jak osoba badana jest odbierana przez otoczenie w codziennym funkcjonowaniu), ocena sytuacji osobistej, rodzinnej, finansowej (ocena jakości życia rodzinnego i sytuacji ekonomicznej), czas wolny (ocena jakości życia wynikającej z posiadania i spędzania czasu wolnego), rozwój własny (ocena zadowolenia z własnego rozwoju), ocena życia seksualnego (ocena zadowolenia z życia intymnego) i ocena generalna jakości życia (ocena ogólna postrzeganej jakości życia). Odrębne pytanie w kwestionariuszu dotyczy nastawienia do przyszłości (wybór obszarów życia, w których osoba badana chciałaby w najbliższym czasie dokonać zmiany). Punkty uzyskane w poszczególnych sferach jakości życia składają się na sumaryczny wynik oceny jakości życia osoby badanej. Kwestionariusz

Temperamentu EAS mierzy 3 cechy temperamentu: emocjonalność (na którą składają się trzy składniki – niezadowolenie, strach i złość), aktywność i towarzyskość. Skala GSES mierzy siłę ogólnego przekonania jednostki co do skuteczności radzenia sobie z trudnymi sytuacjami i przeszkodami.

Wyniki

Do badania przystąpiło 70 kobiet, w tym 35 deklarujących regularną aktywność fizyczną oraz 35 nieaktywnych sportowo. Grupy te nie różniły się istotnie średnim wiekiem: $t(68)=1,15$; $p>0,05$ (aktywne: $M=37,43$, $SD=8,89$; nieaktywne: $M=35,31$, $SD=6,24$). W pierwszej kolejności sprawdzono, czy kobiety uprawiające sport różnią się od kobiet nieaktywnych fizycznie oceną jakości życia i jej sfer oraz poczuciem własnej skuteczności. Przeprowadzono szereg analiz testami t-Studenta (Tabela 2).

Tabela 2. Porównanie ocen jakości życia i poczucia własnej skuteczności kobiet aktywnych i nieaktywnych sportowo.

Zmienne	M		SD		df	t Studenta	d Cohena
	Aktyw.	N.aktyw.	Aktyw.	N.aktyw.			
zdrowie fizyczne	16,49	14,80	1,67	2,13	68	3,69***	0,89
samopoczucie	41,31	36,06	4,91	5,43	68	4,25***	1,03
codzienne funkcjonowanie	29,23	26,17	3,19	3,61	68	3,75***	0,91
ocena sytuacji rodzinnej, osobistej, finansowej	20,40	18,26	3,27	3,70	68	2,57*	0,62
czas wolny	17,63	14,43	3,64	4,65	68	3,20**	0,78
rozwój własny	16,51	14,37	3,07	3,14	68	2,88**	0,70
ocena ogólna jakości życia	8,37	7,89	1,14	1,25	68	1,70 n.i.	---
jakość życia - wynik sumaryczny	149,94	131,97	14,73	16,55	68	4,80***	1,16
poczucie własnej skuteczności	3,27	2,96	0,46	0,32	60,47	3,28**	0,80

Uwaga. *** $p<0,001$; ** $p<0,01$; * $p<0,05$; n.i. – wynik nieistotny statystycznie $p>0,05$.

Okazało się, że kobiety uprawiające sport mają wyższe poczucie własnej skuteczności oraz wyżej oceniają wszystkie sfery własnej jakości życia niż kobiety nieaktywne sportowo. Nieistotny wynik porównań uzyskano jedynie w przypadku oceny ogólnej jakości życia, jednak kierunek zależności jest także w tym wypadku podobny, tj. wyższą średnią ocenę zanotowano w grupie kobiet aktywnych sportowo. W oparciu o wartość d Cohena wnioskować można o największej sile efektu aktywności sportowej dla sumarycznej oceny jakości życia, samopoczucia i codziennego funkcjonowania.

W kwestionariuszowym pomiarze jakości życia jednym z zadań dla osób badanych było wskazanie 2-3 obszarów, które w przyszłości chciałyby zmienić na lepsze. Wśród odpowiedzi zaznaczyć można było takie obszary, jak: zdrowie, kondycja fizyczna, warunki mieszkaniowe, sytuacja finansowa, sytuacja rodzinna, sytuacja osobista, sytuacja zawodowa, kontakty z bliskimi, znajomymi i przyjaciółmi, czas wolny na relaks, odpoczynek, czas spędzany z partnerem, rodziną, czas poświęcany własnym pasjom oraz brak potrzeby zmian („nie chcę niczego zmieniać”). Na podstawie udzielonych odpowiedzi sprawdzono, czy kobiety aktywne i nieaktywne sportowo różnią się wskazywanymi obszarami potrzebnych zmian. Zanotowano tylko jedną istotną statystycznie różnicę między grupami – dotyczyła ona kondycji fizycznej ($\chi^2=4,69$; $p<0,05$). Ponad 57% kobiet nie uczestniczących w regularnych zajęciach sportowych chciałaby poprawić swoją kondycję fizyczną, podczas gdy wśród kobiet ćwiczących jedynie co trzecia. W pozostałych obszarach nie stwierdzono istotnych statystycznie różnic pomiędzy badanymi grupami, chociaż warto podkreślić, że wszystkie panie chciałyby mieć więcej czasu na relaks i odpoczynek. Był to obszar zmiany najczęściej wskazywany przez kobiety ćwiczące (37,1%) i drugi w kolejności u kobiet niećwiczących (40%).

Następnie, stosując analizę mediacyjną w regresji dokonano weryfikacji pośrednictwa poczucia własnej skuteczności we wpływie aktywności fizycznej na jakość życia. W pierwszym kroku sprawdzono, czy relacja między zmienną niezależną aktywność fizyczna a zmienną zależną jakość życia jest istotna. Otrzymano wynik istotny statystycznie: $F(1,68)=23,03$; $p<0,001$; $\beta =0,50$; $p<0,001$. Tym samym potwierdzono, że obecność aktywności fizycznej silnie dodatnio wpływa na ocenę jakości życia. Następnie dokonano weryfikacji relacji zmiennej niezależnej aktywność fizyczna ze zmienną pośredniczącą (mediatorem) – poczuciem własnej skuteczności. Także w tym wypadku otrzymano wynik istotny statystycznie: $F(1,68)=10,77$; $p<0,01$; $\beta =0,37$; $p<0,01$. Oznacza to istnienie dodatniej pozytywnej relacji między aktywnością

fizyczną i poczuciem własnej skuteczności. Ostatnim krokiem analizy był model uwzględniający zarówno zmienną niezależną (aktywność fizyczną) jak i mediator (poczucie własnej skuteczności), będący predyktorem zmiennej zależnej jaką jest jakość życia. Model ten (zob. Tabela 3) okazał się istotny statystycznie ($F(1,68)=22,16$; $p<0,001$). W porównaniu z wyżej opisanym modelem, wpływ aktywności fizycznej na jakość życia zmalał po wprowadzeniu mediatora ($\beta =0,35$; $p<0,001$). Obserwacje te potwierdzono istotnymi wartościami testów statystycznych: Sobel=2,62, $p=0,009$; Aroian=2,57, $p=0,010$; Goodman=2,66; $p=0,008$. Wyniki te wskazują na istnienie mediacyjnej roli poczucia własnej skuteczności pomiędzy aktywnością fizyczną a oceną jakości życia: aktywność fizyczna poprzez podnoszenie poczucia własnej skuteczności może mieć pozytywny wpływ na subiektywną ocenę jakości życia.

Tabela 3. Współczynniki regresji dla zmiennej niezależnej aktywność sportowa oraz mediatora poczucie własnej skuteczności w modelu regresji liniowej ze zmienną zależną jakość życia.

Zmienne	<i>B</i>	<i>Błąd standardowy</i>	β	<i>t</i>	<i>p</i>
Aktywność fizyczna	12,554	3,645	0,351	3,444	0,001
Poczucie własnej skuteczności	17,558	4,368	0,410	4,020	0,000

Kolejnym obszarem badań było sprawdzenie, czy kobiety aktywne i nieaktywne fizycznie różnią się cechami temperamentu w rozumieniu Bussa i Plomina, tj. emocjonalnością, w której skład wchodzi: niezadowolenie, strach i złość oraz aktywnością i towarzyskością. Wykonano analizy z wykorzystaniem testów t-Studenta. Otrzymano tylko jedną różnicę istotną statystycznie: $t(68)=2,33$; $p<0,05$; d Cohena=0,57. Kobiety aktywne fizycznie charakteryzują się istotnie niższym niezadowoleniem ($M=2,26$, $SD=0,79$) niż kobiety nieaktywne fizycznie ($M=2,71$, $SD=0,80$). Zaobserwowano również trzy różnice międzygrupowe, dla których wartość testu t-Studenta nie osiągnęła progu istotności statystycznej, ale znalazła się w obszarze tendencji statystycznej. Mianowicie, (1) kobiety aktywne fizycznie osiągnęły niższy wynik ($M=2,51$, $SD=0,56$) na wymiarze emocjonalności niż kobiety nieaktywne fizycznie ($M=2,76$, $SD=0,54$) przy $t(68)=1,89$; $p=0,06$; d Cohena=0,46; (2) kobiety aktywne fizycznie miały niższy poziom

strachu ($M=2,46$, $SD=0,70$) niż kobiety nieaktywne fizycznie ($M=2,73$, $SD=0,62$) dla $t(68)=1,72$; $p=0,09$; d Cohena= $0,42$; oraz (3) kobiety aktywne fizycznie charakteryzowały się wyższą ($M=3,58$, $SD=0,75$) aktywnością niż kobiety nieaktywne fizycznie ($M=3,24$, $SD=0,81$) dla $t(68)=1,80$; $p=0,08$; d Cohena= $0,44$.

Następnie sprawdzono czy na podstawie zmiennych wyjaśniających (predyktorów): cech temperamentu i aktywności sportowej można przewidywać zmienną wyjaśnianą, czyli ocenę jakości życia. Wykonano wielozmiennową analizę regresji liniowej metodą krokową. Najlepiej dopasowanym modelem ($F(2,69)=33,43$) okazał się model złożony z dwóch predyktorów: niezadowolenia ($\beta=-0,523$, $t(67)=5,82$, $p<0,001$) i aktywności fizycznej ($\beta=0,354$, $t(67)=3,94$, $p<0,001$) i wyjaśniał aż 49% zmiennej zależnej – oceny jakości życia. Można więc stwierdzić, że jakość życia zwiększa się wraz ze spadkiem niezadowolenia jako cechy temperamentalnej i wraz z obecnością aktywności fizycznej.

Dyskusja

Wyniki analiz statystycznych dały podstawę do wnioskowania o tym, że kobiety regularnie uczestniczące w zajęciach sportowych wyżej oceniają jakość swojego życia we wszystkich jej sferach, za wyjątkiem ogólnej oceny jakości życia, a także, że kobiety te wyżej oceniają poczucie własnej skuteczności, niż kobiety nieaktywne sportowo. Ciekawe jest to, że oprócz tradycyjnie wyżej ocenianego zdrowia fizycznego, co znajduje potwierdzenie w wielu badaniach [4], kobiety ćwiczące wyżej oceniały także swoje samopoczucie i codzienne funkcjonowanie oraz rozwój własny i czas wolny. To może oznaczać, że poprzez uprawianie sportu, kobiety realizują swoje pasje i zainteresowania, a także mają poczucie, że w ten sposób się rozwijają. Jednym z elementów oceny jakości życia było wskazanie kilku obszarów, które w przyszłości osoby badane chciałyby zmienić na lepsze. Okazało się, że istotna różnica wystąpiła w obszarze kondycji fizycznej. Możliwe, że samo wzięcie udziału w badaniu mogło przyczynić się do tak wysokich wskazań, ponieważ uświadomiło lub przypomniało kobietom niećwiczącym znaczenie aktywności sportowej dla poprawy ich kondycji fizycznej. W przypadku zaś kobiet ćwiczących, możemy mieć do czynienia z efektem konfirmacji, który polega na tym, że wszelkie zmiany kobiety ćwiczące mogą tłumaczyć pozytywnym wpływem sportu. Jednak tak samo prawdopodobne jest, że wynika to z ich większej świadomości swoich możliwości fizycznych lub z wyznaczonych celów związanych z dalszym rozwojem, co jest spójne z wyższym wynikiem w

obszarze rozwoju własnego. Ponadto, poczucie własnej skuteczności kobiet aktywnych sportowo okazało się istotnie wyższe niż nieaktywnych, co można wyjaśnić tym, że kobiety aktywne sportowo realizują w ten sposób swoje cele sportowe, zwiększa się także ich odpowiedzialność za swoje zdrowie. Wyznaczając sobie kolejne cele, mają możliwość doświadczenia ciągłości procesu rozpoczynającego się od podjęcia decyzji, zaplanowania działań, poprzez podjęcie wysiłku, pokonanie przeszkód aż do osiągnięcia celu i poczucia satysfakcji. Powtarzalność takich sekwencji w obszarze aktywności sportowej może zwiększyć motywację i wiarę w sukces oraz zmniejszyć opór przed stawianiem nowych celów, także w innych obszarach. Wyższe poczucie własnej skuteczności kobiet aktywnych sportowo można też wytłumaczyć odwrotnie, to jest - osoby o wysokiej skuteczności w różnych obszarach życia, chętniej i częściej stawiają sobie cele także w obszarze sportu i je osiągają. Niniejsze badanie potwierdziło również pośrednią rolę poczucia własnej skuteczności, które w połączeniu z aktywnością fizyczną wpływa na jakość życia. Dobre samopoczucie i osiągnięcia uzależnione są od oczekiwań ludzi na ten temat [21]. Każdy człowiek ma określone zdanie na temat swoich możliwości, zdolności do działania w przyszłości. Uprawianie sportu pozwala doświadczyć jednostce pozytywnych zmian: nabycie lub poprawa umiejętności wykonania ćwiczenia, poprawa sprawności fizycznej, wzrost motywacji do dalszych działań i wiary w możliwość osiągnięcia nowych celów. Takie doświadczenia poprawiają spostrzeganie własnej skuteczności. To pozwala lepiej myśleć o sobie i efektywniej uzyskiwać cele i dobra, które są ważne dla dobrostanu jednostki. Jednak należy pamiętać, że sport to jeden z obszarów życia obok aktywności zawodowej, społecznej oraz życia osobistego. Zatem, aby wzrosła ocena jakości życia w związku z uprawnianiem sportu, ten obszar życia musi być wyjątkowo ważny dla danej jednostki. W przeciwnym razie sport pozostanie tylko rodzajem „odskoczni” od trosk życia codziennego. Nie bez znaczenia dla jakości życia pozostaje także temperament, co zostało potwierdzone w prezentowanych tu rezultatach analiz. Z dotychczasowych badań wynika, że wysoka ekstrawersja i niska neurotyczność to cechy sprzyjające zwiększeniu satysfakcji z życia i sportu [22]. Także w tym badaniu kobiety ćwiczące cechowały się niższą emocjonalnością, szczególnie w obszarze niezadowolenia, w porównaniu do niećwiczących. Być może jest to związane z tym, że osoby ćwiczące najczęściej uczestniczyły w zorganizowanych zajęciach sportowych, gdzie kolektywnie w grupie kilkunastu-, kilkudziesięciu- kobiet wykonywały rytmiczne ćwiczenia aerobowe przy muzyce, o średnim nasileniu, bez poczucia rywalizacji, co poprawiało ich nastrój i zadowolenie. Zgodnie z teorią

dystrakcji [23] angażujące ćwiczenia fizyczne pozwalają zapomnieć o bieżących problemach czy zmartwieniach, co chwilowo poprawia nastrój. Liczne badania potwierdzają także, że po intensywnej aktywności fizycznej, zwiększa się wydzielanie endorfin, które odpowiadają za poczucie spokoju i odprężenia oraz hamują wydzielanie hormonu stresu. Jednak badane kobiety aktywne sportowo uczestniczyły w zajęciach od wielu lat (średnio 5 lat), a więc można podejrzewać, że w ich przypadku poprawa nastroju i zmniejszenie negatywnych emocji to raczej efekt długotrwały [1]. Z opisywanych tutaj wyników analiz wnioskować można, że niskie niezadowolenie i aktywność fizyczna łącznie w dużym stopniu modyfikują ocenę jakości życia. Skłonność do niezadowolenia, strachu czy złości (negatywne emocje) obniżają motywację i wiarę we własne możliwości, zatem jednostka niechętnie będzie podejmować nowe zadania obawiając się porażki. Z drugiej strony nie podejmując się nowych zadań, nie ma okazji do doświadczania pozytywnych wyników swoich działań. Takie osoby będą raczej skupiać się na unikaniu przewidywanych trudności niż na proaktywnym pokonywaniu ich. To, co należy podkreślić z uwagi na przyjętą metodę badania poprzecznego, to że niemożliwe jest jednoznaczne określenie kierunku wzajemnego wpływu badanych zmiennych. Prawdopodobne jest, że osoby rozpoczynające swoje zaangażowanie w aktywność fizyczną już na wstępie charakteryzują się lepszą oceną jakości życia, poczucia własnej skuteczności czy też pozytywną emocjonalnością w porównaniu z osobami, które ćwiczyć nie zamierzają. Z drugiej strony jednak, w dostępnej literaturze istnieją badania longitudinalne wskazujące na to, że po programie ćwiczeń wzrasta samopoczucie, zmniejsza się lęk, smutek czy gniew [1].

W dobie starzejącego się społeczeństwa, powiększającej się skłonności do depresji i ograniczonych środków na ochronę zdrowia, sport może być odpowiedzią na wiele pojawiających się problemów. Obecnie nikt nie kwestionuje dobroczynnego wpływu sportu na zdrowie fizyczne. Nasze samopoczucie po treningu zyska zarówno z powodu poprawy kondycji fizycznej, jak również dzięki satysfakcji jaką niesie uczestnictwo w kolektywnych zajęciach sportowych, w miłej atmosferze i poczuciu wspólnoty.

Wnioski

- 1) Jakość życia oraz poczucie własnej skuteczności kobiet regularnie uczestniczących w zajęciach sportowych jest istotnie wyższa niż kobiet nieaktywnych sportowo.

- 2) Aktywność fizyczna poprzez podnoszenie poczucia własnej skuteczności może mieć pozytywny wpływ na subiektywną ocenę jakości życia.
- 3) Kobiety aktywne fizycznie charakteryzuje niższe niezadowolenie jako cecha temperamentu w porównaniu z kobietami nieaktywnymi fizycznie.
- 4) Subiektywna ocena jakości życia kobiet wzrasta wraz ze spadkiem niezadowolenia jako cechy temperamentu oraz wraz z obecnością aktywności fizycznej.

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DIGITAL-BASED INTERVENTIONS FOR TEAMS IN THE WORKPLACE

Submitted by: Davide Giusino

PhD Coordinator

Prof. Elisabetta Crocetti

Tutor

Prof. Luca Pietrantoni

Final exam Academic Year 2023-2024

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Introduction

Rationale for studying team digital interventions at work

Nowadays, most of the work performed in organisations is based on teams (Allen & Hecht, 2004; Barnes & Hollenbeck, 2009; Driskell et al., 2018; Salas et al., 2018). Teams can be defined as distinguishable collections of two or more individuals that interact socially, dynamically, and interdependently, either face-to-face or virtually, with the aim of achieving a common objective that is relevant to the organisation (Allen & Hecht, 2004; Costa et al., 2014; Mathieu et al., 2017; 2018). These individuals are assigned roles or functions and share responsibility for specific outcomes (ibidem). Consistently, the term teamwork refers to the process through which team members collaborate to accomplish the given tasks (Driskell et al., 2018) and enact integrative activities that translate team inputs (e.g., team behaviours) into team outputs (e.g., team effectiveness) via team processes (e.g., team collaboration).

From a psychological perspective, working in groups presents a number of socio-emotional and competence-related benefits (Allen & Hecht, 2004), such as fulfilment of social needs (belonging, affiliation, social comparison), positive attitudes (job satisfaction, mental health, decreased fatigue and stressfulness), reduction of uncertainty, expanded attributional opportunities, high performance illusions, and feelings of superior individual performance. Moreover, from a business-oriented perspective, previous authors (e.g., Mathieu et al., 2017; Salas et al., 2018) have argued that teams constitute a natural response to the increasingly complex, diverse, interdisciplinary modern-day world of work. Indeed, teams are needed when the tasks at hand are complex and require wide-ranging expertise, when the task outcomes affect a number of individuals that share responsibility for it, and when representing diverse constituencies and stakeholders is needed (Barnes & Hollenbeck, 2009; Edmondson, 2002) – all of which are commonplace situations in today's world of work. Mathieu and colleagues (2017) hold that “as work tasks became increasingly higher in scope and complexity, they often demand the specialised skills of more than one person” (p. 456). Interdisciplinary teams include experts having unique expertise, and their collaboration is instrumental in generating new insights into existing problems and new solutions, so they act as catalysts for innovation and learning (Edmondson, 2002). As a result, teams, once relegated to specific projects or departments, are now ubiquitous, being present in every work domain, including hospitals, schools, offices, safety, and security (Salas et al., 2018), and functioning as a cornerstone of productivity and innovation (Grote & Kozlowski, 2023). Furthermore, over the past decade, there has been a growing recognition in the scientific literature about the importance of effective teamwork and team collaboration within workplaces, as they are crucial factors that can lead to success across various industries. For instance, strong teamwork skills (e.g., communication, collaboration, and coordination) in organisations have been linked to a range of positive outcomes,

both at the employee and at the company level, including individual performance, helping behaviours, reduced absence, improved work attitudes, decreased turnover intentions, reduced depression, increased customer satisfaction, and improved organisational safety (Mathieu et al., 2017). In this regard, Driskell and colleagues (2018) cite Cartwright and Zander's (1953) statement about the importance of scientific investigation into teamwork since teams are ubiquitous, mobilise powerful forces producing important effects, these forces can result in positive and negative consequences, and understanding team dynamics allows to deliberately enhance positive consequences.

In recent decades, recognising the paramount role of teamwork in the workplace, a variety of interventions have been designed and tested to enhance the quality and effectiveness of team collaboration (McEwan et al., 2017). Traditional evidence-based strategies for team development and improvement, as described by Lacerenza and colleagues (2018) as well as by Grote and Kozlowski (2023), have predominantly centred on in-person team training, team-building exercises, and post-task debriefing. While these interventions have shown some effectiveness, they have often overlooked the social dynamics and interrelationships that can influence teamwork patterns within teams, for instance communication, which is regarded as a key teamwork competence (*ibidem*). Moreover, digital interventions have recently emerged as a potential alternative to foster positive workplace outcomes, however little is still known about digital interventions in enhancing teamwork rather than individual-level outcomes. The digital age has introduced an array of tools and interventions that promise to enhance teamwork (e.g., Koh et al., 2020). While Salas and colleagues (2018) advocated that “we need to increase the use of technology for team interventions” (p. 599), digital-based interventions, encompassing several technologies such as collaboration platforms, project management tools, and communication applications, have taken center stage in organisational strategies worldwide. Yet, their potential and effectiveness within team environments remain subjects of ongoing exploration, with the need for scientific investigation underscored by the profound impact they have on the dynamics, performance, and well-being of employees in the contemporary workplace. Particularly, no studies have conducted evaluations of real-world or context-specific effectiveness of digital-based interventions for team in the workplace. That is, studies have not delved into the underlying mechanisms or the reasons why such interventions achieved their intended outcomes for the targeted participants.

The present doctoral dissertation describes four studies conducted as part of a three-year research project aiming to investigate digital-based interventions for teams in the workplace. In the pursuit of understanding their implications, this research project has ventured to unravel the complexities and nuances of this type of interventions. The fundamental premise behind this research is rooted in the realisation that the workplace of today stands as an intricate ecosystem where diverse teams navigate a maze of digital tools to achieve common objectives. This necessitates a closer look into the effects of digital-based interventions on team dynamics,

communication patterns, and performance outcomes. As organisations continue to adopt these technologies, the need for empirical evidence regarding their implications becomes pressing. The rationale for this investigation is not solely driven by the pervasiveness of digital tools in the contemporary workplace but also by the potential for their misuse or misalignment with organisational goals, leading to unintended consequences. An extensive body of literature suggests that the inappropriate use of technology may lead to information overload, reduced face-to-face interactions, feelings of isolation, and diminished work-life balance (e.g., Grant et al., 2018). Conversely, when harnessed effectively, digital-based interventions can promote collaboration, knowledge sharing, task coordination, and overall well-being (e.g., Majchrzak et al., 2018). This research project seeks to provide a comprehensive understanding of how teams interact with digital interventions and how these interactions, in turn, influence team processes and performance. By doing so, it positions itself as a relevant contribution to the growing body of knowledge surrounding digital-based interventions for teams in the workplace and aims to offer actionable insights to organisations striving to optimise their team's digital work environment.

In the pages that follow, after a description of the overarching theoretical framework this project was based on, the dissertation delves into the four studies conducted. Each study is presented with its background, methodological approach, empirical findings, and theoretical and practical implications of the research. Through rigorous analysis and synthesis of both qualitative and quantitative empirical data, this work strives to illuminate the complexities, challenges, and opportunities that characterise the intersection of teams, digital interventions, and the modern workplace.

Overarching framework

The doctoral research project reported on in the present dissertation was based on four main theoretical pillars, namely (1) Lacerenza and colleagues' (2018) stepwise framework for team interventions, (2) the IGLO model (Day & Nielsen, 2017; Nielsen & Christensen, 2021; Nielsen et al., 2017), (3) the Job Demands-Resources model (Bakker & Demerouti, 2017; 2018; Schaufeli, 2017), and (4) the realist evaluation approach (Nielsen & Abildgaard, 2013; Nielsen & Miraglia, 2017; Nielsen & Randall, 2013; Nielsen & Shepherd, 2022; Roodbari et al., 2021a, 2021b, 2023). Each theoretical framework is illustrated in detail in the following paragraphs.

Stepwise Framework for Team Interventions

First and foremost, this project followed Lacerenza and colleagues' (2018) stepwise framework for team interventions. In their attempt to provide evidence-based recommendations regarding how to design, deliver, implement, and increase the

effectiveness of team development interventions, Lacerenza and colleagues (2018) introduced a comprehensive framework that can be used as a structured approach to enhance teamwork within organisations. Resonating with more general models of participatory workplace interventions (e.g., Nielsen et al., 2010; Nielsen & Noblet, 2018; von Thiele Schwarz et al., 2021), this framework comprises four main phases, offering a systematic path to facilitate team development and improvement. The first phase of the process involves extensively reviewing experimentally supported evidence from previous academic literature, in order to know which solutions and strategies work according to science. The second phase corresponds to conducting a thorough needs assessment to identify specific areas of concern or improvement within the team; the authors recommend conducting a need analysis in order to provide diagnostic feedback and ensure that stakeholders' expectations align with the goals of the training. The third phase is the implementation of team interventions and focusses on putting the intervention plan into action, utilising various strategies such as team training, team-building activities, and debriefing sessions. Fourth, an integral aspect of the framework is the evaluation phase, where the effectiveness of the intervention is assessed; this evaluation not only gauges the impact of the intervention on teamwork but also informs any necessary adjustments or refinements.

As it constitutes a valuable tool for organisations seeking to systematically improve team dynamics, ultimately leading to enhanced teamwork, team collaboration and team performance, Lacerenza and colleagues' (2018) framework was deployed throughout the four studies that the present research project was composed of. Each study correspond at least one of the phases of Lacerenza and colleagues' (2018) framework and contributes a unique facet to the overarching exploration of digital-based interventions for teams in the workplace. Thus, Study 1 represented a foundational review of the already existing body of knowledge about digital workplace interventions; while this initial phase did not directly involve the implementation of a digital intervention for teams, it was crucial in establishing the groundwork by synthesising previous evidence and providing context for the subsequent studies. Study 2 performed an assessment of team-level needs, thus facilitating the fine-tuning of the forthcoming digital intervention and serving as a preparatory stage to ensure that the subsequent implementation would be tailored to address the requirements of the teams under investigation. Study 3 and Study 4 both concentrated on the implementation and evaluation of a digital-based team intervention, each offering distinct perspectives. While Study 1 and Study 2 did not directly address digital interventions for teams, they were preparatory with respect to the subsequent Study 3 and Study 4, which rather delved into issues related to a specific intervention aiming to promote work communication within teams. Also, whereas implementation and evaluation is addressed in both Study 3 and Study 4, they present different ways of investigating the intervention's factors of effectiveness.

Each study included in the present dissertation received ethical approval by the Bioethics Committee of the Alma Mater Studiorum – University of Bologna (Prot. n. 0185076) and complied with the Declaration of Helsinki (World Medical

Association, 2013). In each study, workers were given an informed consent form detailing participation procedure, study contents, data collection purposes, future data dissemination modalities, participants' rights, and addressable contacts. Participation was voluntary and could be withdrawn at any time without consequences.

IGLO Model

From an ecological perspective, the workplace can be viewed as a system made of different sub-systems, where various patterns of relationships between workers and different working environments occur. Also, workers' outcomes can be seen as embedded in such a system. Consequently, psychosocial interventions should be developed at all systemic levels of the workplace to address potential sources of either good or poor workers' outcomes (e.g., Bakker & Demerouti, 2018; Chen et al., 2018; Martin et al., 2016; Teoh et al., 2020). Specifically, sources of positive or negative work outcomes can exist at four levels, such as the individual (I), the group or work team (G), the leader (L), and the organisation (O). These levels are framed as the IGLO model (Day & Nielsen, 2017; Nielsen & Christensen, 2021; Nielsen et al., 2017). At the individual level, work outcomes can derive from work-specific cognitive, affective, and behavioural factors or resources, for instance, work-related self-efficacy and job crafting. At the group level, work-related psychological states and behaviours can be associated with colleagues' support and workgroup climate and subsequent interventions can be carried out such as team coaching (Clutterbuck, 2010). The leader level encompasses predictors of work-related outcomes like line managers' knowledge, skills, abilities, attitudes, behaviours, and support. Finally, at the organisational level, human resources management practices and policies, job design, and occupational health services can play a meaningful role in promoting or hindering workers' psychological and behavioural outcomes. To summarise, the individual level has to do with personal variables, the group level is about team states, processes, and dynamics, the leader level refers to characteristics of and actions implemented by managers, and the organisational level points to how both the work and the working environment are designed, managed, and organised.

In the present research project, the IGLO model was used in (1) Study 1, to classify digital workplace interventions systematically retrieved from previous literature according to the four levels of the model (Peláez Zuberbühler, Giusino et al., under review), and (2) Study 2, to assess intervention needs at the different workplace levels as part of a broader multilevel workplace needs assessment exercise (Giusino et al., 2022a; 2022b). However, for the purposes of the present dissertation's main topic, which is digital-based interventions for teams in the workplace, the group level of analysis and intervention is the most relevant. Therefore, Study 1 especially highlights findings regarding group-level digital interventions, whereas Study 2 mostly focuses on needs identified at the team level of analysis.

Job Demands-Resources Model

The Job Demands-Resources (JD-R) model (Bakker & Demerouti, 2017; 2018; Schaufeli, 2017) conceives the work environment as a potential source of either positive or negative workers' outcomes depending on how the work environment is designed, organised, and managed. According to this framework, the work environment can be considered as a constellation of job demands and resources, which differently influence workers' psychological states and behaviours. On the one hand, job demands refer to physical, psychological, social, or organisational aspects of the job that require physical or psychological efforts from the worker. Examples may be emotional demands, team conflict, heavy workload, time pressure. As such, job demands can be understood as risk factors for workers. Nevertheless, job demands have recently been differentiated into hindering job demands and challenging job demands (Van den Broeck et al., 2010), where the former hinder the optimal functioning of the individual and the latter stimulate positive work outcomes. On the other hand, job resources correspond to physical, psychological, social, or organisational aspects of the job, that workers can use to counterbalance the costs implied by job demands in terms of physical, cognitive, and emotional energy. Examples may be personal protective equipment, safety devices, cognitive and behavioural patterns, job autonomy, skill variety, performance feedback, support from colleagues or supervisors, role clarity, job control, adequate pay, job security, career opportunities. In addition, recent studies (e.g., Chen et al., 2018) have integrated job resources with personal resources from the positive psychological tradition – such as, for instance, resilience, adaptability, flexibility, optimism, self-efficacy, hope, psychological capital. Job resources are intrinsically motivating and may help workers fulfil their basic needs, achieve work-related goals, and positively influence their personal growth and development. Although both job demands and job resources can independently impact workers' psychological states and behaviours, job resources may buffer job demands by enabling workers to cope with job demands. In this framework, negative work outcomes result from an imbalance between job demands and resources; when job demands exceed resources, poor work outcomes may show up. Specifically, the JD-R model postulates two distinct processes leading to workers' outcomes; through the impairment process, high job demands are causally linked to negative outcomes over time, while through the motivation process, high job resources result in positive outcomes.

In the present research project, the JD-R model was used in Study 2, where a workplace needs assessment exercise was performed to identify job demands and job resources at the different IGLO levels (Giusino et al., 2022a; 2022b). However, consistent to the present dissertation's main topic, which concerns team interventions, Study 2 mostly focuses on job demands and job resources identified at the group level of analysis. For the purposes of this project, the main aim of Study 2 was indeed to check for the organisation-intervention fit (Andersen et al., 2021; Peters et al., 2020)

and ensure the meaningfulness of implementing digital-based interventions to foster teamwork within the targeted organisation.

Realist Evaluation

This project deployed a realist approach towards the evaluation of workplace interventions (Nielsen & Abildgaard, 2013; Nielsen & Miraglia, 2017; Nielsen & Randall, 2013; Nielsen & Shepherd, 2022; Roodbari et al., 2021a, 2021b, 2023). With this, the project intended to refer to a whole corpus of theoretical frameworks and models, such as, Nielsen and Abildgaard's (2013) research-based framework for the evaluation of both intervention process and effects, Nielsen and Randall's (2013) model of process evaluation, the Integrated Training Transfer and Effectiveness Model (Nielsen & Shepherd, 2022), or the Integrative Process Evaluation Framework (Nielsen et al., 2023). Overall, realist evaluation frames the causality process occurring during a workplace intervention through the linkages between the changes in certain mechanisms that happen in the context where the intervention takes place, which, in turn, lead to specific outcomes. In other words, in workplace interventions, the context in which the intervention takes place, influences its working mechanisms, which in turn trigger intervention outcomes (Nielsen & Abildgaard, 2013; Nielsen & Randall, 2013; Roodbari et al., 2021a, 2021b, 2023). Here, the term "context" refers to the conditions under which an intervention is effective. According to Nielsen & Abildgaard (2013), context can be further subdivided into omnibus context and discrete context. Omnibus context is made up of those impacting external factors that exist before and regardless of the intervention (e.g., organisational culture and climate), while discrete context refers to everything that occurs during the actual implementation of the intervention (e.g., pandemics, financial crisis, mergers). Mechanisms correspond to those elements or "ingredients" that make an intervention work, for instance, particular intervention contents, specific activities and exercises, or special devices or tools deployed. Finally, outcomes correspond to the improvements in working conditions and workers' well-being or performance that can be observed, or the effects that were aimed to be produced. So, if certain contextual conditions are present, then certain intervention working mechanisms are activated and, as a result, the intervention brings about the changes it aims to achieve. Therefore, in addition to outcomes, context factors and working mechanisms should be considered when evaluating interventions, as they may influence intervention outcomes, facilitate or hinder the interventions' effectiveness, and explain the success or failure of an intervention. In this sense, the realist evaluation goes beyond the traditional pre-post randomised controlled approach towards the workplace interventions' effectiveness (Nielsen & Miraglia, 2017).

At its very core, realist evaluation tries to provide an answer to the question of "what works for whom, and under which circumstances" (Nielsen & Miraglia, 2017; Roodbari et al., 2023). This is done by investigating the so-called context-mechanisms-

outcomes (CMO) configurations, expressed through the formulation “If... then... as a result” or other similar formulations depending on whether, in statistical terms, they refer to moderation, mediation, or other models, as illustrated later in this paper. The CMO framework allows for an in-depth examination of the contextual factors, underlying mechanisms, and outcomes associated with an intervention, acknowledging the complex and context-dependent nature of real workplace settings in which interventions take place. The realist approach to workplace interventions evaluation allows to delve into the underpinning interventions’ mechanisms, to understand what elements work and why, in which circumstances, and for whom they are particularly successful.

This approach allows achieving an understanding of the working mechanisms of effective interventions, and how they succeed or fail in specific settings. It also involves understanding the context in which the intervention is implemented, identifying the mechanisms by which the intervention produces change, and examining the outcomes of the intervention. For instance, the effectiveness of digital interventions in the workplace can be influenced by the organisational culture and technological infrastructure, which are part of the context (Armaou et al., 2019). The mechanisms may include user engagement with the digital tool, the delivery method of the intervention (e.g., mobile app, web-based platform), and the contents’ relevance to the users’ needs. The outcomes may then be evaluated regarding improvements in psychological and behavioural outcomes at work. This holistic approach ensures that the evaluation of digital interventions is not just focused on the outcomes but also considers the underlying processes and environmental factors that contribute to these outcomes.

One last relevant aspect of the realist evaluation approach concerns intervention recipients’ perceptions. According to recent models for evaluating workplace interventions (Nielsen and Abildgaard, 2013; Nielsen and Randall, 2013) as well as empirical studies in the same research strand (e.g., von Thiele Schwarz et al., 2021), it is crucial to understand and address the perceptions by recipients of the dimensions of workplace interventions. Intervention recipients’ perceptions refer to their attitudes towards the intervention and its various aspects, such as its dimensions, qualities, features, elements, components, and ingredients. These aspects may encompass content, structure, facilitators, design, relevance, usefulness, objectives, and more. Persson et al. (2012) suggest that each worker understands their work environment uniquely. Individuals interpret cues from the environment, developing a shared understanding based on common experiences and conditions. This concept can also be applied to workplace interventions. When workers participate in an intervention together, they may develop a collective perception of the intervention, influenced by cues from one another. In particular, in team interventions, these shared perceptions may shape their views on the positive or negative aspects of the action or initiative. By considering individual and team perceptions, researchers and practitioners can better understand and explain the overall impact of workplace interventions. Specifically, recipients’ perceptions should be an integral part of

workplace interventions' evaluation because they are key underlying mechanisms for the effectiveness of the intervention. Thus, an essential part in evaluating workplace interventions should be measuring change in employees' perceptions of the intervention, and their expectations that the intervention can bring about changes (Nielsen and Randall, 2012).

To the author's knowledge, realist approaches towards the evaluation of digital-based workplace interventions were not frequently mentioned in literature – Havermans and colleagues' (2018) process evaluation of a digital platform-based implementation strategy aimed at work stress prevention in healthcare constitutes a rare exception – and had not yet been applied to investigate the effectiveness of group-level digital interventions. In the present research project, the realist evaluation approach was used in (1) Study 1, to explore the working mechanisms and the contextual conditions for effectiveness of workplace digital interventions systematically retrieved from previous literature (Peláez Zuberbühler, Giusino et al., under review), (2) Study 3, to longitudinally examine recipients' perceptions of a digital team coaching intervention using social network visualisation (Giusino et al., 2023), and (3) Study 4, to test two CMO configurations of the same group-level digital intervention (Giusino et al., in preparation).

Study 1. Digital Interventions at Work: A Multilevel Realist Review

Abstract. Digital interventions (DIs) at work have proven effective in many instances, but knowledge regarding their working mechanisms and the contextual conditions under which these interventions are most effective is still scarce. Also, it is unclear at which level of the workplace ecological system they have been implemented the most. To fill these gaps, a realist synthesis with a multilevel IGLO approach was conducted to explore how, why, and under which circumstances DIs may promote, prevent or reduce workers' psychological and behavioural outcomes. Forty-four DI studies were gathered through a systematic electronic search. For the analysis, context factor, mechanisms and outcomes were extracted from the selected studies. Findings suggested some relevant context factors to consider when implementing DIs at work, and working mechanisms that can trigger the study outcomes. In addition, results showed that, out of 42 digital workplace interventions reviewed, 40 interventions were implemented at individual level, whereas two occurred at leader level. None was implemented at the group level. This was deemed as an input to more research into group-level digital workplace interventions.

Background

With the recent progress in information and communication technologies and increasing digitalisation of work (Cijan et al., 2019), psychosocial digital interventions (DIs) have emerged as a noteworthy opportunity for organisations (Armaou et al., 2019; Stratton et al., 2017; 2022). Generally defined as planned, structured, science-based actions or initiatives aiming to achieve a certain goal by exploiting the potential offered by digital technologies (e.g., Giusino et al., 2021), such interventions may correspond to either interventions that were originally designed to occur in physical presence and subsequently adapted to the digital formats offered by online teleconferencing platforms, or to interventions explicitly meant to be available for computer or smartphone apps only (ibidem). In particular, DIs such as smartphones, websites, or text messaging are delivered via digital technologies to provide effective, safe, and scalable interventions to benefit people at work, whether used in conjunction with, or independently of, other type of people services (Murray et al., 2016). DIs can be used to promote behaviours, improve work-related outcomes, with multiple aims, including sharing experiences with others, changing perceptions and cognitions, assessing and monitoring specific behaviours, and improving communication between customers and professionals (ibidem). Previous literature reviews on DIs have shown that these interventions can be considered primary or secondary

prevention strategies (Armaou et al., 2019), referring to an intervention that aims to prevent exposure to psychologically harmful working conditions – that is, primary prevention –, or to give workers the appropriate skills to cope with or adapt to the working environment – that is, secondary prevention.

Research has highlighted several advantages that DIs offer over traditional face-to-face ones, thanks to their technical features. In their systematic review of the published literature, Griffiths and colleagues (2006) found that reasons for internet delivery of healthcare interventions included reduced costs, increased convenience for the users, overcoming isolation of users, need for timely information, reduced stigma, and increased user and supplier control. A subsequent review article by Baños and colleagues (2022) also mentioned reduction of stigma, geographical accessibility, and temporal flexibility as further advantages of digital interventions. Molino and colleagues (2020) reported that cost and time savings, as well as increased employee satisfaction are among the benefits of using DIs during a pandemic outbreak.

Previous meta-analyses and systematic reviews have consistently shown how DIs may produce benefits for workers, such as increased psychological well-being and work effectiveness (Carolan et al., 2017), reduced mental health conditions (i.e., stress, depression, and anxiety; Heber et al., 2017; Stratton et al., 2017; 2022), and moderate treatment effects on stress, insomnia, burnout, depression, well-being, and mindfulness (Phillips et al., 2019). Focussing more specifically on smartphone-based interventions for the promotion of employees' well-being, the systematic review by Paganin and Simbula (2020) found a lack of theoretical background, reliable study design, and the prevalence of physical health interventions, also underlining the importance of user engagement for intervention effectiveness. A systematic review of randomised controlled trials by Moe-Byrne and colleagues (2022) found promising results regarding presenteeism, sleep, stress, and somatic symptoms.

As it can be concluded based on previous research findings, DIs can be effective in promoting positive work outcomes or preventing/reducing negative outcomes at work. However, attention should not only be focused on the positive aspects of digital interventions, such as their benefits and advantages over traditional interventions. A more balanced presentation of this type of interventions should acknowledge their potential challenges or limitations. In this regard, mixed findings in the field of digital interventions have been shown. For instance, Philippe and colleagues (2022) conducted a systematic and meta-review of article reviews evaluating digital health interventions. The authors demonstrated the overall beneficial effects of digital interventions on various outcomes. Nonetheless, outcomes varied substantially based on intervention features and implementation methodology. These results highlight the need for more research to advance our understanding of these interventions, clarifying essential moderating and mediating factors, such as processes and implementation features.

Previous studies have rarely investigated the mechanisms that make an intervention work as intended in certain contextual conditions. Issues related to the adoption of DIs by employees and the mechanisms that can facilitate their

implementation and positive impact in the workplace may pose a considerable challenge in evaluating their effectiveness. Therefore, there is a critical need in literature to systematically analyse and synthesise the current state of the art of empirical evidence of DI studies at work to understand and report whether the interventions work, what makes them work, for whom, and under which circumstances, following a realist approach towards evaluating workplace interventions.

This study was based on a realist framework to assess workplace interventions (Nielsen & Miraglia, 2017; Nielsen & Shepherd, 2022; Roodbari et al., 2021a, 2021b, 2023). This approach focusses on understanding the causal process underlying workplace interventions by examining the connections between changes in specific mechanisms within the intervention context, leading to distinct outcomes. In essence, this approach

highlights that the context of a workplace intervention influences its operational mechanisms which, in turn, determine the intervention outcomes. The term context refers to the conditions that impact the effectiveness of an intervention and under which working mechanisms are triggered. Nielsen and Abildgaard (2013) further categorised context into omnibus and discrete. Omnibus context comprises pre-existing external factors that influence the intervention, such as organisational culture and climate and working conditions – e.g., work demands and resources –, while discrete context encompasses factors occurring during implementation of the intervention, such as pandemics, financial crises, mergers, organisational restructuring, downsizing, and budget cuts (Nielsen & Randall, 2013). Mechanisms, on the other hand, are the key components or ingredients that enable the intervention to function, including specific intervention content, activities, exercises, or tools deployed. Mechanisms can be classified into (1) process mechanisms, referring to the design and implementation of the interventions, for instance, transfer of training, peer or manager support, (2) content mechanisms, referring to the nature of the changes focussed on the content of action plans, for instance, changes in the work procedures to reflect teamwork, and (3) perceptions, referring to the perceptions of the participants regarding the process and content mechanisms, for instance, changes in their attitudes and abilities (Nielsen & Miraglia, 2017). Lastly, outcomes represent the observable improvements in working conditions and psychological or behavioural outcomes of employees, as well as the intended effects of the intervention. Consistent with psychosocial interventions literature, outcomes can be categorised into proximal, indicating the direct and causally closest outcomes of an intervention – e.g., changes in workers' knowledge, attitudes, and behaviours in relation to working conditions – and distal, indicating the indirect and causally far effects – e.g., job satisfaction, subjective well-being, and performance; De Angelis et al., 2020; Nielsen & Miraglia, 2017; Roodbari et al., 2021a, 2021b). If specific contextual conditions are present, the corresponding intervention mechanisms are activated, leading to the desired changes. Thus, when evaluating workplace interventions, it is crucial to consider not only outcomes, but also contexts and working mechanisms. These elements can influence

the effectiveness of interventions and provide explanations for their impact on workers and work-related outcomes.

Moreover, many calls have been made in literature, advocating for a multilevel approach towards evaluating workplace interventions (e.g., Martin et al., 2016). From an ecological standpoint, the workplace can be seen as a complex system consisting of various interconnected components. These components include the relationships between workers and different work environments. Psychosocial interventions should be implemented at different levels of the workplace system to address factors that can influence workers' psychological states and behavioural outcomes either positively or negatively. The workplace levels can be categorised in various manners, depending on the model that is taken as a reference. In the present study, the IGLO model (Day & Nielsen, 2017; Nielsen & Christensen, 2021; Nielsen et al., 2017) was adopted as an analytical framework for the characteristics of digital interventions at work. In addition, the model was used to classify context factors and working mechanisms of workplace digital interventions at individual, group, leader, and organisational level.

Consistent with realist evaluation and the multilevel approach, this study (Peláez Zuberbühler, Pietrantonio, Mazzetti, De Angelis, Giusino et al., under review¹) aimed to review the current literature to explore and identify how DIs produce their effects on mental health and well-being at work, in which contexts, and for which group of employees. The specific objective of this review was to extract information regarding context factors, mechanisms, and outcomes of digital psychosocial interventions at work. This knowledge might be highly beneficial for researchers and practitioners in designing, implementing, and evaluating DIs at work, as it facilitates the understanding of how, why, and under what circumstances DIs may improve employees' psychological states and behaviours (Roodbari et al., 2021a, 2021b). This review then provides valuable information about the peculiarities of those DIs that are successful at work, and the conditions and mechanisms that ensure that the DIs achieve the intended outcomes. Findings might also help to clarify research areas that facilitate the use of DIs and their implementation in organisations, as well as to formulate research questions for future studies.

Methods

The current systematic review was conducted in line with the realist synthesis approach (Nielsen & Miraglia, 2017; Pawson et al., 2005; Wong et al., 2013). In contrast to other types of review – e.g., meta-analyses – that focus on intervention effectiveness, the realist review aims to explain causal interactions on how and why observed outcomes occur, by reporting empirical evidence about contextual factors, working mechanisms, and outcomes. A systematic realist synthesis combines the principles

¹ Currently resubmitted after major revision and awaiting reviewer scores from *Organizational Psychology Review*. <https://journals.sagepub.com/home/opr>

and rigor of the systematic review methodology (Perestelo-Pérez, 2013; Petticrew & Roberts, 2008) with the realist evaluation framework (Nielsen & Abildgaard, 2013; Nielsen & Miraglia, 2017; Nielsen & Randall, 2013; Nielsen & Shepherd, 2022; Roodbari et al., 2021a, 2021b, 2023). In particular, realist synthesis provides a nuanced way of understanding how interventions work in different contexts. Compared to a general overview of available interventions, the main added value of a realist synthesis is that it explicitly addresses causality, going beyond merely describing the presence or absence of an intervention and its outcomes, as is the case in a traditional systematic review. Instead, realist synthesis aims to identify the context, the mechanisms triggered by the intervention and the observed outcomes. This approach is in line with the general aim of realist synthesis to uncover the underlying mechanisms that influence the outcomes of interventions in diverse and dynamic settings. In addition, realist synthesis also enables exploration of the role of context in determining the effectiveness of interventions. This contextual sensitivity is crucial for understanding the variability of the effects of interventions in different contexts. In general, this approach builds on the recognition that the functioning of interventions often involves a non-linear process, so it relies on the interaction between context, mechanisms, and outcomes. In doing so, it overcomes the descriptive nature of traditional systematic reviews. Furthermore, a standard review of the literature would make it possible to systematise which interventions work and which do not, while realist synthesis also provides specific insights at the operational level to further investigate, as it makes it possible to tailor interventions to specific contexts and optimise their effectiveness. For all these reasons, it was concluded that a general and descriptive review of digital interventions to promote work outcomes might be simpler but would not be as valuable in terms of novelty and input, as it would not go significantly beyond what is already known in the academic literature.

In the realist review methodology, the search process is characterised by its iterative and interactive nature, engaging in a dynamic back-and-forth movement between the literature and the research questions. This iterative process often leads to the development of search strategies and terms as the researcher's understanding deepens, as described by Pawson and colleagues (2005) and Nielsen & Miraglia (2017). Typically, a realist review search involves several different steps. The present review progressed following four steps inspired by the Realist And Meta-narrative Evidence Syntheses: Evolving Standards (RAMESES) quality and publication standards (Wong et al., 2013), namely (1) defining research questions, (2) searching for evidence, (3) selecting and appraising the studies, and (4) extracting, analysing, and synthesising the findings. Each of these steps is illustrated in detail in the following paragraphs.

The realist review approach has previously been used by researchers to understand the implications of contexts and mechanisms on the outcomes of other forms of health promotion interventions in the organisational psychological field (e.g., Adnan et al., 2022; Dickson et al., 2022; Gray et al., 2019; Micklitz et al., 2021; Roodbari et al., 2021a, 2021b; Sinclair et al., 2021). The usefulness of this theoretical framework is also underpinned by the studies that rely on its implementation to assess face-to-

face intervention designed to promote workers' mental health (e.g., Abildgaard et al., 2020; Higgins et al., 2015) and the effectiveness of digital tools in promoting mental health and well-being in the general population (e.g., Andersen et al., 2022). However, this study constituted one of the first attempts to apply the realist methodology to DIs for mental health and well-being at work. This study represents an application of realist review methods specifically to DIs in the context of workplace mental health. The insights gained from this study were expected to shed light on aspects that facilitate the use of DIs and their implementation within organisations. Additionally, the findings were expected to assist in formulating pertinent research questions for subsequent studies in this field.

Defining Research Questions

The overarching research question for this realist review was "What works, how, why, for whom, and in what contexts in relation to DIs at work?". To guide the review, the three specific research questions of this review were (1) "what are the context factors that impact the effectiveness of DIs at work?", (2) "what are the mechanisms through which workplace DIs bring about changes?", and (3) "what are the outcomes of DIs in terms of improved mental health or reduced well-being at work?".

Searching for Evidence

To identify relevant studies to include in the review, a systematic search was conducted (Perestelo-Pérez, 2013; Petticrew & Roberts, 2008) and different databases were consulted, such as PsycNet, Scopus, Web of Science and PubPsych, which are known to ensure good coverage of high-quality peer-reviewed articles (Mingers & Leydesdorff, 2015). Papers published in English, Spanish, and Italian were selected, and the search was limited to articles published between 2011 and 2022 in peer-reviewed scientific journals, while excluding conference papers, abstracts, doctoral theses, books, and unpublished research. The search for scientific articles was done by six authors belonging to independent research teams. The search terms were focused on titles, abstracts, and keywords, using the Boolean operators' combination ("OR", "AND"). The keywords used for this search were "digital" OR "digital-based" OR "digital based" OR "smartphone*" OR "smartphone-based" OR "smartphone based" OR "app" OR "app-" OR "app based" OR "app based" OR "web*" OR "web-based" OR "web based" OR "computer" OR "computer-based" OR "computer based" OR "on-line" OR "on line" OR "online" OR "on-line-based" OR "on line-based" OR "on line based" OR "on-line based" OR "internet" OR "internet-based" OR "internet based" OR "desktop*" OR "desktop-based" OR "desktop based" OR "game-based" OR "game based" OR "video-assisted" OR "video assisted" OR "video-based" OR "video based" / AND "intervention*" OR "training*" OR "program*" / AND "mental health" OR "m-health" OR "mhealth" OR "e-mental health" OR "e mental health" OR

“wellbeing” OR “wellbeing” OR “psychological” OR “psychosocial” / AND “work*” OR “organisation*” OR “organization*” OR “occupation*” OR “employee*” OR “manager*” OR “leader*” OR “team*” OR “job*” OR “compan*” OR “enterprise*”. Using this search strategy, 3604 records were yielded. Last search was run in November 2022.

Selecting and Appraising the Studies

To be eligible for inclusion, studies needed to meet the four following criteria. First, studies in the review needed to provide empirical evidence, as the aim was to gather knowledge capable of eventually informing evidence-based practice (American Psychological Association, 2006). Quantitative and qualitative studies were included, as well as meta-analyses and systematic reviews focussing on empirical studies. Second, studies had to be conducted in work settings, whereas studies conducted outside the workplace or involving the general population or students were excluded. Third, the digital interventions had to focus on mental health and well-being at work, thus including digital interventions focussing on mental disorders and/or positive states of psychological well-being (Leka et al., 2015), or work-related issues laying a positive motivational path towards a state of fulfilment and goal achievement (Bakker & Demerouti, 2018). Accordingly, only studies promoting well-being or preventing, reducing, or managing distress – e.g., stress, anxiety, depression indicators – were included. Tertiary level prevention with a strong rehabilitative or treatment orientation were excluded from the original review’s main scope. Fourth, studies had to use or focus on digital interventions, being delivered via the internet, mobile technology, or a computer program. Diagnosis or assessment-based digital technologies without suggesting or implementing any action plan for improvement were excluded.

A spreadsheet was used to collect the titles and references of the articles screened in this first process. After removing 428 duplicates from the initial 3604 records, 3176 papers remained for further screening. Next, an abstract screening was conducted, considering the four inclusion criteria, resulting in 152 papers. The following analytical step involved careful reading of the full texts and selection of the studies according to the inclusion and exclusion criteria. This left a total of 81 papers eligible for the review. The studies were screened independently by six authors (i.e., reviewers) following the search strategy and eligibility criteria. Discrepancies in the screening were resolved through discussion. Afterwards, using the open-source Rayyan software (Ouzzani et al., 2016), two other researchers (i.e., judges) settled the doubts regarding the studies in disagreement between the reviewers, and other four authors jointly discussed the final list and agreed on the final number of studies to be included in the review. This resulted in a final sample of 44 studies (Althammer et al., 2021; Avey et al., 2022; Bégin et al., 2022; Bormann et al., 2017; Bostock et al., 2019; Cantarero et al., 2021; Carissoli et al., 2015; Carolan et al., 2017; Cieslak et al., 2016;

Ebert et al., 2015; Ehrlich, 2022; Hammer et al., 2011; Hirshberg et al., 2022; Hosseinzadeh Asl, 2022; IJntema et al., 2021; Imamura et al., 2015, 2016; Keller et al., 2016; Keng et al., 2022; Knox & Franco, 2022; Kriakous et al., 2021; Lennefer et al., 2019, 2020; Li et al., 2021; Makowska-Tłomak et al., 2022; Nadler et al., 2020; Neumeier et al., 2017; Oliver & MacLeod, 2018; Ouwenel et al., 2013; Pandya, 2021; Paterson et al., 2021; Phillips et al., 2014; Pospos et al., 2018; Purdie et al., 2022; Querstret et al., 2016; Shann et al., 2019; Shiotsuki et al., 2017; Smith et al., 2020; Tonkin et al., 2018; Uglanova & Dettmers, 2022; Vanhove et al., 2016; Wang et al., 2021; Weber et al., 2019; Zhang et al., 2022). The fact that the final number of studies considered was reduced from 3604 to 44 articles can be thought as reflecting the soundness and clarity of the established criteria for study selection, resulted in a precise framework for the included DIs.

The quality appraisal of the 44 studies was conducted in accordance with RAMESES realist synthesis methodology. The relevance of the studies was assessed by examining whether they provide a description of the DIs' context factors, working mechanisms and mental health and well-being outcomes. This quality assessment was important for the analysis and synthesis of the findings. However, it was not considered an exclusion criterion for the review. To respond to the research questions, it was necessary to analyse the state of the art of the current DIs for mental health and well-being at work, in addition to the strengths and limitations of the selected studies on whether considering or not contexts, mechanisms and/or outcomes. This analysis may help identify research gaps and define research questions for future studies.

Figure 1 shows the flow diagram, which represents the search and retrieval process.

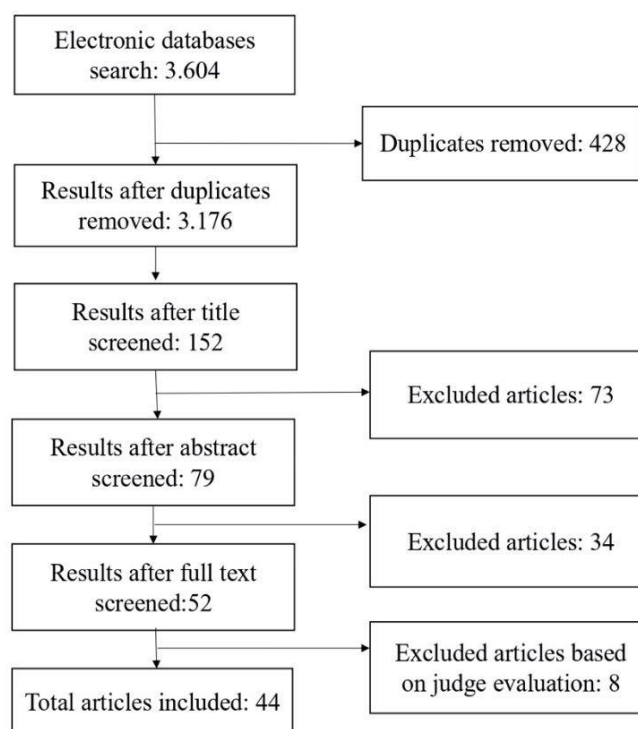


Figure 1. Flow diagram of the literature search process

Extracting, Analysing, and Synthesising the Findings

Five researchers participated in the data extraction process, including coding, analysis, and synthesis. A Microsoft Excel codebook was used to extract the information from all the studies included. The data was organised into the following themes, namely characteristics of the studies (including year, country, main theoretical framework, sample, method, design, and analysis), characteristics of the interventions (including format, level, structure, and contents), context factors, mechanisms, and outcomes. The context, mechanisms, and outcomes were either explicitly proposed in the studies, or implicitly extracted by the authors of the current review. Finally, these three elements were categorised based on the levels of analysis provided by the IGLO model (Day & Nielsen, 2017; Nielsen & Christensen, 2021; Nielsen et al., 2017).

Results

The current section reports the main findings of the present study, regarding the characteristics of the selected studies, the characteristics of the retrieved digital workplace interventions, and their context factors, working mechanisms, and outcomes.

Characteristics of the Studies

Of the 44 studies, 18 single studies were conducted in Europe (six from Germany, six from United Kingdom, two from Poland, two from The Netherlands, one from Italy, one from Turkey), 9 in the United States, 7 in Asia (three from China, three from Japan, one from Singapore), 3 were multinational, and 2 were from Oceania (Australia and New Zealand). The articles included a wide range of participants from different sectors and occupational backgrounds. The sample sizes varied across studies, ranging from 20 to 1236 participants, with mean sample size of 202.69 ($SD = 268.01$). The age of participants ranged from 19 to 70 years old. Only 25 studies reported the gender, 23 studies have predominantly female participants (ranging from 45.7% to 100%). All the papers were published after 2011. Most of these articles reported a quantitative approach ($n = 39$), a small number were meta-analysis ($n = 2$), one article was systematic review, one article was scoping review, and one article used a mixed-method approach combining quantitative and qualitative methods. Regarding the design of the 39 quantitative studies, 25 were randomised controlled trials with pre-, post-intervention and follow-up evaluation, 4 were controlled trial without randomisation, 2 were randomised controlled trials with longitudinal evaluation (pre-, post-, follow-up, and 12 months after baseline), 1 was a diary study, 1 was quasi-experimental with pre-test and post-test with a control group, 1 was non-randomised and non-waiting list (pre-, post-, follow-up at 4 months), and 1 was cross-sectional.

The type of analyses of the researches was very varied, ranging from ANOVAs ($n = 9$), t -tests ($n = 6$), one-way analysis of variance ($n = 5$), MANOVAs ($n = 4$), linear mixed regression modelling ($n = 4$), hierarchical linear modelling ($n = 4$), linear-mixed effects models ($n = 2$), structural equation modelling ($n = 2$), mixed model ($n = 1$), random effects models ($n = 1$) and, cross-lagged designs ($n = 1$).

Characteristics of the Digital Interventions

Selected studies reported interventions showing a variety of formats. Almost all the interventions entailed by studies were completely digital in nature, that is, making use of digital tools such as internet browsers, smartphone apps, and instant chat platforms. Nonetheless, two studies (Ehrlich, 2022; Hammer et al., 2011) sought comparisons between the online and the face-to-face version of two distinct interventions.

The specifically digital format of the reported interventions varied across studies. Twenty-two interventions were purely web-based, meaning that they deployed the internet browser as the main digital place where the intervention was delivered. Web-based interventions included websites and online platforms. Nine interventions were purely app-based, meaning that they were delivered on smartphone, tablet, or computer applications. Four interventions deployed a mixed format with both web- and app-based components. Two interventions made use of instant chat platforms that could be installed both on smartphone and computer. Finally, one intervention deployed a mixed format in which web- and app-based components were complemented by a physical activity tracker. Table 1 summarises the digital format of interventions reported by selected studies.

Table 1. Format of examined digital workplace interventions

<i>Digital format</i>	<i>n</i>	<i>References</i>
Web-based	22	Althammer et al., 2021; Bormann et al., 2017; Cieslak et al., 2016; Ebert et al., 2015; Hosseinzadeh Asl, 2021; Keller et al., 2016; Knox & Franco 2022; Ijntema et al., 2021; Imamura et al., 2015, 2016; Li et al., 2021; Makowska-Tłomak et al., 2022; Nadler et al., 2020; Neumeier et al., 2017; Oliver & MacLeod, 2018; Ouweneel, 2013; Paterson et al., 2021; Phillips et al., 2014; Querstret et al., 2016; Shann et al., 2019; Tonkin et al., 2018; Uglanova & Dettmers, 2022
App-based	9	Avey et al., 2022; Bostock et al., 2019; Carissoli et al., 2015; Hirshberg et al., 2022;

		Keng et al., 2022; Pandya, 2021; Purdie et al., 2022; Smith et al., 2020; Weber et al., 2019
Web- and app-based	4	Bégin et al., 2022; Lennefer et al., 2019; Pospos et al., 2018; Shiotsuki et al., 2017
Chat-based	2	Wang et al., 2021; Zhang et al., 2022
Web, app, and activity tracker	1	Lennefer et al., 2020

Note. n = 38

There was also some extent of variability in the structure and contents of the reported interventions. The total length ranged from three weeks to three months. Also, the number of course sessions ranged from three to eight. Intervention activities included psychoeducational and informational reading materials, instructions for practical exercises (e.g., meditation), interactive games, pre-recorded listening pieces (e.g., audio guides, relaxing music), asynchronous videoclips and animated clips, synchronous videoconferences, questionnaires, polls, quizzes, essay forms for writing, downloadable documents, and stress and physical activity tracking based on wearable devices. Intervention facilitators were present in a minority of interventions, where sessions were delivered in real time via videoconferencing platforms or other virtual meeting formats, such as for instance in instructor-led mindfulness and on-demand online psychological support. However, most digital interventions were self-delivered. Most of the interventions were digital as per their original design. However, there were also digital adaptations of previously validated face-to-face interventions, especially in the case of mindfulness-based interventions (e.g., Althammer et al., 2021) and interventions that were translated online due to Covid-19 (e.g., Knox & Franco, 2022).

Almost all the DIs followed a scientific theoretical framework, the most predominant were based on mindfulness (Kabat-Zinn, 2015) with 13 studies, eight studies were based on cognitive-behavioural therapy (Beck, 1993), three on self-determination theory (Deci & Ryan, 2012), three on positive psychology (Seligman, 2012), two on the conservation of resources theory (Hobfoll, 1989), two on self-compassion (Neff, 2003), two on the job demands-resources model (Bakker & Demerouti, 2017; 2018). The rest of the studies are based on the broaden-and-build theory of positive emotions (Fredrickson, 2001), change theory (Lyubomirsky et al., 2005), and social support theory (Lakey & Cohen, 2000).

The content of almost all the reported interventions exclusively referred to psychosocial issues, such as factors of positive and/or negative mental health and well-being at work, or indicators of psychosocial well-being and/or ill-being at work. There was only one study (Lennefer et al., 2020) integrating elements of physical health and physical activity in an intervention touching upon dimensions of workplace mental health, such as job control, self-efficacy, emotional strain, and negative affect.

Regarding the IGLO levels at which the examined digital interventions were implemented (Day & Nielsen, 2017; Nielsen et al., 2017; Nielsen & Christensen, 2021),

40 interventions took place at the individual level, whereas the remaining 2 interventions took place at the leader level. This means that the reviewed digital interventions mostly aimed to address cognitive, affective, behavioural, and personal skills and abilities of individual workers – e.g., mindfulness, resilience, self-efficacy, self-compassion –, while only a huge minority of interventions aimed to tackle leaders' knowledge, skills, abilities, attitudes, or behaviours – e.g., supervisor support to employee work-family balance. Particularly, digital interventions mostly aimed to promote either individual- or leader-level job resources and positive aspects of work rather than reducing job demands and negative aspects of work.

Context Factors

All the methodological sections of the selected papers mentioned at least one omnibus context factor related to the general setting where the digital interventions took place. In most studies, these factors referred to participants' job positions or to the industrial sector of the involved organisation. Only a small number of studies explicitly tested the influences among context factors, working mechanisms and intervention outcomes, whereas in most cases information about implicit contextual conditions in which the interventions occurred had to be extracted.

At the individual level of analysis, context factors included participants' job positions, gender, age, previous levels of health and well-being, previous knowledge regarding intervention contents, and personal resources. Employees from selected studies represented a wide range of occupational sectors, encompassing healthcare (e.g., Bormann et al., 2017; Keng et al., 2022; Know & Franco, 2022; Kriakous et al., 2021; Pospos et al., 2018; Purdie et al., 2022; Wang et al., 2021; Zhang et al., 2022), education (Ebert et al., 2015; Hirshberg et al., 2022), social work (Hosseinzadeh Asl, 2021), self-employment (Neumeier et al., 2017), IT (Imamura et al., 2015), and manufacturing (Shirotsuki et al., 2017). Studies mainly included female participants over the age of 18. Only two studies found significant effects of gender and age, tested as control variables. Pandya (2021) found a smartphone app-based intervention to be more effective among males in reducing emotional exhaustion and depersonalisation and increasing resilience at post-test. Keller and colleagues (2016) found a web-based intervention to be more effective among older participants in increasing self-efficacy at follow-up. Several studies adopted pre-intervention levels of participants' health and well-being as an inclusion criterion, looking at baseline indicators such as burnout (Wang et al., 2021), exposure to traumatic events at work (Cieslak et al., 2016), work-related stress (Bostock et al., 2019; Carolan et al., 2017), digital stress during Covid-19 (Makowska-Tłomak et al., 2022), work-related affective rumination (Querstret et al., 2016), depression (Carolan et al., 2017; Phillips et al., 2014), and insomnia and psychological detachment from work (Carolan et al., 2017; Ebert et al., 2015). Other studies specifically focussed on non-clinical populations and healthy employees (Bostock et al., 2019; Imamura et al., 2015; Li et al., 2021; Shirotsuki et al., 2017; Weber

et al., 2019). Few studies tested the effects of pre-intervention levels of participants' health and well-being on intervention outcomes. Generally, employees reporting poorer levels of health and well-being at pre-test benefited significantly more from digital interventions (Carissoli et al., 2015; Hammer et al., 2011; Lennefer et al., 2019), but Avey and colleagues (2022) also found positive effects of higher levels of health and well-being at pre-test on employees' post-intervention resilience. Similarly, several studies adopted participants' pre-intervention knowledge about intervention contents as an exclusion criterion, especially prior experience with meditation practice before participating in mindfulness-based digital interventions (Hirshberg et al., 2022; Keng et al., 2016; Li et al., 2021; Purdie et al., 2022; Querstret et al., 2016). On the other hand, Nadler and colleagues (2020) found an online mindfulness training to be beneficial in terms of increased well-being – that is, higher resilience and positive mood, lower stress and negative mood – for both participants with and without prior meditation experience. Further, Makowska-Tłomak and colleagues (2022) suggested that prior use of information and communication technologies might be a relevant contextual factor when it comes to employees' engagement in digital interventions. For instance, during Covid-19 pandemic employees might have been reluctant to participate in additional online activities to prevent digital overload (*ibidem*). Finally, employees' personal resources was the last individual-level context factor identified. This encompassed readiness for change before starting the digital intervention (Makowska-Tłomak et al., 2022) and previous attitudes towards spirituality before taking part in a mantra repetition digital program (Bormann et al., 2017). Althammer and colleagues (2021) found lower work-family segmentation preferences to predict better intervention outcomes. Li and colleagues (2021) found no differential effects of pre-intervention trait self-compassion during a self-compassion digital intervention.

At the group level, context factors included peer support and type of interaction. Regarding the former, Ouweneel and colleagues (2013) found that lack of support from colleagues negatively impacted participants' work engagement after an online positive psychology intervention to promote engagement at work. Regarding the latter, Cantarero and colleagues (2021) found that employees interacting with other people via phone or internet during Covid-19 pandemic reported, as compared to employees interacting in person, higher levels of basic psychological needs satisfaction and well-being as a result of an online intervention.

At the leader level, manager support emerged as a relevant context factor. In this regard, Tonkin and colleagues (2018) found that the uptake of and engagement with an online gamified intervention was higher in organisations where senior managers encouraged employees to participate and provided useful resources during the implementation process. Similarly, Shann and colleagues (2019) showed the influence of supervisory support on transfer of a digital leadership intervention.

Finally, at the organisational level, context factors included organisational culture, economic incentives, organisational changes, and societal environment. Regarding indicators of organisational culture, several studies reported that employee participation in digital workplace interventions was voluntary, which can be

conceived as a discrete context factor. Testing voluntary participation's effect on intervention outcomes, Neumeier and colleagues (2017) found that self-selected employees were more motivated and reported larger gains from an online well-being programme. Organisational culture also encompassed omnibus context factors such as collective readiness, organisational capability to address digital interventions, and existing workplace mental health strategies, which can affect transfer of training (Shann et al., 2019). Economic incentives were leveraged to motivate employee participation in interventions, including gift vouchers (Hammer et al., 2011; Neumeier et al., 2017; Tonkin et al., 2018), money (Keng et al., 2022; Smith et al., 2020), continuing education credits, and credits for the company wellness store (*ibidem*). Moreover, organisational changes were reported to impact digital interventions' processes and outcomes. Shann and colleagues (2019) found that a shift in government and political priorities affected training transfer during a digital leadership intervention. Ijntema and colleagues (2021) found that resilience after an online coaching-based intervention was positively affected by changes in governmental policies related to working conditions during the process of a merger. With regard to the broader societal environment, several studies referred to Covid-19 pandemic outbreak during implementation of digital interventions in the workplace (Cantarero et al., 2021; Ehrlich, 2022; Hirshberg et al., 2022; Hosseinzadeh Asl, 2021; Keng et al., 2022; Knox & Franco, 2022; Makowska-Tłomak et al., 2022; Zhang et al., 2022).

Working Mechanisms

Twenty-nine studies mentioned working mechanisms of digital workplace interventions. Of these, 20 tested working mechanisms' effects on intervention outcomes. Working mechanisms were identified at the individual and the organisational level, whereas no mechanisms could be retrieved at the group and the leader level of analysis.

At the individual level, process mechanisms included type of DI usage, frequency of practice, implementation adherence, training transfer, modality, and duration. As an example of type of usage, Pandya (2021) found that using a smartphone meditation app once or twice a day, perusing both videos and learning sessions, and self-practicing daily, resulted in lower emotional exhaustion and depersonalisation and higher personal achievement and resilience post-test. Frequency of practice was measured in terms of amount of time using web or mobile app interventions (Bormann et al., 2017; Bostock et al., 2019; Keng et al., 2022; Purdie et al., 2022; Tonkin et al., 2018) and number of activities per week (Ebert et al., 2015). Implementation adherence was mainly operationalised as dose delivered versus dose received and measured in terms of number of attended sessions (Bostock et al., 2019; Knox & Franco, 2022; Weber et al., 2019) or modules (Ebert et al., 2015; Oliver & MacLeod, 2018). In all studies, higher implementation adherence predicted post-intervention improvements in mental health and well-being outcomes. Training

transfer was also found to be an important facilitator of workplace digital interventions' effectiveness. Vanhove and colleagues (2016) showed that greater opportunities to bring learnt skills back to everyday work led to improvements in resilience intervention outcomes. Similarly, Shann and colleagues (2019) reported that a set of context factors – i.e., collective readiness, attitudes and stigma, organisational changes – impacted training transfer which, in turn, impacted mental health and depression after an online leadership intervention. Few studies addressed modality as a working mechanism of digital interventions in the workplace. Vanhove and colleagues (2016) found that a computer-based delivery format of a resilience-building programme was less effective than its group-based classroom format in triggering well-being outcomes. Carolan and colleagues (2017) conducted a systematic review of the literature and found that studies that utilise secondary modalities for delivering the DIs and engaging users – i.e., e-mails and text messages, SMS – and use elements of persuasive technology – i.e., self-monitoring and tailoring – may achieve greater engagement and adherence, which lead to increased psychological well-being and work effectiveness. Regarding duration, Carolan and colleagues (2017) also found that a shorter intervention timeframe – from 6 to 7 weeks – led to higher engagement and adherence as compared to digital workplace interventions of longer duration.

Perception mechanisms included change in attitudes and relevance of intervention content. Shann and colleagues (2019) showed that the sustainability of attitude change and the content of an online leadership intervention being considered relevant by its recipients helped participants to overcome workplace stigma and improve mental health. Moreover, Querstret and colleagues (2016) mentioned mechanisms of changes, which can be classified as content mechanisms. Specifically, the authors focussed on facets of mindfulness – that is, acting with awareness, describing, nonjudging, and nonreacting – during an internet-based instructor-led mindfulness intervention for work-related rumination, fatigue, and sleep. Increased levels of acting with awareness explained the intervention outcomes (*ibidem*). Task crafting could be conceptualised as a working mechanism in the study by Uglanova and Dettmers (2022) regarding a web-based job crafting intervention, but no significant results about it were found.

Finally, at the organisational level, a process mechanism could be identified, namely external support, relating to how workplace interventions are designed, organised and managed. This variable referred to whether facilitator's guidance or supervision was provided during the intervention. Carolan and colleagues (2017) suggested that interventions that achieve the greatest engagement and adherence offer some form of guidance, such as therapist, coach, a coordinator or member of staff, and clinical psychologist. Later, Ijntema and colleagues (2021) tested the coach-client working relationship's strength and found that it was related to most of the immediate effects of a web-based resilience-building programme. The intervention seemed most effective for employees who experienced a stronger coach-client working relationship.

Outcomes

Workplace digital interventions' outcomes could mainly be identified at the individual level of analysis. Outcomes could be categorised into proximal versus distal outcomes, as well as into well-being versus ill-being outcomes.

Individual-level proximal well-being outcomes mostly included mindfulness (Keng et al., 2022), resilience (Zhang et al., 2022), self-compassion (Li et al., 2021), self-efficacy (Ouweneel et al., 2013), and purpose of life (Hirshberg et al., 2022; Ijntema et al., 2021). Furthermore, distal well-being outcomes at the individual level mostly included psychological well-being (Neumeier et al., 2017), general health (Lennefer et al., 2019), positive affect (Ouweneel et al., 2013), satisfaction (Keng et al., 2022) and job performance or work effectiveness (Carolan et al., 2017; Vanhove et al., 2016). Individual-level proximal ill-being outcomes mainly included perseverative thinking (Hirshberg et al., 2022; Ebert et al., 2015; Querstret et al., 2016) and negative affect (Lennefer et al., 2020). Furthermore, distal ill-being outcomes at the individual level mostly included stress (e.g., Carissoli et al., 2015), anxiety (e.g., Zhang et al., 2022), depression (e.g., Imamura et al., 2015), and burnout (e.g., Li et al., 2021).

Social support (Paterson et al., 2021) could be identified as a group-level proximal outcome of digital interventions in the workplace. No leader- or organisational-level outcomes could be identified. Table 2 summarises the main findings from the present study, namely context factors, working mechanisms and outcomes of digital workplace interventions, classified according to the IGLO model (Day & Nielsen, 2017; Nielsen et al., 2017; Nielsen & Christensen, 2021).

Table 2. Context factors, working mechanisms and outcomes at IGLO levels of digital workplace interventions

	<i>I</i>	<i>G</i>	<i>L</i>	<i>O</i>
Context	Job position Gender Age Previous health and knowledge	Peer support Interaction	Manager support	Culture Incentives Changes Society
Mechanisms	DI usage Frequency of practice Adherence Transfer Modality Duration Attitude change			External support

	Content relevance	
	Health and well-being	
Outcomes	Work effectiveness	Social support
	Stress, anxiety, depression, burnout	

Note. I = individual. G = group. L = leader. O = organisation

Discussion

The aim of this systematic review was to explore how, why, and under what circumstances digital interventions in the workplace are effective. In other words, the study sought to identify the contextual factors that might influence workplace digital interventions' effectiveness, the working mechanisms that need to be triggered for the intervention to work as intended, and the work-related outcomes that such interventions produce. It did so by adopting a realist CMO evaluation approach (Nielsen & Abildgaard, 2013; Nielsen & Miraglia, 2017; Nielsen & Randall, 2013; Nielsen & Shepherd, 2022; Roodbari et al., 2021a, 2021b, 2023) as well as an IGLO multilevel perspective (Day & Nielsen, 2017; Nielsen & Christensen, 2021; Nielsen et al., 2017), integrated into one analytical framework. As a result, several omnibus and discrete context factors, process, perceptions and content mechanisms, and proximal and distal well-being and ill-being outcomes were extracted from the selected studies, along with a number of varied digital interventions applied in workplace settings.

Despite variables being identified at all IGLO levels, most of them corresponded to individual characteristics, attitudes, psychological states, and behavioural outcomes. This probably reflects a more generalised tendency of research in work and organisational psychology towards predominantly focussing on individual issues and the individual as its essential unit of study (Anseel et al., 2018). However, the findings from the present study also pointed out to the importance of interpersonal dimensions such as, for instance, peer and manager support as relevant group- and leader-level context factors for workplace digital interventions' effectiveness (e.g., Ouweneel et al., 2013; Shann et al., 2019; Tonkin et al., 2018), as well as the relationship between the recipients and the facilitators as a process mechanism leading to desirable outcomes of digital interventions in the workplace (e.g., Carolan et al., 2017; Ijntema et al., 2021). On the one hand, these results are consistent with previous literature highlighting that peer and manager support positively affect workplace interventions' effectiveness (Christensen et al., 2019; Helland et al., 2021; Nielsen et al., 2010, 2023; Roodbari et al., 2021a, 2021b). Support from peers and managers facilitate the effectiveness of workplace interventions, as they provide

workers with the necessary social resources that sustain their participation in the interventions as well as the application of what they acquired during the intervention into their everyday work (*ibidem*). On the other hand, the present results are in line with previous empirical studies showing the positive effects of consultant support on organisational interventions' outcomes (e.g., Jenny et al., 2015; Niks et al., 2018; Roodbari et al., 2021a, 2021b).

In general, based on the present study's findings, it can be stated that digital interventions prove promising in reducing job demands and promoting job resources as conceived by the JD-R model (Bakker & Demerouti, 2017; 2018; Schaufeli, 2017), thus providing benefits to workers. Particularly, by fostering resources, DIs contribute to enhancing individuals' ability to cope with stress, adapt to challenging situations, and maintain a positive mindset, thereby promoting overall well-being at work.

This study has limitations which should be considered when interpreting its results. First, potential publication bias could not be controlled, as positive and significant findings are more likely to be published, thus leading to an overestimation of intervention effectiveness. Second, only English, Italian and Spanish studies could be included due to the authors' limited language skills, which may have results in the omission of relevant information from studies in other languages. Similarly, grey literature was not included in the study, but this decision was made to maximise high-quality scientific standards of reviewed evidence. Third, as contemporary technological developments continue advancing at an increasingly rapid pace, the review might have failed to capture the most recent workplace digital interventions. Fourth, as most context factors and working mechanisms were not explicitly investigated as such in the selected studies, some extent of subjectivity has to be taken into account regarding the conceptualisation of relevant variables identified throughout the review. Moreover, there was a causality problem to be taken into account in the selected articles, given the predominance of cross-sectional studies. The causal relationships in DIs were multifaceted and often not directly observable in the reviewed studies. Finally, the variability across studies, evidenced by differences in intervention methods, outcome metrics, and participant demographics, presented a significant impediment to the effective comparison and synthesis of research findings. This heterogeneity, spanning a range of study foci from mindfulness to job crafting, self-determination theory, and self-efficacy, complicated the interpretation of results and the conclusions that could be drawn.

This study also has strengths. Most of all, a systematic literature review is a useful tool for both researchers and practitioners, as it provides a comprehensive and rigorous examination of the available evidence by deploying a structured and predefined approach to searching, selecting and evaluating relevant studies. By following a predetermined set of criteria, this review method minimises bias and enhances the reliability and validity of the findings. Additionally, the inclusion of a wide range of studies allowed for a holistic understanding of digital interventions in the workplace, capturing their diverse forms and manifestations across different workplace settings. Particularly, the current review was conducted integrating

elements of both systematic and realist review methodologies. A hybrid approach was adopted, combining the structured rigor of systematic reviews (Perestelo-Pérez, 2013) with the flexible, theory-driven nature of realist reviews (Nielsen & Miraglia, 2017; Pawson et al., 2005; Wong et al., 2013). This combination was strategically chosen to mitigate the potential limitations of a purely realist synthesis approach, thereby enhancing the robustness and scope of the research.

On this basis, implications for both theory and practice can be considered. In terms of theoretical advancements, the findings from this study contribute to knowledge about and operationalisation of context factors, working mechanisms and outcomes of digital interventions in the workplace. While confirming some of the findings from previous reviews on context factors and working mechanisms of traditional organisational interventions (e.g., Dieleman et al., 2009; Gray et al., 2019; Roodbari et al., 2021a, 2021b), the present study moves beyond by applying the realist evaluation approach into digital-based interventions at work. Particularly relevant to future research, the identification of context factors, working mechanisms and outcomes allows the possibility of developing theories and empirically testable models to explain how different mechanisms of workplace digital interventions lead to various outcomes in several contextual conditions (Pawson et al., 2005; Roodbari et al., 2021a, 2021b). Overall, extracting frequent working mechanisms within the reviewed studies might help formulating context-mechanism-outcome (CMO) configurations that could be tested via moderation or mediation analysis. These theories and models may be used by occupational researchers to advance the knowledge and understanding of what makes digital interventions work for whom in what circumstances, as well as by practitioners for the design, implementation, and evaluation of digital interventions in the organisational field. As for practical implications, the insights gained from this review can inform evidence-based decision-making, guide the development of effective interventions, and ultimately contribute to the enhancement of work environments. Applying a realist perspective provides an opportunity to gain a better understanding of how different DIs can improve work outcomes, under which circumstances, and for whom, thus maximising their impact. As such, this study provides an important potential contribution to practitioners for the design, implementation, and evaluation of future workplace digital interventions. Through an attempt to combine the realist evaluation framework and a multilevel analytical perspective, the review aimed not merely to provide a body of empirical evidence on the effectiveness of digital interventions, but rather to specify how interventions are affected by contextual factors and working mechanisms to lead to specific outcomes. Practitioners and organisational managers could use this knowledge when planning new DIs and/or selecting existing ones to improve workers' outcomes. In other words, this approach allows for a more nuanced understanding of how interventions can be tailored to different contexts. This adaptability is critical when selecting DIs, as it enables the customisation of strategies to better meet the diverse needs of users. The additional insights gained from the application of such knowledge can inform decision-makers about the likely

effectiveness of a digital intervention in specific contexts, supporting context-sensitive and evidence-based decision-making in the selection of interventions.

Conclusion

It is encouraging that an increasing number of studies are providing evidence regarding workplace digital interventions, thus offering reliable solutions and strategies to organisations willing to act upon the working environment and workers. The present review enhances the practical knowledge related to such a type of interventions by identifying the contextual conditions under which they can be most effective, and the working mechanisms that can produce their intended outcomes. Knowledge is added not only to claim whether workplace digital interventions are effective, but also how and why, thus contributing to the overall realist evaluation literature.

Nevertheless, this study sheds light on several gaps still remaining in the literature, which can constitute an agenda for future research into digital-based interventions in the workplace. Most of all, studies should investigate more leader-, group-, and organisational-level digital interventions. Regarding the IGLO model, 40 interventions from the present review were implemented at the individual level, whereas two occurred at the leader level. Particularly relevant to the topic of the present dissertation, no intervention was implemented at the group level. Furthermore, studies should deploy qualitative and mixed method designs to reach a more comprehensive view of the factors that may affect intervention effectiveness. The incorporation of qualitative and mixed method designs could provide a more multifaceted and in-depth understanding of the contextual factors that influence digital interventions' effectiveness. These methods can help identify which contextual factors may either support or inhibit the mechanisms leading to the desired outcomes. Also, studies should be conducted including not only working mechanisms already suggested by previous research – e.g., acceptability of the intervention, consultant integrity, quality of implementation, opportunities to transfer and integrate training into everyday work –, but also mechanisms that take into account the peculiar digital nature of interventions – e.g., usability, human-technology interaction.

All of the above was deemed as an input to more targeted research into group-level digital workplace interventions. Such research was carried out throughout the studies that are introduced in the next chapters of the present dissertation.

Study 2. Job Demands and Resources at the Group Level: A Multi-Source Qualitative Needs Assessment Exercise

Abstract. Psychosocial interventions addressing workers should build upon an exhaustive understanding of job demands and job resources in the workplace at all the levels they might unfold, namely the individual, the group, the leader, and the organisation. This study draws upon a multilevel workplace needs assessment exercise performed within three different departments of a large healthcare institution and involving both managers and employees. It aims to illustrate the job demands and resources in the targeted organisation, differentiate among workers' mental models of their working conditions, and discuss the research and practical implications of such findings. Particularly relevant to the main topic of the present dissertation, resources and demands were found at the group level of analysis. Resources included mutual support, trust, cohesion, diverse expertise, and inter-professional cooperation. Communications, interactions, and information exchanges were reported as demands needing improvements at team level. Communication about organisational initiatives to support workers was reported as improvable. In one department, communication between doctors was described as fragmented. In another department, lack of training in communication skills was mentioned. This evidence supported the meaningfulness of implementing digital-based interventions to foster teamwork within the targeted organisation.

Background

When psychosocial interventions are introduced in organisations, an exhaustive understanding of the work environment is needed (e.g., Di Tecco et al., 2020; Fridrich et al., 2015; Ramos et al., 2020). This can be achieved through a workplace needs assessment exercise. Consistent with JD-R and IGLO, such assessment should encompass, on the one hand, both job demands and job resources and, on the other hand, cover all levels of the workplace system. So, since they can allow the identification of multilevel sources of either good or bad workers' outcomes within a given working environment – i.e., barriers to positive outcomes at work/major causes of negative work-related outcomes, and positive aspects of work/major causes of positive work-related outcomes –, JD-R and IGLO can be deemed as flexible and easy-to-use instruments, not only to perform actual interventions, but also to conduct workplace needs assessment exercises. In this regard, an integration between JD-R and IGLO can be achieved by making the individual, group, leader, and organisational levels serve as a classificatory framework for job demands and job

resources. Ultimately, this framework can be deployed as a guide for a needs assessment exercise in the workplace.

Workplace assessment activities are useful to tailor subsequent interventions according to the specific workers' needs. However, these needs may depend on how workers perceive their work environment. According to Persson et al. (2012), each worker has their mental model about their work environment. That is, people take cues from the work environment and make sense (e.g., Weick, 1995) and develop a certain understanding of it. This understanding is collectively shared because of common conditions that people may find themselves in together. For instance, one worker's type of perception of a given work environment may depend on the position the worker occupies within the work environment – e.g., in the hierarchy, in the organisational chart – as well as on local working conditions. Similarly, a perceptual distance phenomenon (Gibson et al., 2001, 2009) may occur, whereby managers and employees do not always interpret a given work situation in the same way. By taking cues from each other at work, people develop shared mental models about their working conditions. So, these appraisals influence what workers see as being a job demand or a job resource. Therefore, it is important to gather workers' mental models of their working environment and to identify which aspects of the work environment they share according to the positions they find themselves in or local conditions. In this vein, workplace needs assessment exercises should be properly contextualised (Nielsen et al., 2014; Vignoli et al., 2017). This means conducting them within the field of homogeneous work environments – e.g., same roles, same physical space, same set of activities, same cultural setting, same working patterns, and so on –, as each work environment is likely to show its own characteristics, peculiarities, and idiosyncrasies, and therefore is not necessarily comparable to others, as much as the knowledge retrieved in one work environment is not necessarily generalisable to another one. In addition, needs assessment activities should adopt a multi-source approach to combine information from different actors and acquire a comprehensive picture thanks to the triangulation of perspectives on the targeted assessment issues.

To capture workers' mental models, assessment activities need to involve workers directly (Nielsen et al., 2021). The bottom-up, participatory approach is a guiding principle in this regard. It is one of the most critical success factors of assessment and intervention activities and it consists of the direct participation of relevant employees and stakeholders throughout the whole process. According to Nielsen et al. (2010), the importance of employee participation is because it can (1) help optimise the fit with the local organisational context, (2) be considered an intervention, and (3) facilitate the intervention process. In a bottom-up perspective on workers' outcomes promotion, employees should not be seen as passive subjects, but rather as active actors able to change their work environment. Such an approach is able to (1) ensure the use of relevant stakeholders' local knowledge of what the key issues are concerning job demands and job resources, (2) show what changes need to be made and how, and (3) ensure stakeholders feel valued, empowered, and looked after. By using a participatory approach, workers and their managers collectively gain

resources, knowledge, and skills to identify workplace problems, develop solutions, and implement changes to improve their working conditions (Nielsen et al., 2013). Thus, employees and managers are to be considered key informants along the assessment activities (Christensen et al., 2019; Tafvelin, 2018).

The present chapter draws upon a broader multilevel workplace needs assessment exercise performed within a healthcare setting (Giusino et al., 2022a; 2022b²). The overarching aim was to collect suggestions to inform the subsequent design, development, and implementation of multilevel interventions, actions, and initiatives addressing healthcare workers' outcomes. However, for the purposes of the present dissertation's main topic, only job demands and job resources at the group level of analysis are reported, consistent with Lacerenza et al.'s (2018) indication that, when interventions are delivered to teams, needs of the teams are to be exhaustively understood first. The ultimate aim was to verify the organisation-intervention fit (Andersen et al., 2021; Peters et al., 2020) and ensure the relevance of the subsequent implementation of team-level digital-based interventions in the targeted workplace settings. The chapter presents the main group-level findings from the needs assessment exercise and discusses a list of key insights from performing the field experience. The importance of locally assessing work-related issues by means of a contextualised, bottom-up, participatory approach is argued. Consistently, the focus of the needs assessment was on three hierarchical positions (i.e., senior managers, middle managers, employees) from three departments of the targeted healthcare organisation, since (1) the implementation of subsequent interventions, actions and initiatives was planned in each of these departments, (2) this was thought to enable comparisons across both hierarchical positions and departments as well as to (3) clarify whether mental models of job demands and job resources differed across hierarchical positions and/or departments. Thus, along the chapter, study results are synthesised and organised according to hierarchical roles and departments. Finally, practical implications and recommendations for future research are discussed.

Methods

The workplace needs assessment exercise, designed to capture job demands and job resources, was carried out in three departments of a large healthcare institution in Northern Italy. The institution is one of Italy's largest public healthcare organisations in terms of size and care complexity. Its jurisdiction includes 46 municipalities on approximately 3.000 square kilometres, encompassing over 870.000 inhabitants, of which over 23% is over 65 years old, 8% is over 80 years old, and 11% is made of foreign residents. The organisation is divided into six territorial districts, extending across the metropolitan area, and is composed of six hospital departments, four

² Preliminary findings from these published studies are also included in the European Association of Work and Organisational Psychology's EAWOP 2022 Conference Legacy Document. <http://www.eawop.org/legacy-of-eawop-2022-glasgow-conference>

territorial department, and five support departments. It has nine district clinics, a growing number of healthcare facilities for older adults, and outpatient clinics spread throughout the whole metropolitan area. It employs over 9.000 professionals, more than 1.300 of which are physicians and 5.100 are care workers.

In the present study, the three targeted departments will be named Department A, Department B, and Department C to preserve privacy and anonymity. Each department is different from the others not only in terms of size and discipline, but also because of unique work and organisational cultures. First, Department A counts around 600 employees and most of the clinical activities performed here by healthcare workers are characterised by emergency and urgency. These activities are multidisciplinary and take place in different buildings spread over and beyond the metropolitan area. Second, Department B counts around 600 employees, and it is a multidisciplinary medical institute whose clinical activities – e.g., prevention, diagnosis, and non-surgical treatment of several diseases – are quite routine as compared to Department A. These activities are also spread over the urban territory, such that Department B may include both people working in central hospitals and people working in peripheral hospitals. Third, Department C counts about 300 employees and it consists of a both clinical and research institute, where the medical and the academic mindsets intertwine. Given its monodisciplinary focus and strive for scientific excellence, Department C is considered as more of a specialised hospital. Different from the other two departments, all activities performed by Department C's healthcare workers take place in one building only. Despite the General Director of the main healthcare organisation being the same for all three departments, the management style may differ significantly across them, also due to the presence of different senior and middle managers. Also, the three department do not necessarily show the same working patterns, processes, and procedure, as well as they are not necessarily equipped in the same way in terms of technical, financial, and human resources. All these differences between the three departments legitimated the need for differently investigating working conditions and healthcare workers' mental models in each of them.

The workplace needs assessment methodology comprised four main parts, such as (1) a quali-quantitative contextual measurement, aiming to capture the extent to which management was committed to dealing with workers' issues, what sort of policies, practices, and programmes were in place in the healthcare setting and how they were perceived, (2) semi-structured individual interviews with middle and senior managers, aiming to understand middle and senior managers' experiences, ideas and perspectives around the needs for interventions to improve workers' outcomes, (3) focus groups with employees, aiming to gain mutual knowledge of psychosocial factors affecting psychological and behavioural outcomes at work, and (4) an action plan workshop with a Steering Committee composed of main organisational stakeholders, aiming to identify strategies and interventions needed to improve workers' outcomes in each department. In the targeted organisation, middle managers corresponded to coordinators of work teams within the hospital

departments – for instance, they might be head nurses managing other nurses or head physicians managing other physicians. In contrast, senior managers corresponded to directors or heads managing all the employees of the hospital departments. Available participants were invited to participate by the manager of Health and Safety unit at the hospitals, via either emails or direct contacts in the workplace. To get on the same page about the topic and to make the research protocol following our theoretical model, before each data collection session, all participants were shown a subtitled cartoon video (NTNU Lectures, 2016) providing an easily accessible description of the JD-R model. Also, at the beginning of each session, all participants were given an oral explanation of the IGLO framework. Then, they were encouraged to answer our exploratory questions by keeping in mind the integration between JD-R and IGLO.

Contextual Measurement: Organisational Perspective

Contextual measurement consisted of a survey to be completed by a small group of employee representatives and investigating three thematic areas, namely (1) description of policies, programs, and practices within the organisation, (2) perception of policies, programs, and practices, and (3) management support, commitment and priority, and organisational communication, involvement, and participation. The instrument was composed of 7 open-ended questions for part (1). Eleven Likert-type items were used for part (2) and inspired from the Workplace Integrated Safety and Health (WISH) assessment by Sorensen et al. (2018) and López Gómez et al. (2021). Twelve Likert-type items were used for part (3) and inspired from the Psychosocial Safety Climate (PSC-12) assessment by Hall et al. (2010). The Health and Safety Manager, the Workers Safety Representative, and each Director of the three departments contributed to completing the survey.

Qualitative text data went through full NVivo content analysis (Bazeley & Jackson, 2013), while quantitative answers were used to complement the summaries of qualitative findings.

Semi-Structured Individual Interviews: Managerial Perspective

For senior and middle managers, we administered online semi-structured individual interviews via a computer-based teleconferencing platform compliant with the General Data Protection Regulation (EU) 2016/679 (GDPR). Twenty-one one-hour individual interviews were completed. This flexible strategy was deemed appropriate to managers due to high unpredictability of their work schedules. Validity of this technique is supported by recent literature (Howlett, 2021). In Department A, two senior managers and three middle managers were interviewed ($n = 5$). In Department B, four senior managers and six middle managers were interviewed ($n = 10$). In Department C, two senior managers and four middle managers were interviewed ($n = 6$). The interviews investigated (1) perceptions, knowledge, and attitudes, (2)

hindering and facilitating aspects for the middle/senior managers' role, (3) needs towards creating a sustainable workplace, (4) barriers and triggers related to implementing interventions, and (5) proposals to create and implement workplace initiatives successfully. Interviews were conducted between September and October 2020 by two trained researchers, who committed to a strict code of ethical scientific and professional conduct whereby they should not disclose any sensitive information they might be aware of regarding the interviewed persons.

Interviews were audio-recorded. Recorded data were transcribed verbatim with any identifying information anonymised. One researcher cleaned data by formatting raw data files in a common format. Four native Italian speaker researchers performed deductive content analysis via NVivo version 1.3.1 software (Bazeley & Jackson, 2013). The output of the analysis was then directly translated into English by one bilingual native Italian and English-speaking researcher and approved by three other English-only-speaking researchers. Qualitative content analysis of text data implies the "systematic classification process of coding and identifying themes or patterns" (Hsieh & Shannon, 2005; p. 1278). It provides a flexible and practical technique to investigate human perspectives into matters of health and illness (Hsieh & Shannon, 2005). Particularly, deductive analysis starts from models or frameworks that determine the initial coding scheme or categories, as well as key themes and relationships among them, and are used as a guide for the coding process (Hsieh & Shannon, 2005; Thomas, 2006). The goal is to provide support or to expand existing theories or conceptualisations. In our study, the JD-R/IGLO integrated theoretical framework provided the initial coding scheme. Thus, content analysis was performed by searching for job demands and resources at individual, group, leader, and organisational level – although only group-level job demands and resources are reported here. If new themes emerged as subcategories to predefined codes, we deployed a more inductive analytical procedure (Braun & Clarke, 2006). No interrater reliability index calculation was deemed necessary as the starting theoretical framework determined agreement on identified codes since early process phases.

Focus Groups and Cognitive Mapping: Employee Perspective

We held in-person two-hour focus groups (Woodyatt et al., 2016) with employees at devoted meeting rooms at participants' workplaces. The focus groups were conducted by two trained researchers in September 2020. Four focus groups with 27 healthcare professionals were conducted. In Department A, two focus groups took place with a total of three doctors, eight nurses, and four healthcare assistants ($n = 15$). In Department B, one focus group was conducted with two doctors, three nurses, and one healthcare assistant ($n = 6$). In Department C, one focus group was conducted with six nurses and two healthcare assistants ($n = 8$), whereas doctors could not participate. The focus groups investigated (1) perceptions, knowledge, and attitudes, (2) hindering

and facilitating working conditions, and (3) needs towards creating a better workplace.

Trust-based measures were taken by focus group facilitators by informing participants that their information would be kept confidential by the study researchers. They also explained the importance of every participant's compliance with maintaining privacy and confidentiality about what was discussed in the groups. Each focus group participant was advised to respect the confidentiality of what was shared by the other members during the focus group. Researchers declared they would expect to collect various perspectives, so there would be no correct or wrong version. Researchers encouraged participants' willingness to share opinions by creating a convivial meeting climate. All participants were granted equal opportunity to contribute to the discussion and could decide not to intervene at any time.

The same procedure and data analytical approach as the managerial interviews was followed. Focus groups were audio-recorded and transcribed qualitative text data went through NVivo deductive content analysis.

Each focus group included a cognitive mapping exercise to gather healthcare workers' reflections on how job demands and job resources at different IGLO levels interact with each other. To ensure all employees having the same understanding about workplace issues and workers' psychological states and behavioural outcomes, the exercise started with an explanatory video of the JD-R model. Then, participants were given five minutes to individually note up to three keywords reflecting the current main issues they perceived in their workplace. With this reflection in mind, participants were asked to fill in green post-it notes with at least three job resources and red post-it notes with at least three job demands. This first part of the exercise was carried out without seeing the cognitive map not to bias participants' ideas with predefined categories, but to make them think freely. Subsequently, the facilitator introduced the actual cognitive map. On the map, some gears illustrated how the IGLO levels interact with each other in the workplace system. There were three smaller gears for the individual (I), the group (G), and the leader (L), placed within a larger gear for the organisation (O). Each gear had some example work-related categories attached to its teeth, which the facilitator provided some brief explanation about. However, there was also room for additional categories, if needed, which the facilitator could fill in by the unmarked teeth on the gears. Employees were instructed to place their post-it notes on the most suitable category. The facilitator could assist participants by discussing the reported job demand or job resource. Post-it notes that could not be placed clearly could be parked on a "P" area and discussed further later. Finally, participants' statements were discussed collectively in the group. The facilitator sorted participants' statements according to the map categories to check whether some were misplaced and/or pertained to multiple categories. "Parked" statements were discussed to identify the best fit on the map. The facilitator moved the green and red post-it notes around the map and drew relationships, if any, with a marker pen. (S)he summarised the most important discussion points and asked if participants had anything to add before concluding the meeting. In this way, the

facilitator guided the meeting to a harmonious and productive closure, ensuring that all important facets of the discussion had been explored and that every participant had the opportunity to enrich the collective understanding. The collaborative process, punctuated by the movement of colorful post-it notes, resulted in a visually captivating and intellectually stimulating meeting.

Results

The current section reports the group-level job demands and resources as expressed by each of the various groups of stakeholders involved in the workplace needs assessment activities at the targeted healthcare organisation, namely the employee representatives, the senior and middle managers, and the employees.

Group-Level Demands and Resources as Reported by Employee Representatives

Despite work was not reported to be systematically and preventatively designed, organised, and managed with the explicit aim of promoting healthcare workers' psychological states and behavioural outcomes – with the only exception being the organisation of work shifts –, some structured practices could be identified at the team level of analysis. Monthly team meetings were reported to be held whereby employees can discuss both work-related issues and concerns. These meetings were reported to help conflict management thanks to adopting a mediating leadership style. In addition, a “feedback meeting” was mentioned as part of a more extensive performance evaluation system and reported as an occasion for employees to express concerns. Interviewees agreed about internal organisational communication being a need to address for the organisation to be able to support workers.

Group-Level Demands and Resources as Reported by Senior Managers

In Department A, a positive team climate was reported as a group-level resource, as indicated by effective teamwork, cooperation, collaboration, cohesion, open communication, and ability to manage interpersonal conflicts. Especially cooperation among and between senior and middle managers was mentioned, as it was stated,

“through working groups, through meetings, which we do periodically, through observation, we try to understand how to strengthen the service and individuals”.

Nevertheless, work-related problems were reported to be perceived as an individual weakness, which may be due to some degree of stigma, that is, a group-level demand. A desire for more open and inclusive organisational communication was also expressed.

In Department B, at the group level, the main resource reported was the availability of multidisciplinary teams composed of diverse expertise, skills, and abilities, which were deemed necessary to deal with patients' medical complexity. Team cohesion and open attitudes towards mental health in the workplace were also mentioned. Still, communication between doctors was described as fragmented.

In Department C, interpersonal conflicts, both among peers and different hierarchical roles, were reported as a group-level job demand, whereas no group-level job resources were mentioned. Interestingly, a supportive, empowering, and intellectually stimulating managerial style – that is, a leader-level job resource – was self-reported as fostering team cohesion, which is a relevant team-level outcome. Also, several demands attached to leaders' role were reported as preventing leaders from always ensuring a positive team climate. These findings show how the level of the group and that of the leader are closely intertwined, as there is no team without a leader as well as there is no leader without a team of followers.

Table 3 summarises the group-level job demands and resources in the healthcare institution as reported by senior managers.

Table 3. Group-level job demands and resources as reported by senior managers

<i>Department</i>	<i>Demands</i>	<i>Resources</i>
A	Stigma towards workers' issues	Positive team climate
B	Fragmented communication between doctors	Multidisciplinary teams Team cohesion Open attitudes
C	Interpersonal conflicts	

Note. A = multidisciplinary clinical activities in emergency. B = prevention, diagnosis, and non-surgical treatment. C = clinical and research institute

Group-Level Demands and Resources as Reported by Middle Managers

In Department A, at the group level, exchange of positive feedback within teams was reported among job resources. Among job demands, interpersonal conflict was linked to unfair career opportunities and lack of recognition from the management.

In Department B, a positive team climate was reported as a group-level resource, as indicated by cohesion, support, collaboration, trust, and good group communication. For instance, it was stated,

"[...] being able to feel like someone whose point of view is being asked is already a great openness".

In Department C, positive team climate was reported as group-level job resource, as indicated by cohesion, openness to problems, and quality teamwork. For instance, it was stated,

"There is a beautiful environment and exchange between specialists, which allows us to grow together".

On the other hand, concern was expressed that such a cohesive, established, and long-lasting team may reveal counterproductive over time when faced with needs for change or adaptations. For instance, it was stated,

"When I came to run this facility, I felt as if I was dealing with people who were all very rigid and reluctant to change".

Lack of communication skills training was also mentioned. Table 4 summarises the group-level job demands and resources in the healthcare institution as reported by middle managers.

Table 4. Group-level job demands and resources as reported by middle managers

<i>Department</i>	<i>Demands</i>	<i>Resources</i>
A	Interpersonal conflicts	Mutual positive feedback
B		Positive team climate
C	Excessive cohesiveness Lack of communication skills training	Positive team climate

Note. A = multidisciplinary clinical activities in emergency. B = prevention, diagnosis, and non-surgical treatment. C = clinical and research institute

Group-Level Demands and Resources as Reported by Employees

In Department A, at the group level, teamwork was referred to as a fundamental job resource. For instance, it was stated,

“My motivation for this work stays high only because of the group support”.

Doctors and nurses listed peer mutual support, listening in difficult working conditions, organisational citizenship behaviours, strong sense of community, positive work climate, and team cohesion as crucial group-level job resources. Nevertheless, interpersonal conflicts, blaming attitudes, and disrespectful behaviours could be identified as group-level job demands. Figure 2 and Figure 3 show the cognitive maps from the first and second focus group at Department A. The maps allowed to gather links and interactions among job demands and job resources at different IGLO levels, but only group-level findings are highlighted here, consistent with the topic of the present dissertation. As it is shown, an individual-level job resource such as initiative for discussing workplace issues was reported to lead to a group-level job resource such as positive relationships with colleagues. Also, at the group level, team cohesion was mentioned as a possible resource to increase the sharing of both positive and negative experiences at work.

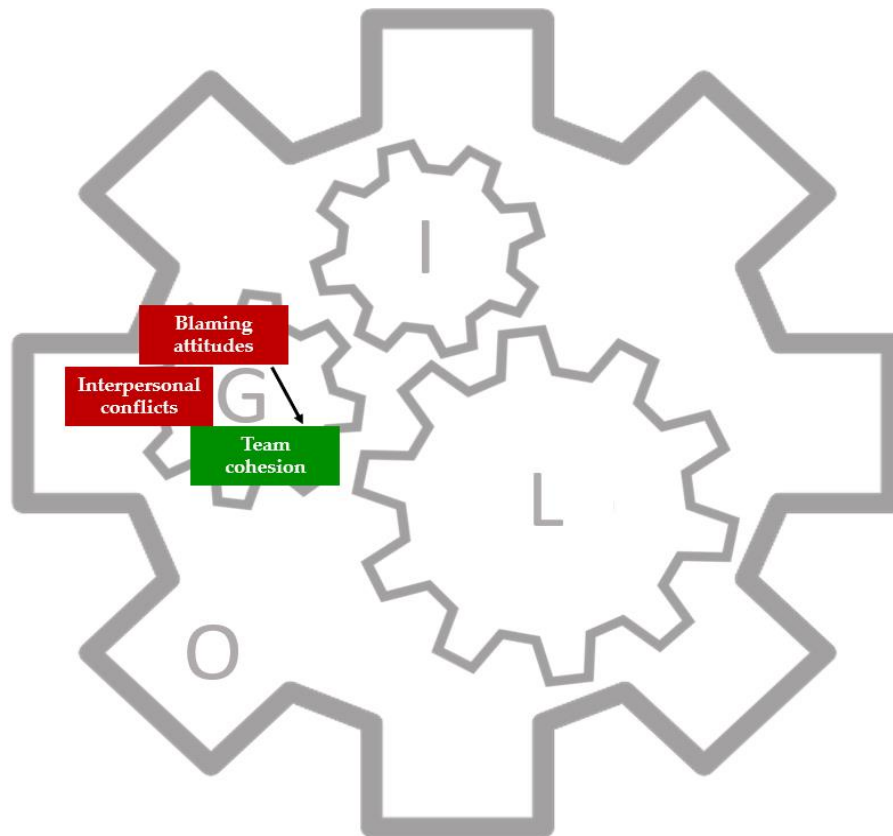


Figure 2. Cognitive map from first focus group at Department A

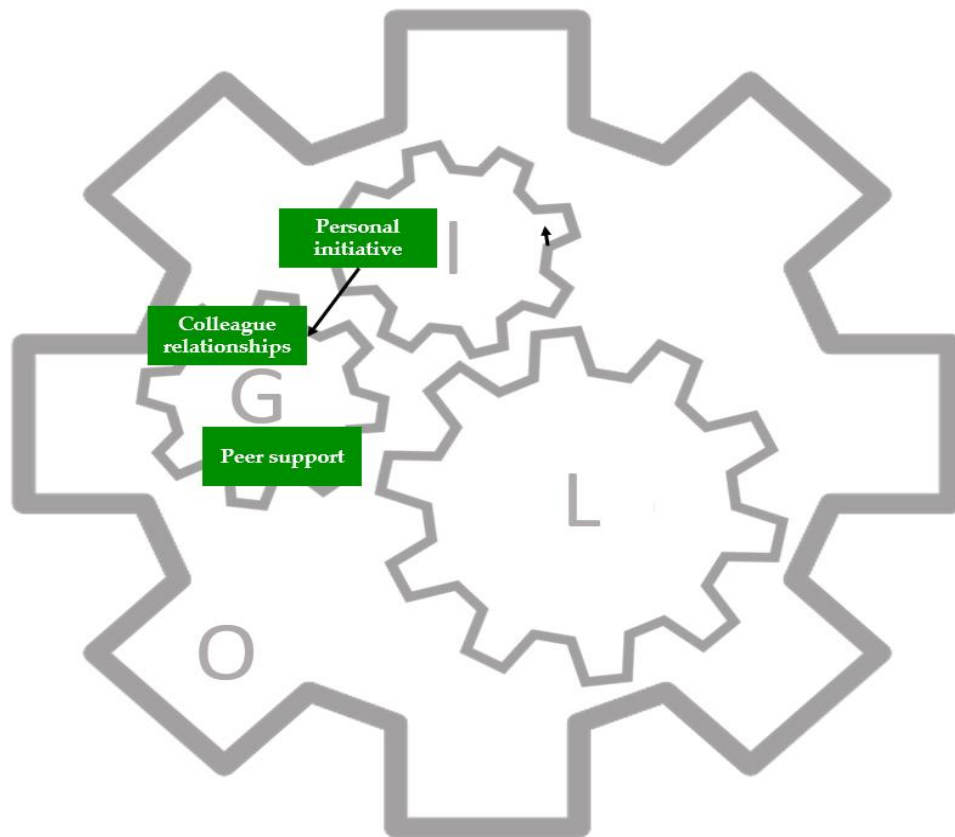


Figure 3. Cognitive map from second focus group at Department A

In Department B, team cohesion, effective cooperation, and positive relationships with colleagues were listed as group-level job resources. Figure 4 shows the cognitive map from the focus group held at Department B. With regard to group-level job demands and resources and their linkages to other IGLO levels, an organisational-level job demand such as high workload was described as threatening team cohesion and positive relationships with colleagues. High workload was in turn traced back to understaffing, which is another organisational-level job demand.

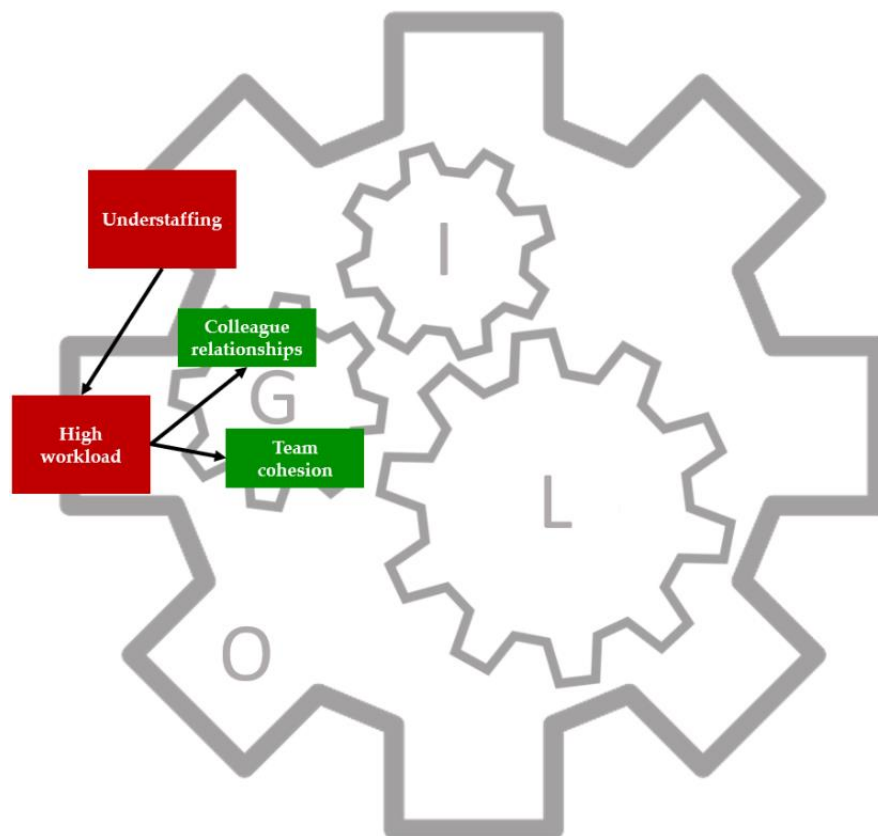


Figure 4. Cognitive map from focus group at Department B

In Department C, at the group level, team cohesion, team support, and positive team climate were listed among job resources. On the other hand, ineffective communication processes were also reported as a job demand. For instance, it was stated,

“There is a lack of communication on how information is received and how it is given”.

Figure 5 shows the cognitive map from the focus group held at Department C.

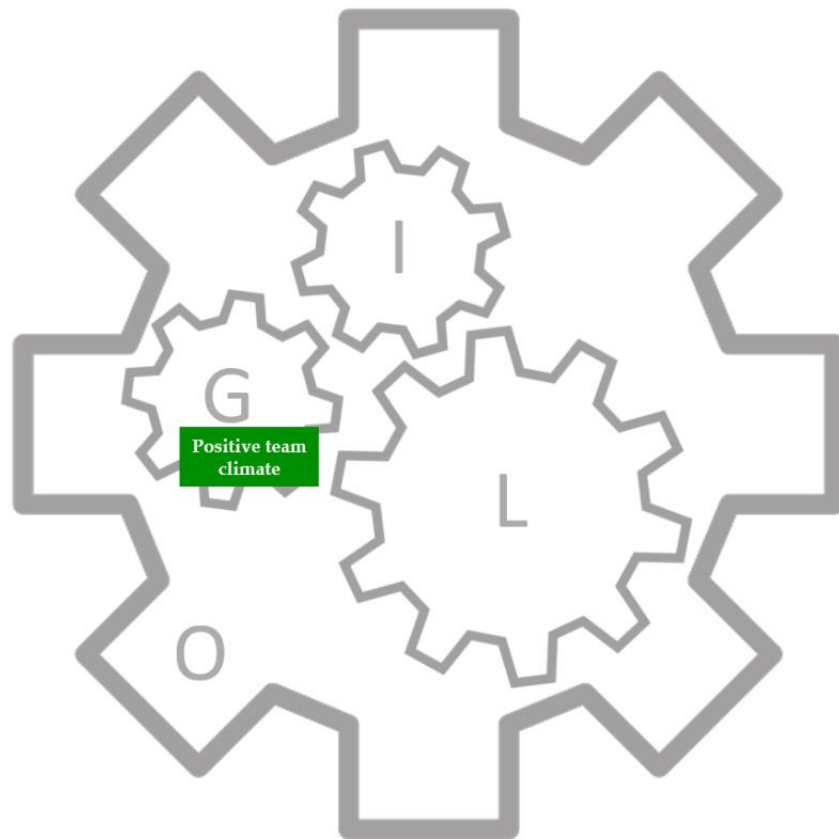


Figure 5. Cognitive map from focus group at Department C

Covid-19-Related Group-Level Job Demands and Resources

The workplace needs assessment exercise took place during the Covid-19 pandemic period. Thus, participants inevitably mentioned additional job demands and resources that were specifically related to the impact that the pandemic outbreak was having on healthcare workers and work at the targeted healthcare organisation. As for group-level job demands, social distancing was reported to hamper both formal and informal team interactions and communications, as it was reported by one middle manager,

“Certainly, one negative aspect of Covid is the distance [...]. We’re a close working group, and we like to work close together, we like to collaborate, we like to exchange opinions, and [...] we like to stay [...] close together to talk. Distance works against us because we can’t “huddle together” anymore, so we can’t even take a break [...]. This is a very negative aspect of Covid [...]. This has had a great impact [...], this crisis of the relational aspect”.

Information exchange had been made difficult by social distancing despite ICTs. Mention was made of communication about organisational changes. Besides functional aspects, social distancing was reported to make it harder to keep good team climate. In this regard, one senior manager indicated that

“Even just trivially, the meeting [...]. We’re talking about how to keep the climate, how to act discussions [...]. Think of the difficulty we have, as departments, as coordination, in reaching the staff. We used to have the departmental meeting, everyone came, you used to talk and, at least, you managed to do it. Now [...] the meetings are all in small groups, but this means that the coordinator must speak six times because, at each shift change, you take the group off in the morning, on in the afternoon, and you have the meeting [...]. Or, trivially, it was sometimes a moment of pause – no? – to say, “come on, it’s the birthday of [...], we’ll stop for a moment in the kitchen”. It became those ten minutes of breath of oxygen that, every now and then, are good for you. We can’t do all this anymore”.

Social distancing was sometimes reported to make teams’ life even more complicated due to lack of proper workspaces to comply with it. Furthermore, at group level, most resources were reported, namely, mutual support, increased cohesion, solidarity, teamwork, and inter-professional cooperation. One nurse referred to group-level resources as follows,

“We all started from the same base, and this created wonderful groups, because there was daily talking and discussion”.

A shared perception was expressed about Covid-19’s positive effect on interpersonal relationships within workgroups. Feeling “all in the same boat” facilitated dealing with the pandemic situation. This emerged, for instance, from the following words of one middle manager,

“a marvellous atmosphere was created that I’d never have thought possible [...] and a kind of solidarity and extremely positive atmosphere was created despite the heavy workload [...] because we felt very close to each other [...]. The [...] infectious disease [...] meant that this new group immediately came together [...]. With Covid, a great deal of solidarity was created, [...] there was a coexistence of a couple of months where everyone appreciated each other, was well integrated”.

A similar report was provided by another middle manager, as follows,

“certainly, working closely with everyone made this period less burdensome [...]. It wasn’t easy, but the key element that led us, however, to make stress a source of wellbeing, was the collaboration [...]. Our working together has made us feel less alone, that’s for sure [...]. The team collaboration [...] made everything flow spontaneously”.

The pandemic situation was reported to have transformed previous inter-professional conflicts into a collaborative, cohesive, and mutually supportive climate. One middle manager reported that

“the way of looking at each other has changed [...] There is a mutual esteem that there was not before; so, one has seen you working there, you have seen him working there, you have seen the shifts – and you look at them in a different way. And that’s nice”.

Another middle manager reported,

“we all worked and there was no longer the professor, the first-level manager, the nurse”.

Nonetheless, employees reported the above group aspects were already forgotten after the first Covid-19 wave, but that improvements at the group level were needed anyway. As the groups already started to feel less cohesive and collaborative, participants stated that the Covid-19 experience should teach everybody much about teamwork.

Discussion

Psychosocial interventions addressing workers should build upon an exhaustive understanding of causes of negative work outcomes and factors of positive psychological states and behaviours among workers. Also, interventions should reduce job demands and promote job resources. As a mean to the design and development of tailored interventions, workplace needs assessment exercises should follow the same logic. Particularly, participatory, bottom-up assessment activities are required to directly ask workers what they feel about their working environment, as this cannot be known a priori. The present chapter drawn upon a workplace needs assessment exercise (Giusino et al., 2022a; 2022b) involving senior managers, middle managers, and employees from three departments of a large healthcare institution in Northern Italy. Healthcare workers’ perceptions of local working conditions could be gathered via semi-structured individual interviews and focus groups. The workplace needs’ assessment exercise allowed to gather job demands and job resources that healthcare workers perceived at the group level, identify similarities and differences in perceptions – i.e., shared versus diverging mental models – of working conditions across the considered hierarchical positions and departments, and inform a tailored action plan to enhance teamwork within the targeted organisation.

The contextualising, bottom-up, participatory approach towards the workplace needs’ assessment exercise allowed to gather the extent to which different key stakeholders in different departments agreed or disagreed about major issues. That is, it was possible to capture similarities and differences in both the nature and the content of reported issues across hierarchical positions and healthcare departments. Ultimately, such an approach allowed to verify whether needs from senior managers versus middle managers versus employees – both within and between departments – , as well as those from Department A versus Department B versus Department C – both within and between hierarchical positions –, aligned or misaligned. In other

words, a picture of shared versus diverging mental models of working conditions within the targeted organisational contexts across the considered hierarchical positions and departments could be taken thanks to collecting a variety of views, perceptions, and needs. So, senior managers tended to agree about teamwork – e.g., team cohesion, peer support – as a group-level job resource. However, this was not the case for senior managers in Department C, who consistently reported frequent interpersonal conflicts as a group-level job demand. Middle managers from all three departments agreed about teamwork – e.g., team cohesion, peer support, positive social climate – as a group-level job resource. Nonetheless, middle managers from Department C reported concerns about excessive team cohesion in face of needs for change and adaptation. Also, middle managers from Department B mentioned effective team communication as a group-level job resource, whereas middle managers from Department C reported lack of training in communication skills as a group-level job demand. Finally, all employees mentioned positive social climate, team support, and team cohesion as crucial group-level job resources – and this finding was consistent with both senior and middle managers' reports. Nevertheless, employees from Department A and Department C mentioned interpersonal conflicts and ineffective team communication as group-level job demands. In addition, employees from all three departments agreed about ineffective leadership as a leader-level job demand, which was also associated with interpersonal conflicts and poor communication. Overall, shared versus diverging mental models of local working conditions could be gathered across both hierarchical positions and departments. All healthcare workers tended to agree about effective teamwork as a crucial group-level job resource. Ultimately, workplace needs' assessment findings might vary within the same healthcare organisation, as well as within the same hierarchical position – depending on healthcare departments – and within the same healthcare department – depending on hierarchical positions. Differences in results from assessment exercises may depend on perceptions of local working conditions. This was the case in the herein described workplace needs' assessment exercise since each department the assessment activities took place in corresponded to a unique work and organisational culture – e.g., workers providing healthcare versus research services, workers in central versus peripheral hospital, and so on. Thus, such differences reflect the organisational complexity of the targeted public healthcare institution, which is to be considered when performing workplace needs' assessment exercises.

Moreover, the needs' assessment exercise allowed to inform a tailored action plan to enhance work outcomes within the targeted organisational contexts. To ensure fit between the organisational context and subsequently implemented interventions (Peters et al., 2020), and thanks to the participatory approach, healthcare workers themselves offered concrete, practical, and applicable suggestions for improving their work environments. These suggestions varied according to shared or diverging mental models about local working conditions. For instance, suggestions for group-level interventions included the improvement of interdepartmental communication and training programmes to support open dialogue between managers and

employees. Employees from Department C also suggested regular meetings to share objectives, work plans, and updates. Psychological support services were suggested to improve healthcare workers' communication skills.

As a final phase of the workplace needs' assessment process, an action plan was developed. Healthcare workers' needs and suggestions were reported to the project's Steering Committee, which was composed of key stakeholders from each targeted hospital department. The aim was to define consistent interventions for each department. Factors to promote job resources – i.e., elements to preserve – and factors to reduce job demands – i.e., elements to improve – were identified. In Department A, elements to preserve included team cohesion and support. On the other hand, elements to improve included intra-team and inter-departmental team building. In Department B, elements to preserve included shared vision and identity, team cohesion, and team support. On the other hand, elements to improve included communication skills – i.e., between healthcare workers and customers, as well as between managers and employees. Finally, in Department C, elements to preserve included intergenerational collaboration, whereas elements to improve included team building.

Despite recruiting healthcare workers from different hierarchical positions and hospital departments may have ensured a sufficient degree of data triangulation (Ramos et al., 2020), also allowing to take the perceptual distance phenomenon into account (Gibson et al., 2001, 2009), one main limitation of the present study concerns a potential drawback of the deployed recruitment and data collection strategy, whereby ingroup bias might have occurred within focus groups, thus jeopardising the quality of collected information. For instance, doctors, nurses, and healthcare assistants in Department B seemed to express consistent opinions depending on the belonging occupational group. Therefore, the role of the facilitator in clarifying expectations related to the needs assessment is crucial; they should encourage participants' willingness to reveal their sincere viewpoints by creating a climate of psychological safety within a comfortable and convivial meeting environment. Regarding another limitation, although adopting the JD-R/IGLO analytical framework should have provided researchers with a shared mental model ensuring consistency of findings, these derive from an interpretive process, which is inherent to qualitative research and might be biased toward adopted theories (Hsieh & Shannon, 2005). Deductive coding might mitigate subjectivity-related inconsistencies across researchers, which more likely occur in inductive coding where no initial framework is adopted. Also, multiple data sources – i.e., managers and employees from three hospital areas – might enhance the credibility of our analysis (Hsieh & Shannon, 2005); in fact, the wide variety within our sample is why we define our study as “multi-source.” Recall and self-report bias (Stone et al., 2002) might have occurred, and generalisability remains questionable.

This study has also a number of strengths. Most of all, in-depth qualitative methods proved effective to capture both risk and protective factors of psychological states and behavioural outcomes in the given work environment. Also, involving

organisational representatives and stakeholders within the action plan phase allowed them to have their say about future initiatives based on the results from the needs assessment, which can maximise the feasibility and monitorability of implemented actions.

As for the research and practical implications, occupational scientists and practitioners are hereby provided with a usable assessment methodology to gather job demands and job resources that workers might experience. The methodology considers the local conditions workers may find themselves working in or their hierarchical positions within their peculiar work environment. In general, the described procedure might inform the design and implementation of workplace needs' assessment activities in workplace settings, but also the development of interventions based on suggestions for improvement from participants.

Conclusion

To summarise, resources and demands were found at the group level of analysis. Resources included mutual support, trust, cohesion, diverse expertise, and inter-professional cooperation. Particularly regarding Covid-19, working for a common good such as patients' health was associated with motivation and lowered inter-professional conflict. Arguably, perceiving a common "enemy" or "threat" and the shared goal of fighting it determined social recategorisations whereby ingroup and outgroup have come to feel as one. This seems consistent with Makowiecki et al. (2020) stating that "war is happiness, in the sense that increased trust, friendship and collaboration in the fight" (pp. 35–36). On the other hand, communications, interactions, and information exchanges were reported as demands needing improvements at team level, despite increased deployment of ICTs. Communication about organisational initiatives to support workers was reported as improvable. Communication between doctors was described as fragmented. Lack of training in communication skills was also mentioned.

Taken together, this evidence supported the meaningfulness of implementing digital-based interventions to foster teamwork within the targeted organisation. The implementation was then carried out in a subsequent study (Giusino et al., 2023), which is introduced in the next chapter of the present dissertation.

Study 3. Digital Team Coaching for Workplace Communication Using Social Network Visualisation: Longitudinal Evaluation of Recipients' Perceptions

Abstract. The purpose of this study was to describe the implementation of a digital-based team coaching intervention aimed at improving team communication in the workplace through social network visualisation. The study examined recipients' perceptions of the intervention at two time points and assessed the temporal stability of various factors, including the intervention's integrity, design, transferability, acceptance and the usability of the adopted visualisation tool. The moderating role of digital usability was also evaluated. Four team coaching sessions were delivered to 62 participants from seven teams across three departments within a large public healthcare organisation in Northern Italy. Perceptions of the intervention dimensions were collected after the second and fourth sessions. Results indicated that, at both time points, recipients appreciated the intervention's integrity and usability more than its design, transferability and acceptance. Furthermore, no significant changes in recipients' perceptions were observed over time. The transferability of the intervention was significantly associated with its acceptance, but only when the usability of the digital tool was high. The study underscores the potential of integrating specific techniques such as Sociomapping and coaching within organisations, encouraging more research and development in these areas. The study also emphasises the critical role of usability and integrity in digital-based team coaching interventions, suggesting that high-quality, user-friendly tools not only lead to initial effectiveness but also sustain positive impacts over time, while also increasing transferability and acceptance.

Background

Teamwork and team communication are critical factors for successful team performance in the workplace and vital outcomes of team-level workplace interventions, as they significantly contribute to team effectiveness (Salas et al., 2018). Effective communication can be characterised by four primary attributes, such as clarity, timing, relevance and frequency (Franc et al., 2019). By focusing on these aspects, team-level workplace interventions, such as team coaching leveraging social network analysis and visualisation, can offer valuable strategies for its optimisation, foster teamwork and communication, ultimately promoting team effectiveness and organisational success (Bahbouh, 2012; Bahbouh & Lasker, 2014; Bahbouh & Willis, 2022).

Healthcare, in particular, exemplifies a domain where teamwork and communication are considered critical for patient safety and team performance (Rosen et al., 2018; Shoukat et al., 2022). Healthcare is a complex, demanding and diverse field requiring interdisciplinary collaboration. Team-based work plays a pivotal role in making informed decisions that draw upon a wide range of expertise (Barnes and Hollenbeck, 2009) and performing tasks requiring multiple individuals' specialised skills (Mathieu et al., 2017). Past research highlights the importance of communication as a critical determinant of team effectiveness in healthcare settings (Ervin et al., 2018; Fowler et al., 2021; Hopkinson et al., 2021; Molleman et al., 2010). Fostering effective communication within healthcare teams is thus crucial not only for team success but also for overall organisational performance.

Digital technologies have expanded the scope of interventions, giving rise to digital-based team coaching interventions in the workplace. Digital workplace interventions can be defined as structured, planned and science-based actions aiming to promote desirable work outcomes by exploiting the potential offered by digital technologies. It may be adaptations of traditional, in-person interventions facilitated through online teleconferencing platforms or designed exclusively for computer or smartphone applications (Baños et al., 2022). Although research has shown that digital interventions can effectively promote desirable workplace outcomes (Phillips et al., 2019), and despite promising preliminary evidence regarding the effectiveness of digital-based team coaching interventions in various industries (Bernardová, 2012; Franc et al., 2019; Tetour, 2019), much of the existing literature has primarily focused on individual-level implementation and evaluation of digital workplace interventions. This is especially true regarding digital-based team coaching interventions based on social networks and sociometric analysis (Bahbouh, 2012; Bahbouh & Lasker, 2014; Bahbouh & Willis, 2022). This leaves a gap in our understanding of the potential benefits and challenges associated with team-level digital-based team coaching interventions, presenting an opportunity for further investigation and development in this emerging field.

Recently, the main frameworks for the evaluation of both the process and effects of workplace interventions (Nielsen & Abildgaard, 2013; Nielsen & Randall, 2013) have underlined the importance of taking into consideration the perceptions of the workplace actors involved in interventions, including the recipients of the intervention itself. These models argue that recipients' perceptions should be integral to workplace interventions' evaluation as they are vital mechanisms for their effectiveness. Recipients' perceptions constitute underlying psychological aspects that may explain workers' behavioural reactions to the intervention activities and, as such, may facilitate or hinder the effectiveness of the intervention itself, thus contributing to its success or failure. Particularly, Nielsen and Randall (2013, p. 608) stated that "an important part of [...] evaluation should be the measurement of change in employees' knowledge of the intervention, their expectations that the intervention can bring about changes".

In light of the above, this study (Giusino et al., 2023³) represented the first attempt to monitor aspects of the process, in particular, the perceptions of the recipients over time, of a digital-based team coaching intervention designed to enhance team communication through social network visualisation and team coaching techniques in the healthcare sector. The aim was to test whether this monitoring can provide an explanatory framework for the observed results and act as a catalyst for future outcome evaluation studies. This intervention was implemented in a large public healthcare organisation in Northern Italy as part of a broader project focused on evaluating organisational interventions (De Angelis et al., 2020). The study had three primary objectives. The first objective was to assess recipients' perceptions of dimensions that are relevant to digital team interventions according to previous literature (Broetje et al., 2022; Holton et al., 2000; Martin et al., 2020; Vuori et al., 2012; Yelon et al., 2004; Zhou et al., 2019), specifically the usability, the transferability, the integrity, the training design and the acceptance of the intervention. The second objective was to evaluate the temporal stability of recipients' perceptions, explicitly examining whether and how these perceptions evolved during the intervention implementation. The third objective was to investigate the role of the usability of the digital tool as a moderator in the relationship between perceptions of the intervention and its overall acceptance.

From Traditional Face-to-Face to Digital-Based Team-Level Interventions

Traditional face-to-face team interventions in healthcare have generally yielded positive results over recent decades, particularly those aimed at enhancing team communication. Numerous studies have demonstrated the efficacy of such interventions in improving various aspects of healthcare delivery (Hung et al., 2020; Kilpatrick et al., 2020; McCulloch et al., 2011; Prewett et al., 2013; Sacks et al., 2015). A systematic review by McCulloch and colleagues (2011) found some evidence of benefit on healthcare staff attitudes, teamwork quality, technical performance, efficiency, and medical error rates from studies with intensive team communication training programmes. Prewett and colleagues (2013) concluded that a low-cost, one-day teamwork training delivered to medical residents and encompassing guided discussions for feedback, positively affected behavioural choices for teamwork in the trauma room. A meta-analysis by Sacks and colleagues (2015) concluded that communication-based interventions prove promising to improve patient outcomes, healthcare efficiency, and surgical culture. Hung and colleagues (2020) found that multidisciplinary team discussions resulted in survival benefit for patients with stage III non-small-cell lung cancer. Finally, in a systematic review of brief team training interventions conducted in acute care in-patient settings, Kilpatrick and colleagues

³ Preliminary findings from this published study have also been presented at the XXX National Congress of the Italian Association of Psychology, held in Padua, Italy, in September 2022.
<https://aipass.org/xxx-congresso-aip-plenario-padova-27-30-settembre-2022/>

(2020) found that three to four 30- to 60-minute training sessions spread out over several weeks with structured facilitation and debriefing appeared to improve non-technical skills, for instance, communication. Taken together, these studies suggest that even brief, well-structured interventions can have a significant impact on team communication and overall performance in healthcare settings. In other words, traditional in-person team communication interventions have demonstrated considerable success in fostering team communication within the healthcare sector. These interventions, ranging from intensive team programs to brief targeted sessions, have been linked to improvements in staff attitudes, teamwork quality, technical performance, healthcare efficiency and patient outcomes. As the healthcare landscape continues to evolve, it is essential to build upon these findings and explore innovative ways to enhance team communication and performance further (Larson and DeChurch, 2020).

Digital-based interventions have become powerful tools for enhancing team communication across various industries. By leveraging cutting-edge technology, these interventions facilitate more effective, efficient and adaptable communication strategies, which are critical for team success. Relative to traditional face-to-face interventions, digital workplace interventions generally come with peculiar challenges, including, for example, users' engagement and adherence, ethics, privacy and data protection (Baños et al., 2022; Phillips et al., 2019). On the contrary, digital interventions for teamwork can offer unique advantages over traditional face-to-face interventions, such as increased geographical and temporal accessibility, cost-effectiveness, personalisation and attractiveness (ibidem). Thus, digital interventions constitute a promising avenue for the future of workplace interventions. In this context, several studies have explored the impact of digital-based interventions on team communication in different sectors, such as military aviation (Bernardová, 2012), private enterprises (Franc et al., 2019), the hospitality industry (Tetour, 2019) and undergraduate business education (Willox et al., 2023). Taken together, these studies suggest that digital-based team communication interventions can enhance information sharing and situational awareness, improve clarity, timing, relevance and frequency of communication among team members, promote greater collaboration, problem-solving and overall productivity, and streamline communication processes, reduce misunderstandings and enhance overall team performance.

The increasing adoption of e-health practices (Eikey et al., 2015), along with the surge in remote teamwork following the Covid-19 pandemic (Newman & Ford, 2021), has led to a rise in online communication within the healthcare sector. This shift underscores the need to explore the potential of technology in facilitating healthcare team interventions, explicitly focussing on digital-based approaches to improve team communication. There is a notable gap in the current research landscape regarding digital-based interventions targeting team communication in healthcare settings.

Team Coaching Interventions Based on Social Network Analysis and Visualisation

Innovative team coaching techniques and tools are revolutionising the way organisations foster collaboration, communication and performance among their teams. In *The Team Coaching Casebook*, Clutterbuck and colleagues (2022) provided a comprehensive analysis of the power and impact of team coaching in organisational settings, emphasising the importance of a customised approach based on individual team dynamics and challenges. Techniques assessing team strengths and weaknesses provide coaches with invaluable insights into areas for development. Team coaching tools facilitate creativity and problem-solving by engaging team members in hands-on, collaborative activities or navigating interpersonal conflicts and promoting understanding among team members. Furthermore, digital technologies are being integrated into team coaching practices, enabling interactive experiences that foster team bonding and enhance learning. These innovative approaches not only drive team performance but also contribute to creating a culture of continuous learning, adaptability and organisational change capability (Supriharyanti & Sukoco, 2023).

In this framework, team communication interventions grounded in social network analytical theory and methods (Wasserman and Faust, 1994) leverage digital tools for collecting, processing and visualising complex relational data. In Bahbouh's (2012) seminal handbook, the theory and technique of an innovative digital-based approach to improving team communication by leveraging sociometric analysis, social network visualisation and team coaching were presented. Sociometric analysis and social network visualisation help identify group communication patterns, isolated members and subgroups, and pinpoint the influencers. For example, when teams and their managers understand who from the team feels isolated or there are smaller cliques within the group, they can take actions to work on it. However, these approaches focus on the as-is situation. Using valid and reliable team communication measures, such as clarity, timing, relevance and frequency of communication with each team member, these digital tools use algorithms to create graphical representations, known as "sociomaps", depicting the current and desired communication structures within the team (Rozehnalová, 2013). The structure of each intervention session comprises five sequential steps, such as (1) data collection, (2) team sociomap presentation, (3) team coaching, (4) creation of action plans and commitment, and (5) review of action plans and commitments. The intervention is based on team coaching workshops, defined as direct interactions with a team, intended to help members make coordinated and task-appropriate use of their collective resources in accomplishing the team's work (Clutterbuck et al., 2022). In these team interventions, the sessions and team coaching activities are guided by the digital visualisation of sociomaps to stimulate team reflexivity and self-awareness. The insights from the visualisation then help formulate individual and team action plans that lead to the desired and effective communication state within the team.

Therefore, the sociomaps help not only understand the current dynamics in the group but also provide visualisation of the desired situation and enable monitoring of the group dynamics over time. The visual clarity of sociomaps visualisation – 3D color-coded map of group dynamics – makes the trends and patterns visually apparent and more actionable than traditional sociograms. Still, sociomapping, sociometric analysis or social network visualisation can bring benefits to the group mainly when used with interpretation and potentially with team coaching sessions – face-to-face or online. This approach aims to foster better communication, collaboration and understanding among team members in various organisational settings. By combining digital software-based social network analysis and team coaching strategies, Bahbouh (2012) addressed communication gaps and inefficiencies within teams, ultimately enhancing overall team performance and satisfaction. This intervention not only highlights the importance of understanding the underlying structure of team communication networks but also emphasises the role of tailored coaching in addressing individual and collective needs.

Later, Zakharchyn and Kosmyrna (2015) confirmed the benefits of using sociometric analysis techniques for organisations aiming to optimise employee behaviour and overall team performance. Despite such preliminary findings, the present contribution aims to advance the evidence related to this type of approach to improving team communication via social network analysis and team coaching by monitoring aspects related to the implementation of the intervention, that is, the recipients' perceptions of the intervention itself and of the actors involved – e.g., the facilitator or the coach. Back in 2009, Baron and Morin (2009) underscored the importance of coaching relationships in leadership coaching, emphasising that the quality of these relationships is a critical factor in achieving desired outcomes. Applying the significance of coaching relationships to team coaching, similar principles can be observed. In the context of team coaching, the quality of relationships between the coach and the team members, as well as among team members themselves, plays a crucial role in the success of the intervention. By establishing an environment where team members feel comfortable sharing their thoughts, concerns and feedback, the coach can facilitate meaningful discussions and encourage collective problem-solving. In team coaching, the role of the coach is to nurture and support the development of both individual and collective competencies, aiming to improve overall team performance.

Perceptions of the Dimensions of the Team Coaching Intervention

According to the realist evaluation approach and recent models and empirical studies about evaluating workplace interventions (Nielsen & Abildgaard, 2013; Nielsen & Randall, 2013; von Thiele Schwarz et al., 2021), it is important to comprehend the way individuals perceive the various dimensions of interventions. These perceptions, encompassing attitudes toward the intervention's content, structure, facilitators,

design, relevance, and more, are pivotal in evaluating their effectiveness. Each employee interprets their work environment uniquely, forming a shared understanding rooted in common experiences and conditions, which extends to workplace interventions as well. When employees engage in interventions as a team, their collective perceptions are influenced by interactions with one another, considerably influencing their perspectives on the intervention's positive or negative aspects. Therefore, it is recommended to both researchers and practitioners to incorporate recipients' perceptions as a key element in the evaluation of workplace interventions. These perceptions act as underlying mechanisms that play a pivotal role in determining the intervention's effectiveness. Consequently, a crucial component of evaluating workplace interventions should involve measuring changes in employees' perceptions of the intervention and their expectations of its ability to bring about positive changes. This approach underscores the importance of considering individual and collective viewpoints in evaluating the overall impact of workplace interventions.

Numerous studies (Broetje et al., 2022; Holton et al., 2000; Martin et al., 2020; Vuori et al., 2012; Yelon et al., 2004; Zhou et al., 2019) have underscored the importance of various factors associated with the implementation of interventions, including their transferability, integrity, design, acceptance and usability. These elements are pivotal in comprehending the underlying reasons for an intervention's effectiveness and its potential for broader application. Specifically, a thorough understanding of these aspects allows for a more nuanced analysis than merely assessing the effectiveness of an intervention pre- and post-implementation. It enables an examination of the components that may have been instrumental during the intervention, leading to a shift in perception among the participants.

Transferability can be defined as the extent to which intervention recipients think that the knowledge and skills that they learn during interventions are transferrable to the real-world workplace setting (Yelon et al., 2004). Yelon and colleagues (2004) emphasised the importance of transferability in the context of interventions, particularly in relation to their effectiveness and long-term impact. A successful intervention should not only facilitate learning and improvement within the context of the program but also enable participants to transfer these gains to their daily work. The integrity of an intervention refers to the extent to which the facilitator's behaviour was positive, rewarding and relevant to the recipients' participation (Vuori et al., 2012). When interventions are implemented with integrity, the intended benefits are more likely to be realised as participants receive the full range of intended support, guidance and resources. Design can be defined as the degree to which intervention recipients perceive that the intervention has been designed and delivered to give them the ability to transfer learning to the job and that intervention instructions match job requirements. Holton and colleagues (2000) stressed the importance of well-designed training interventions for ensuring effectiveness and impact.

When considering digital-based team intervention tools, it is essential to also recognise the role of acceptance of digital tools in determining the success of these interventions. It can be defined as the recipients' experience that the intervention met their expectations and needs. Specifically referring to an internet-based team development tool deployed among nurses, Broetje et al. (2022) argued that understanding the acceptance and other recipients' attitudes toward digital team interventions is critical to their successful implementation, uptake, adoption and use. Thus, these authors highlight the importance of investigating recipients' perceptions and acceptance, specifically when evaluating digital-based team-level interventions.

Finally, potential factors worth examining include the usability of the digital tool. Usability can be understood in terms of the quality of the recipients' experience with the technological platform where the digital interventions take place, entailing dimensions such as aesthetics, feedback, interactivity, functionality and other design elements (Zhou et al., 2019). Zhou and colleagues (2019) conducted a study to evaluate the usability of a mobile health social network analysis tool designed to enhance communication and collaboration among healthcare professionals. The digital tool aimed to facilitate information sharing and teamwork in healthcare settings, ultimately improving patient outcomes. The authors found that the tool was well-received by healthcare professionals and was efficient in helping users identify communication patterns and collaborate more effectively with their colleagues. The usability of the platform used in a team-based intervention is crucial for ensuring its recipients' better acceptance of the intervention. When a platform is highly usable, users are more likely to engage with the intervention, increasing adoption and compliance (Cruz Zapata et al., 2015; Kumar & Mohite, 2018). First, a user-friendly platform minimises frustration and barriers to use, allowing participants to focus on the intervention's content and objectives. Second, efficient and effective platforms enable users to achieve their goals within the intervention more easily. Third, when participants enjoy using the platform and find it beneficial, they are more likely to share their positive experiences with their colleagues, creating a ripple effect that can further enhance the acceptance of the intervention within the organisation. However, little is known about how the usability of digital platforms interacts with other dimensions of digital-based team communication interventions and its potential role as a moderator.

Methods

This study was part of the European project H-WORK (De Angelis et al., 2020), funded by the EU-H2020 research and innovation framework. The project aimed to design, implement and validate multilevel workplace interventions. The study received ethical approval and adhered to standard requirements. Data collection occurred between March 2021 and January 2022. To maintain anonymity, participants created

a personal code for each completed survey. The researchers' contact information was shared with participants to address any questions or concerns.

Implementation and Recipients of the Intervention

The intervention implementation followed a workplace needs assessment and stepwise framework for team interventions (Lacerenza et al., 2018), ensuring organization-intervention fit (Andersen et al., 2021; Peters et al., 2020). Team-level communication, interactions and information exchanges were identified as areas needing improvement – see Study 2 in the present dissertation as well as Giusino and colleagues (2022a, 2022b). A steering committee was established, as recommended by Nielsen et al. (2013), which included the health and safety manager, workers safety representative, directors of involved hospital departments and nursing manager.

Recipients were recruited through voluntary subscriptions to the team coaching intervention course, with information provided by their health and safety manager. Inclusion criteria required participants in each intervention edition to be members of the same team. Teams were usually co-located teams that only attended the digital team intervention. As healthcare workers, they do not usually work in a remote mode. The intervention was delivered in remote format, on a videoconferencing platform, because of Covid-19-related social distancing public health and safety measures during the implementation period.

The intervention consisted of four team sessions, with two-month intervals between sessions. The first session was 3 hours long, while the remaining sessions were 2 hours each. Sessions were led by two trained professionals, one as the main facilitator and the other as an assistant. The first session introduced participants to the intervention framework and key concepts, such as team communication and effectiveness. Using visualised sociomaps, a team discussion on current and desired communication helped formulate improvement strategies. The second session discussed team communication and developing action plans for desired changes. The third session provided feedback and evaluated progress while also enhancing meta-communication skills. Participants shared thoughts on behaviours or work situations to improve communication effectiveness within the team and individual feedback on colleagues' communication styles. The fourth and final session was a debriefing based on team coaching principles. Team communication measures were collected at each session to generate updated sociomaps, allowing for comparisons and enriching discussions.

Figure 6 displays an example of sociomaps used in the implementation of the intervention for the same team. The sociomap visually represents the interconnectedness of team members based on their communication about work-related topics. The positions of individuals on the map indicate their existing or desired communication patterns. For instance, communication frequency is represented by the proximity of team members on the map; the closer they are, the

more frequent their interactions are or are intended to be. Each individual's height and colour on the sociomap signify their average communication intensity within the team. A higher elevation, marked by red, does not inherently indicate a positive characteristic, just as a lower height, denoted by blue, does not necessarily suggest a negative quality. However, individuals in red may experience communication overload, while those in blue might be insufficiently engaged with the team. Ultimately, the interpretation relies on each person's role within the team and their perception of their position on the sociomap.

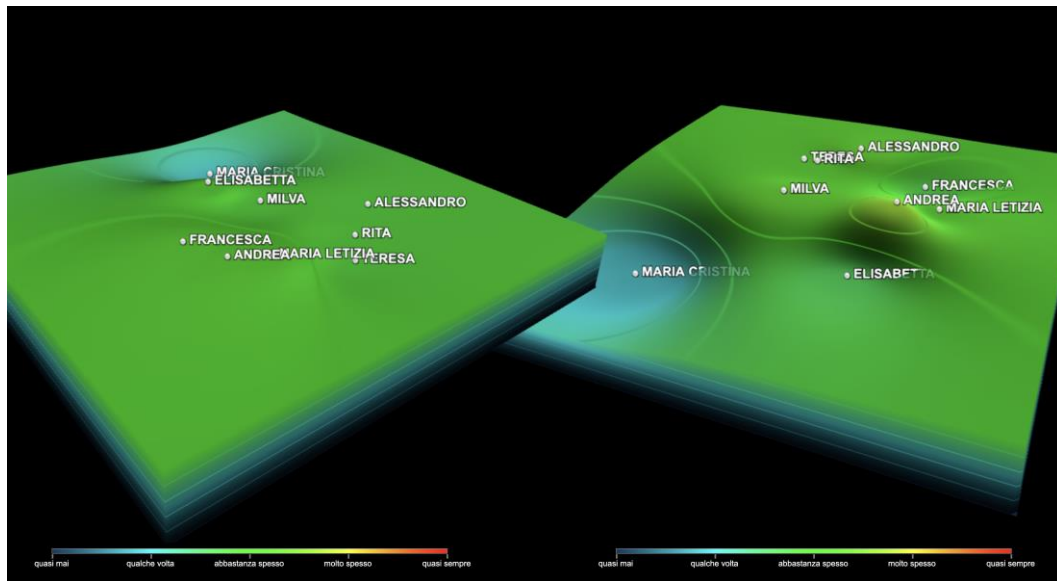


Figure 6. Example of a “sociomap” of the same team in different sessions

The intervention was delivered to seven teams across three departments (Department of Emergency, Department of Medicine and Department of Neuroscience) within a large public healthcare organisation in Northern Italy. Three teams belonged to the Department of Emergency (Team 2, Team 4 and Team 5; 26 individual participants, 41.9%), two teams were from the Department of Medicine (Team 1 and Team 3; 16 individual participants, 25.8%) and two teams were from the Department of Neuroscience (Team 6 and Team 7; 20 individual participants, 32.3%). Each team comprised 6 to 12 members. Team 1 had 7 members (11.3%), Team 2 had 10 members (16.1%), Team 3 had 9 members (14.5%), Team 4 had 6 members (9.7%), Team 5 had 10 members (16.1%), Team 6 had 8 members (12.9%), and Team 7 had 12 members (19.4%).

In total, 62 participants attended at least one of the four team coaching sessions. Specifically, 4 participants (6.5%) completed one session, 12 participants (19.4%) completed two sessions, 16 participants (25.8%) completed three sessions and 30 participants (48.4%) completed all four scheduled sessions. Although the entire team agreed to participate in the intervention, not all members completed all sessions.

Archival data from organisational records regarding participants' sociodemographic and job-related information were provided by the contact persons

from the targeted organisation during the design phase. Twenty-seven participants were nurses (43.5%), 13 were physiotherapists (21%), 8 were healthcare assistants (12.9%), 5 were doctors (8.1%), 4 were speech therapists (6.5%), 3 were healthcare technicians (4.8%), 1 was an educator (1.6%) and 1 was an ambulance driver (1.6%). Three participants (4.8%) were identified as head nurses and 1 participant (1.6%) as a head doctor, while 58 participants (93.5%) held no leadership roles. Organisational tenure ranged from 0 to 35 years ($M = 15.15$, $SD = 10.49$), and ages ranged from 29 to 65 years ($M = 46.9$, $SD = 9.44$). Forty-six participants were female (74.2%) and 16 were male (25.8%).

Measures

Measures were collected one week after the second session (T1) and one week after the fourth and last session (T2). Items referenced the team using “we” (Chan, 1998) and were administered in Italian. Five dimensions of the intervention were measured as follows.

Usability. Six items adapted from Zhou and colleagues (2019) were used to assess the usability of the digital tool. Example items include, “The digital tool was easy to use” and “Overall, I am satisfied with this digital tool”. Response options ranged from “1 = strongly disagree” to “5 = strongly agree.” Cronbach’s α was .90 at T1 and .76 at T2.

Transferability. Three items from Yelon and colleagues (2004) measured the perception of the transferability of the intervention. An example item is, “The skills we developed during the team intervention will help us in our work”. Response options ranged from “1 = strongly disagree” to “5 = strongly agree”. Cronbach’s α was .84 at T1 and .90 at T2.

Integrity. Six items adapted from Vuori and colleagues (2012) assessed the perceptions of the intervention’s integrity. Example items include, “Did the facilitators make you feel like your participation was valued?” and “Did you find group discussions useful?”. Response options ranged from “1 = not at all” to “5 = all the time”. Cronbach’s α was .83 at T1 and .82 at T2.

Design. Four items from Holton and colleagues (2000) measured the perception of the intervention’s design. An example item is, “The activities and exercises helped us apply learning on the job”. Response options ranged from “1 = strongly disagree” to “5 = strongly agree”. Cronbach’s α was .76 at T1 and .90 at T2.

Acceptance. Three items from Martin and colleagues (2020) assessed the acceptance of the intervention. An example item is, “I would recommend the team intervention to others in a similar situation”. Response options ranged from “1 = to a very low extent” to “5 = to a very high extent”. Cronbach’s α was .88 at T1 and .78 at T2.

Sociodemographic and job-related information was collected through the same

questionnaire. At the end, respondents were asked to create a unique ID code to maintain anonymity while allowing tracking of individuals' answers across different data collection time points.

Data Analysis

The Statistical Package for the Social Sciences (SPSS) statistics software version 25 was used to perform statistical analysis. Frequencies were run to gather information about the sample. Descriptives were run to calculate skewness and kurtosis to test the assumption of normality of difference scores between the observations of continuous variables aimed to be compared. Following George and Mallery (2010), if values of skewness or kurtosis were between -2 and +2, the distribution was assumed to be normal. The parametric paired-samples *t*-test, assuming normal data distribution, was conducted to compute mean scores and investigate statistically significant within-subjects within-time differences between variables whose difference scores were normally distributed. The nonparametric Wilcoxon signed-rank test, which does not assume normal data distribution, was conducted to compute mean scores and investigate significant within-subjects within-time differences between variables whose difference scores were not normally distributed. To calculate the effect size of statistically significant differences in paired-samples *t*-test, Cohen's *d* was computed by dividing the mean difference by the standard deviation of the difference (Cohen, 1998). A repeated-measures *t*-test was performed to investigate significant within-subjects differences across time in variables with normally distributed difference scores, whereas the Wilcoxon test was conducted to explore within-subjects differences across time in variables with not normally distributed difference scores. The paired-samples *t*-test and Wilcoxon test were preferred to repeated-measures analysis of variance and Friedman test, respectively, because they are considered to have less error risks when two observations are compared instead of more. The use of the Wilcoxon test was not generalised to both normally and not normally distributed variables as nonparametric tests ensure less statistical power when applied to normal data, whereas the precise identification of the actual existence of meaningful statistically significant differences was one main goal of this study. However, during researchers' exploration of the data set, results did not change substantially when applying paired-samples/repeated-measures *t*-test to not normal data nor when applying the Wilcoxon test to normally distributed data. The average measure intraclass correlation coefficient (*ICC*; Shrout & Fleiss, 1979) was calculated as an index of inter-rater reliability to assess the level of agreement among team members in their subjective evaluations at both data collection time points. Due to one team having only one participant providing valid data, the intraclass correlation coefficient could be computed for a sample of six teams. Finally, correlation analysis was performed before moderation analysis, which was conducted using the PROCESS macro for SPSS.

Results

The current section reports the main findings of the present study, regarding the recipients' perceptions of the team coaching intervention dimensions, the temporal stability of recipients' perceptions, and the usability of the digital tool as a moderator.

Recipients' Perceptions of the Team Coaching Intervention Dimensions

At T1, 33 recipients (72.7% females, 36.4% nurses, $M_{\text{age}} = 46.3$, $M_{\text{tenure}} = 15.4$) completed the questionnaire, yielding a 53% response rate. At T2, 29 recipients (72.4% females, 37.9% nurses, $M_{\text{age}} = 49.5$, $M_{\text{tenure}} = 16.4$) completed the questionnaire, yielding a 46% response rate. At T1, inter-rater reliability was statistically significant at $p < .05$ for four teams out of six, such as Team 1 ($ICC = .99$, $n = 4$), Team 5 ($ICC = .93$, $n = 6$), Team 6 ($ICC = .87$, $n = 3$) and Team 7 ($ICC = .98$, $n = 8$), thus indicating a high level of consistency of evaluations within the teams after the second intervention session. Similarly, at T2, inter-rater reliability was statistically significant at $p < .05$ for five teams out of six, such as Team 1 ($ICC = .97$, $n = 4$), Team 3 ($ICC = .85$, $n = 5$), Team 5 ($ICC = .81$, $n = 5$), Team 6 ($ICC = .98$, $n = 5$) and Team 7 ($ICC = .98$, $n = 6$), thus indicating a high level of consistency of evaluations across the teams after the fourth and last intervention session.

Table 5 displays the results from paired-sample t -tests at T1. Significant differences were observed between integrity ($M = 3.97$, $SD = .56$) and design ($M = 3.70$, $SD = .53$); $t(32) = 2.65$, $p = .012$. In addition, a significant difference was found between integrity and transferability ($M = 3.65$, $SD = .55$); $t(32) = 3.12$, $p = .004$. Finally, a significant difference was noted between integrity and acceptance ($M = 3.61$, $SD = .69$); $t(32) = 3.43$, $p = .002$. The Wilcoxon signed-rank test for usability and acceptance, where the difference score was not normally distributed, revealed no statistically significant difference between their mean scores ($Z = -1.782$, $p = .075$). Effect size of statistically significant differences was between small and medium. These findings suggest that, at T1, recipients appreciated the intervention's integrity significantly more than its design, transferability and acceptance.

Table 5. Paired-samples t -test of recipients' perceptions of intervention at T1

	95% CI				t	df	p	d
	M	SD	Lower	Upper				
Integrity – Usability	.192	.70	-.056	.440	1.578	32	.124	.27
Integrity – Design	.273	.59	.063	.482	2.648	32	.012	.46
Integrity – Transferability	.323	.59	.112	.534	3.121	32	.004	.54
Integrity – Acceptance	.359	.60	.145	.572	3.427	32	.002	.59
Usability – Design	.081	.80	-.204	.366	.578	32	.567	.10
Usability – Transferability	.131	.81	-.155	.417	.935	32	.357	.16
Design – Transferability	.051	.45	-.108	.209	.650	32	.520	.11

Design – Acceptance	.086	.66	-.148	.320	.747	32	.461	.13
Transferability – Acceptance	.035	.55	-.158	.229	.372	32	.712	.06

Note. *M* = mean. *SD* = standard deviation. *CI* = confidence interval. *df* = degrees of freedom. *d* = effect size

Table 6 presents the results from paired-samples *t*-tests at T2. Significant differences were found between usability (*M* = 4.03, *SD* = .58) and design (*M* = 3.70, *SD* = .70); $t(28) = 2.16, p = .039$. In addition, a significant difference was observed between usability and acceptance (*M* = 3.51, *SD* = .65); $t(28) = 3.85, p = .001$. Other significant differences were identified between integrity and design; $t(28) = 2.63, p = .014$, and between integrity and acceptance; $t(28) = 3.92, p = .001$. Finally, significant differences were found between acceptance and both transferability [$t(28) = 3.15, p = .004$] and design [$t(28) = 2.16, p = .039$]. The Wilcoxon signed-rank test for integrity and transferability, where the difference score was not normally distributed, indicated a statistically significant difference between their mean scores ($Z = -2.166, p = .03$). Effect size of statistically significant differences was between medium and large. These findings suggest that, at T2, recipients appreciated the intervention's usability and integrity significantly more than its design, acceptance and transferability.

Table 6. Paired-samples *t*-test of recipients' perceptions of intervention at T2

	95% CI				<i>t</i>	<i>df</i>	<i>p</i>	<i>d</i>
	<i>M</i>	<i>SD</i>	Lower	Upper				
Usability – Integrity	.037	.65	-.212	.285	.303	28	.764	.05
Usability – Transferability	.251	.69	-.013	.514	1.949	28	.061	.36
Usability – Design	.328	.82	.017	.639	2.161	28	.039	.40
Usability – Acceptance	.526	.74	.246	.806	3.852	28	.001	.71
Integrity – Design	.291	.60	.065	.518	2.631	28	.014	.48
Integrity – Acceptance	.490	.67	.234	.745	3.921	28	.001	.73
Transferability – Design	.078	.44	-.088	.244	.957	28	.347	.17
Transferability – Acceptance	.276	.47	.096	.455	3.147	28	.004	.58
Design – Acceptance	.198	.49	.011	.386	2.165	28	.039	.40

Note. *M* = mean. *SD* = standard deviation. *CI* = confidence interval. *df* = degrees of freedom. *d* = effect size

Temporal Stability of Recipients' Perceptions

Nineteen recipients (78.9% females, 47.4% nurses, $M_{\text{age}} = 49$, $M_{\text{tenure}} = 17.9$) completed the questionnaire at both T1 and T2. The repeated-measures *t*-test revealed no statistically significant effects of time on examined perceptions with normally distributed difference scores, such as design [$t(18) = .84, p = .407$] and integrity [$t(18) = .63, p = .535$]. Similarly, the Wilcoxon signed-rank test indicated no statistically significant effects of time on transferability ($Z = .709, p = .478$), acceptance ($Z = -1.803$,

$p = .071$) and usability ($Z = -.400$, $p = .689$). These findings suggest that recipients' perceptions of the team coaching intervention dimensions remained consistent over time throughout the implementation process.

Usability of the Digital Tool as a Moderator

A moderation analysis was conducted to investigate whether the usability of the digital tool moderated the relationship between the perceptions of the team coaching intervention dimensions – i.e., integrity, design and transferability – and the overall acceptance of the digital intervention at the end of the intervention, meaning after the fourth and last session.

Table 7 presents the mean scores and the intercorrelations of recipients' perceptions at T1 and T2, both at individual and team levels of analysis with aggregated data. Individual-level correlation analysis showed positive and statistically significant – either $p < .05$ or $p < .01$ – associations among almost all the recipients' perceptions of the intervention dimensions both at T1 and T2, with Pearson's r ranging from .41 to .78. Only the perception of the usability of the digital tool did not appear to be associated with the other perceptions of the intervention dimensions. Some main differences can be observed in the team-level correlation analysis, where integrity at T1 did not appear to be associated with other perceptions of the intervention dimensions, and usability at T2 appeared to be negatively associated with transferability ($r = -0.75$, $p < .05$) and acceptance ($r = -0.85$, $p < .05$).

Table 7. Correlations among perceptions of intervention dimensions at T1 and T2

<i>Individual level</i>									
	M_{T1}	SD_{T1}	M_{T2}	SD_{T2}	1	2	3	4	5
1.Usability	3.78	.61	4.03	.58	-	.27	.17	.07	.18
2.Integrity	3.97	.56	4.00	.59	.28	-	.64*	.58*	.42**
3.Transferability	3.65	.55	3.78	.58	.02	.42*	-	.78**	.71**
4.Design	3.70	.53	3.70	.70	.01	.41*	.65*	-	.73*
5.Acceptance	3.61	.69	3.51	.65	.16	.55**	.63**	.44*	-
<i>Team level</i>									
	M_{T1}	SD_{T1}	M_{T2}	SD_{T2}	1	2	3	4	5
1.Usability	3.79	.27	4.00	.18	-	-.52	-.75*	-.70	-.85*
2.Integrity	3.98	.38	3.99	.32	.07	-	.87*	.84*	.64
3.Transferability	3.70	.28	3.82	.34	-.02	.70	-	.88*	.86*
4.Design	3.78	.26	3.76	.50	.10	.41	.85*	-	.80*
5.Acceptance	3.66	.34	3.60	.38	-.09	.71	.84*	.64	-

Note. Correlations at T1 are reported in the lower semi diagonal, whereas correlation at T2 are reported in the higher semi diagonal. M = mean. SD = standard deviation. * = $p < .5$. ** = $p < .01$

Then, moderation results revealed that only the relationship between transferability and acceptance was moderated by usability ($n = 19$). Particularly, there was a significant interaction effect of transferability and usability ($B = .47, p < .05$) on acceptance. Figure 7 shows that the transferability of the intervention was significantly associated with the acceptance of the intervention, but only when the usability of the digital tool was high [$R^2 = .50, F(3,15) = 6.21, p < .01$]. This finding suggests that the individual perception of the ease of use and effectiveness of the digital platform play a crucial role in facilitating the successful implementation of interventions, ensuring that participants can apply the learned skills in their work environment, ultimately leading to greater acceptance of the intervention.

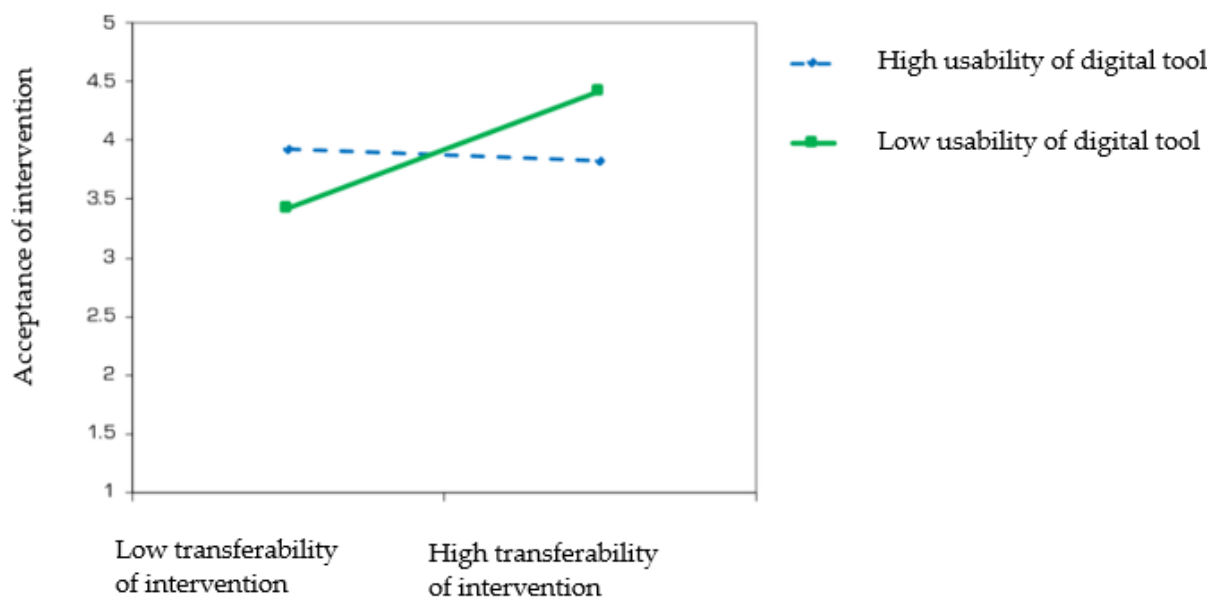


Figure 7. Usability of the digital tool as moderator

Discussion

This study (Giusino et al., 2023) investigated specific process dimensions of a digital-based intervention that was implemented in a hospital with the aim of fostering communication among team members. The study contributes to the literature on team communication interventions by providing insights into the relevance of monitoring aspects of the implementation, and particularly, the recipients' perceptions of the intervention and of the actors involved, such as transferability, design, integrity, acceptance and usability of the intervention. Although there is a growing literature on digital interventions for group communication, there is a lack of research on how the implementation process affects the relevance and acceptance of these interventions. These process measures are crucial for understanding why a given digital intervention may be relevant in a specific working environment and how to ensure that the digital intervention can be perceived as useful and that its contents can be transferred to the

workplace. Moreover, in this case, digital intervention relies on specific intervention techniques such as sociomapping and coaching, thus providing interesting insights into the further development of these techniques as intervention mechanisms in healthcare organisations. The importance of these observations is amplified given the current changes within hospital environments, notably the consolidation of healthcare processes across different departments and their digital transformation, alongside the consequences of the Covid-19 pandemic on healthcare personnel and facilities in both rural and urban regions (Knop et al., 2021).

In this perspective, we examined recipients' perceptions of a digital team coaching intervention at two time points – T1 and T2 – and assessed the temporal stability of these perceptions. At T1, 33 recipients participated, while 29 participated at T2, with 53% and 46% response rates, respectively. The results indicated that at both time points, recipients appreciated the intervention's integrity and usability more than its design, transferability and acceptance. Furthermore, no significant changes in recipients' perceptions were observed over time.

The first aspect to discuss relates to the preference of team coaching intervention participants for usability and integrity. From this perspective, the scarcity of medical professionals, increasingly burdened with more complex and managerial responsibilities, has spurred the development of a broader organisational strategy. This strategy delegated traditional tasks to other team members, such as nurses and healthcare assistants (Knop et al., 2021). Although the roles of general practitioners and nurses are converging in some areas, the success of this multi-actor approach hinges on several factors, including effective communication among team members (Mohr et al., 2011). The facilitators may have also been crucial in nurturing a group dynamic that encouraged team members to discuss communication patterns openly. Essentially, the ability to openly discuss these aspects in sessions led by a supportive coach, who can foster an open and stigma-free atmosphere, and through a user-friendly digital solution, may shed light on the reasons behind these preferences. Reflecting and contemplating better communication through graphical maps could help clarify role definitions, streamline task delegation and ultimately lead to better patient outcomes. These aspects align with previous studies demonstrating the importance of a training facilitator's skills in promoting an open atmosphere. The quality of questions, attention to detail and linking to practical examples are all pivotal for participant engagement and learning outcomes (Wavre & Kuknor, 2023). Defining learning curves – e.g., current and desired communication and action plans – through graphical outputs, which can be reviewed and discussed in groups, might have been instrumental in leveraging peers' and facilitators' feedback to create a supportive learning environment. In the present study, such elements linked to the facilitator's skills and the type of atmosphere the participants perceived during the coaching sessions are among the most relevant online. Future studies that want to investigate the impact of online training coaching sessions should continue to include these process aspects in their evaluations.

As a result, aspects such as design, transferability and acceptance, although positively evaluated, become secondary in importance. For instance, elements of the intervention's design may have been seen as less important because it often focuses on aesthetic aspects, which might be perceived as secondary to the actual content and function of the intervention. While a visually appealing design can enhance user experience, it might not have been considered as crucial as the core components of the intervention. Recipients might have encountered difficulties seeing how the intervention could be generalised or adapted to their specific situations, which could have led them to value transferability less than other aspects. Furthermore, acceptance might not have been highly valued because recipients may have focused more on the immediate experience of engaging with the intervention rather than considering their overall acceptance. Alternatively, recipients may have had mixed opinions about the intervention, which could have contributed to lower average scores for acceptance.

Another aspect to discuss concerns how the digital tool was perceived as highly usable, with a user-friendly and intuitive interface that greatly enhanced the user experience and reduced the learning curve and potential frustrations. Participants quickly became familiar with the digital tool, including how to read the maps and develop improvement actions consistent with the workplace. The efficient and effective engagement with the content is attributed to the digital tool, which increases satisfaction and motivation, ultimately resulting in better outcomes. The high usability of the digital tool is a crucial factor that should not be overlooked, as it can significantly affect the success of the intervention.

One potential explanation for the favorable perceptions of the team coaching intervention among recipients may be related to the uses of visualisation tools, which have been identified as critical mechanisms for effective workplace interventions (Abildgaard & Nielsen, 2018; von Thiele Schwarz et al., 2017). Recipients may have appreciated the graphical indicators used by the digital tool, such as colours, heights and distances, which could aid in comprehending sociomaps and provide a visual representation of abstract concepts, such as communicating with colleagues. Furthermore, the digital nature of the team coaching intervention may have provided benefits or affordances that would not be as readily accessible in other forms of team interventions, such as group scores and visualisation, as well as the ability for individuals to participate remotely if they are not physically present during the sessions.

Our moderation analysis indicated that the usability of the digital tool moderated the relationship between a vital team coaching intervention dimension – i.e., transferability – and the overall acceptance of the digital intervention. This result suggests that when the digital tool is easy to use and efficient, participants are more likely to perceive the intervention as transferable to their work environment, leading to greater acceptance. One possible explanation is that a user-friendly digital tool enhances the participants' experience, allowing them to focus on the content and applicability of the intervention rather than being distracted or frustrated by technical difficulties. Put differently, the innovative element, namely, the ability to visualise the

geographic distribution of communication among team members, and to decipher the patterns and volume of information shared within the team through colours, heights and distances on the geographic plane, can, if easily comprehended, facilitate a guided group discussion on a topic as intricate and elusive as communication between colleagues and superiors. The digital tool enables a more effective visualisation of fairly complex information, allowing participants to grasp the content more thoroughly and, consequently, contemplate strategies for transferring actions to the workplace to address any identified deficiencies. This increased focus on the intervention's content may help participants better understand and apply the learned skills in their work setting, ultimately improving their perception of the intervention's transferability. This result highlights the importance of investing in developing and improving digital tools that are both user-friendly and effective in facilitating the successful transfer of learned skills to the workplace. By ensuring that digital tools are easy to use and support the intervention's objectives, professionals can lead to greater acceptance and satisfaction among participants. This, in turn, can contribute to improved team dynamics, increased productivity and enhanced well-being in the workplace.

The findings of this study have several implications. From a theoretical point of view, it contributes to the existing literature on digital interventions for group communication by shedding light on the importance of process dimensions, such as recipients' perceptions of the intervention and its various components – e.g., transferability, design, integrity, acceptance and usability – and the implementation process. It emphasises the need for further research on the effects of implementation processes on the relevance and acceptance of digital interventions. In addition, it highlights the potential of combining specific intervention techniques, like sociomapping and coaching, as intervention mechanisms in healthcare organisations, thus encouraging further development and research on these techniques. From a practical perspective, the results highlight the importance of prioritising the usability and integrity of a digital-based team coaching intervention. Participants perceived these aspects more favorably than design, transferability and acceptance. Therefore, developers and facilitators of such interventions should focus on creating user-friendly digital tools and maintaining consistency throughout the intervention process to ensure its effectiveness. Second, the temporal stability of recipients' perceptions suggests that once a team coaching intervention is well received, its positive impact can be sustained over time. This underlines the value of investing in developing high-quality interventions that meet the needs and expectations of participants from the outset. Finally, the study reveals a significant association between transferability and acceptance of the intervention when the usability of the digital tool is high. This implies that when a digital tool is easy to use, participants are more likely to perceive the intervention as applicable to their work context and be open to adopting the changes it proposes. Consequently, designers and practitioners may prioritise the development of user-friendly digital tools to enhance the likelihood of successful implementation and adoption of team communication interventions.

The study's limitations should be recognised, including the small sample size, the lack of a control group and the reliance on self-reported evaluations. The study focussed solely on recipients' perceptions of the intervention without measuring its effects on team communication patterns. Thus, while the findings offer valuable insights into the perceived qualities and dimensions of the intervention, they do not provide direct evidence of its impact on the intended outcomes. Another limitation is that the generalisability of the results is limited to the specific context of digital interventions aimed at improving team communication in healthcare settings. Therefore, caution should be taken when applying the findings to other workplaces or organisations.

Despite these limitations, the study provides valuable recommendations for managers and practitioners seeking to implement digital-based workplace interventions to improve team communication. The study also contributes to filling a gap in the literature by exploring the potential of a digital-based intervention combined with an online coach involved with the goal of promoting better communication among team members, which represents a novel approach compared to traditional in-person interventions. Moreover, it sets the stage for future research on the implementation processes of digital interventions and their effects on recipients' perceptions and the interventions' relevance and acceptance.

Conclusion

The implementation of the team coaching intervention at the targeted healthcare organisation could be deemed as successful. Overall, findings suggested that recipients' perceptions of intervention characteristics were positive. Also, recipients' perceptions did not change from second to fourth session, suggesting they remained stably satisfied with the intervention over time. However, this study only focussed on recipients' perceptions as explanatory process factors of intervention effectiveness, whereas the realist evaluation approach includes more, such as contextual conditions and working mechanisms. These types of process factors were then investigated in a separate study (Giusino et al., in preparation), which is introduced in the next chapter of the present dissertation.

Study 4. Digital Team Coaching Intervention Using Social Network Visualisation: Evaluation of Two CMO Configurations

Abstract. Digital workplace interventions using social network visualisation and team coaching hold potential to promote teamwork within organisations. However, knowledge regarding the contextual conditions and the explanatory mechanisms of their effectiveness is still scarce. To contribute filling this gap, this study aimed to conduct the realist evaluation of a digital team coaching intervention using social network visualisation. Two context-mechanism-outcome (CMO) configurations were elaborated. The first CMO configuration corresponded to the hypothesis of a moderating role of transfer and action plan implementation in the relationship between manager support and teamwork, team coordination, interpersonal conflict at work, and team performance. The second CMO configuration corresponded to the hypothesis of a mediating role of transfer and action plan implementation in the relationship between peer support towards training transfer and the same outcome variables. Main findings revealed that the positive relationship between manager support and team coordination and team performance was weaker when participants reported higher levels of implementation of action plans developed as part of the intervention. Peer support towards training transfer had positive direct effects on transfer, teamwork, team coordination, and team performance. It could be concluded that teams implementing action plans developed during the intervention might need less support from immediate managers in order to coordinate collective efforts and accomplish good collective performance. Moreover, peer support towards training transfer may constitute a relevant contextual factor contributing to the intervention effectiveness. Practitioners willing to implement the examined intervention might design the implementation environment according to this study results.

Background

In the last decades, considering the importance of teamwork in the workplace, a number of interventions have been developed and tested with the aim of promoting teamwork quality and effectiveness. Traditional evidence-based approaches to team development and improvement (Grote & Kozlowski, 2023; Lacerenza et al., 2018) have primarily focused on in-presence team training, team building, and team debriefing. Team training refers to a structured approach to improving the teamwork skills, with a focus on the shared understanding of roles, responsibilities, and goals. Team

building has a stronger focus on improving the relationships within the team, encompassing, for instance, activities and exercises to promote trust, respect, and mutual understanding among team members. Team building corresponds to a process of reflection and discussion among the team members after a task or event to identify what went well, what could have been improved, and how to apply lessons learned to future situations. While these interventions have demonstrated some effectiveness (McEwan et al., 2017), they have often neglected the complex social dynamics and interrelationships that might influence teamwork patterns within teams. In this regard, for example, effective and high-quality team communication in the workplace – meaning, communication that is clear to the receivers, and timely and relevant to the tasks it refers to – has to be regarded as an impactful variable, as it has been associated with improved team performance in terms of innovation, budget performance, efficiency, and goal achievement (Bui et al. 2019).

In recent years, digital interventions have emerged as an alternative powerful tool to foster desirable workplace outcomes. Digital interventions can be defined as planned, structured, science-based actions or initiatives aiming to achieve a certain goal – e.g., fostering teamwork – by exploiting the potential offered by digital technologies (e.g., Stratton et al., 2022). They may correspond to either interventions originally designed to occur in physical presence and subsequently adapted to the digital formats offered by online teleconferencing platforms, or interventions explicitly meant to be available for computer or smartphone apps only. Whereas the majority of the results about digital interventions' effectiveness in promoting desirable workplace outcomes mostly refer to individual-level employee well-being and effectiveness (*ibidem*), less is known about the effectiveness of digital interventions in improving teamwork. In this realm, one innovative approach is the use of social network visualisation, coupled with group-level coaching, to promote effective teamwork within organisations (Bahbouh, 2012; Bahbouh & Lasker, 2014). By leveraging digital technologies, social network visualisation can provide a deeper understanding of the social dynamics occurring within workplace teams, so that targeted interventions can be designed to address teamwork challenges effectively. The use of social network visualisation as a digital intervention tool offers unique opportunities to identify teamwork patterns, map existing and potential relationships, and pinpoint areas where teamwork breakdowns may occur. The visualisation provides valuable insights into the structure and dynamics of social networks within teams, enabling stakeholders to identify key individuals, opinion leaders, and potential bottlenecks in the team workflow (Bahbouh & Lasker, 2014). Moreover, group-level team coaching can be incorporated into the intervention (Bahbouh & Willis, 2022), so that workers can receive tailored feedback, guidance, and support to improve their teamwork skills, enhance communication, collaboration, and coordination, and foster a culture of learning and knowledge sharing.

To date, very few studies are available regarding the effectiveness of group-level coaching-based digital interventions utilising social network visualisation within workplace team settings. Bernardová (2012) carried out a study within the military

aviation sector, highlighting the potential of social network analysis and visualisation as a tool for psychodiagnostics. Using social network analysis tools, Franc and colleagues (2019) found a positive effect of quality and frequency of team leaders' communication on sales team performance in a large financial institution. Findings from Tetour (2019) supported the effectiveness of social network visualisation as an intervention tool among teams in the hospitality sector. Finally, Zakharchyn and Kosmyna (2015) argued in favour of the application of sociometric methods to model optimal personnel behaviour within private enterprises. Despite the value held by each of these studies, to the best of our knowledge none of them has performed evaluations of real-world or contextualised effectiveness of interventions based on social network visualisation and team coaching. That is, the available research has not taken into account their explanatory mechanisms or the reasons why such interventions worked as intended for the targeted recipients. This study (Giusino et al., in preparation⁴) aimed to fill this gap by deploying a realist evaluation approach to workplace interventions' effectiveness, in particular, a digital team coaching intervention using social network visualisation to promote desired and effective teamwork patterns (Bahbouh, 2012; Bahbouh & Lasker, 2014; Bahbouh & Willis, 2022). This approach can indeed provide insights into when and how interventions work.

The objective was to contribute with an evidence base that can guide future implementation efforts from practitioners in organisations willing to promote the quality of teamwork. This contribution appears to be especially relevant in light of the fact that nowadays teamwork and leadership are performed virtually more and more (Bell et al., 2023), with remote work, hybrid work, and other new trending forms of work organisation being accelerated after the Covid-19 pandemic.

This study deployed a realist approach towards the evaluation of workplace interventions (Nielsen & Abildgaard, 2013; Nielsen & Miraglia, 2017; Nielsen & Randall, 2013; Nielsen & Shepherd, 2022; Roodbari et al., 2021a, 2021b, 2023) and aimed to investigate the contextual factors, the underlying mechanisms, and the connected outcomes of a digital team coaching intervention that utilises social network visualisation. It did so by developing and testing two specific context-mechanism-outcome (CMO) configurations that were considered relevant to the implemented intervention. On the basis of the intervention's contents and goals, relevant outcomes of the intervention were considered in this study, such as improved teamwork and team coordination, reduced interpersonal conflict at work, and improved team performance. Here, these variables are defined as follows. Teamwork can be defined as the process through which a group of individuals working together towards a common goal puts collective efforts to produce higher performance than the sum of individual inputs (Salanova et al., 2006). Team coordination refers to the process of synchronising and integrating the efforts of team members to achieve a

⁴ Preliminary findings from this study have been presented at the XXI Congress of the European Association of Work and Organisational Psychology, held in Katowice, Poland, in May 2023. <https://eawop2023.org>

common goal (Salanova et al., 2011). Interpersonal conflict at work indicates negative interactions between individuals in the workplace (Friedman et al., 2006), such as arguments with colleagues or being treated poorly by a supervisor. Finally, team performance refers to the ability of a group of individuals to work together effectively towards a common goal or objective (Callea et al., 2014; Dawson et al., 2006). The final selection of the relevant intervention's outcomes to be investigated in this study is the result of extensive discussions among the authors, informed by their expertise as well as knowledge on the intervention's contents and goals and psychometric measures available within the extant academic literature.

CMO Configuration #1: The Moderating Role of Transfer and Action Plan Implementation between Manager Support and Team Outcomes

In line with previous studies (Christensen et al., 2019; Helland et al., 2021; Nielsen et al., 2010), one important contextual factor that is known to positively affect workplace interventions' effectiveness is manager support. Manager support can be defined as the extent to which the immediate manager of a team looks after their team members, asking whether team members have problems at work, helping to make team members' work easier, and listening to team members when they have problems (Holton III et al., 2000). Here, manager support is conceived as an omnibus context factor – that is, existing in the workplace regardless of the intervention being implemented. Within the realist evaluation literature, for instance, the systematic review by Roodbari and colleagues (2021a, 2021b) and the Integrative Process Evaluation Framework by Nielsen and colleagues (2023), the support from the immediate manager is an example of a context factor that facilitates the effectiveness of workplace interventions, as it provides workers and teams with the necessary social resources that sustain their participation in the intervention itself as well as the application of what they acquired during the intervention into their everyday work – managers, for instance, can support the use of new skills and knowledge in the team. In addition, the influence of manager support on intervention effectiveness can be optimised when it interacts with two well-known intervention's working mechanisms, namely transfer and action plan implementation. Here, transfer is defined as the degree to which intervention recipients transfer their learning to the actual work context, using the knowledge gained in the intervention, managing to apply the intervention contents in their everyday work, and transferring the skills learned in the intervention back to their actual job (Grohmann & Kauffeld, 2013). On the other hand, action plan implementation is defined as the extent to which intervention recipients implement the action plans developed throughout the intervention, continuously working and revisiting them (ibidem). Nielsen and Shepherd's (2022) Integrated Training Transfer and Effectiveness Model and Nielsen and colleagues' (2023) Integrative Process Evaluation Framework suggested that, for workplace interventions to be effective, participants need to transfer their learning to

the work context and implement the action plans that they developed as part of the intervention activities. To summarise, the realist evaluation perspective claims that workplace interventions are effective when recipients have support from their managers, transfer new skills and knowledge to workplace, and implement action plans in order to make the intervention work as intended.

On the basis of the above, and considering the key outcomes identified for the digital team coaching intervention being investigated, the following statement is proposed.

CMO configuration #1: If intervention recipients perceive high levels of support from their immediate manager, as a result they will show better teamwork and team coordination, less interpersonal conflict at work, and better team performance at post-intervention, when they transfer learnings and implement the developed action plans to a greater extent.

Previous research provides extensive empirical evidence that manager support can improve teamwork and team outcomes. Nielsen and Randall (2009) stressed the importance of managers' active support when implementing teams in the workplace. Gilley and colleagues (2010) concluded that effective managers that exhibit certain skills and behaviours (i.e., involving employees in decision-making, coaching others, communicating effectively, motivating others, and helping employees grow and develop) can build teams and significantly improve team performance, with support from managers being an important factor in successful teamwork. Consistently, Nielsen and colleagues (2010) found that training managers may enhance the effects of implementing teamworking. Given such a corpus of evidence referring to the positive relationship existing between manager support and desirable team outcomes, this study assumed that a positive relationship occurs between manager support, on the one hand, and teamwork (TW), team coordination (COR), and team performance (TP), on the other hand. Also, a negative relationship was assumed between manager support and interpersonal conflict at work (ICW), which constitute an undesirable team outcome. Within this frame, a moderating role of transfer and action plan implementation was hypothesised, as shown in Figure 8.

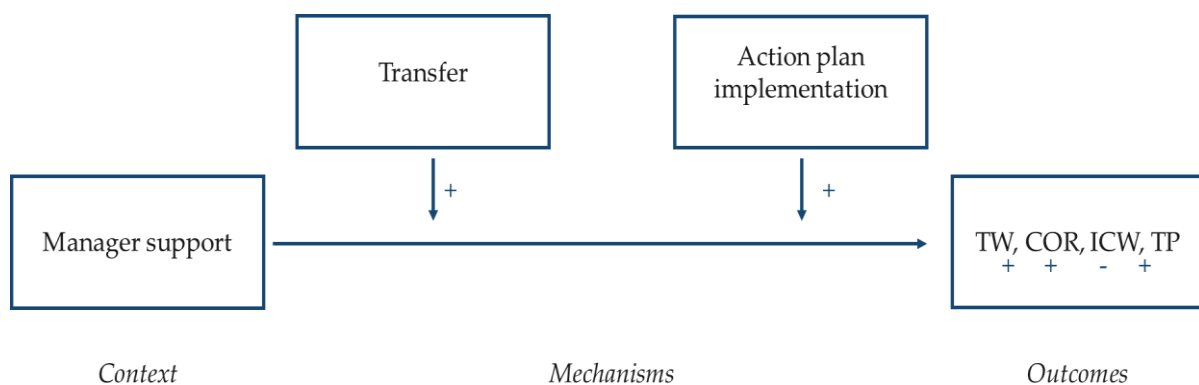


Figure 8. Hypothesised moderation model

CMO Configuration #2: The Mediating Role of Transfer and Action Plan Implementation between Peer Support Towards Training Transfer and Team Outcomes

Another important contextual factor that facilitates workplace interventions' effectiveness is peer support towards training transfer (Nielsen et al., 2023). Peer support towards training transfer can be defined as the extent to which intervention recipients' colleagues support the use of new skills and knowledge acquired as part of the intervention activities, appreciating, encouraging and expecting the efforts to apply them into everyday work (Holton III et al., 2000). Here, peer support towards training transfer is conceived as a discrete context factor – that is, existing in relation to the intervention that is being implemented in the workplace. Nielsen and colleagues' (2023) Integrative Process Evaluation Framework suggested that, for workplace interventions to be effective, recipients' peers and colleagues need to support the use of new skills and knowledge as well as the implementation of action plans developed during the intervention itself. Thus, if recipients' colleagues support the transfer and action plan implementation of the intervention, then the recipients themselves are able to transfer and implement the intervention, which would in turn maximise the intervention's effectiveness in generating the expected outcomes, as also suggested by Nielsen and Shepherd's (2022) Integrated Training Transfer and Effectiveness Model. In other words, if recipients actually transfer and implement the intervention during their everyday work, this might be because of their peers' support to do so.

On the basis of the above, and considering the key outcomes identified for the digital team coaching intervention under examination, the following statement is proposed.

CMO configuration #2: If intervention recipients perceive high levels of peer support to apply action plans that they developed throughout training sessions to everyday work life, then they are more likely to make transfer and implementation of such action plans, and as a result they will show better teamwork and team coordination, less interpersonal conflict at work, and better team performance at post-intervention.

This study explored the role of transfer and action plan implementation as explanatory mechanisms within the relationship between peer support towards training transfer and team outcomes, such as teamwork (TW), team coordination (COR), interpersonal conflict at work (ICW), and team performance (TP). A mediating role of transfer and action plan implementation was hypothesised, as it is shown in Figure 9.

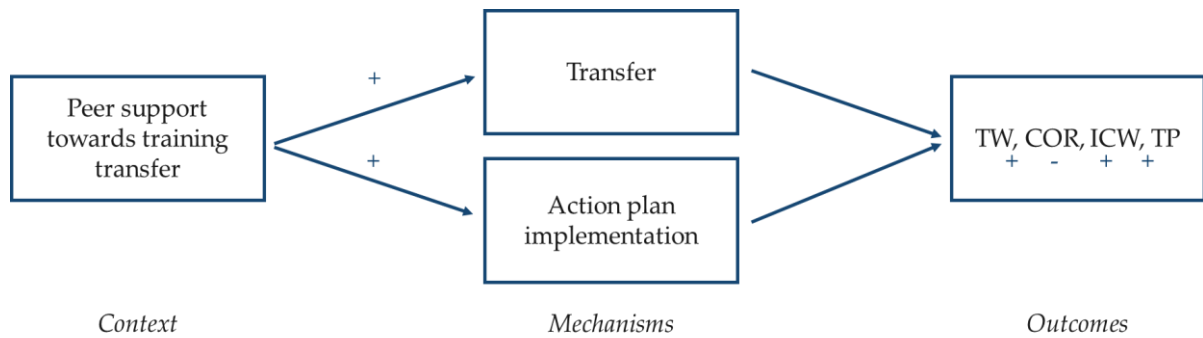


Figure 9. Hypothesised mediation model

Methods

The implemented intervention (Bahbouh, 2012; Bahbouh & Lasker, 2014) is a group-level digital software-based intervention that serves the analysis and promotion of desired and effective teamwork patterns. According to the main classifications of workplace interventions' levels (i.e., individual, group, leader, and organisation; Nielsen et al., 2018), group-level workplace interventions refer to formal or informal, planned, structured, science-based, behavioural, or psychological actions or initiatives aiming to bring about changes in team states, processes, and dynamics, such as communication, cohesion, mutual support among colleagues, work group social climate, or collective decision-making and performance (ibidem). The intervention under study falls within the group level.

The number of intervention sessions usually ranges from three to four, with each of the sessions lasting from one and a half to three hours and being structured according to five subsequent phases, such as (1) online survey-based relational data collection about current and desired teamwork patterns, (2) visualisation of the sociomaps that the software generates based on participant's responses to the survey – e.g., current and desired frequency of communication, current and desired quality of interaction, stress levels among the team members, sources of stress, and so on, (3) team discussion and team coaching activities that are guided by the visualisation of the sociomaps, (4) action plan development based on the resolutions made through the team coaching activities, and (5) review of the developed action plans.

The term "sociomap" here refers to landscape-like graphical representations of specific teamwork patterns.

This intervention is both a diagnostic and an intervention tool, including action plan development, aiming to promote team awareness, team dynamics, and team patterns. Its success implies a collective effort within the targeted teams. By visualising social networks and providing targeted coaching activities, this intervention seeks to promote efficient collaboration and improve overall team performance. Based on social network analytical theory and methods (Bahbouh, 2012; Bahbouh & Lasker, 2014), the delivered intervention deploys a software for collection, elaboration, and visualisation of complex relational data. By processing participants' responses to valid

and reliable Likert-type single-item questions regarding teamwork (Rozehnalová, 2013), its algorithm generates graphical representations, named sociomaps, of current and desired teamwork structures existing within the team. The intervention is based on team coaching workshops, defined as direct interactions with a team intended to help members make coordinated and task-appropriate use of their collective resources in accomplishing the team's work (Bahbouh & Willis, 2022). These workshops are guided by the visualisation of sociomaps to stimulate team reflexivity and self-awareness, with an aim to formulate individual and team action plans, solutions, and strategies that would lead to the desired and effective communication flow within the team. As in the example in Figure 10, the landscape-like sociomap shows how team members are interconnected based on their perceptions of teamwork variables in their team. Mutual positions among members reflect their current or desired pattern. Individual height and colour represent the average pattern intensity of the given person in the team. During sessions, participants are provided with an overview of the intervention's framework and key concepts, like teamwork and team effectiveness, by means of slideshows. Guided by the visualisation of the generated sociomaps, workshops are conducted based on team coaching principles. A collective discussion is set up on current and desired teamwork, with the aim to formulate strategies for improvement. The workshops aim to discuss the situation of the team and to develop an action plan about how to modify aspects of teamwork that team members desired to change. Particularly, participants are offered the opportunity to share their thoughts about behaviours or work situations that should stop, start, or continue to enhance teamwork effectiveness within the team. Also, each participant can provide individual feedback to colleagues about their work style, what they do well and what they can improve. The teams develop action plans during the sessions and progress is monitored against the developed action plan and is discussed within the workshops. Sessions include topics such as team dynamics, importance and quality of teamwork, and mutual feedback and progress. The last session is usually a final debriefing, which briefly retraces previously addressed milestones and also includes a concluding purpose sharing to maintain the communicative efficiency achieved. Teamwork measures are collected at each session to generate updated sociomaps and allow comparisons among sociomaps from different sessions to enrich the shared reflections and discussions.

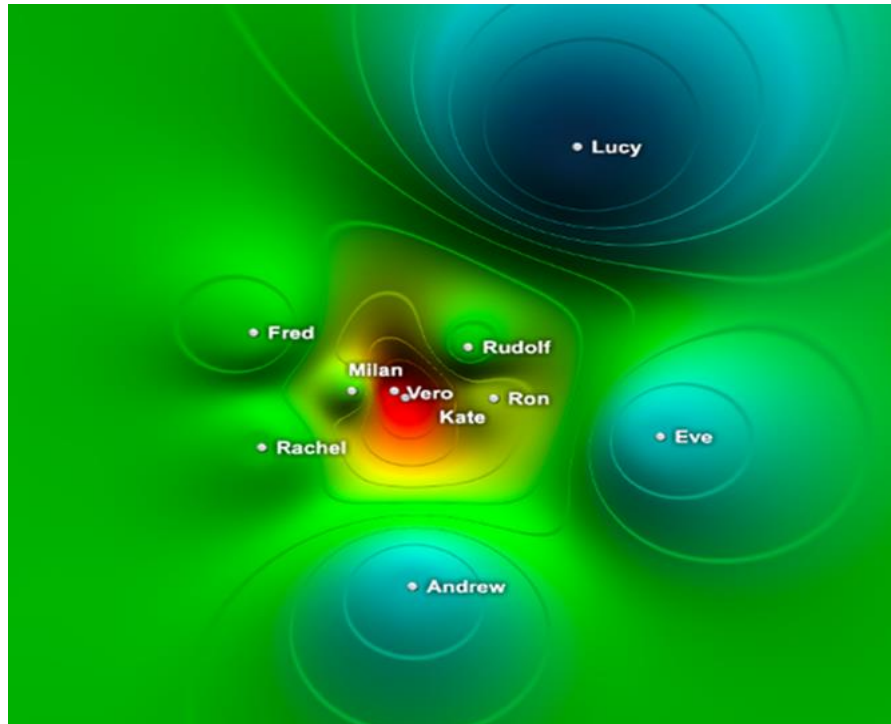


Figure 10. Example “sociomap”

Intervention Implementation

In this study, the intervention implementation was part of a broader research and innovation project about multilevel interventions to promote mental health in the workplace (De Angelis et al., 2020). The project aimed to design, develop, implement, and validate assessment, intervention, and evaluation toolkits to promote mental health in the workplace across Europe. Following a team analysis as indicated by Lacerenza and colleagues' (2018) stepwise framework for team interventions in order to ensure the intervention fit (Andersen et al., 2021; Peters et al., 2020) and identify key team-level needs to be addressed – see Study 2 in the present dissertation as well as Giusino and colleagues (2022a, 2022b) – the intervention was introduced across four distinct organisations based in two European countries, namely Italy and the Czech Republic. Sessions were delivered by trained psychologists from academic research groups. The Italian segment of this intervention was conducted within two disparate organisations; a sizable public healthcare organisation and a modestly scaled enterprise specialising in private education. The healthcare organisation, owing to its structure, facilitated the intervention between May and December 2021 – one session every two months and a half, four sessions in total – via an online platform distributed across seven teams emanating from three individual departments. Most participants in this organisation were women (74%) and the average age was 47 years, with team sizes ranging between 6 and 13 members, thereby resulting in an aggregate of 62 participants. The private education enterprise held the intervention online between February and April 2022 – one session per month, three sessions in total – within seven

separate departments, each forming a team of 6 to 23 individuals, totalling 101 participants. This subset of participants comprised entirely of women, with an age demographic concentrated between 25 and 34 years. Similarly, in the Czech Republic, two distinct corporate organisations facilitated the intervention; one being in the retail sector, the other in the IT industry. The retail sector organisation opted for an online intervention delivery method, implemented between May 2021 and February 2022 – one session every four months, three sessions in total –, thereby engaging 15 teams from six departments. These teams, comprised of 4 to 17 members each, amassed a total of 118 participants, of which 41% were women, falling within the age bracket of 30 to 40 years. The IT organisation, in contrast, adopted a hybrid delivery format – two online sessions, one in-person session – for the intervention, implemented between March 2022 and January 2023 – one session every three months, three sessions in total –, reaching out to 11 teams – 3 to 11 members per team – from 11 departments, with a collective total of 104 participants. This participant subset included a minimal female representation (15%), with no data available regarding the age demographic. In all involved organisations, intervention recipients were recruited based on voluntary subscription upon information from their managers. For each intervention group, the only inclusion criterion was that participants had to come from the same team or work process – not necessarily the whole team, but also part of it. As recommended by Nielsen and colleagues (2013), feasibility of the intervention project was ensured by establishing a Steering Committee in each organisation. Its members planned and implemented a structured organisational communication strategy and facilitated employee participation by encouraging their teams to partake in the intervention. Researchers maintained constant communication with contact persons from the targeted organisations to ensure a smooth conduction of activities.

To summarise, the intervention extended its reach to a total of 40 teams from 27 unique departments, involving team sizes varying between 3 and 23 members. Consequently, the intervention engaged a total of 385 participants, with a dominant representation of female individuals within the age spectrum of 25 to 47 years.

Measures

Data collection was survey-based, as data were collected through online questionnaires, distributed with the help of contact persons or project representatives at the targeted organisations. Before filling out the questionnaires, participants could read a detailed Informed Consent Form explaining the aim of the study and their rights as participants of the research. This study received ethical approval and complied with standard requirements.

Data were collected longitudinally with a multi-wave research design across three time points. Context measures – i.e., manager support and peer support towards training transfer – were collected three months after the start of the intervention (T1). Working mechanisms – i.e., transfer and action plan implementation – were measured

six months later than context measures (T2), during the intervention period. Finally, outcome variables – i.e., teamwork, team coordination, interpersonal conflict at work, and team performance – were measured at a post-intervention data collection time point (T3) occurring three months later than collection of working mechanisms. Time lags between data collection points were conditional on the research design of the broader project that this study was part of. Time lag between the end of the intervention and T1 aimed to allow sufficient time for the participants to experience and perceive the support provided by their managers and peers. Time lag between T1 and T2 aimed to permit participants to engage with the training materials, apply their learning, and implement action plans. Finally, the purpose of time lag between T2 and T3 was to allow sufficient time for the intervention to produce, manifest, and stabilise its intended outcomes.

Quantitative psychometric measures were collected for context factors, working mechanisms, and outcome variables. Where not otherwise specified, Italian and Czech translations of the validated psychometric scales deployed were carried out by trained researchers for administration at the targeted organisations, following a conventional back-translation procedure.

Context factors were measured as follows.

Manager support. Three items based on Holton III and colleagues (2000) were developed, for instance, “Our immediate manager asks us if we have problems or troubles at work”. Response options ranged from “1 = strongly disagree” to “5 = strongly agree”. Cronbach’s α for reliability was .94.

Peer support towards training transfer. Three items based on Holton III and colleagues (2000) were developed, for instance, “We appreciate each other’s efforts to use new skills we have learned in training”. Response options ranged from “1 = strongly disagree” to “5 = strongly agree”. Cronbach’s α for reliability was .90.

Working mechanisms were measured as follows.

Transfer. Three items based on Grohmann and Kauffeld (2013) were developed, for instance, “In our everyday work, we often use the knowledge we gained in the training”. Response options ranged from “1 = strongly disagree” to “5 = strongly agree”. Cronbach’s α for reliability was .88.

Action plan implementation. Three items based on Grohmann and Kauffeld (2013) were developed, for instance, “We have implemented the planned action plans”. Response options ranged from “1 = strongly disagree” to “5 = strongly agree”. Cronbach’s α for reliability was .86.

Outcomes were measured as follows.

Teamwork. Three items from Salanova and colleagues (2006) were used, for instance, “My work team has clear working objectives”. Response options ranged from “0 = never” to “6 = always”. Cronbach’s α for reliability was .72.

Team coordination. Three items from Salanova and colleagues (2016) were used, for instance, “We co-ordinate with one another to complete the necessary tasks”. Response options ranged from “0 = never” to “6 = always”. Cronbach’s α for reliability was .79.

Interpersonal conflict at work. Nine items from Friedman and colleagues (2000) were used. This is a two-dimensional scale, encompassing task conflict and role conflict. Four items were from the task conflict sub-scale, for instance, “How often do you work with disagree about opinions regarding the work being done?”. Five items were from the role conflict sub-scale, for instance, “There are often feelings of hostility among parties”. For the purposes of this study, interpersonal conflict at work was computed as a single variable. Response options ranged from “1 = not at all” to “5 = a lot”. Cronbach’s α for reliability was .95.

Team performance. Fifteen items from the Aston Team Performance Inventory by Dawson and colleagues (2006) were used. This is a four-dimensional scale, encompassing team support, team autonomy, team reflexivity, and team participation. Three items were from the team support sub-scale, for instance, “Team members are generally warm and supportive to each other”. Four items were from the team autonomy sub-scale, for instance, “In this team we set our own goals”. Four items were from the team reflexivity sub-scale, for instance, “The methods used by the team to get the job done are often discussed”. Finally, four items were from the team participation sub-scale, for instance, “Everyone in the team contributes to decision making”. For the purposes of this study, team performance was computed as a single variable. For the Italian organisations involved, the Italian validation by Callea and colleagues (2014) was used, whereas a translation was created by trained researchers for the Czech organisations. Response options ranged from “1 = completely disagree” to “5 = completely agree”. Cronbach’s α for reliability was .81.

Participants

The final set of data that could be used to conduct the analyses planned in this study included a sample composed of 317 respondents (response rate = 82%). Of these, 101 (31.9%) were from the private IT Czech organisation, followed by the private retail Czech organisation ($n = 100$, 31.5%), the Italian educational organisation ($n = 69$, 21.8%), and the large public healthcare organisation in Italy ($n = 47$, 14.8%). One-hundred and twelve were males (35.3%) and 73 were females (23%). The majority (34.7%) was between 25 and 34 years old, holding a master or equivalent level of education (20.5%), a permanent full-time employment (45.1%), between 3 and 4 years of organisational tenure (18%), and with no leadership responsibilities (42.9%).

Data Analysis

Quantitative statistical analysis was performed merging data from all the four organisations involved in the study. Descriptives were run to obtain frequencies, means, and standard deviations. The internal consistency of scales was assessed using Cronbach’s α . Pearson’s r was calculated to conduct correlation analysis, aiming to explore the relationships among the variables under examination. These analyses

were performed using the IBM SPSS Statistics software version 23. Then, moderation analysis was conducted to test CMO configuration #1 (H1), whereas mediation analysis was conducted to test CMO configuration #2 (H2). These analyses were performed using the Jamovi software version 2.3.26. Eight moderation models and eight mediation models were run, as many as there were combinations among the hypothesised predictors – i.e., contextual factors –, moderators or mediators – i.e., working mechanisms –, and outcomes. The threshold of acceptable statistical significance was set at $p < .05$.

Results

The current section reports the main findings of the present study, regarding the correlation analysis, the moderation analysis which was performed to test CMO configuration #1 (H1), and the mediation analysis which was performed to test CMO configuration #2 (H2).

Correlation Analysis

Correlation analysis, as shown in Table 8, highlighted statistically significant bivariate relationships between manager support and each of the intervention's outcomes, such as, teamwork ($r = .56, p < .01$), team coordination ($r = .41, p < .01$), interpersonal conflict at work ($r = -.32, p < .05$), and team performance ($r = .52, p < .01$). Also, peer support towards training transfer was statistically significantly associated with transfer ($r = .44, p < .01$), action plan implementation ($r = .38, p < .01$), and teamwork ($r = .35, p < .05$). Finally, action plan implementation was statistically significantly associated with teamwork ($r = .41, p < .05$), team coordination ($r = .41, p < .05$), and team performance ($r = .45, p < .01$).

Table 8. Correlation matrix for the study variables

<i>Variables</i>	<i>n</i>	<i>M</i>	<i>SD</i>	1	2	3	4	5	6	7	8
1. Manager support	128	3.76	1.05	-							
2. Peer support	128	3.58	.81	.37**	-						
3. Transfer	78	3.10	.82	.07	.44**	-					
4. Implementation	76	3.04	.81	.09	.38**	.44**	-				
5. Teamwork	102	4.51	.81	.56**	.35*	-.01	.41*	-			
6. Coordination	102	4.44	.84	.41**	.21	.10	.41*	.63**	-		
7. Interpersonal conflict	99	1.94	.73	-.32*	-.13	-.16	-.32	-.57**	-.59**	-	
8. Team performance	105	3.89	.68	.52**	.25	.12	.45**	.62**	.65**	-.61**	-

Note. n = sample size. M = mean. SD = standard deviation. * = $p < .05$ (2-tailed). ** = $p < .01$ (2-tailed)

Moderation Analysis

As shown in Table 9, two statistically significant interactions were found.

Table 9. Estimates from the moderation models

	<i>B</i>	<i>SE</i>	<i>Z</i>	<i>p</i>
<i>Outcome: Teamwork</i>				
Manager support * Transfer	-.17	.23	-.76	.44
Manager support * Implementation	-.27	.15	-1.75	.08
<i>Outcome: Team coordination</i>				
Manager support * Transfer	-.45	.25	-1.78	.07
Manager support * Implementation	-.48	.17	-2.81	.00
<i>Outcome: Interpersonal conflict at work</i>				
Manager support * Transfer	-.05	.28	-.18	.85
Manager support * Implementation	.31	.19	1.63	.10
<i>Outcome: Team performance</i>				
Manager support * Transfer	-.13	.28	-.47	.63
Manager support * Implementation	-.40	.18	-2.21	.02

Note. *SE* = standard error. *Z* = moderator's value

First, the model including manager support as a predictor, action plan implementation as a moderator, and team coordination as an outcome variable was statistically significant. There was a statistically significant interaction ($p < .05$) between manager support at T1 and action plan implementation at T2 ($B = -.48$, $Z = -2.81$) in the model including team coordination at T3 as an outcome variable. The effect of manager support on team coordination was statistically significant ($p < .001$) at low levels of action plan implementation ($B = 1.14$, $Z = 4.39$), as shown in Figure 11.

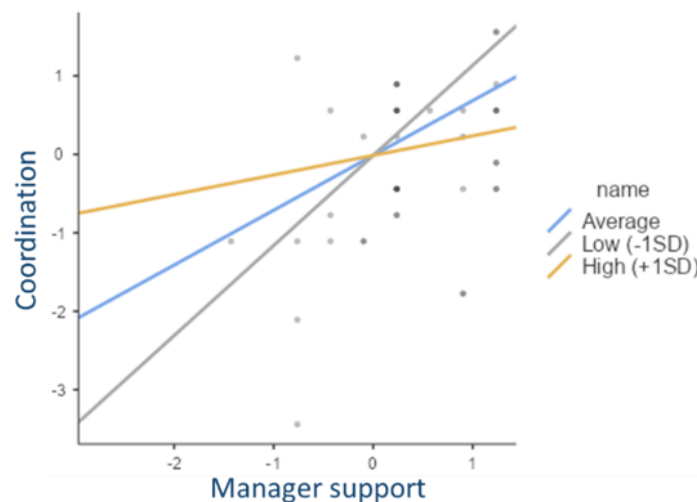


Figure 11. Moderation effect of action plan implementation in the relationship between manager support and coordination

Second, the model including manager support as a predictor, action plan implementation as a moderator, and team performance as an outcome variable was statistically significant. There was a statistically significant interaction ($p < .05$) between manager support at T1 and action plan implementation at T2 ($B = -.40$, $Z = -2.21$) in the model including team performance at T3 as an outcome variable. The effect of manager support on team performance was statistically significant ($p < .001$) at low levels of action plan implementation ($B = .94$, $Z = 3.69$), as shown in Figure 12.

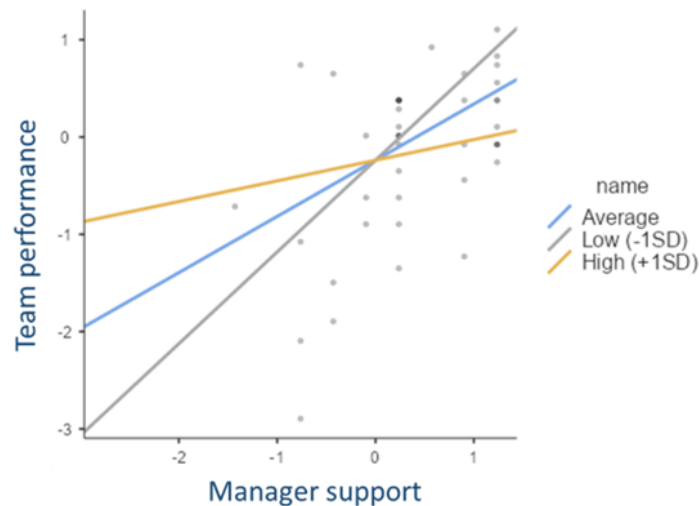


Figure 12. Moderation effect of action plan implementation in the relationship between manager support and team performance

So, two models turned out to be statistically significant, while six interactions were not significant. Thus, no statistically significant interaction effects were found in the majority of the hypothesised moderation models – i.e., six out of eight.

Overall, these findings partially supported CMO configuration #1.

Mediation Analysis

As shown in Table 10, no statistically significant indirect effects were found in any of the hypothesised mediation models – i.e., eight out of eight.

Table 10. Estimates from the mediation models

<i>Effect</i>	<i>B</i>	<i>SE</i>	<i>Z</i>	<i>p</i>
<i>Peer support, Transfer, Teamwork</i>				
Indirect	-.27	.17	-1.58	.11
Direct	1.29	.28	4.58	.00
Total	1.01	.29	3.46	.00
<i>Peer support, Implementation, Teamwork</i>				
Indirect	.09	.11	.81	.41

Digital interventions for teams

Direct	.91	.30	3.01	.00
Total	1.01	.29	3.46	.00
<i>Peer support, Transfer, Coordination</i>				
Indirect	-.23	.17	-1.34	.18
Direct	1.05	.34	3.11	.00
Total	.82	.33	2.47	.01
<i>Peer support, Implementation, Coordination</i>				
Indirect	.11	.13	.855	.39
Direct	.70	.34	2.04	.04
Total	.82	.33	2.46	.01
<i>Peer support, Transfer, Interpersonal conflict at work</i>				
Indirect	-.00	.13	-.05	.95
Direct	-.44	.35	-1.25	.20
Total	-.45	.32	-1.39	.16
<i>Peer support, Implementation, Interpersonal conflict at work</i>				
Indirect	-.12	.13	-.93	.35
Direct	-.32	.33	-.97	.32
Total	-.45	.32	-1.39	.16
<i>Peer support, Transfer, Team performance</i>				
Indirect	-.06	.14	-.47	.63
Direct	.87	.35	2.43	.01
Total	.80	.33	2.43	.01
<i>Peer support, Implementation, Team Performance</i>				
Indirect	.19	.15	1.23	.21
Direct	.61	.32	1.86	.06
Total	.80	.33	2.43	.01

Note. SE = standard error. Z = mediator's value

However, statistically significant direct effects were found within the positive relationships between peer support towards training transfer at T1 and transfer at T2 ($B = .56$, $Z = 2.14$, $p < .05$, $n = 46$, $r = .44$), peer support towards training transfer at T1 and teamwork at T3 ($B = 1.29$, $Z = 4.58$, $p < .001$, $n = 42$, $r = .35$), peer support towards training transfer at T1 and team coordination at T3 ($B = 1.05$, $Z = 3.11$, $p < .01$, $n = 42$), and peer support towards training transfer at T1 and team performance at T3 ($B = .87$, $Z = 2.43$, $p < .01$, $n = 43$).

Overall, these findings partially supported CMO configuration #2.

Discussion

The aim of this study (Giusino et al., in preparation) was to conduct a realist evaluation (Nielsen & Abildgaard, 2013; Nielsen & Miraglia, 2017; Nielsen & Randall, 2013; Nielsen & Shepherd, 2022; Roodbari et al., 2021a, 2021b, 2023) of a digital team

coaching intervention using social network visualisation (Bahbouh, 2012; Bahbouh & Lasker, 2014; Bahbouh & Willis, 2022), which was implemented across four organisations from two European countries. Building on the CMO configurations approach and the question of “what works for whom, and under which circumstances” (Nielsen & Miraglia, 2017; Roodbari et al., 2021a, 2021b, 2023), the empirical analysis attempted to identify under which contextual conditions and for which recipients the intervention was most effective in producing the expected outcomes. To do so, the moderating role of transfer and action plan implementation – conceived as important working mechanisms – was investigated in the relationship between manager support – conceived as an omnibus context factor – and a range of key intervention outcomes, such as teamwork, team coordination, interpersonal conflict at work, and team performance. Moreover, the mediating role of transfer and action plan implementation was investigated in the relationship between peer support towards training transfer – conceived as a discrete context factor – and the same outcome variables, which were considered to be relevant with respect to the intervention’s contents and goals.

Although the expectations were only partially supported, the main findings nevertheless provided some interesting and useful insights that are worth discussing. First, moderation analysis highlighted significant effects of manager support on team coordination and team performance, thus supporting the role of manager support as a contextual factor that is relevant to the intervention’s effectiveness, as also argued by previous studies (e.g., Christensen et al., 2019; Helland et al., 2021; Nielsen et al., 2022). However, the effects of manager support on team coordination and team performance were significant at low levels of action plan implementation. This suggests that, among workers that participated in the delivered intervention, the positive relationship between manager support, on the one hand, and team coordination and team performance, on the other hand, was stronger when action plans developed throughout the intervention were being implemented less. Conversely, such relationship becomes weaker when the intervention participants report higher levels of action plan implementation. In other words, when participants report higher levels of action plan implementation, the influence of manager support on team coordination and team performance tends to disappear. One possible interpretation of this result is that teams implementing action plans developed during the intervention might need less support from their immediate managers in order to coordinate their collective efforts and to accomplish good collective performance. It might be speculated that the implementation of the action plans for improvement of teamwork, crafted in a shared fashion within the participative environment of the workshops held as part of the intervention, might help the teams to “emancipate” from the necessity of receiving support from their immediate managers when having to deliver good team coordination and good team performance. In this sense, the action plans’ implementation might be thought of as contributing to the autonomy and independence of the teams from the support of their immediate managers. After all, the team coaching workshops aim typically to increase autonomy of the team, as

all team members are equally handled and invited to speak out and participate in the team action planning. This process typically promotes their engagement and ownership. However, for future research, it would also be interesting to investigate what role the managers had regarding the action plans in each team – information that was not available for this study. It might be that, in some teams, managers were responsible for setting the plans into action and that thereby is some kind of management support in it. A further explanation may be due to the leadership style of the managers. For instance, the situational leadership theory developed by Hersey and Blanchard (1969) emphasises the need for leaders to adapt their leadership style based on the readiness or maturity of their followers. The theory posits that effective leaders are those who can flexibly adjust their leadership behaviours to meet the evolving needs of their team members. It may be possible that the teams observed in the research had a high readiness level. Or, on the other hand, notwithstanding their readiness levels, their managers could have had a delegating or absent role concerning the implementation of plans related to non-technical skills.

Furthermore, a second interesting result was highlighted by mediation analysis. No significant indirect effect was found in the relationship between peer support towards training transfer and the intervention outcomes – that is, no mediations. However, the findings revealed a few significant direct effects of peer support towards training transfer on transfer, teamwork, team coordination, and team performance respectively. Therefore, transfer and action plan implementation might not necessarily act as explanatory mechanisms within the hypothesised models, but peer support towards training transfer may still constitute a relevant contextual factor contributing to the effectiveness of the examined intervention in determining at least some of the outcomes considered, which is in line with previous literature from the realist evaluation research stream (e.g., Nielsen et al., 2023).

The above results should be taken with caution in light of the limitations of this study. For instance, due to contingencies and diverse organisational needs, some extent of variability occurred in the intervention implementation protocols across the four involved organisations in terms of number of sessions and delivery format. Although implementation differences were slight and did not substantially alter the nature of the intervention, future studies might aim to compare the intervention effects and CMO configurations across different implementation protocols, which may also provide insights into cross-country differences – e.g., between Italy and the Czech Republic. However, this was deemed impossible in this study due to the high turnover and participants' dropout rates across intervention sessions, which resulted in a small sample size and impaired the statistical power of the analyses. However, this issue was difficult to manage since, at least in the large public healthcare institution in Italy, team rotation was a regular organisational procedure that workers had to undergo, whereas both corporate organisations in the Czech Republic went through mergers and acquisitions during the intervention implementation. While it had only minor effects on the set-up of teams in the IT company, it led to major restructuring in the retail company where the majority of teams was reformed. Also,

the generalisability of findings beyond Italian and Czech organisations in the healthcare, retail, IT, and education sectors remains questionable. In terms of external validity, it is also to be considered that the study was conducted during the Covid-19 pandemic, which might have impacted on the way in which workers usually interact. It might be interesting to conduct future studies outside of the pandemic context, but also, in future working life, with more remote work and different ways of organising work, it would be recommended to follow how these affect interactions and outcomes at work. Moreover, the quantitative data that have been used are subjected to self-report bias (Stone et al., 2002) and might provide more narrow insights as compared to nuanced findings that might be looked for by future qualitative studies. In addition, interpersonal conflict at work – a multidimensional construct composed of task conflict and role conflict – and team performance – composed of team support, autonomy, reflexivity, and participation – were computed as single variables and it cannot be excluded that this might have affected the findings. Future studies might delve deeper into this issue by performing differential analyses per each dimension of such constructs, which was however out of the scope of this study. Finally, as mentioned earlier, the digital team coaching intervention was delivered as part of a larger project on multilevel interventions, in the frame of which further interventions were offered during the same time, which might have impacted the results of this study. Nonetheless, the intervention examined here was the only intervention at the group level.

While it has some limitations that should be acknowledged, this study also shows some strengths that are worth highlighting. A first strength of this study resides in its longitudinal design, which allows to make some degree of speculations regarding the causal relationships among the variables investigated. Moreover, in terms of its approach to evaluating workplace interventions, the realist approach presents advantages over the traditional pre-post randomised controlled approach in that it answers not only the question of whether an intervention works – i.e., what the effects are, if any –, but also how it works as intended – i.e., in which context, through which mechanisms, and for which people. This aspect should be of interest to practitioners willing to implement digital team coaching interventions to promote teamwork in their organisations. These professionals are hereby provided with practically relevant knowledge about the omnibus and discrete contextual conditions under which such interventions might be effective – i.e., manager support and peer support towards training transfer – as well as its working mechanisms – i.e., transfer and action plan implementation, so that they can design the implementation environment accordingly, as well as evaluate and adjust the intervention accordingly with the results from the evaluation. For example, when embarking on a digital team coaching intervention project to improve teamwork, practitioners might take particular care of sensitising recipients' managers and peers on the matter in order to ensure their support. Similarly, they might ensure that recipients have the proper opportunities to transfer what they learned during the intervention into their everyday work, as well as to implement the action plans that they developed as part

of it. A further advantage of the realist evaluation, in which no control groups are deployed as this methodological practice is out of scope in this kind of approach, is the prevention of spillover effects between experimental and control groups, as well as the possibility to offer the intervention to all participants interested, not to exclude anyone, thus being less problematic in terms of ethical standards (Nielsen & Miraglia, 2017).

Conclusion

To the best of the authors' knowledge, previous studies have not conducted realist evaluations of digital interventions using social network visualisation and team coaching. Thus, knowledge about the explanatory conditions and working mechanisms involved in the effectiveness of such interventions is still scarce. This study contributed to filling this gap and added to the broader realist evaluation literature and theory. It also aimed to guide future implementation efforts from practitioners. This contribution seems relevant to today's changing world of work, which is more and more relying on the deployment of – either in-presence or remote – teamwork, and where digital interventions might be needed more than ever.

General Discussion

The present doctoral dissertation reported on a three-year research project that investigated a team-level digital-based workplace intervention based on team coaching and social network visualisation (Bahbouh, 2012; Bahbouh & Lasker, 2014). The investigation was carried out through a sequence of four interdependent studies, grounded on Lacerenza and colleagues' (2018) stepwise framework for team interventions, the IGLO model (Day & Nielsen, 2017; Nielsen & Christensen, 2021; Nielsen et al., 2017), the JD-R model (Bakker & Demerouti, 2017; 2018; Schaufeli, 2017), and the realist evaluation approach (Nielsen & Abildgaard, 2013; Nielsen & Miraglia, 2017; Nielsen & Randall, 2013; Nielsen & Shepherd, 2022; Roodbari et al., 2021a, 2021b, 2023). Study 1 was a systematic literature review with a realist synthesis approach about workplace digital interventions at multiple levels, highlighting the need for more research about group-level digital workplace interventions. Study 2 was a qualitative needs assessment exercise that verified the fit between the targeted organisations and the selected intervention. Following the tailored implementation of the intervention, Study 3 analysed recipients' positive perceptions of intervention characteristics, with usability and integrity being appreciated the most, and acceptability being appreciated the least. While the intervention was considered usable and recipients felt valued during sessions, training did not fully meet their expectations. Also, recipients' perceptions did not change from second to fourth session, suggesting they remained stably satisfied with the intervention over time. Finally, Study 4 tested two relevant CMO configurations and suggested that teams implementing action plans developed during training might need less support from immediate managers to coordinate collective efforts and accomplish collective performance. Moreover, peer support towards training transfer was confirmed as a relevant contextual factor contributing to intervention effectiveness. Overall, this multifaceted and complex research project offers a nuanced examination of team-level digital interventions within the contemporary workplace, unveiling valuable insights and opportunities for further refinement and application.

Of course, conclusions should not be drawn from this project's findings without considering a number of limitations. The studies were all based on subjective data collected via self-report measurement techniques, whereas no objective indicators of explored variables were utilised, which makes the interpretation of results prone to biases. Also, research and intervention activities were mostly implemented in a healthcare setting, which limits the generalisability of findings. Furthermore, one notable weakness of the project lies in the link between Study 1 and the subsequent studies; while the systematic literature review shows a stronger focus on digital interventions for workplace mental health, the needs assessment exercise and the implementation and evaluation studies concentrate more on team issues and a digital-based intervention to promote team communication at work. However, the scope of this doctoral project was subordinate to the scope of the larger project it was

tied to (i.e., De Angelis et al., 2020), which was an EU-H2020 project on multilevel interventions to promote mental health in small- and medium-sized enterprises and public organisations. In addition, the impact that teamwork and team communication can have on workers' mental health should not be underestimated (e.g., Bronkhorst et al., 2015; Eguchi et al., 2012; Hopkinson et al., 2021). For example, Bronkhorst and colleagues (2015) indicate that group relationships between co-workers are important in explaining the mental health of healthcare workers. Similarly, Eguchi and colleagues (2012) suggest that promotion of communication in the workplace was associated with reduced psychological distress among workers. Finally, Hopkinson and colleagues (2021) have suggested that effective communication in nursing teams is associated with job satisfaction, an indicator of work-related psychological well-being of healthcare workers. Therefore, while the focus of the project oscillates somewhat between workplace mental health and teamwork, the two topics are actually strictly intertwined, and digital interventions addressing the latter in terms of team communication have the potential to affect dimensions of the former as well.

On the other hand, the project has a number of strengths too, that contribute to its academic and practical value. First and foremost, in line with mainstream literature in the field of workplace interventions (e.g., Nielsen et al., 2010; Nielsen & Noblet, 2018; von Thiele Schwarz et al., 2021), the project followed a stepwise framework for team interventions (i.e., Lacerenza et al., 2018), going from the review of previous evidence to the assessment of team needs, and from the tailored implementation of interventions to the evaluation of implemented actions. This approach ensures a structured, well-organised and methodical exploration of team-level digital interventions, addressing unique needs and challenges faced by organisations nowadays. This framework not only offers a clear path for developing and implementing interventions but also aligns the research with the best practices in the field, enhancing its credibility and relevance. Complementary to that, the adoption of a realist approach towards evaluating interventions (Nielsen & Abildgaard, 2013; Nielsen & Miraglia, 2017; Nielsen & Randall, 2013; Nielsen & Shepherd, 2022; Roodbari et al., 2021a, 2021b, 2023) allowed the project to engage with the complexity of real-world contexts and the intricacies of intervention mechanisms, shedding light on the contextual factors that influence the effectiveness of team interventions. This realist lens enhances the project's ability to unearth the underlying mechanisms and reasons behind the outcomes observed, making it insightful and adaptable to practical settings.

Secondly, the deployment of a mixed-method approach, encompassing both qualitative and quantitative techniques for collecting empirical data, as evidenced throughout the four studies, represents a methodological strength. This approach amalgamates the strengths of both qualitative and quantitative research methods, enriching the research comprehensiveness and depth. It allows for a nuanced understanding of the multifaceted dynamics at play within digital workplace interventions. The mixed-method approach also enables the research project to triangulate findings, thereby bolstering the overall robustness and validity of the

research outcomes. In sum, the synergistic use of Lacerenza and colleagues' (2018) stepwise framework for team interventions, a realist perspective and a mixed-method approach collectively underpin the strengths of this doctoral project, ensuring that it stands as a theoretically grounded, methodologically sound, and practically impactful contribution to the understanding and enhancement of digital-based team interventions in contemporary organisations.

More specifically related to the setting where the present research project took place, another strength is the project's prevalent contextualisation within the healthcare domain. Teamwork is topical in healthcare, which is a complex, demanding, diverse, interdisciplinary, critical, and specialised sector. Here, the importance of effective teamwork is understood considering its positive relationships with indicators of patients' health, safety, and well-being (Rosen et al., 2018). In turn, if patients' health and safety is ensured, this allows to avoid unnecessary human and social costs, as well as economic and financial burdens to the wider healthcare system. In addition, healthcare teams have been heavily hit by the recent Covid-19 pandemic outbreak, often in ways that can be described as unprecedented (Zajac et al., 2021). Therefore, conducting empirical research into factors of effective teamwork in the healthcare sector might provide unedited insights of both theoretical and practical relevance. Previous literature has shown the crucial role of communication as a main factor for several indicators of team effectiveness (Marlow et al., 2018) and this has not been different in research about team communication in the healthcare sector (Evin et al., 2018; Fowler et al., 2021; Hopkinson et al., 2021; Molleman et al., 2010). For instance, Hopkinson and colleagues (2021) have suggested that effective communication in nursing teams is associated with quality of patients' treatment, an indicator of job performance in the healthcare domain.

Additional strengths of the presented research reside in the contents of the specific intervention that was implemented throughout Study 3 and Study 4, that is, a team-level digital-based workplace intervention based on team coaching and social network visualisation (Bahbouh, 2012; Bahbouh & Lasker, 2014). First, the intervention taps into relevant dimensions of team communication in the workplace. As argued by previous contributions to the literature (e.g., Franc et al., 2019; Grote & Kozlowski, 2023), communication as a factor of team effectiveness should not be considered *per se*. Rather, it is the quality of communication that determines the extent to which communication impacts the work of the team in a desirable manner. Particularly, high-quality communication is defined in terms of its (1) clarity – that is, communication should be understandable for its receivers, (2) timing – that is, communication should be delivered neither too early nor too late with respect to the tasks it refers to, (3) relevance – that is, communication should constitute a meaningful contribution to those tasks, and (4) frequency – that is, communication should occur neither too often nor too infrequently, in order to avoid, on the one hand, information overload and unmanageability, and, on the other hand, failing to provide timely information about the tasks it relates to. Given the questions administered to generate sociomaps – for instance, “How often do you communicate with your colleagues

about work-related topics?”, “How often would you like to communicate with your colleagues in order to work most effectively?”, “How important is communication with your colleagues to deliver high-quality outputs?”, “Evaluate the quality of work communication, taking into account its relevance, content, and timeliness” –, it can be seen that the implemented intervention addresses not only the quantity but also the quality of communication in teams. Moreover, the use of digital sociomaps itself constitutes an interesting aspect of such intervention. Previous empirical studies (Abildgaard & Nielsen, 2018; von Thiele Schwarz et al., 2017) have shown beneficial impacts of material artifacts and visual devices on intervention effectiveness. Then, the delivered intervention and the presented research extend the scope of these strategies’ potential to digital artifacts, offering insights into the design of workplace digital intervention environments.

Overall, this research project holds several implications both for theory and practice. The project contributes to supporting Lacerenza and colleagues’ (2018) stepwise framework for team interventions, the IGLO model (Day & Nielsen, 2017; Nielsen & Christensen, 2021; Nielsen et al., 2017), the JD-R model (Bakker & Demerouti, 2017; 2018; Schaufeli, 2017), and the realist evaluation approach (Nielsen & Abildgaard, 2013; Nielsen & Miraglia, 2017; Nielsen & Randall, 2013; Nielsen & Shepherd, 2022; Roodbari et al., 2021a, 2021b, 2023) as useful and actionable theoretical frameworks that can be recommended for continued deployment in future research. Also, the project’s findings advance the available knowledge regarding team-level digital-based workplace interventions, particularly focussing on possible boundary conditions of their effectiveness, such as contextual factors and working mechanisms, in spite and because of previous academic literature having paid little attention to the matter. In so doing, the project answers recent literature’s call for performing the evaluation of workplace intervention processes and working mechanisms beyond the mere evaluation of outcomes (ibidem). Realist evaluations of digital-based workplace interventions are not frequently mentioned in literature, with Havermans and colleagues’ (2018) process evaluation of a digital platform-based implementation strategy aimed at work stress prevention in healthcare representing a rare exception. Particularly, such approach had not yet been applied to investigate group-level digital interventions. Therefore, this research contributes to meeting the need for a deeper understanding of what intervention elements work and why, in which circumstances, settings, and for whom they are especially successful or, rather, fail. For the same reason, practitioners – encompassing employers, policymakers, human resource managers, work and organisational psychologists, and various professionals in the workplace – are hereby provided with suggestions on how to design, develop, implement, and evaluate digital-based interventions to promote teamwork.

Finally, despite its valuable contributions, this project still leaves some open questions and thus paves the way for future research in the field of digital-based interventions for teams in the workplace. Building upon the four studies described in this dissertation, future studies might attempt to perform similar research starting

from different methodological choices and explore whether more robust conclusions can be achieved. In particular, reviewing previous literature based on a search string more focussed on teamwork and team interventions, as well as the collection of less subjective and self-reported data, might be considered. Future studies might also try to expand the knowledge gathered throughout this project's studies by, on the one hand, examining additional factors for the effectiveness of the same digital team coaching intervention delivered – for example, recipients' perceptions of different intervention characteristics, the impact of other context factors and working mechanisms –, and, on the other hand, exploring the same factors in relation to a different digital-based team-level intervention. Finally, other organisational sectors and settings and working populations might be targeted in future studies. In conclusion, the present doctoral project not only contributes to filling some gaps from past academic research, but hopefully inaugurates constructive directions for further scientific investigation.

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Investigating the Role of Stress-Preventive Leadership in the Workplace Hospital: The Cross-Sectional Determination of Relational Quality by Transformational Leadership

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Claas Lahmann,
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Christine Allwang,
Technical University of Munich,
Germany

*Correspondence:

Felicitas Stuber
felicitas.stuber@med.uni-tuebingen.de

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Felicitas Stuber^{1*}, Tanja Seifried-Dübon¹, Monika A. Rieger², Contributors of the SEEGEN Consortium, Stephan Zipfel¹, Harald Gündel³ and Florian Junne¹

¹ Department of Psychosomatic Medicine and Psychotherapy, Medical University Hospital Tübingen, Tübingen, Germany,

² Institute of Occupational and Social Medicine and Health Services Research, University Hospital Tübingen, Tübingen, Germany, ³ Department of Psychosomatic Medicine and Psychotherapy, Ulm University Medical Center, Ulm, Germany

Introduction: A good relationship quality between leaders and staff members promotes mental health and prevents stress. To improve the relationship quality, it is important to identify variables which determine relationship quality at the workplace. Therefore, this study aims to identify specific leadership characteristics which support the development of a positive relationship between hospital leaders and staff members.

Methods: A cross-sectional study design was applied. A total number of 1,137 leaders ($n = 315$) and staff members ($n = 822$) of different professions (physicians, nursing staff, therapeutic professionals, administration staff, IT staff, clinical services, office assistants, scientists, others) working at a tertiary hospital in Germany assessed transformational leadership style as a staff-oriented leadership style and leader-member relationship quality by self-report questionnaires [integrative leadership questionnaire (FIF), leader-member exchange (LMX-7) questionnaire]. The data were statistically analyzed by mean comparisons and a multiple linear regression analysis.

Results: Leaders rated their own transformational leadership style ($M = 3.98$, $SD = 0.43$) systematically higher than staff members assessed their leader ($M = 2.86$, $SD = 1.04$). Evaluation of relationship quality showed similar results: leaders evaluated their relationship quality to one exemplary staff member higher ($M = 4.06$, $SD = 0.41$) than staff members rated their relationship quality to their direct leader ($M = 3.15$, $SD = 0.97$). From the staff members' perspective, four sub-dimensions of transformational leadership, that is, "individuality focus," "being a role model," "fostering innovations," and "providing a vision" showed large effect sizes in the regression analysis of relationship quality ($R^2 = 0.79$, $F(14,690) = 189.26$, $p < 0.001$, $f = 1.94$).

Discussion: The results of our study are in line with previous investigations in other working contexts and point to a profession-independent association as the professional

group of participants did not contribute to the variance explanation of the regression analysis. The exploration of potential determinants of relationship quality at work can, for example, support the development of leadership training programs with a focus on transformational leadership style. This might be an opportunity to foster high relationship quality between leaders and staff members and consequently might represent one strategy to prevent stress in the health care sector.

Keywords: transformational leadership, relationship quality, health care sector, staff members, leaders

INTRODUCTION

Considering the maintenance of employees' mental health as an operational task, and thus as a leader's task, has indeed an ethical aspect and is also a legal imperative in Germany. In 2012, the legal obligation of German employers to assess and reduce psychological health risks at the workplace was substantiated by an amendment of the respective German occupational health and safety act (1). Accordingly, the employer has to judge the risk to which employees are exposed to at their workplace including psychological stress at work and to determine which measures of occupational safety and health are necessary to reduce this risk. With regard to psychological stress at work, working conditions as well as social relationships (e.g., workplace bullying and harassment) and the working culture have to be addressed (2), with leadership being one important aspect.

Empirically, leadership has been found to be an important variable in relation to job performance (3, 4) as well as employees' health (5, 6). That is, different leadership styles are differentially associated with employees' job performance and mental health. Destructive leadership is defined as a deleterious behavior against a person and/or an organization in an active or passive way (7). It reduces productivity and has detrimental effects on the health of staff members (8), whereas appreciative leadership behavior leads to a higher work satisfaction (9), higher intention to stay at the present workplace (10), and higher well-being of staff members (11–13) as well as to improvements in leaders' own well-being (14).

A unifying characteristic of all these staff-oriented leadership behaviors is the importance of the relationship between leaders and staff members. A leadership approach that elaborates on this dyadic relation between direct leaders and their staff members is the leader–member exchange (LMX) approach [for an overview, see (15)]. The LMX approach targets the specific and individual dyad between one leader and one staff member. Thus, relationship quality between a leader and his/her various staff members can differ (16) and the development of the dyadic relationship can be described as a continuous process [e.g., Refs. (15, 17)].

A mature relationship has been positively related to several positive health and performance-oriented outcomes for staff members: for example, job performance (18), procedural distributive justice (19), and general job satisfaction (20). On the other hand, mature relationships were negatively related to turnover intention and role conflicts at work (3). Consequently, a mature relationship between leader and staff member is

preferable at the workplace, although high relationship quality can be perceived as a rather abstract construct without clear recommendations on how to establish such relationships on a behavioral basis (21). Thus, research has tried to reveal factors that contribute to a mature leader–staff member relationship at the workplace on the part of staff members and leaders (3, 21). Although good relationships at the workplace are not only stress preventive for staff members but also for leaders, we decided to concentrate in this study on stress preventive implications for staff members (22).

In this study, we focused on behavioral leadership characteristics which have been found to be subject to change (23) and could explain a substantial variance of the quality of the leader–member relationship (3). The leadership style that has been found to be associated with mature leader–member relationships (3) is known as transformational leadership (24). Transformational leadership behavior is an appreciative and toward personal growth-oriented leadership style aiming to motivate staff members through, for example, long-term aims and adjustment of values. It supports staff members to focus not only on individual goals but also on group and organizational goals (25). Transformational leadership comprises six different core behaviors (25–27), which have been labelled as “fostering innovation,” “team spirit development,” “performance development,” “individuality focus,” “providing a vision,” and “being a role model” (28).

Empirically, transformational leadership behavior shows robust relations to performance-oriented and health-oriented outcomes. Specifically, transformational leadership is associated with increased job performance (27, 29), work-related satisfaction, and motivation [e.g., Refs. (29, 30)], attachment to the leader (31), fewer days of absence due to sickness, and fewer critical incidents at the hospital [e.g., Ref. (32)], as well as less perceived stress and higher well-being [e.g., Refs. (33–36)].

Although the association of transformational leadership in general with improved quality of the leader–member relationship (LMX) seems well supported by the current literature (37–39), the specific sub-dimensions of the transformational leadership approach that foster the quality of the leader–member relationship have not been well researched to date. Furthermore, evidence is lacking especially with regard to specific working contexts and professional groups, such as at the workplace hospital.

To explore determinants that could be associated with higher relationship quality between leaders and staff members at the workplace hospital seems to be an important point as

relationship quality between the direct leader and her staff members is one of the few working conditions which can be influenced by leaders and staff members themselves and therefore constitutes an opportunity for stress prevention (33). As the workplace hospital is a psychologically demanding workplace where studies showed an increasing burnout and depression level in physicians (40) and where chronic work overload was also associated with poorer patient care (41, 42), maintaining psychological health, e.g., by strengthening relationship quality is of particular importance. Although professional groups within the workplace hospital differ in their every day work, they are unified by the fact of social interaction and relationships between leaders and staff members. Thus, further research is needed to clarify the specific determinants, as part of the transformational leadership behavior, that lead to improved leader-member relationships at this specific organization [for an overview on the relevance of context see Ref. (43)].

Therefore, this study was conducted in the context of a tertiary hospital in Germany to examine the association between transformational leadership sub-dimensions with the quality of the perceived leader-member relationship.

The study aims to answer the following research questions:

1. How do leaders perceive the quality of their relationship with staff members and vice versa?
2. Does the perception of transformational leadership differ between leaders and staff members?
3. In which way are the sub-dimensions of transformational leadership behavior associated with the quality of leader-member relationships from the view of staff members in the workplace hospital?

MATERIALS AND METHODS

Implementation

A cross-sectional online survey was conducted from May 23, 2018, to July 18, 2018, and was approved by the ethics committee of the University Hospital and Medical Faculty of Tübingen (622/2017BO2) as well as by the chief executive board and the employees' council of the tertiary hospital. Completion time for the online survey was about 10 min. Overall, $N = 10,101$ employees received the survey invitation and the response rate was 11.26%.

Materials

We created an online survey with questions on transformational leadership behavior and relationship quality using validated standardized instruments delivered *via* the Unipark survey software (QuestBack GmbH). Questions on both aspects were asked either from the leaders' or the staff members' perspective. That is, leaders evaluated their own leadership behavior, whereas staff members assessed their direct supervisor. To discriminate participating employees according to their hierarchy level, employees had to define themselves either as leaders or as staff

members. Yet, there was no possibility to assess leaders and their directly associated team due to data protection requirements.

Questionnaire Assessing Transformational Leadership

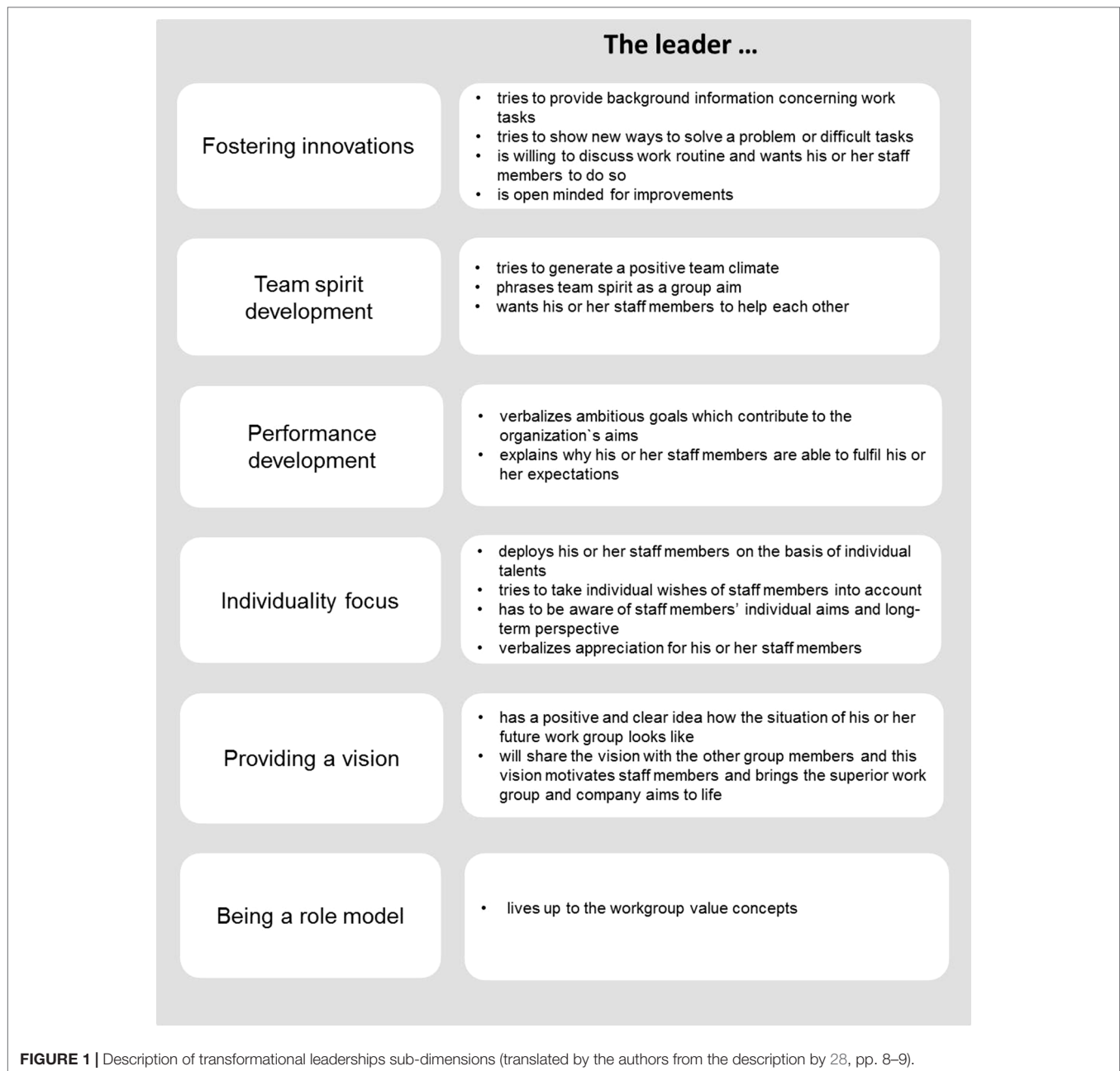
The questionnaire used to assess the sub-dimensions of transformational leadership was the "integrative leadership" questionnaire (Fragebogen zur Integrativen Führung, FIF) (28), a standardized instrument which measures leadership and communication style in four modules. In our survey, we applied transformational leadership as one part of the "integrative leadership" questionnaire. The construct of transformational leadership in the questionnaire draws on the concept of Heinitz and Rowold (26) and Ref. (25), see **Figure 1** for more details). Participants were asked to rate 32 statements using a five-point Likert scale from 1 (agree not at all) to 5 (totally agree). The item ratings can be summarized in six different scale scores or in one overall transformational leadership score. The scales of transformational leadership show a sufficient internal consistency with Cronbach's $\alpha = 0.83$ – 0.92 for the staff members' assessment provided by the manual (28) and Cronbach's $\alpha = 0.86$ – 0.94 for the staff members' assessment by our study. In addition, Cronbach's $\alpha = 0.75$ – 0.83 for the leaders' assessment provided by the manual (28) and Cronbach's $\alpha = 0.67$ – 0.81 for the leaders' assessment by our study. The convergent validity of the transformational leadership scale of the FIF was confirmed by high correlations with the frequently used questionnaire Transformational Leadership Inventory (TLI) (25, 26).

Questionnaire Assessing the Quality of the Leader-Member Relationship

The LMX-7 questionnaire (15, 44) in its German version is based on the LMX model (15) which represents the relationship quality between leaders and staff members. It is a standardized unidimensional scale with seven items. Participants are asked to rate seven questions and statements on a five-point Likert scale from 1 (low relationship quality) to 5 (high relationship quality) either in a version for leaders to assess the relationship quality to one exemplary staff member or in a version for staff members to assess the relationship quality to their direct leader. Graen and Uhl-Bien (15) postulated that the LMX-7 measures the three highly correlated relationship aspects respect, trust, and obligation as one LMX dimension. The ratings of the participants can be summarized and presented through one overall LMX score. The LMX-7 has shown high internal consistency for staff members' ratings (Cronbach's $\alpha = 0.89$ and $\alpha = 0.92$), whereas internal consistency was not reported for leaders' rating (44). In our study, LMX-7 showed an internal consistency of $\alpha = 0.74$ for leaders and $\alpha = 0.93$ for staff members.

Statistical Analyses

For the description of the participants as well as for descriptive specifications of leadership behavior and relationship quality, mean (M), percentage (%) and distribution in the form of standard deviation (SD) were applied. To compare leaders' and staff members' ratings, we used *t*-tests as the data satisfied the



condition of normal distribution. To determine the effect size of mean comparisons, Cohen's d was applied. A result of $d \leq 0.2$ can be interpreted as a small, $d \leq 0.5$ as a medium, and $d \leq 0.8$ as a large effect size (45). Moreover, a multiple linear regression was conducted to explore the association between transformational leadership subdimensions and LMX overall score. Assumptions of multiple regressions (linearity, normality, homoscedasticity, and independence of residuals) were checked, and f was reported for the effect size. A result of $f \leq 0.10$ can be interpreted as a small, $f \leq 0.25$ as a medium and $f \leq 0.40$ as a large effect (46). The level of significance was set for all analyses to $\alpha = 0.05$, and all analyses were conducted by using IBM SPSS version 25. For

multiple comparisons, we adjusted alpha levels by Bonferroni correction. Total scores of transformational leadership behavior and relationship quality were only calculated when no missing values occurred in sub-dimensions. Concerning the linear multiple regression, cases were only included when no values of subdimensions and total scores were missing. As the variable *Professional Group* was categorical with the categories: physicians, nursing staff, therapeutic professionals, administration staff, IT staff, clinical services, office assistants, scientists, and other professions, dummy coding was used for the linear multiple regression. For the baseline group, the category Administration staff was chosen as this professional group was the largest.

A dummy variable is defined in our linear multiple regression as the difference in relationship quality perception for the administration staff and one other professional group [either physicians or nursing staff or therapeutic professionals or IT staff or clinical services or office assistants or scientists or other professions; for a detailed description of dummy coding, see Ref. (47), p. 208–215].

RESULTS

Population

A total of 1,137 employees of a tertiary hospital in Germany participated in the study, with 315 (27.7%) identifying themselves as leaders and 822 (72.3%) as staff members without leadership responsibilities. Of the staff members, 554 (74.8%) were female and 187 (25.2%) were male, whereas in the leader group 174 (59.6%) were female and 118 (40.4%) were male. One hundred four participants provided no information on their gender. For detailed information on the characteristics of the participants, see **Tables 1–2**.

TABLE 1 | Age group frequencies depending on hierarchy level.

Age groups in years	Hierarchical group			
	Staff members		Leaders	
	%	n	%	n
<20–24	3.7	30	0.3	1
25–30	15.1	123	3.5	11
31–35	11.2	91	9.3	29
36–40	12.6	103	12.2	38
41–45	8.6	70	15.1	47
46–50	13.7	112	13.5	42
51–54	15.1	123	17.6	55
>55	20.1	164	28.5	89

%, percent; n, number of participants; n = 6 staff members and n = 3 leaders didn't provide information on their age, N = 1128.

Transformational Leadership Behavior at the Hospital

Leaders (M = 3.98, SD = 0.43, n = 275) and staff members (M = 2.86, SD = 1.04, n = 737) differed significantly in their perception of the total transformational leadership score at their workplace [$t(1,000.31) = -24.21, p < .001, d = 1.23$]. Leaders assessed themselves as leading more transformational than the staff members evaluated their direct leaders. This result was seen for all sub-dimensions as well: leaders rated themselves in all dimensions higher than staff members evaluated their leaders (see **Table 3**). Expect one sub-dimension (performance development) which revealed a medium size effect, all other sub-dimensions showed a high effect size.

LMX at the Hospital

Leaders and staff members perceived the relationship quality between leaders and staff members at the hospital in significantly different ways [$t(1,054.83) = -21.68, p < .001$]. Leaders (M = 4.06, SD = 0.41, n = 293) rated the relationship quality they offered to one exemplary staff members higher than the subordinates rated their relationship quality with their direct leaders (M = 3.15, SD = 0.97, n = 777).

Sub-Dimensions of Transformational Leadership as Potential Determinants of Relationship Quality from a Staff Members' Perspective

Linear multiple regression analysis was applied to assess the extent to which the sub-dimensions of transformational leadership behavior determine the variance of the perceived relationship quality at the hospital from a staff members' perspective. Professional groups of the staff members (see **Table 2**) were also entered as dummy variables into the linear multiple regression to control potential professional related differences in the association of transformational leadership and relationship quality. All assumptions of multiple regression analysis were met, and predictors were all entered simultaneously

TABLE 2 | Proportion of professional groups depending on hierarchy level and depending on professional groups overall.

Professional groups	Hierarchical level				Overall	
	Staff members		Leaders		%	n
	%	n	%	n		
Physicians	53.8	84	46.2	72	13.7	156
Nursing staff	67.6	142	32.4	68	18.5	210
Therapeutic professionals ^a	80.8	59	19.2	14	6.4	73
Administration	70.4	157	29.6	66	19.6	223
IT	78.9	56	21.1	15	6.2	71
Clinical services ^b	72.7	8	27.3	3	1.0	11
Office assistants	89.3	100	10.7	12	9.9	112
Scientists	77.0	87	23.0	26	9.9	113
Others	76.8	129	23.2	39	14.8	168

%, percent; n, number of participants, N = 1137.

^ae.g. physiotherapist, psychotherapist.

^be.g. caretaker service, catering.

TABLE 3 | Leaders' and subordinates' ratings of transformational leadership sub-dimensions.

Sub-dimensions of TFL	Staff members			Leaders			<i>t(df)</i>	<i>p</i>	<i>d</i>
	<i>M</i>	<i>SD</i>	<i>n</i>	<i>M</i>	<i>SD</i>	<i>n</i>			
Fostering innovations	3.15	1.10	811	4.27	0.50	307	(1,082.34) = -23.38	<.001	1.15
Team spirit development	2.84	1.18	805	4.07	0.60	304	(1,017.23) = -22.69	<.001	1.17
Performance development	2.89	1.09	794	3.66	0.73	300	(799.81) = -13.30	<.001	0.77
Individuality focus	2.78	1.22	805	4.02	0.61	307	(1,038.91) = -22.41	<.001	1.14
Providing a vision	2.55	1.17	798	3.57	0.74	306	(872.58) = -17.26	<.001	0.96
Being a role model	3.00	1.30	803	4.35	0.55	299	(1,085.18) = -24.26	<.001	1.18

TFL, transformational leadership; *M*, mean; *SD*, standard deviation; *n*, number of included participants; *t*, *t*-test statistic; *df*, degrees of freedom; *p*, *p*-value; *d*, Cohen's *d*.

into the model (see **Table 4** for correlations of continuous variables). The result of the linear multiple regression analysis is presented below in **Table 5**.

The total variance of relationship quality that could be explained by this model was 79% [$R^2 = 0.79$, $F(14,690) = 189.26$, $p < .001$] which corresponded to a large effect ($f = 1.94$). The sub-dimensions "fostering innovation," "individuality focus," "providing a vision," and "being a role model" were included as significant determinants of the variance explanation. Standardized beta values (β) revealed that on a single factor level the sub-dimensions "individuality focus" and "being a role

model" made the strongest contribution to explain the variance of relationship quality.

DISCUSSION

To our knowledge, this is the first study which investigates sub-dimensions of transformational leaderships and the quality of leader-member relationships across all professions in the workplace hospital from leaders' and staff members' point of view. Leaders and staff members' perception of

TABLE 4 | Intercorrelations of transformational leadership sub-dimensions and relationship quality from a staff members' perspective.

Variables	1	2	3	4	5	6	7
1. Staff members' total LMX	—	0.78***	0.76***	0.66***	0.84***	0.77***	0.80***
2. Fostering innovations		—	0.78***	0.70***	0.75***	0.77***	0.77***
3. Team spirit development			—	0.71***	0.74***	0.74***	0.78***
4. Performance development				—	0.62***	0.75***	0.70***
5. Individuality focus					—	0.74***	0.73***
6. Providing a vision						—	0.78***
7. Being a role model							—

Pearson correlations for staff members ($n = 705$) are presented above the diagonal. *** $p < .001$.

TABLE 5 | Linear multiple regression analysis for staff members' perception of relationship quality.

Sub-dimensions	<i>B</i>	<i>SE(B)</i>	β	<i>t</i>	<i>p</i>	<i>CI(B)</i>
Constant	0.93	0.06	—	14.49	<0.001	0.80–1.05
Admin. vs Physicians	-0.11	0.07	-0.04	-1.70	0.09	-0.24 to 0.02
Admin. vs nursing staff	-0.12	0.06	-0.04	-1.92	0.06	-0.22 to 0.00
Admin. vs Therapeutic professionals	-0.02	0.07	-0.06	-0.32	0.75	-0.17 to 0.12
Admin. vs IT staff	-0.02	0.08	-0.01	-0.24	0.81	-0.18 to 0.13
Admin. vs Clinical services	0.11	0.23	0.01	0.50	0.62	-0.34 to 0.56
Admin. vs Office assistants	-0.08	0.06	-0.03	-1.24	0.22	-0.20 to 0.05
Admin. vs Scientists	-0.04	0.07	-0.01	-0.056	0.58	-0.17 to 0.09
Admin. vs Other professions	0.04	0.06	0.02	0.69	0.49	-0.07 to 0.15
Fostering innovation	0.11	0.03	0.13	3.66	<0.001	0.05–0.17
Team spirit development	0.05	0.03	0.07	1.96	0.05	0.00–0.11
Performance development	0.02	0.03	0.02	0.63	0.53	-0.03 to 0.07
Individuality focus	0.35	0.02	0.43	14.52	<0.001	0.30 to 0.39
Providing a vision	0.08	0.03	0.09	2.62	<0.01	0.02 to 0.13
Being a role model	0.19	0.03	0.25	7.55	<0.001	0.14 to 0.24

B, unstandardized coefficient; *SE*, standard error of *B*; β , standardized coefficient Beta; *t*, *t*-test; *p* = *p*-value; *CI*, confidence interval of *B*, $n = 705$ subordinates; Admin., Administration staff, $R^2 = 0.79$, $F(14,690) = 189.26$, $p < .001$.

transformational leadership and relationship quality at the workplace hospital differed significantly on an overall basis and at a dimensional level. That is, leaders rated transformational leadership behavior and relationship quality higher than the staff members of the same hospital did. Furthermore, the results provide insight into the association between the sub-dimensions of transformational leadership and relationship quality from a staff members' point of view: The sub-dimensions "individuality focus," "being a role model," "fostering innovations," and "providing a vision" explained 79% of the variance of the perceived relationship quality, whereas the professional group of staff members could not contribute to the variance explanation.

When comparing our rating results of transformational leadership to the results of a representative sample of German leaders and subordinates provided by the manual of the questionnaire of integrative leadership (FIF) (28), the ratings of our sample can be located in the lower half of the average range. That is, transformational leadership was perceived as average in our sample with a tendency to lower staff members' ratings.

Relationship quality has been examined with the here used questionnaire LMX-7 in the health sector before (48). Research showed scale values for staff members' perception of LMX relating to their direct leader in the medium range between 3.34 and 3.36 (21) and 3.32 (49). Our results are comparable to these study results with the tendency to lower staff members' ratings parallel to the ratings of transformational leadership. Although our rating results seem at least comparable to other study results, taking into account relationship qualities' impact on staff members' well-being (50) and the potential improvement through transformational leadership with regard to fewer undesirable patient outcomes (e.g., medication errors), more job satisfaction (30), and higher occupational and patient safety culture in hospitals (51) an increasing rate of transformational leadership behavior and relational quality might be seen as desirable for the workplace hospital.

According to the rating discrepancy between leaders and staff members previous studies discussed that employees tended to rate their job performance more positively and less variably in self-assessments compared with other sources (e.g., peers, supervisors, subordinates) because of more indulgence and less discriminant validity (52). This result seems in line with our findings where leaders rated their transformational leadership behavior more positively and had less variance in their assessments than staff members showed in their ratings of transformational leadership behavior of their direct leaders. The ratings of the participating leaders in our sample could be contaminated by social desirability, similar to the results of Sarros et al. (53) who found significant correlations between personality characteristics (e.g. courage, compassion) and social desirability in leaders' self-assessments.

Aside from this potential bias, it is worthwhile to discuss the meaning of such different perceptions of leaders and staff members concerning transformational leadership on an organizational level. Aarons et al. (54) interpreted these different perceptions as clues to the organizational culture

quality. The results of their study showed an association between transformational leadership rating and organizational culture: the higher the rating discrepancy between leaders and staff members, the worse the organizational culture was, especially when leaders rated themselves as better than their staff members did. This shows the need to shorten the rating distance between leaders and staff members, although leaders' and staff members' rating cannot be related to each other directly.

To get a better understanding of what leaders can contribute to relationship quality from a staff members' perspective, we ran a regression analysis with the result that four sub-dimensions of transformational leadership behavior ("individuality focus," "being a role model," "fostering innovations," and "providing a vision") significantly determined the relationship quality between leaders and staff members, whereas the professional group of the staff members did not contribute to the variance explanation. These findings may support the theoretical assumptions and empirical approaches of previous research that transformational leadership is associated positively with the LMX model (15, 20, 24, 38).

To discuss and classify the impact of the four sub-dimensions of transformational leadership on relationship quality a comparison to other study results concerning the dimensions "individuality focus," "being a role model" and "providing a vision" is possible whereas the dimension "fostering innovation" has not been found to determine relationship quality before. That is, the explanation for the impact of the dimension fostering innovation is rather speculative. The effect of the dimension "fostering innovation" could be explained by the health care sector as study context: Employees working there could show a higher affinity to innovations in general as improving patient care through innovative treatment methods can be seen as one important part of medical advance which is important for employees' every day work in the health care sector. Although the association of "fostering innovations" and relationship quality has not been explained explicit yet, this dimension has been associated significantly positive to other staff-oriented variables like job satisfaction, affective commitment and organizational citizenship behavior (28).

The dimensions "individuality focus" and "being a role model" could explain a considerable higher part of variance than "fostering innovations" in the performed regression analysis. Both aspects could be seen as a part of high employee orientation and are in line with other empirical approaches. Deluga (55) examined the relationship of transformational leadership and relationship quality on a sub-dimensional level as well. He found on the basis of the four factorial transformational leadership model (56) the sub-dimension "charisma" [corresponding to parts of the sub-dimension "providing a vision" and "being a role model" in our study; see Ref. (28)] and "individual consideration" [corresponding to the sub-dimension "individuality focus" in our study; see Ref. (28)] as two predictors for relationship quality in the military context. Yukl et al. (57) showed in their study that the transformational leadership sub-dimensions "leading by example" [corresponding to the sub-dimension "being a role

model” in our study; see Ref. (28)] could explain parts of the variance of relationship quality.

Our results revealed comparable sub-dimensions of transformational leadership related to relationship quality for the hospital context as Deluga (55) found for the context of the U.S. Navy. This concordance has been shown despite very different working contexts and thereby could lead to the assumption that the relation of transformational leadership subdimensions and relationship quality could be quite independent of the working context. The idea of generalization is also supported by the result of our regression analysis that the professional group of staff members did not contribute to the variance explanation of relationship quality. The association between transformational leadership and relationship quality is independent of the professional group in our study. Future investigations could examine this aspect further by including first and secondary care hospitals or focusing on other sectors. For example, the economic sector where leaders have more direct access to monetary resources, as studies have shown that transformational leadership style is especially relevant when leaders have no direct access to monetary reward systems (29) and when workplaces are more hierarchically structured (36), which are both applicable for our study as well as for Deluga’s (55) study context but won’t fit to the economic sector in the same way.

Further research is needed to investigate the effect level of sub-dimensions of transformational leadership behavior (e.g. individual level, dyadic level, group level or organizational level). Seltzer and Bass (58) assumed that the sub-dimension “charisma” and thus also the sub-dimension e.g., “providing a vision” mainly have an effect on a dyadic level as well as the outcome variable relationship quality. We assume that “individuality focus” and “being a role model” could also show an effect on a dyadic level as they can be perceived as the relationship-based sub-dimensions of transformational leadership.

Limitations

First, ratings of leaders and staff members cannot be associated directly with each other (the leaders rated by staff members might not be the ones that have participated in the study). That is, it could be possible that the most transformational leaders and the most unsatisfied staff members participated and distorted the survey results in the respective directions. Future studies should aim to enable the connection between a leader’s self-ratings and the ratings of their actual respective staff members. Second, future investigations need to use more than just one measurement method (e.g., self report questionnaires and qualitative data from outside observers). As the exclusive use of self-report tools is an important limitation of our study. Third, we had a low response rate, and participation in our survey was voluntary, which may also have rendered the sample less representative with, for example, the more motivated employees participating. Fourth, the cross-sectional design hinders causal inference from the study results but gave the opportunity to consider the relation of transformational leadership sub-dimensions and relationship

quality without adding any temporal variables in this early stage of study (59). Another point is the high proportion of variance explanation in the regression analysis which could be a hint for overestimation of the relation between the sub-dimensions of transformational leadership and relationship quality although the two constructs can be distinguished by their theoretical background: Whereas transformational leadership focuses on leadership behavior, the model of relationship quality refers to the relationship between leaders and staff members. Despite this potential overestimation, the investigated association can be seen as one important part of relationship quality research besides other examined determinants like subordinates’ characteristics, interactional characteristics, and context variables (60).

To sum up, the hypotheses that can be raised from our results may well justify future studies that employ interventional longitudinal designs to enlighten the effects of transformational leadership on relationship quality as well as the by now theoretical based assumption that there is an opportunity to prevent stress by fostering relationship quality.

Practical Implication

This study explored specific determinants of relationship quality in the workplace hospital to explore opportunities to enhance relationship quality. Based on our results first, leaders should remember that their transformational leadership behavior could have an impact on the relationship quality with their staff members. And that by fostering the relationship quality, an opportunity to prevent stress in their staff members comes along. Second, leaders should get the opportunity to participate in leadership training programs to reflect, develop, and improve their transformational leadership skills. Studies have already shown that transformational leadership can be improved by leadership interventions (61, 62).

The next step should be to assess whether this can lead to a change in perceived relationship quality as an important working condition regarding staff members’ well-being in the health care sector. Besides other important measures (e.g. reduction of high quantitative demands, improving personnel shortage, addressing the hazardous of working with critical ill patients), this ultimately might represent one of the promising strategies to prevent stress-related disorders in the health workforce.

CONTRIBUTORS OF THE SEESEN CONSORTIUM

Eva Rothermund, Nadine Mulfinger, Mark Jarczok, Department of Psychosomatic Medicine and Psychotherapy Ulm University Medical Center, Ulm, Germany; Peter Angerer, Institute of Occupational and Social Medicine University Hospital Düsseldorf, Düsseldorf, Germany; Imad Maatouk, Clinic of General Internal Medicine and Psychosomatic, Heidelberg, Germany; Andreas Müller, Faculty of Educational Science Work and Organizational Psychology, Essen, Germany; Bernd Puschner, Department of Psychiatry and Psychotherapy II Ulm

University and Bezirkskrankenhaus Günzburg, Ulm, Germany; Jochen Schweitzer-Rothers, Institute for Medical Psychology University Hospital Heidelberg, Heidelberg, Germany; Stefan Süß, Department of Business Administration, in particular Work, Human Resource Management and Organization Studies Heinrich-Heine-University Düsseldorf, Düsseldorf, Germany; Ute Ziegenhain, Clinic of Child- and Adolescents Psychiatry/ Psychotherapy University Hospital Ulm, Ulm, Germany.

DATA AVAILABILITY

The implementation of this study had to be approved by the chief executive board and the employees' council of the tertiary hospital. This approval required that raw data were only made available to direct project associates.

ETHICS STATEMENT

The study was approved by the ethics committee of the University Hospital and Medical Faculty of Tübingen (622/2017BO2) named Ethik-Kommission an der Medizinischen Fakultät der Eberhard-Karls-Universität und am Universitätsklinikum Tübingen and was carried out in accordance with the recommendations of the ICH-GCP-guidelines, Declaration of Helsinki. All subjects gave written informed consent in accordance with the Declaration of Helsinki.

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AUTHOR CONTRIBUTIONS

FS, TS-D, and FJ planned and conducted the study. MR, SZ and HG as well as the contributors of the SEESEN Consortium gave feedback and support during the writing process of the manuscript.

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Conflict of Interest Statement: The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Mental wellbeing at work

NICE guideline

Published: 2 March 2022

www.nice.org.uk/guidance/ng212

Your responsibility

The recommendations in this guideline represent the view of NICE, arrived at after careful consideration of the evidence available. When exercising their judgement, professionals and practitioners are expected to take this guideline fully into account, alongside the individual needs, preferences and values of their patients or the people using their service. It is not mandatory to apply the recommendations, and the guideline does not override the responsibility to make decisions appropriate to the circumstances of the individual, in consultation with them and their families and carers or guardian.

All problems (adverse events) related to a medicine or medical device used for treatment or in a procedure should be reported to the Medicines and Healthcare products Regulatory Agency using the [Yellow Card Scheme](#).

Local commissioners and providers of healthcare have a responsibility to enable the guideline to be applied when individual professionals and people using services wish to use it. They should do so in the context of local and national priorities for funding and developing services, and in light of their duties to have due regard to the need to eliminate unlawful discrimination, to advance equality of opportunity and to reduce health inequalities. Nothing in this guideline should be interpreted in a way that would be inconsistent with complying with those duties.

Commissioners and providers have a responsibility to promote an environmentally sustainable health and care system and should [assess and reduce the environmental impact of implementing NICE recommendations](#) wherever possible.

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This guideline replaces PH22.

This guideline is the basis of QS147.

Overview

This guideline covers how to create the right conditions for mental wellbeing at work. It aims to promote a supportive and inclusive work environment, including training and support for managers and helping people who have or are at risk of poor mental health.

Who is it for?

- Employers
- Senior leadership and managers, including supervisors of volunteers
- Human resource teams
- Employees, self-employed people and volunteers
- Local and regional authorities
- Professional and employee-representative organisations
- All those with a remit for workplace health (including occupational safety and health professionals and occupational health teams)
- Members of the public

Recommendations

Making decisions using NICE guidelines explains how we use words to show the strength (or certainty) of our recommendations, and has information about prescribing medicines (including off-label use), professional guidelines, standards and laws (including on consent and mental capacity), and safeguarding.

The recommendations in this guideline apply to micro, small, medium-sized and large organisations equally, although some recommendations may need to be tailored to specific organisations and circumstances.

1.1 Strategic approaches to improving mental wellbeing in the workplace

- 1.1.1 Adopt a tiered approach to mental wellbeing in the workplace by using organisational-level approaches as the foundation for good mental wellbeing (the first [bottom] tier), followed by individual approaches (the second [middle] tier) and targeted approaches (the third [top] tier).
- 1.1.2 Adopt a preventive and proactive strategic approach to mental wellbeing at work in your organisation. Take into account:
- workplace culture
 - workload
 - job quality and role autonomy
 - concerns that employers and employees may have about mental health, including stigma.

See also the section on organisation-wide approaches.

- 1.1.3 Proactively promote mental wellbeing by ensuring that it is embedded in the

overall business strategy of all organisational policies and practices. Take into account the recommendations in the [section on supportive work environment](#).

- 1.1.4 Ensure that a [stress](#) risk assessment is carried out for each role as required by the [Health and Safety at Work etc Act 1974](#), for example using the [Health and Safety Executive's risk assessment template](#):
- If any risks are identified, take proactive steps to reduce the risks and their negative impacts.
 - If a high-risk role is indicated, see the [section on organisational-level approaches for high-risk occupations](#).
 - Discuss with employees as necessary and feed back the results of the assessment to them.
- 1.1.5 Ensure that systems are in place to provide support for employees for whom external factors are influencing their mental wellbeing. See the [section on training and support for managers](#).
- 1.1.6 Monitor and evaluate the support you provide at least on an annual basis using a relevant evaluation tool. [Public Health England's evaluation in health and wellbeing](#) provides a list of resources and summarises what they are used for.
- 1.1.7 When measuring mental wellbeing, use a validated measure of mental wellbeing, for example the [government's voluntary reporting on disability, mental health and wellbeing: a framework to support employers \(for large employers\)](#), [What Works Wellbeing's workplace wellbeing questionnaire](#) or [Warwick Medical School's Warwick-Edinburgh Mental Wellbeing Scales \(WEMWBS\)](#).

For a short explanation of why the committee made these recommendations and how they might affect practice, see the [rationale and impact section on strategic approaches to improving mental wellbeing in the workplace](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review A: organisational universal-level approaches](#)
- [evidence review B: manager interventions](#)
- [evidence review C: targeted organisational-level approaches](#)
- [evidence review D: universal individual-level approaches](#)
- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work](#)
- [evidence review G: expert testimony](#).

1.2 Supportive work environment

1.2.1 Foster a positive, compassionate and inclusive workplace environment and culture to support [psychological safety](#) and mental wellbeing by:

- ensuring active leadership and management support and engagement
- increasing [mental health literacy](#)
- encouraging and facilitating peer support (for example, using mental health champions and peer mentoring or 'buddying')
- supporting people who manage and support employees
- encouraging employees to recognise and take action to prevent discrimination in the workplace, for example by establishing and supporting [staff networks](#)
- being aware that mental wellbeing in the workplace also depends on factors

beyond the workplace itself (such as physical health, domestic relationships, home environment and financial circumstances) and also on societal discrimination (such as racism, homophobia and sexism)

- promoting good communication and engagement with employees
- including mental health awareness in manager training (see the [section on training and support for managers](#)).

1.2.2 Develop policies, processes and ways of working with staff that are equitable and inclusive, and that encourage a fair and supportive workplace environment and culture, in order to maximise employee wellbeing. Take into account:

- legal obligations (such as the [Equality Act 2010](#) and [Health and Safety at Work etc Act 1974](#))
- statutory requirements (such as the [ACAS codes of practice](#))
- employer-led strategies or interventions (such as anti-bullying, work-life balance, confidentiality and flexible working).

1.2.3 Offer employees a private space and protected time to engage with interventions, taking into account the need for confidentiality.

1.2.4 Ensure that all employees have the opportunity and the means to access interventions (such as private access to the internet and IT equipment for remotely delivered interventions).

See also the [sections on organisational commitment, senior leadership and leadership style of line managers in the NICE guideline on workplace health: management practices](#), and the [section on workplace culture and policies in NICE's guideline on workplace health: long-term sickness absence and capability to work](#).

For a short explanation of why the committee made these recommendations and how they might affect practice, see the [rationale and impact section on supportive work environment](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review A: organisational universal-level approaches](#)
- [evidence review B: manager interventions](#)
- [evidence review C: targeted organisational-level approaches](#)
- [evidence review D: universal individual-level approaches](#)
- [evidence review E: targeted individual-level approaches](#)
- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work](#).

1.3 External sources of support

- 1.3.1 Use external expertise in the local authority (see the [section on local and regional strategies and plans](#)), Department for Work and Pensions and other agencies (for example, from the voluntary, charity and social enterprise sector or chambers of commerce) to access support for employees, including action plans and toolkits (for example, from [What Works Wellbeing](#), [Business in the Community](#), [Mind](#) and the [Health and Safety Executive](#)).
- 1.3.2 Make employees aware that if they have mental health problems, they can use the [Department for Work and Pensions' access to work mental health support service](#). NHS and social care staff can use the [staff mental health and wellbeing hubs](#) (and other national health and wellbeing support offers).
- 1.3.3 Use local and national resources, and advice from a variety of evidence-informed sources, such as the local [Improving Access to Psychological Therapies](#) services offer, the employee's GP, professional bodies, unions and trade organisations (for

example, [Federation of Small Businesses](#), [ACAS](#) and the [Chartered Institute of Personnel and Development \[CIPD\]](#)).

See also the [section on early intervention in NICE's guideline on workplace health: long-term sickness absence and capability to work](#).

For a short explanation of why the committee made these recommendations and how they might affect practice, see the [rationale and impact section on external sources of support](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review A: organisational universal-level approaches](#)
- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work](#)
- [evidence review G: expert testimony](#).

1.4 Organisation-wide approaches

- 1.4.1 Involve employees and workplace representatives in identifying and minimising sources of stress at work. (See also the [section on job design in NICE's guideline on workplace health: management practices](#).)
- 1.4.2 Consider using workplace accreditations or charters, such as guidance to improve the organisation-wide workplace environment and culture (for example, the [Workplace Wellbeing Charter](#), [Mindful Employer](#) and [Mind's Workplace Wellbeing Index](#)).
- 1.4.3 Tailor interventions to meet the needs of the organisation and its employees (for example, according to the industry sector or the size of the organisation).
- 1.4.4 Refer to existing guidance and best practice on job quality, work design and organisation to identify and reduce work stressors, such as [Health and Safety](#)

[Executive's Management Standards for work-related stress](#), [Mindful Employer](#) or COVID-19-specific advice (for example, from the [CIPD](#)).

- 1.4.5 Consider using staff surveys or other engagement approaches, for example working with employee representative organisations (such as trade unions or staff networks), to determine whether tailored solutions are needed to improve mental wellbeing in the workplace (for example, [What Works Wellbeing's employee wellbeing snapshot survey](#)).
- 1.4.6 Consider giving all employees free access to an employee assistance programme and occupational health services, and raise awareness of them if they are offered (for small and medium-sized enterprises). (See also the [section on making this guideline relevant for small and medium-sized enterprises \[including micro-enterprises\]](#)).
- 1.4.7 Have a plan for responding to unexpected traumatic events affecting employees, such as the death of a colleague, a pandemic or a terrorist attack. This should include supporting people socially and with their mental wellbeing. For example, see the [UK Health Security Agency's course on COVID-19: psychological first aid](#) or [NHS England and NHS Improvement's guidance on responding to the needs of people affected by incidents and emergencies](#).

See also the [section on monitoring and evaluation in NICE's guideline on workplace health: management practices](#).

For a short explanation of why the committee made these recommendations and how they might affect practice, see the [rationale and impact section on organisation-wide approaches](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review A: organisational universal-level approaches](#)
- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work](#)
- [evidence review G: expert testimony](#).

1.5 Training and support for managers

1.5.1 Offer systematic support for managers. Include training, and regular refresher training, in:

- line management
- communication skills (the ability to listen, communicate clearly, understand and empathise).

1.5.2 Equip managers with the knowledge, tools, skills and resources to:

- improve awareness of mental wellbeing at work
- promote mental wellbeing and prevent poor mental wellbeing
- improve employees' understanding of and engagement in organisational decisions
- improve communication between managers and employees.

This should include managing people remotely.

1.5.3 When offering mental health training for managers, consider including:

- how to have a conversation on mental wellbeing with an employee, including at times of crisis
- information about mental wellbeing
- how to identify early warning signs of poor mental wellbeing
- resources on mental wellbeing, including knowing where to go for further help or support in complex situations
- awareness of the stigma associated with poor mental wellbeing
- ongoing management and monitoring of mental wellbeing in the workplace
- topics suggested by managers.

- 1.5.4 Ensure that all managers have time to attend relevant training sessions.
- 1.5.5 Empower managers to make necessary adjustments to workload or work intensity for their employees, for example flexible or hybrid working.
- 1.5.6 Encourage managers to address their own mental health needs as well as those of their employees, for example by peer-to-peer support for managers on mental wellbeing.
- 1.5.7 Consider a group approach to deliver mental health training. Training could be delivered either face to face or using online formats.
- 1.5.8 Evaluate how mental health training for managers affects employee outcomes (for example, by surveying employees and managers or focus groups) and feed the results back into future training and strategy.

See also the [section on training in NICE's guideline on workplace health: management practices](#).

For a short explanation of why the committee made these recommendations and how they might affect practice, see the [rationale and impact section on training and support for managers](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review B: manager interventions](#)
- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work](#)
- [evidence review G: expert testimony](#).

1.6 Individual-level approaches

- 1.6.1 Do not use individual-level approaches to replace organisational strategies for

reducing work stressors, or for the main purpose of increasing productivity.

- 1.6.2 Encourage managers to create opportunities for fostering good relationships with (and between) employees, for example by socialising with them at work (in person or virtually). Create opportunities to talk with them about their general health and wellbeing.
- 1.6.3 Encourage managers to discuss mental wellbeing with employees, and employees to discuss any mental wellbeing concerns they may have with their manager or another relevant person (for example, another manager, a mental wellbeing champion or a union representative):
- Use these conversations to identify and understand any sources of stress.
 - Agree whether any additional support is needed and what this might be (see the [section on external sources of support](#)).
 - Agree steps to minimise work-related stressors (see the [section on approaches for employees who have or are at risk of poor mental health](#)).
- 1.6.4 Offer all employees (or help them to access) mindfulness, yoga or meditation on an ongoing basis. This can be delivered in a group or online, or using a combination of both.

See also the [section on supporting employers in NICE's guideline on physical activity in the workplace](#).

For a short explanation of why the committee made these recommendations and how they might affect practice, see the [rationale and impact section on individual-level approaches](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review C: targeted organisational-level approaches](#)
- [evidence review D: universal individual-level approaches](#)
- [evidence review E: targeted individual-level approaches](#)
- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work](#).

1.7 Approaches for employees who have or are at risk of poor mental health

- 1.7.1 Ensure that confidentiality is discussed when talking with someone about their mental health (see [recommendation 1.2.2](#)), and be clear about when confidentiality will and will not be respected (that is, when the employee is considered at risk to themselves or others because of their mental health).
- 1.7.2 Offer organisational support to employees identified as having or being at risk of poor mental health. This may include flexible working hours; changes to the job, workplace or culture to minimise any risks to mental wellbeing; or maintaining supportive line management relationships. (See [recommendation 1.5.5](#) and [recommendation 1.6.3](#).) Remind them that they can visit their GP for further assessment and support.
- Consider working with them to create a wellness action plan (see [Mind's guides to wellness action plans](#)).
 - Assess whether this has highlighted if changes need to be made at an organisational level.

1.7.3 Discuss with the employee if they would like to:

- have further support and, if so, whether they prefer a particular type of support
- have ongoing regular, confidential discussions about their mental health support needs.

1.7.4 For employees who want further support, offer (or provide access to):

- cognitive behavioural therapy sessions **or**
- mindfulness training **or**
- stress management training.

If employees choose not to have an intervention now, tell them that the offer will still be available in the future if they reconsider.

1.7.5 Reassure colleagues that it is their choice whether to continue with an intervention or restart it at any time.

See also the [sections on sustainable return to work and reducing recurrence of absence in NICE's guideline on workplace health: long-term sickness absence and capability to work.](#)

For a short explanation of why the committee made these recommendations and how they might affect practice, see the [rationale and impact section on approaches for employees who have or are at risk of poor mental health](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review C: targeted organisational-level approaches](#)
- [evidence review D: universal individual-level approaches](#)
- [evidence review E: targeted individual-level approaches](#)
- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work](#).

1.8 Organisational-level approaches for high-risk occupations

The following recommendations are aimed at organisations, workplaces or workforces where employees are likely to experience traumatic events in the normal course of their work (such as the emergency services). See also [recommendation 1.1.4](#).

- 1.8.1 Regularly review organisational-level policies and protocols on how to support employees in high-risk occupations after an occupational traumatic event. Use data such as reasons for absence and staff turnover to ensure that support is targeted in the right way.
- 1.8.2 Ensure that practice is consistent with established best practice (for example, [Mind's Blue Light Programme](#)).
- 1.8.3 Offer task-focused skills training (for example, through imagery, simulation and skills training) before deployment for employees in high-risk occupations (such as emergency services) to ensure that they have the skills needed to deal with predictable and stressful occupational events.

See also [NICE's guideline on post-traumatic stress disorder](#).

For a short explanation of why the committee made these recommendations and how they might affect practice, see the [rationale and impact section on organisational-level approaches for high-risk occupations](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review D: universal individual-level approaches](#)
- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work](#).

1.9 Engaging with employees and their representatives

- 1.9.1 Work with employees and their representative organisations (for example, trade unions and staff networks; see [recommendation 1.2.1](#)) to consult about how, when and where mental wellbeing interventions are offered and delivered, for example through staff surveys.
- 1.9.2 Take account of the following potential barriers and facilitators when consulting with employees about interventions:
- workplace culture (including the concern that raising issues can impact negatively on staff roles or job security)
 - workload
 - concerns that employers and employees may have about mental health, including stigma, and how this may affect their ability to discuss any difficulties or engage with certain forms of support
 - timing of the intervention and the option of delivering it in and outside the workplace and work hours

- specific needs and preferences of employees
- specific reasonable needs of the employing organisation.

1.9.3 Ensure that factors associated with an employee such as contract type, income level, protected characteristics and job role are not barriers to accessing interventions. Do this by:

- monitoring intervention uptake and identifying groups where uptake is relatively low
- having a mechanism to identify, understand and overcome barriers to participating in the intervention.

For a short explanation of why the committee made these recommendations and how they might affect practice, see the [rationale and impact section on engaging with employees and their representatives](#).

Full details of the evidence and the committee's discussion are in [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work](#).

1.10 Local and regional strategies and plans

These recommendations are for local and regional authorities.

- 1.10.1 Take a leadership role in championing mental wellbeing and preventing poor mental wellbeing at work as part of the local authority role in public health and wellbeing.
- 1.10.2 Engage with local and regional employers, employee representatives, chambers of commerce, local enterprise partnerships and voluntary, charity and social enterprises to develop and promote health and wellbeing strategies to include mental wellbeing at work.
- 1.10.3 Integrate mental wellbeing at work into local and regional public health activities

and strategies.

- 1.10.4 Raise awareness among the general public and employers of the importance of mental wellbeing at work, for example through social media.
- 1.10.5 Identify and address local barriers and facilitators to employer engagement with local mental wellbeing at work initiatives. This could include, for example, working with employers to ensure they know about resources or services that can help them improve the mental wellbeing of their employees and minimise the resource impact that this will have, especially for micro, small and medium-sized enterprises.
- 1.10.6 Offer support to help local employers improve the mental wellbeing and prevent poor mental wellbeing of their employees. This support could include advice on enablers of mental health and on developing action plans towards accreditation (see [recommendation 1.4.2](#)) or setting up a [Local Workplace Health Accreditation Scheme](#).
- 1.10.7 Curate or work with local business support organisations to list local and national sources of support for employers and employees, such as [Mind](#), [Mental Health at Work](#), the [Department for Work and Pensions' access to work mental health support service](#) and, for NHS and social care staff, [staff mental health and wellbeing hubs](#) (and other national health and wellbeing support offers).
- 1.10.8 Explore and evaluate the value of incentives or pilot incentive programmes to promote uptake of support and encourage employers to participate in accreditation schemes (see [recommendation 1.4.2](#)).
- 1.10.9 Use contracting and ethical procurement arrangements to strongly encourage supply chain organisations to promote mental wellbeing among their employees (for example, public sector organisations must use the [Public Services \[Social Value\] Act 2012](#)).

For a short explanation of why the committee made these recommendations and how they might affect practice, see the [rationale and impact section on local and regional strategies and plans](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work](#)
- [evidence review G: expert testimony](#).

1.11 Making this guideline relevant for small and medium-sized enterprises (including micro-enterprises)

- 1.11.1 Business owners and owner-managers should address their own mental health needs as well as those of their employees.
- 1.11.2 Take a preventive approach to mental wellbeing at work, for example using mental health and communication skills training to foster positive mental wellbeing, as well as tackling poor mental wellbeing. Refer to the [Mental Health at Work website](#) for curated resources and toolkits on how to improve the mental wellbeing of your employees.
- 1.11.3 Seek advice and support from local authorities; local enterprise partnerships; voluntary, charity and social enterprises; trade unions and other bodies; and, for NHS and social care staff, [staff mental health and wellbeing hubs \(and other national health and wellbeing support offers\)](#), on how to prevent poor mental wellbeing in your employees, and how to support employees through mental ill health.
- 1.11.4 Think about signing up to the [Mental Health at Work Commitment](#) to help achieve better mental health outcomes for employees.

- 1.11.5 Think about accessing employee assistance programmes and occupational health services. See the [Department for Work and Pensions' access to work mental health support service](#) as an example of a low-cost service (see recommendation 1.11.2).

For a short explanation of why the committee made these recommendations and how they might affect practice, see the [rationale and impact section on additional approaches for small and medium-sized enterprises](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review D: universal individual-level approaches](#)
- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work](#)
- [evidence review G: expert testimony](#).

Terms used in this guideline

This section defines terms that have been used in a particular way for this guideline. For other definitions, see the [NICE glossary](#) and the [Think Local Act Personal's Care and Support Jargon Buster](#).

Employee

Everyone aged 16 or over in full- or part-time employment, including people on permanent, training, temporary or zero-hours contracts, those who are self-employed and volunteers.

Mental health literacy

A person's knowledge and beliefs about mental health problems and how to look after their own mental health. It includes knowing how mental health problems are managed and treated, how to seek information about them and how to recognise them.

Organisation

For the purposes of this guideline, organisation refers to any size of workplace, including micro, small and medium-sized enterprises.

Psychological safety

A person's desire and need to feel comfortable and safe in the workplace to express themselves and communicate openly.

Role autonomy

A person's ability to influence what happens in their work environment, in particular to influence matters that are relevant to their personal goals and the way in which they carry out their work.

Staff networks

Groups of employees who come together in a safe environment for discussion and support, and from which they can be a voice for change in the workplace. This includes raising awareness of issues in the wider organisation. They are commonly groups of people who identify as an under-represented group or who have a protected characteristic in the [Equality Act 2010](#).

Stress

The [Health and Safety Executive \(HSE\) guide on working together to reduce stress at work](#) defines stress as the adverse reaction people have to excessive pressures or other types of demand placed on them.

Recommendations for research

The guideline committee has made the following recommendations for research.

Key recommendations for research

1 Individual-level interventions

What is the long-term effectiveness and cost effectiveness of universal individual-level interventions on mental wellbeing in different types of organisations?

For a short explanation of why the committee made the recommendation for research, see the [rationale section on individual-level approaches](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review D: universal individual-level approaches](#)
- [evidence review E: targeted individual-level approaches](#)
- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work](#).

2 Organisational-level approaches for all organisations

What is the long-term impact and cost effectiveness of employee assistance programme provision on mental wellbeing?

For a short explanation of why the committee made the recommendation for research, see the [rationale section on organisation-wide approaches](#).

Full details of the evidence and the committee's discussion are in [evidence review A: organisational universal-level approaches](#).

3 Training for managers

What is the long-term effectiveness (more than 6 months) of manager training on employee mental wellbeing?

For a short explanation of why the committee made the recommendation for research, see the [rationale section on training and support for managers](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review B: manager interventions](#)
- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work](#).

4 Approaches for micro, small and medium-sized enterprises

What are the specific needs of different kinds of micro, small and medium-sized enterprises (SMEs) in promoting mental wellbeing in the workplace, including organisational, targeted and individual level approaches?

For a short explanation of why the committee made the recommendation for research, see the [rationale section on additional approaches for small and medium-sized enterprises](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review D: universal individual-level approaches](#)
- [evidence review E: targeted individual-level approaches](#)
- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work](#).

5 Core outcomes for study reporting

Which mental wellbeing and productivity outcomes should be used in a core outcome set for research into workplace mental wellbeing?

For a short explanation of why the committee made the recommendation for research, see the [rationale section on strategic approaches to improving mental wellbeing in the workplace](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review A: organisational universal-level approaches](#)
- [evidence review B: manager interventions](#)
- [evidence review C: targeted organisational-level approaches](#)
- [evidence review E: targeted individual-level approaches](#)
- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work](#).

Other recommendations for research

6 Supportive work environment

What are the views of organisations about the benefits of investing in mental wellbeing?

For a short explanation of why the committee made the recommendation for research, see the [rationale section on supportive work environment](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review A: organisational universal-level approaches](#)
- [evidence review B: manager interventions](#)
- [evidence review C: targeted organisational-level approaches](#)
- [evidence review D: universal individual-level approaches](#)
- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work](#).

7 Identifying employees at risk of poor mental wellbeing

What tools (for example, wellbeing surveys) can be used to identify employees at risk of poor mental wellbeing rather than mental ill health?

For a short explanation of why the committee made the recommendation for research, see the [rationale section on approaches for employees who have or are at risk of poor mental health](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review A: organisational universal-level approaches](#)
- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work](#).

8 Needs of different employee groups

What specific needs of employees from different groups (such as income levels, ethnic

groups, male or female groups, and age groups) need addressing to facilitate access to individual-level interventions?

How effective are individual-level interventions across different groups (such as income levels, ethnic groups, male or female groups, and age groups)?

For a short explanation of why the committee made the recommendations for research, see the [rationale section on individual-level approaches](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review C: targeted organisational-level approaches](#)
- [evidence review D: universal individual-level approaches](#)
- [evidence review E: targeted individual-level approaches](#)
- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work](#).

9 Approaches for micro, small and medium-sized enterprises

What is the long-term effectiveness of universal individual-level interventions in different kinds of SMEs?

For a short explanation of why the committee made the recommendation for research, see the [rationale section on making this guideline relevant for small and medium-sized enterprises \(including micro-enterprises\)](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review D: universal individual-level approaches](#)
- [evidence review E: targeted individual-level approaches](#)
- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work.](#)

10 Addressing study reporting

What are the key characteristics of an organisation and its employees that need to be included in reporting research into workplace mental wellbeing?

For a short explanation of why the committee made the recommendation for research, see the [rationale section on strategic approaches to improving mental wellbeing in the workplace](#).

Full details of the evidence and the committee's discussion are in:

- [evidence review A: organisational universal-level approaches](#)
- [evidence review B: manager interventions](#)
- [evidence review C: targeted organisational-level approaches](#)
- [evidence review E: targeted individual-level approaches](#)
- [evidence review F: barriers and facilitators to the implementation and delivery of interventions to improve and protect mental wellbeing at work.](#)

Rationale and impact

These sections briefly explain why the committee made the recommendations and how they might affect practice.

Strategic approaches to improving mental wellbeing in the workplace

Recommendations 1.1.1 to 1.1.7

Why the committee made the recommendations

The committee noted that the range and nature of workplaces and organisations (especially in terms of the number of employees) made it challenging to make generic recommendations for all organisations. They agreed that some organisations might need to tailor the recommendations to make them relevant.

The committee agreed on the need for organisations to embed strategic approaches to mental wellbeing into their organisational policies and practices, based on their understanding that this approach was fundamental to mental wellbeing at work and based on the evidence and expert testimony.

The committee recognised that the ability of organisations to promote and support mental wellbeing is negatively affected by health inequalities and by work stressors such as bullying, poor communication, job insecurity (including zero-hours contracts), workload, monotony, isolation and poor prospects. In contrast, it is enhanced by good job quality (including a fair wage), role autonomy, organisational fairness, respect, recognition, peer support and clear communication. The committee were clear that some of these factors were outside of the remit of this guideline.

A wide range of evidence from the UK showed that organisation-wide interventions may help to improve mental wellbeing and stress outcomes for employees, and may also benefit employers. The committee agreed that, in their experience, 1 of the key foundations of mental wellbeing in the workplace was an organisational commitment to it, together with a recognition that mental wellbeing is a spectrum and poor mental wellbeing

is not a weakness. Organisations that took a strategic and whole-hearted approach to improving mental wellbeing from the top down tended to have the most success. The committee agreed that this was best demonstrated by organisations with a preventive and proactive approach to mental wellbeing – that is, they proactively took steps to promote mental wellbeing rather than simply tackling poor mental wellbeing. They agreed that because these organisational-level responses were so important, the first recommendations in the guideline should establish that organisation-wide strategic approaches were the foundation of good mental wellbeing at work, and that individual and targeted approaches could enhance these but were not a substitute for them. In the committee's experience, it is unlikely that individual or targeted interventions would be successful without an organisation-wide approach, but it noted the temptation for organisations to start with individual approaches because it seems easier.

The committee also agreed with expert testimony on the impacts of the pandemic on mental wellbeing in the workplace that highlighted the need to view mental wellbeing as equally important to physical wellbeing in the workplace and to take it into account when drafting policies or introducing new practices.

Expert testimony about the long-term impacts of the COVID-19 pandemic on mental wellbeing in the workplace highlighted that employers of all sizes are legally required to carry out a stress risk assessment for each role (and record it if they have more than 5 employees) under the Health and Safety at Work etc Act 1974. The committee saw this as a good opportunity for organisations to identify risks to employees' mental wellbeing, and subsequently take steps to reduce stressors.

The committee also highlighted that employees may have poor mental wellbeing as a result of external factors that are beyond the control of the employer, such as caring responsibilities, health concerns and discrimination (such as homophobia or racism). But they agreed that it is important for organisations to ensure that they provide additional support to groups affected by these issues.

The committee also agreed, based on their experience, that it is important for any interventions to be evaluated and monitored as part of an ongoing strategy of employee engagement, and that validated measures of wellbeing need to be part of this process.

The committee noted that further research is needed to understand how data and outcomes could best be used to improve mental wellbeing in the workplace. In particular, research could investigate which outcomes would be useful in a core outcome set for

research into workplace mental wellbeing, and to understand how more detailed reporting of the nature of organisation and employee characteristics can be included in research into workplace mental wellbeing (see the [recommendations for research on core outcomes for study reporting](#) and [addressing study reporting](#)).

How the recommendations might affect practice

The use of tiered approaches to support mental wellbeing at work reflects best practice and therefore would not have a large resource impact in organisations that have already adopted best practice. This may have more of an impact for other organisations. Smaller organisations may not always have the resources to offer all aspects of the tiered approach. They can best use resources by concentrating on ensuring that they have an organisation-wide approach in place. Improving employee wellbeing might lead to less absenteeism and presenteeism and may improve staff retention and productivity.

[Return to recommendations](#)

Supportive work environment

[Recommendations 1.2.1 to 1.2.4](#)

Why the committee made the recommendations

The committee agreed that overall, a supportive, inclusive work environment and climate is crucial for good mental wellbeing in the workforce. Social interactions, including those between managers and employees, play an important role in this.

Having the right policies can help to create a supportive workplace environment and culture. It can help to ensure that leadership is supportive and engaged, that there are effective peer support networks and there is good organisational-wide mental health literacy. A supportive work environment can be achieved by adhering to existing legal obligations (such as health and safety) and statutory requirements (such as the ACAS codes of practice), and engaging with employees to draft and refine policies such as anti-bullying.

Organisations can also promote mental wellbeing interventions by reducing any potential barriers to using them and supporting employees to access them. This would embed the

importance of mental wellbeing into the organisational culture (see the [section on engaging with employees and their representatives](#)).

Despite the lack of strong evidence for leadership interventions, the committee were confident that management buy-in is important for promoting the wellbeing of employees. They cited the [government review Thriving at work](#), which provides an evidence-based whole-settings approach to improving mental wellbeing, including the importance of leadership, culture and effective people management.

The committee noted that there was little evidence on the views of organisations about mental wellbeing. (See the [recommendation for research on supportive work environment](#).)

How the recommendations might affect practice

The recommendations reflect current good practice around communication across the organisation, active leadership involvement and engagement with employees. The committee noted that many organisations would already have structures in place like those they recommended, and that to align them with the recommendations would not have a large resource impact. The committee agreed that for organisations that were resource poor and that had not previously invested in a supportive work environment, the use of freely available external resources could help minimise costs.

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External sources of support

[Recommendations 1.3.1 to 1.3.3](#)

Why the committee made the recommendations

The committee agreed that supporting mental wellbeing in the workplace might be particularly challenging for organisations with limited resources, especially micro, small and medium-sized ones. They agreed that it was important to help employers find external, low-cost or free resources to support them in promoting mental wellbeing.

Based on expert testimony from a mental health and productivity pilot and their experience, the committee agreed that it was the responsibility of employers and local and

regional authorities to be aware of sources of support available in their area. These sources could be national, local or within the organisation, for example an occupational health service or an employee assistance programme (if the organisation has these). By being able to direct employees to these, employers will be helping and supporting employees by providing them with tools and resources.

The Department for Work and Pensions' access to work mental health support service has guidance to help employers understand mental ill health and how to support employees with mental health concerns. It can also support employees by offering eligible employees an assessment to find out their needs and help them develop a support plan.

How the recommendations might affect practice

The recommendations encourage employers to use expertise and resources that are external to their organisation when appropriate. These sources of support are freely available and will provide employers with information and resources to support their employees. They will also help organisations who are committed to improving the mental wellbeing of their employees to manage the resource impact.

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Organisation-wide approaches

[Recommendations 1.4.1 to 1.4.7](#)

Why the committee made the recommendations

Evidence from the UK showed that organisation-wide interventions may help to improve mental wellbeing and stress outcomes for employees and may also benefit employers. Although the evidence had some limitations in terms of its quality, the committee concluded that the work environment can be improved in 2 ways: employers can work with their employees to identify work stressors and put in place solutions to deal with these stressors, or employers can use evidence-based methods that are specifically tailored to their organisation. The committee clarified their use of the word 'stress' to be the adverse reaction people have to excessive pressures or other types of demand placed on them.

Based on the evidence and their experience, the committee strongly advised that

organisational-level approaches are the best starting point when considering strategies to improve mental wellbeing at work. These approaches demonstrate a commitment to mental wellbeing at work, which is essential for encouraging employees to take up interventions. The committee emphasised that individual-level approaches are not a suitable alternative to organisational-level approaches because these are less likely to be effective on their own. They noted that because of the variation in the size and structure of workplaces, many interventions might need to be tailored to match the specific workplace in which they were going to be delivered. This could be done through staff surveys, for example.

The committee heard expert testimonies about insights from the [Thriving at Work Leadership Council](#), the mental health and productivity pilot, participatory organisational interventions, and prevention and management of work-related stress and mental ill health. They agreed with the experts that striving to attain workplace charters or accreditation was a useful way for organisations to work with external bodies to improve mental wellbeing and make their organisation a more attractive place to work. These would also allow employers to access external support and advice about improving and maintaining mental wellbeing at work. The committee also agreed that existing guidance, such as the [Health and Safety Executive's Management Standards for work-related stress](#), would be useful.

The committee agreed that employee assistance programmes and occupational health services are good options for supporting employees' mental wellbeing, although they noted that there was a lack of evidence about how effective they are. They agreed with expert testimony about major challenges to small and medium-sized enterprises in improving the mental wellbeing of staff, what they can do to improve staff mental wellbeing and that low-cost schemes such as [Mindful Employer](#) may be useful for smaller organisations with limited resources. The committee also discussed that some employees in occupations that are not generally considered high-risk may be exposed to traumatic events at work – for example, because of a pandemic or terror attack – and that employers need to have plans in place to support employees in case such events do occur.

The lack of published evidence about the effectiveness of employee assistance programmes led the committee to make a [recommendation for research on organisational-level approaches for all organisations](#) because evidence on this could help NICE to make more specific recommendations on this topic in future.

How the recommendations might affect practice

The recommendations reflect good practice in communication across the organisation, in active leadership involvement and in engaging with employees. There should not be an extensive resource impact because the recommendations involve adhering to existing best practice.

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Training and support for managers

[Recommendations 1.5.1 to 1.5.8](#)

Why the committee made the recommendations

The committee agreed it was important that all line managers received training and support. They considered that this was good practice in all industries and all sizes of organisation, and that managers benefit in terms of their mental wellbeing from feeling skilled to perform their line management duties.

There was a range of evidence showing that manager training interventions were effective (especially in terms of outcomes for the manager who had been trained) or had no effect, although the committee noted that much of this evidence was uncertain. There was no evidence of unintended consequences associated with these interventions. There was some higher quality evidence that manager training can help to increase managers' knowledge of how to reduce stigma, and also help to increase their confidence in identifying and supporting employees who may be at risk of poor mental health. This agreed with the experience of the committee, who found that training managers in mental health awareness can increase their knowledge and willingness to discuss mental health.

Reducing stigma and equipping managers with skills to have conversations with employees about mental health is likely to facilitate conversations between managers and employees about any concerns about their mental wellbeing. This makes it more likely that managers can support employees with mental health issues. Providing managers with skills to discuss mental wellbeing improves the relationship between manager and employee so that they can identify and reduce work stressors. The evidence also showed that increasing managers' knowledge leads to more employees using the support services on offer. Although the evidence was not strong in this area, the committee agreed some

points that would be a useful core for the content of mental health training for managers. They also agreed it was important for people to be able to suggest topics they thought were important.

The committee agreed with the qualitative evidence that manager training interventions delivered in groups had added benefit because they allow managers to learn from each other and to reinforce best practice. Therefore, the committee agreed, based on their experience and the evidence, that it can be helpful to offer group training as part of mental health awareness training. But they acknowledged that this might not be possible, for example in smaller organisations. Expert testimony about major challenges to small and medium-sized enterprises in improving the mental wellbeing of staff and what they can do to improve staff mental wellbeing, managing mental health in the workplace during and after COVID-19, and committee experience, highlighted that managers have additional pressures related to their role, and that delivering any training in a group format would provide peer support. The committee also discussed that because of the increased pressures faced by managers, it is important that they are supported by human resources, occupational safety, and health and wellbeing professionals.

The committee discussed expert testimony about managing mental health in the workplace during the COVID-19 pandemic and about the likely long-term impacts of COVID-19 on mental wellbeing in the workplace. It helped them to make recommendations about the content of management training to support mental wellbeing. They agreed that managers should be empowered to make reasonable adjustments to the workload or intensity to reduce stressors for employees. This would give employees relief from work stress sooner because requests would not need to be escalated before support could be given. This expert testimony also highlighted the value of peer-to-peer support for managers, which the committee agreed matched their own expert experience.

The committee further discussed that although there was some data on the effectiveness of the interventions reviewed in terms of employee outcomes, overall, there was insufficient data and they had to extrapolate from their expertise and experience along with the small amount of evidence they had. They suggested that this may be because interventions had a short follow up of 3 months. This might be sufficiently long to show a difference in manager outcomes, but it may not be long enough to show a change in employee outcomes, including mental wellbeing. Therefore, the committee agreed that further research is needed on employee outcomes with longer follow ups (see the recommendation for research on training for managers).

How the recommendations might affect practice

The committee agreed that most organisations with a management structure would have some form of manager training programme, and that these recommendations reflect good practice in training managers. If the recommendations are built into those existing training structures, the committee agreed the resource impact of these recommendations would be small. For other organisations, the committee agreed that buying in external training could be expensive, but that training costs could be minimised by using free training resources.

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Individual-level approaches

[Recommendations 1.6.1 to 1.6.4](#)

Why the committee made the recommendations

The committee agreed that within the 3-tiered approach covered in the [section on strategic approaches to improving mental wellbeing in the workplace](#), it was important for organisations to prioritise an organisational approach to improving mental wellbeing in the workplace. The committee emphasised that individual-level approaches are not a suitable alternative to organisational-level approaches because these are less likely to be effective on their own. So, individual approaches need to be additional to organisational approaches and not a substitute for them.

The committee recognised the importance of good relationships between managers and employees, and of employees being able to approach managers to discuss any concerns. Making opportunities – for example, for small talk – to develop good relationships could help with this. This would help employees to discuss issues they may have outside work and it may help to identify support that could be put into place. The committee also highlighted that in some cases, a manager could have a negative impact on an employee's mental wellbeing. Therefore, mechanisms are needed for employees to discuss any concerns with an appropriate person.

The committee saw evidence on a range of individual-level interventions that aimed to improve mental wellbeing in an unselected population. They were clear that these were not a substitute for organisational-level approaches. The evidence they were presented with

had some limitations, but the committee agreed that mindfulness, meditation and yoga were most effective overall in reducing job stress and mental health symptoms and having a positive effect on employee mental wellbeing. The evidence showed that these interventions were effective when delivered either in a group or online. The committee decided that employees should be able to choose how the interventions are delivered (see the [section on engaging with employees and their representatives](#)).

The committee noted a lack of evidence about the long-term effectiveness of universal individual-level interventions in all organisations (see the [recommendation for research on individual-level interventions](#)) and a lack of evidence about the specific needs of different groups, for example different age groups or employees from different cultural backgrounds that prevented them from making specific recommendations about this. (See the [recommendations for research on the needs of different employee groups](#).)

How the recommendations might affect practice

The committee recognised that many small businesses would not have the resources to provide mindfulness, yoga or meditation interventions, but noted that there are free or low-cost options for all of these, which would only need signposting by employers.

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Approaches for employees who have or are at risk of poor mental health

[Recommendations 1.7.1 to 1.7.5](#)

Why the committee made the recommendations

The committee raised concerns that managers may face difficulties around confidentiality if they think that an employee is at risk of harming themselves or others. To reduce the burden placed on individuals, the committee decided that organisations should have clear policies on confidentiality.

The committee discussed that a preventive approach is important for reducing poor mental wellbeing. But they acknowledged that some employees will already have poor mental health and others will be at increased risk of poor mental health. Therefore, these

employees should be offered support. The committee suggested that, although there was no specific evidence for them, wellness action plans were likely to be a useful way to open a dialogue between managers and employees about mental health concerns and what support could be put in place to help employees. They could also help to highlight needs for organisational change.

The evidence agreed with the committee's collective experience and showed that cognitive behavioural therapy, mindfulness and stress management were effective in improving mental wellbeing outcomes in employees with poor mental health. However, there was more limited evidence for cognitive behavioural therapy than for the other 2 options. They noted that if treatment is commissioned by the employer, they are required to check that the provider has the necessary qualifications and is accredited and regulated by relevant professional organisations to offer the interventions.

The committee noted that there was a potential resource impact for offering these and that for smaller organisations, free or low-cost options existed (for example, online resources such as the local [Improving Access to Psychological Therapies scheme](#)). The committee thought it was important that employees were made aware of the option to not have an intervention and to take up an offer of it at a later date, or to stop an intervention at any time and restart it later. This avoids employees feeling pressured to start or continue an intervention. They also agreed it was important that employers recognise that an employee may prefer a particular type of intervention, possibly because of their previous experiences with interventions.

The committee noted the lack of evidence about which strategies can be used to identify employees at risk of poor mental wellbeing. (See the [recommendation for research on identifying employees at risk of poor mental wellbeing](#).)

How the recommendations might affect practice

The recommendations reflect good practice around managing and supporting employees. The committee noted a potential resource impact to implement interventions, both in terms of work hours and financial resources. But this could be limited by using free resources.

They noted that there may be a resource impact to offering flexible working hours, job changes or other organisational support to people at risk of poor mental health but assessed that this would often be very low.

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Organisational-level approaches for high-risk occupations

[Recommendations 1.8.1 to 1.8.3](#)

Why the committee made the recommendations

If the psychosocial risk assessment (see the [section on strategic approaches to improving mental wellbeing in the workplace](#)) for a role indicates that it is high risk, it is important that organisations have additional processes in place to support employees. The committee agreed that it was important to make sure these processes conformed to best practice in the field, and from their experience, they were able to identify Mind's Blue Light Programme as an example of best practice.

There was good evidence showing that when police and healthcare professionals were given the skills to deal with stressful occupational events through task-focused skills training (including imagery and simulation), mental health symptoms were reduced, and mental wellbeing and quality of life improved. Based on this evidence, the committee decided that organisations should provide task-focused skills training for employees in high-risk occupations. They also recommended, in line with the evidence, that employees in high-risk occupations are offered support after a traumatic event.

The committee noted that there were exceptional circumstances, for example, the COVID-19 pandemic, which could cause stressful occupational events more widely (for example, some people might find homeworking or social distancing in the workplace stressful).

How the recommendations might affect practice

All high-risk occupations should already have policies and procedures in place on how to deal with predictable and stressful occupational events. These recommendations will not affect the resources needed for this.

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Engaging with employees and their representatives

Recommendations 1.9.1 to 1.9.3

Why the committee made the recommendations

The committee suggested that consulting employees about the type and format of organisational approaches and individual interventions offered would help employers understand what good mental wellbeing would look like in their organisation. It would also help them to tailor their approach to the needs of their employees and the organisation. They believed that this would give employers the opportunity to raise awareness about why the interventions are being implemented, which could improve employee support for them.

The committee discussed that by providing interventions during the working day, employers would give employees a beneficial break from work and send a clear message about the importance of mental wellbeing. However, organisations should be flexible because employees may also prefer to access interventions outside work hours. The committee also noted that people will have different preferences about how they learn. For example, some employees would benefit from a group setting, whereas others would prefer one-to-one or online interventions. This highlights the importance of engaging with employees to ensure that their needs are considered, and that if online interventions are offered, digital exclusion does not prevent any employee from accessing the intervention.

The committee also discussed, based on their experience, that the effectiveness of certain interventions may be different for different groups. Factors may include socioeconomic factors such as low income and whether people have disabilities, work in urban or rural locations or have good digital access. The committee agreed that staff surveys and consultation could be used to regularly monitor intervention accessibility.

How the recommendations might affect practice

The committee discussed the resource implications of these recommendations but overall did not think they would be significant in most cases. They noted that some organisations may not be able to provide interventions during work hours for financial reasons. In these cases, it may be better to provide interventions outside work hours rather than not making them available at all. They also noted that some organisations may not have the space to provide certain interventions on site, and this may affect the type or format of intervention

offered.

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Local and regional strategies and plans

[Recommendations 1.10.1 to 1.10.9](#)

Why the committee made the recommendations

The committee discussed that local and regional authorities should be role models in ensuring that their own workplaces actively promote mental wellbeing, given their role in public health. The committee highlighted that many local and regional authorities already have strategies in place to improve physical wellbeing in their population, and that these could be expanded to include mental wellbeing as part of a more holistic approach to wellbeing. This includes working with employers to ensure they know about resources or services that can help them improve the mental wellbeing of their employees and minimise the resource impact that this will have, especially for small and medium-sized enterprises and micro-enterprises. This could be done together with local enterprise partnerships and chambers of commerce. This can also be tailored to the needs of each organisation.

The committee heard expert testimony about a mental health and productivity pilot that included the possibility of local authorities using financial incentives to encourage employers to think about job quality and wellbeing in their workplaces. The committee discussed this and also discussed that in times of financial hardship, there may be non-financial incentives that are more achievable for local authorities. Because there was limited evidence about this, the committee agreed that it would not be appropriate to recommend incentive schemes, but that any authorities interested in them could introduce them as a pilot or as an evaluation.

Local and regional authorities will have ethical procurement frameworks in place, and a duty under the Social Value Act to consider wider social, economic and environmental factors during procurement. Therefore, the committee suggested that local and regional authorities could consider how organisations in their supply chains value job quality and mental wellbeing in the workplace.

How the recommendations might affect practice

Local and regional authorities already have schemes in place to help employers improve mental wellbeing in their workplaces, including the [Department for Work and Pensions' access to work mental health support service](#). Many sources of support have already been curated by organisations such as [Mind](#) and [Business in the Community](#), and local and regional authorities would only need to signpost employers to these.

The committee were aware that some local authorities may be having funding difficulties and did not want to place too much of a burden on them. Expert testimony about a mental health and productivity pilot discussed the possibility of local authorities using financial incentives to encourage employers to think about job quality and wellbeing in their workplaces. However, the committee noted that local and regional authorities may be able to provide other forms of incentives that do not need extra funds, for example dedicated advice and guidance.

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Making this guideline relevant for small and medium-sized enterprises (including micro-enterprises)

[Recommendations 1.11.1 to 1.11.5](#)

Why the committee made the recommendations

The committee heard from expert testimony from the [Thriving at Work Leadership Council](#) that many business owners are at risk of poor mental health and exhaustion as a result of the pandemic. They noted that leaders need to ensure that they also consider their own mental wellbeing.

The committee noted that a lot of the evidence was from larger organisations, and that small and medium-sized enterprises (SMEs) are likely to have fewer resources to help them address mental wellbeing in the workplace, such as occupational health and human resource professionals. The committee discussed that taking a preventive approach to ensuring good mental wellbeing could avoid problems later on (for employees and for the organisation). They agreed that employers could find a lot of guidance and resources on

how to do this through the [Mental Health at Work website](#), and [Health and Safety Executive resources](#).

Public bodies such as local authorities and local enterprise partnerships should also be able to signpost employees of SMEs to information on how to prevent poor mental wellbeing at work and promote positive mental wellbeing, as well as signposting them to resources and services to support employees with poor mental health, such as the [Department for Work and Pensions' access to work mental health support service](#). The committee suggested that SMEs may also want to sign up to the [Mental Health at Work Commitment](#), which is a framework to help organisations improve the mental wellbeing of their employees.

The committee agreed that further research into SMEs was needed – particularly on the specific needs of SMEs for implementing individual-level interventions and the long-term effectiveness of universal individual-level interventions in both larger organisations and SMEs. So they made a [recommendation for research on specific needs of different kinds of micro, small and medium-sized enterprises \(SMEs\) in promoting mental wellbeing in the workplace, including organisational, targeted and individual-level approaches](#) and a [recommendation for research on long-term effectiveness of universal individual-level interventions in different kinds of SMEs](#). These would enable NICE to make more specific recommendations for SMEs in future.

How the recommendations might affect practice

The committee were aware that, compared with larger organisations, SMEs may face additional constraints in terms of time and resources. The recommendations reflect ways that smaller organisations can look after the mental wellbeing of their workforce, without needing too much time or specialist knowledge about mental wellbeing.

The committee also discussed that employee assistance programmes and occupational health services may be a useful way of helping employees, and that smaller organisations may benefit from free or low-cost services such as the Department for Work and Pensions' access to work mental health support service or [Mindful Employer](#).

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Context

This guideline is for all people aged 16 or over in full-time or part-time employment, including those on permanent, training, temporary or zero-hours contracts, and those who are self-employed and volunteers.

This guideline has been updated because NICE identified new evidence that could affect the recommendations.

Despite evidence that better mental wellbeing and job satisfaction are associated with increased workplace performance and productivity, the [government review Thriving at work](#) estimates that 15% of UK workers have an existing mental health condition. Poor mental wellbeing costs employers in the UK an estimated £42 billion to £45 billion annually through presenteeism, sickness absence and staff turnover ([Deloitte \[2020\] Mental health and employers: refreshing the case for investment](#)).

The total annual cost of poor mental wellbeing to the government, including NHS costs, benefit provision and tax revenue losses, is between £24 billion and £27 billion. Lost output costs the economy between £74 billion and £99 billion (Thriving at work). Changes to workplaces and working patterns as a result of the COVID-19 pandemic have had a large impact on working practices and organisational cultures; however, it is unclear what the longer-term effects of this will be.

Workplace policies and activities to promote and protect employee mental wellbeing vary widely. Mental wellbeing has been described as 'feeling good and functioning well', reinforcing that mental wellbeing is on a spectrum and positive mental wellbeing is not just the absence of symptoms of poor mental health. Consequently, the aim of interventions should not just be to prevent poor mental health, but instead should promote positive mental wellbeing.

The Department for Work and Pensions reports that most employers have basic health and wellbeing policies, including at least 1 covering flexible working, sick pay or injury training ([Department for Work and Pensions \[2014\] Health and wellbeing at work: a survey of employees](#)). Larger and public sector organisations are more likely to offer at least 1 of the following: health screening, occupational health services, independent counselling or stress management.

Finding more information and committee details

To find NICE guidance on related topics, including guidance in development, see the [NICE topic page on mental health and wellbeing](#).

For full details of the evidence and the guideline committee's discussions, see the [evidence reviews](#). You can also find information about [how the guideline was developed](#), including [details of the committee](#).

NICE has produced [tools and resources to help you put this guideline into practice](#). For general help and advice on putting our guidelines into practice, see [resources to help you put NICE guidance into practice](#).

Update information

March 2022: This guideline updates and replaces NICE's guideline on mental wellbeing at work, published in November 2009.

Minor changes since publication

April 2022: In recommendations 1.3.2, 1.10.7 and 1.11.3 we clarified that social care staff can use the staff mental health and wellbeing hubs.

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How Time Pressure Matter University Faculties' Job Stress and Well-Being? The Perspective of the Job Demand Theory

Zhong Chen¹, Tzaichiao Lee², Xianghua Yue^{3*} and Jie Wang^{4,5}

¹School of Business, Xiamen Institute of Technology, Xiamen, China, ²Chongqing Metropolitan College of Science and Technology, Chongqing, China, ³School of Economics and Management, Xiangnan University, Chenzhou, China, ⁴Guangdong University of Finance and Economics, Guangzhou, China, ⁵Stamford International University, Bangkok, Thailand

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Ku Ho Lin,
National Chung Hsing University,
Taiwan

Reviewed by:

Sophia Shi-Huei Ho,
University of Taipei, Taiwan
Lin Wang,
Foshan University, China

*Correspondence:

Xianghua Yue
460022688@qq.com

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The work environment of employees has been greatly affected by the COVID-19 pandemic, and many limitations and risks can be seen until now. In addition to employees in firms, the faculty in colleges and universities also suffer from pressure and face challenges. For the purpose of performance assessment and promotion, the faculty not only needs to teach students, but also assumes the time pressure from academic research. This study discusses the process in which the faculty's subjective well-being is affected, in an effort to learn about the job demands of the faculty under the work environment with a high level of time pressure, and the effect of these time-related job demands on their psychological health. In this study, 347 valid questionnaires were collected from universities in coastal areas of the Chinese mainland. The results show that time-related job demands have a positive impact on time pressure; time pressure has a negative impact on subjective well-being; and time-related self-efficacy can significantly mediate the relationship among time-related job demands, time pressure, and subjective well-being. On this basis, this study proposes its theoretical and practical implications.

Keywords: job demand theory, psychological health, subjective well-being, time-related job demand, time-related self-efficacy

INTRODUCTION

In face of the increasingly grave and fickle COVID-19 pandemic, countries all over the world have to adapt their isolation policies to changes from time to time. In this context, the faculty members are facing increasingly intensified time pressure, resulting in the growing importance of discussion on time in workspace (Orlikowski and Yates, 2002). To adapt to the work pace during the pandemic, faculty members have to change their teaching and research contents (Parker et al., 2014), accompanied by the tight and fast requirements for academic work. These work-related requirements enable many faculty members to perceive the time urgency, thus bringing about negative consequences (Major et al., 2002; Chong et al., 2020; van Antwerpen and Scholtz, 2021). Since the time-related job demand is the major source of time pressure, many scholars often mix it with time pressure or directly use time pressure as the dimension related to time in job demands. Häusser et al. (2010) indicated that one of the most important

job demands is time pressure, except for the workload. Demerouti et al. (2001) also presented that time pressure is one of the most important job demand dimensions, except for the physical workload and work environment. As a result, following these prior arguments (Schaufeli et al., 2008; Crawford et al., 2010), this study focuses on the time-related job demand and deems it as the important source of time pressure for faculty members to discuss the resulting consequences, i.e., job stress, and the impact of this consequence on psychological health and life of faculty members.

As defined in the job demand-resource (JD-R) theory (Demerouti et al., 2001; Bakker and Demerouti, 2007), the job demand is the primary predictor of negative consequences of job stress (van der Wal et al., 2018; Han et al., 2020; Wang and Chen, 2022). Specifically, the time-related job demand is one of the pressure sources that have the most negative effect on job attitude and well-being (Bakker and Demerouti, 2007), especially on job stress (e.g., Kleiner and Wallace, 2017; Chong et al., 2020). Job stress, as a common pressure indicator in the JD-R theory, can result in negative effects featuring emotional exhaustion, negative attitude, and loss of career self-efficacy (Maslach et al., 2001). According the view of health impairment process in the JD-R theory, job stress is not only significantly affected by job demands of different types (Llorens et al., 2006; Hakanen et al., 2008; Crawford et al., 2010; Bottiani et al., 2019), but also plays a mediating role in the relationship between job demands and outcome variables. Thus, this study aims to discuss the mediating effect of job stress, and its impact on psychological health and life of faculty members.

The time in workspace has been widely discussed in the academic community. Recent studies also highlight that the time-related job demands can bring about adverse outcomes to physical and psychological health of employees (Demerouti et al., 2001; Häusser et al., 2010). However, there is no sufficient empirical research on verifying the health loss progress from the perspective of the time in workspace, and most of such research only verify the impact of time-related job demand on job stress and job burnout, but few of them has discussed how the time-related job demands affect outcome variables of employees through job stress. Most empirical studies verifying the JD-R theory integrate job demands at multiple dimensions into a single, overall job demand (Demerouti et al., 2001; Llorens et al., 2006; Hakanen et al., 2008) and discuss a single outcome factor such as job performance. In the context of tension and anxiety resulting from the COVID-19 pandemic, the psychological health becomes particularly significant because of the direct relation between psychological attitude and behaviors. Although prior studies of psychological health factors took workers as the research object, variables are still mainly well-being (Van der Doef and Maes, 1999; Bakker and Demerouti, 2007; Xu et al., 2021), such as life satisfaction and job satisfaction. However, few studies focus on the psychological health impairment process and discuss the impact of job demands on psychological health achieved through job stress while exploring these associations. Generally speaking, no investigation has ever been conducted based on a complete psychological health impairment process, despite some studies have explored

the impact of job burnout of nursing staff on their intra-role performance (e.g., Parker and Kulik, 1995; Bakker and Heuven, 2006). In consideration that psychological health is a key outcome indicator in the field of human resource management, this study aims to examine the effect of job stress in connection with time-related job demand by taking well-being and life satisfaction as an outcome variable.

Unlike the previous JD-R model (Demerouti et al., 2001; Han et al., 2020) that emphasizes job resources such as social support, job autonomy, and job feedback, Bakker et al. (2014) contended that the most significant breakthrough of the JD-R theory in recent years is its inclusion of the concept of individual resource. Individual resource is a positive self-assessment correlated with resilience. It refers to the perception of an individual against his/her capability of controlling and affecting the environment (Hobfoll et al., 2003) and emphasizes the importance of self-efficacy (Zhao et al., 2021; Cao et al., 2022). As reported by previous studies, such positive self-assessment can mitigate the impact of time-related job demands on outcomes of job stress (e.g., anxiety, depression, and job burnout), in addition to predicting goal setting, motivation, performance, job and life satisfaction, and other desired job outcomes (Judge et al., 2004; Xanthopoulou et al., 2013) helps to adapt to the work requirements and environment.

Based on the above statements, this study will make several contributions as following: (1) applying the JD-R theory to faculties in higher education institutions and discussing the effect of time-related job demand; (2) verifying the development and establishment of job stress and the relevance between job stress and subjective well-being; and (3) introducing the concept of positive psychology into JD-R model to discuss the moderating effect of time-related self-efficacy among time-related job demands, job stress, and subjective well-being.

LITERATURE REVIEW

JD-R Theoretical Model

More than 15 years have been passed since the JD-R model (Demerouti et al., 2001) was completely published. As hundreds of studies of the model were conducted, it has been applied in thousands of organizations (Bakker and Demerouti, 2017; Wang and Chen, 2022). There are too numerous retrospective and meta-analysis studies to mention one by one (e.g., Crawford et al., 2010; Nahrgang et al., 2011; Bakker et al., 2014; Bakker and Demerouti, 2017), and considerable research findings have been achieved. A majority of these research findings focus on the dual process: health impairment process and motivation process, and the interaction effect between them (Bakker and Demerouti, 2017). According to the JD-R model of early version (Demerouti et al., 2001), work environments can be discussed from two aspects: job demands and job resources, and specific job demands and resources will further be concentrated on according to the research purpose and background. Job demand means the part into which physical and mental efforts need to be continuously put, which may involve physical, psychological, social, and organization dimensions. That is why it is correlated

with physical and/or psychological costs. Job resource may also involve physical, psychological, social, and organizational dimensions. Furthermore, job resource is able to (1) play a role in achieving the job-related goals, (2) reduce the job demand and the physical and mental costs resulting from job demand, and (3) motivate individuals to grow (Wang and Chen, 2022). The two job characteristics will lead to two kinds of psychological process. According to the psychological health impairment process, high-level job demands requiring continuous efforts may wear out employees' resources, leading to the exhaustion of mental energy and psychological health problems. At the level of mental health, the well-being is taken as a key outcome variable in this study. Subjective well-being usually refers to a state in which a person is dominated by positive emotions and less pessimistic emotions occur during a certain period (Walsh et al., 2018; Peng, 2022). Some scholars argued that well-being is the subjective emotional experience from an individual, which represents people's demands and values by virtue of real life.

In addition, as illustrated by the motivational process, the likelihood of accessing job resources will lead to organizational commitment and job engagement (Han et al., 2020). The role of job resources as intrinsic or external motivation will motivate employees to achieve their goals; employees gain a sense of satisfaction and achievement, resulting in more dedication to their work (Wang et al., 2021). The findings of some empirical studies (Hakanen et al., 2006; van der Wal et al., 2018; Bottiani et al., 2019) also verified that some job resources such as support or coaching may give rise to job engagement (Han et al., 2020). In addition to the direct impact, the JD-R model also proposes based on the interaction hypothesis that job resources can be a cushion for the relationship between job demands and negative job outcomes. Under demanding working conditions, instructors with higher temporal self-efficacy are able to deal with more job demands and the resulting stress after effect, thus mitigating negative outcomes that may originally result from high job demands (Bakker et al., 2005; Xanthopoulou et al., 2007; Bakker et al., 2010; Cao et al., 2022).

Job Stress

In the context of the high time-related job demand, job stress may lead to emotional exhaustion of employees and the negative view on things. This will bring about tiredness, boredom, and lack of enthusiasm and sense of achievement, and throw employees into a dilemma featuring excessive impairment of resources and incapability of responding to job demands (Bakker and Demerouti, 2007). Fatigue and sense of incapability caused by job stress will reduce the available energy and subjective effort that put into work by employees (Zhao et al., 2022). As a result, employees may have no sufficient energies to maintain a good job performance (Gaillard, 2018). The intense job stress will also undermine employees' confidence in solving job-related problems, thus producing the learned helplessness against job dilemmas (Zhao et al., 2022). If employees feel tired from the job stress, employees are able to be trapped in a vicious circle that depletes their motivations to seek for support to change their situation, leading to the reduction of job performances (Han et al., 2020).

Studies of psychological health impairment in different types of organizational environment came to a similar conclusion that the intense time pressure caused by time-related job demands would deplete employees' physical and mental energies and cause health problems, thus adversely affecting individuals' life and job-related well-being (Toppinen-Tanner et al., 2002; Chong et al., 2020). Demerouti et al. (2001) conducted a study of 109 German nurses. They found that the time pressure dimension in job demands can effectively predict the job stress of nurses and affect their life satisfaction. This shows that job burnout is also supposed to play a mediating role in the effect of time-related job demands on employees' subjective well-being. Some other empirical studies also indicated that the job-related time pressure facing nurses is closely correlated with their physical and psychological impairment status and even affect their performance in attending to patients (Teng et al., 2010). Zhao et al. (2022) reported that job stress, as the source of employees' negative cognition and psychological state, may cause poor working conditions and psychological issues, thus reducing the subjective well-being perceived by employees. To sum up, as illustrated by the health impairment process in the JD-R theory, efforts required to meet job demands can generate certain physical and/or psychological costs among employees, which will further impair their physical and psychological resources and have a negative effect on job outcomes (Bakker and Demerouti, 2007; Bottiani et al., 2019). As a result, this study assumes that time-related job demands would affect instructors' subjective well-being through the mediating effect of job stress. Thus, H1 is developed as follows:

H1: Job stress mediates the relationship between time-related job demand and subjective well-being.

Time-Related Job Demand

Job burnout has been an important concern since the beginning of the 20th century, and any people who work with others, without limitation to specific races or groups, cannot escape from job burnout (Demerouti et al., 2003; Han et al., 2020; Sumner and Kinsella, 2021). Job burnout is defined as a chronic fatigue syndrome, featuring cynical conviction, negative attitude, and reduced professional efficacy in work (Maslach et al., 2001). Lee and Ashforth (1996) found in their meta-analysis research that job burnout is closely related with many types of job demands and often appears when organizations ask employees to inject too much physical and psychological resources. Crawford et al. (2010) further verified in their meta-analysis research that both challenge and hindrance demands are positively associated with job burnout. Job burnout also plays a vital role in the JD-R theory (Bakker and Demerouti, 2007). As seen from previous studies of the health impairment process "job demand-stress-job outcome" in the JD-R model, job burnout is a very common mediator for stress (e.g., Hakanen et al., 2008; Bottiani et al., 2019). When job demands are beyond an extent that is bearable, employees will be unable to work in a positive attitude, leading to emotional exhaustion and negative view on things, tiredness, boredom, and lack of

enthusiasm and sense of achievement. Such status of loss in physical and psychological resources is known as the job burnout, i.e., a dilemma where resources are excessively depleted and job demands cannot be met (Bakker and Demerouti, 2007). This will further harm other physical and psychological health indicators and job attitudes. Llorens et al. (2006) conducted a cross-cultural sampling from 657 Spanish employees and 477 Dutch employees to verify the mediating role of job burnout in the health impairment process. The results showed that job burnout always significantly mediates the relationship between job demands and organizational commitment, no matter which country or occupational environment is involved, what kind of methods is used to collect information (*via* computer or paper test), and which measuring tool is used to evaluate key variables in the JD-R model. Hakonen et al. (2008) designed a panel study using the two-wave three-year follow-up data of 2,555 Finnish dentists. The results also verified the mediating role of job burnout in the health impairment process and the casual relationship among variables involved in this process. All in all, empirical studies confirmed that job burnout plays a mediating role in the health impairment process.

However, prior studies of JD-R model paid more attention to workload or emotional demand but neglected the key role of time-related indicators in measuring job demands (McCauley et al., 1994; Bakker and Demerouti, 2017). It has been early found that time is the primary source of job-related stress (Cooke et al., 2019; Huang et al., 2021) and that time-related job demands (e.g., high work pace) leave employees with less time to complete job-related tasks (Kinicki and Vecchio, 1994), resulting in impairment of physical and psychological energies, loss of sense of achievement and self-efficacy, lack of enthusiasm for work, and generation of job burnout (Bottiani et al., 2019).

Empirical studies also presented that time-related job stress will lower the subjective well-being (Huang et al., 2021). A study on 800 female employees demonstrates that there is a high probability to cause health problems, as well as intense anxiety, depression, and sense of incapability when job demands include a high work pace. Chong et al. (2020) conducted a four-week follow-up study for accountants, covering the peak working days (the end of a month) and common days. The results revealed that the time pressure caused by job demands will indeed intensify negative emotions and emotional exhaustion over time, and the two factors are exactly the primary source of job stress. Thus, H2 is developed as follows:

H2: Time-related job demand is positively correlated with job stress.

Time-Related Self-Efficacy

The JD-R theory highlights the role of job resources in early time (Bakker et al., 2010), but the focus is shifted to the key role in individual resources in recent years. Besides, many studies believed that both individual resources and job resources are key factors to drive motivations and reduce stress (Bakker et al., 2005; Bakker and Demerouti, 2007, 2017). The belief and attitude deriving from such resources can be regarded as

self-efficacy. The interaction hypothesis of JD-R model presents that both individual resources and job demands have an interactive effect on employees' physical and psychological health or job outcomes (Bakker and Demerouti, 2007). There are two mechanisms. First, individual resources are supposed to mitigate the negative impact of job demands on stress because such resources can help employees cope better with job demands and reduce physical or psychological impairment. Second, the importance of individual resources will become prominent when there are higher job demands, so job demands will also enhance the positive impact of individual resources on motivations. In other words, employees with higher self-efficacy are less affected by stress perception and its consequences caused by job demands, and highly motivated to complete job tasks than those with lower self-efficacy (Zhao et al., 2021; Cao et al., 2022).

This study focuses on the time in workspace to discuss the impact of time-related job demands on job performance through job burnout. Thus, time-related individual resources are selected. Bakker and Demerouti (2007) defined individual resources as people's belief into what extent they can control their work environment. As a result, this study argues that employees' belief in controlling their work time is also an important individual resource, i.e., time-related self-efficacy. Time-related efficacy (Macan et al., 1990; Peeters and Rutte, 2005) refers to the extent to which employees perceive that they can control time and the self-assessment that they can arrange and complete their work before deadline (Häfner and Stock, 2010; Zhao et al., 2021).

It can be found from relevant studies that the time-related self-efficacy of employees in the job context mainly stems from the overall assessment of their own time management ability (Claessens et al., 2004). Employees with the ability to set goals and priorities will have a high degree of time-related self-efficacy. Thus, employees can respond to the job demand for fast work pace with a schedule or time plan, so that they can mitigate job burnout caused by time-related job demands without consuming too many individual resources. In addition, the time-related self-efficacy is also the self-efficacy for the work time management (Cao et al., 2022). Therefore, even in the context of high-level time-related job demands, employees still believe that they are able to meet the challenges and do a good job in scheduling and time management. Employees with high time-related self-efficacy believe that they are able to work following the time schedule and better control the work time (Häfner and Stock, 2010; Zhao et al., 2021), which helps reduce the likelihood of producing job burnout due to tight schedules. As a result, the positive effect of time-related job demands on job burnout will be weakened when employees perceive the high time-related self-efficacy. Given this, H3 is developed as follows:

H3: Time-related self-efficacy will negatively mediate the relationship between time-related job demands and job stress.

In face of the job demand for fast work pace, employees are more confident in manage their work time and pay more

attention to time schedule and plan if they have a positive self-assessment on their time management ability. As a result, the time management efficiency will be enhanced (Häfner and Stock, 2010) to deal with the stress that may be caused by time-related job demands and lower the probability of negative outcomes. Time-related efficacy can mitigate the effect that time-related job demands exert *via* the mediating role of job burnout on job performances, or in other words, slow down the overall health impairment process (Bakker and Demerouti, 2017) caused by time-related job demands, in addition to mediating the relationship between time-related job demands and job burnout. Empirical studies also demonstrate that individual resources can significantly slow down the overall health impairment process. Based on the above statement, H4 is developed as follows:

H4: Time-related self-efficacy will negatively mediate the relationship between job stress and subjective well-being.

According to the above hypotheses, the research framework is shown in **Figure 1**.

METHODOLOGY

Sampling

In this study, data were collected by questionnaires from employees of organizations in different sectors, with a view to enhance the generalization of research findings. Researchers collected data by recruiting one supervisor and one subordinate, attempting to avoid the possibility of common method variance (Podsakoff et al., 2003). Respondents were mainly recruited from three channels: (1) reference from relatives or friends; (2) enterprises in previous industry-university cooperation project; and (3) on-the-job students in colleges and universities. Potential respondents or enterprises were first contacted *via* telephone or face-to-face interview to inform them of the research purpose and process of this study. In this study, we conducted a survey using the purposive sampling. The purposive sampling can be implemented based on the individuals'

subjective judgment and allows us to select samples that are most suitable for the purpose of the study. However, this sampling method has several disadvantages. Vulnerability to errors in judgment by researchers, low level of reliability and high level of bias, and inability to generalize research findings are three main disadvantages. To avoid these disadvantages, some conditions were set during sampling in this study, which makes the collected samples better conform to the sample reliability and to improves the generalization of the study. The results indicate that subjects do not have a significant impact on the research variables, so there is no need to include an independent variable in subsequent analyses. In the questionnaire, participants were informed of the research purpose, research ethics, and low risks, and the questionnaire information was processed in an anonymous way. After confirming their participation willingness, the paper questionnaires were posted to them or delivered to them in person, and they were also asked to hand some questionnaires over to their colleagues randomly. After completion, respondents mailed questionnaires back to researchers in sealed envelopes to further ensure the confidentiality of data. Each questionnaire is designed with numbers, which allows researchers to match questionnaires of supervisors to their subordinates after questionnaires are taken back. The unmatched questionnaires were deemed as invalid ones, and those not completed or with obviously inappropriate answers were screened out. The remaining valid questionnaires were included in the statistical analysis procedure. A total of 353 out of 500 questionnaires were withdrawn, and 347 were ultimately judged as valid ones, accounting for 70.16% of total questionnaires. The sample background is shown in **Table 1**.

Furthermore, the study collects information from the same respondents in form of a single questionnaire, which may lead to the common method bias (CMB). In this study, the single factor verification from Harman is adopted and all the measured items are analyzed by the non-rotating matrix. The analysis results demonstrate that there are nine factors, of which the eigenvalue is greater than 1, and the explanatory variance of factor 1 is 33.23% that could not explain most of the variance. Therefore, it can be concluded from the verification results that there is no common method bias in this study.

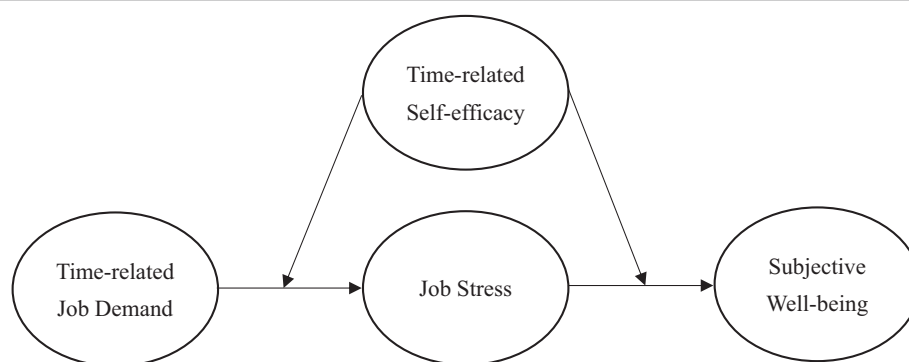


FIGURE 1 | Research framework.

TABLE 1 | Demographic characteristics of research samples.

Variable	Size	Percentage (%)	Variable	Sample	Percentage (%)
Gender			Managerial Position		
Male	209	60.2	Yes	142	40.9
Female	138	39.8	No	205	59.1
Age			University type		
30–40	135	38.9	Research orientation	63	18.2
40–50	143	41.2	Teaching orientation	83	23.9
50–60	59	17.0	Teaching and Research both	201	57.9
More than 60	10	2.9	Marital Status		
Affiliate			Married	259	74.6
Lecturer	149	42.9	Unmarried	84	24.2
Associate	183	52.7	Divorced	4	1.2
Professor					
Professor	15	4.3			

TABLE 2 | Measurement.

	1	2	3	4	5	6	7
1. TWD	0.920						
2. Workload	0.347	0.722					
3. IS	0.432	0.713	0.759				
4. Emotional	−0.249	0.013	−0.111	0.836			
5. Psychological	−0.192	0.172	0.038	0.739	0.819		
6. Social	−0.192	0.141	0.033	0.694	0.706	0.851	
7. TSE	−0.056	0.303	0.165	0.537	0.617	0.547	0.882
α	0.939	0.815	0.877	0.855	0.836	0.873	0.857
AVE	0.846	0.521	0.576	0.699	0.671	0.724	0.777
CR	0.956	0.867	0.905	0.903	0.891	0.913	0.913

TWD, Time-related Job Demand; IS, insufficient resources; and TSE, Time-related self-efficacy.

Instruments

Previous studies often include Karasek's (1985) job content instrument and time-related items in their questionnaires when discussing time-related job demands or measuring time pressure in job demands, which may lead to the problem of single item (Wanous and Reichers, 1996). In order to measure time-related job demands in a more comprehensive manner, this study adopts three items for the variable "tightness orientation" in Schriber and Gutek's (1987) work time scale. For job stress, this study adopts the scale revised by Tongchaiprasit and Ariyabuddhiphongs (2016), which owns two measuring dimensions of workload and insufficient resources, as well as 13 measuring items, such as "Lack of feedback on performance," "Insufficient management support," and "Poor communication between staff." Subjective well-being adopted the scale revised by Keyes (2005), which owns three measuring dimensions of emotional, psychological, and social well-being, as well as 12 measuring items. For time-related self-efficacy, this study adopts the scale revised by Macan (1994) from the one developed by Macan et al. (1990), and there are five items in total in this scale, such as "I am clear about time required to complete different tasks" and "I think I can control my time."

RESULTS

Measurement Evaluation

The Confirmatory factor analysis (CFA) was performed to examine the internal and external reliability of the constructs. In particular, the instruments' Cronbach alpha (α), factor loadings, composite reliability (CR), and average variance extracted (AVE) were computed. All scores obtained during the CFA were recorded above the least acceptable value, as shown in **Table 2**. Moreover, the satisfactory results were also computed in the case of each subgroup. Our results show the CFA of the measurement models, indicating that each construct has good convergent validity. The square root of AVE for each latent construct (see **Table 2**) is greater than its cross-correlation with other constructs, confirming discriminant validity. In **Table 2**, many correlation coefficients are lower than 0.8, although some of them are higher. This indicates that there is no multicollinearity among items in the scale.

Path Analysis

After completing the confirmatory factor analysis, we conducted the path analysis and tested the structural model. The path model was tested through structural equation modeling (SEM)

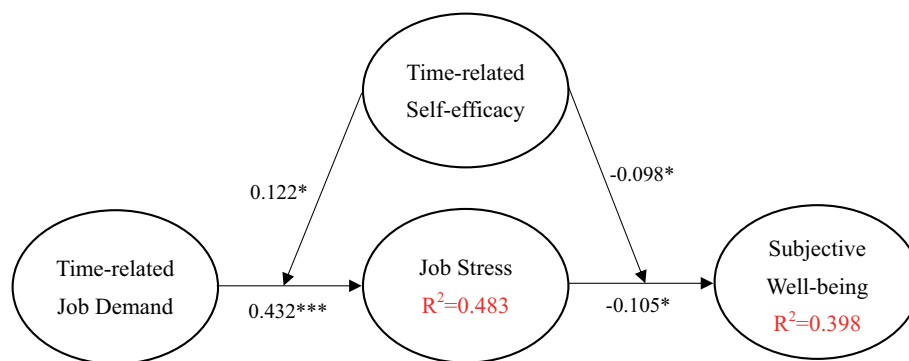


FIGURE 2 | Structural model. * $p < 0.05$; *** $p < 0.001$.

TABLE 3 | Results of hypotheses testing.

Paths	Coefficients	t-value	Results
H1: Time-related Job Demand → Job stress	0.432	8.303	Confirmed
H2: Job stress → Subjective Well-being	-0.105	2.221	Confirmed
H3: Time-related Job Demand*Time-related Self-efficacy → Job stress	0.122	2.322	Confirmed
H4: Job stress*Time-related Self-efficacy → Subjective Well-being	-0.098	1.985	Confirmed

analysis to assess direct and indirect effects. Partial least squares structural equation modeling (PLS-SEM) was adopted to construct the structural model. Compared with CB-SEM, PLS-SEM is more suitable for this study including when the research objective is exploratory research for theory development; when the analysis is for a prediction perspective; when the structural model is complex; when the structural model includes one or more formative constructs; when the sample size is smaller due to a small population; when distribution is lack of normality; and when research requires latent variable scores for consequent analyses. The above reasons provide supports to consider the PLS is an appropriate SEM method for a study. In this study, the SRMR value was 0.048 (< 0.08) and the NFI was 0.932 (> 0.90) and the dULS $<$ bootstrapped HI 95% of dULS and dG $<$ bootstrapped HI 95% of dG indicating the data fits the model well. Most results were significant at the $p < 0.01$ level. The empirical results are presented in **Figure 2**.

As indicated in the **Figure 2**; **Table 3**, job stress appeal has a negative significant effect on well-being ($\beta = 0.195$, $p < 0.01$); thus, H1 is supported. Moreover, time-related job demand appeal has a positively significant effect on job stress. Overall, the figure shows that job stress has a full mediating effect on the relationship between time-related job demand and subjective well-being; thus, H2 is supported. Our findings also evidence that time-related self-efficacy significantly and positively moderates the relationships of time-related job demand to job stress ($\beta = 0.246$, $p < 0.001$); thus, H3 is supported. Similarly, results indicate that time-related self-efficacy significantly and negatively moderates the relationships of job stress to subjective well-being; thus, H4 is supported. The Stone-Geisser Q2 values obtained through the blindfolding procedures for subjective well-being ($Q2 = 0.543$) and job performance ($Q2 = 0.325$) were

larger than zero, supporting the model has predictive relevance (Hair et al., 2017).

The interactions among time-related job demand, job stress, and time-related self-efficacy are significant for job stress and subjective well-being. To show the moderating effects among these relationships clearer, we plotted these significant interactions and indicated that time-related job demand and job stress significantly predicts employees' job stress and subjective well-being only when their time-related self-efficacy is high, as shown in the simple slope chart in **Figure 3**.

CONCLUSION

Discussion

Unlike previous studies of JD-R model that focus on job characteristics such as workload and emotional demand in job demands, this study puts emphasis on the time-related dimensions that are seldom involved in discussions, especially for university faculty. This study adds an element, i.e., positive psychology, to the JD-R model to mediate the relationship between variables and discuss their effect on subjective well-being of the faculty from the positive cognition belief. Particularly, considering changes to job forms in the context of pandemic, the tightness orientation is designed as a main variable in this study to discuss the impact of time-related job demands on the psychological health impairment of teachers and the mediating role of time-related self-efficacy in the psychological health impairment process. The results show that time-related job demands can mediate the impairment on the subjective well-being through job stress. Furthermore, this study verifies the

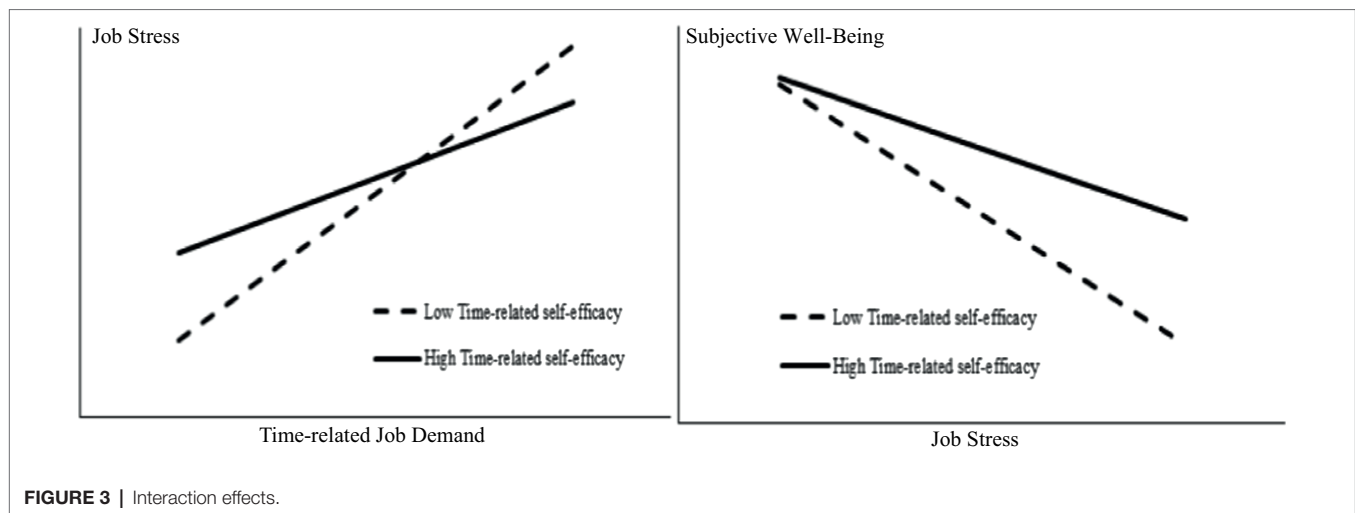


FIGURE 3 | Interaction effects.

mediating role of teachers' time-related self-efficacy in the relationship among time-related job demands, job stress, and subjective well-being. This study expands the research domain of JD-R model in terms of the subject of work time and offers preliminary empirical evidence.

Different from previous studies of job characteristics and job demands (e.g., Häusser et al., 2010; Bakker and Demerouti, 2017; Wang et al., 2021), this study discusses time-related demands (i.e., tightness orientation) from negative psychological features such as anxiety at work and in life in the context of pandemic. The results demonstrate that time-related job demands do have an impact on teachers' psychological health, and thus lower their subjective well-being. It is also found that time-related job demands are significantly correlated with job stress, which is consistent with arguments of other scholars. Thus, it can be seen that the strict job demands in terms of teaching and academic performance will increase the job stress of the faculty, thus intensifying their feeling of uncertainty in workplaces and affecting the effectiveness of school governance. That is, in a fast-paced and uncertain work environment, job demands will increase teachers' job stress, covering teaching quality, academic performance, promotion demands, etc.

Given the importance and difference of varied job demands for employees' job attitudes and performance, many scholars are attempting to clarify the efficacy of job demands in more detail. Specifically, time-related job demand is undoubtedly important, as a job demand dimension that has the greatest impact on employees in addition to workload (Häusser et al., 2010). Apart from job stress and subjective well-being verified in this study, other scholars also found that the high work pace, one of time-related job demands, also has an adverse effect on physical health. The reason may be that the high time-related demands, especially the repeated and tight work pace (Bosch et al., 2011), change the extent, frequency, and/or duration of teachers exposed to general risk factors such as the high-risk pandemic context. This is consistent with views of prior studies. Besides, it is necessary to explore the potential positive effect of job demands, and it is also worth examining

in depth the potential positive effect of time-related job demands on teachers' job attitudes and performance, and even the buffering effect of perceived stress on stress (Bakker and Demerouti, 2017). It is found from research results that the intense job stress caused by time urgency may cause the faculty has to sacrifice rest time to deal with teaching and academic work. This will lead to a low life and job satisfaction, and further reduce the subjective well-being of the faculty.

Moreover, prior studies of job resources and job control mostly highlighted the importance of the individual's sense of subjective control over the outside world. But the research focus is often put on the degree of job control or autonomy in job characteristics, which cannot become individual resources that are applicable for all kinds of job situation and job characteristic (Hobfoll, 2002; Bakker and Demerouti, 2007). This study examines teachers' self-efficacy on work time and contends that teachers' perceived time control is an important individual resource. The results of this study present that time-related job demands have a significant effect on psychological health impairment and can affect the subjective well-being through job performance. On the contrary, the highly perceived time control will not only reduce the impact of time-related demands on job stress, but also lower the negative effect of job stress on subjective well-being, thus enhancing teachers' psychological health. As in previous studies, this result also verifies the importance of time-related self-efficacy and its positive correlation with job satisfaction and subjective well-being (Claessens et al., 2004; Xu et al., 2021; Cao et al., 2022).

Implications

Although the results verify that time pressure caused by time-related job demands will lead to psychological health impairment, the job demand for high work pace can hardly be avoided in the organizational context nowadays, especially in the context of COVID-19 pandemic. Thus, appropriate adjustment of performance evaluation, diversified communication with teachers, or enough time given teachers for preparation and planning

may turn teachers' subjective assessment of time-related job demands into a challenging stressor. Thus, this study suggests that colleges and universities should take initiatives to communicate with the faculty to obtain feedback on their self-assessment of time pressure and rebuild the subjective perception over time-related job demands through in-depth conversation with teachers, thus avoiding psychological impairment process and unnecessary negative consequences. However, in the self-statement of psychological status from students, errors may occur. Considering research ethics, if the actual psychological status of students is assessed, it may be easier to understand the connection between job demands, stress, and subjective well-being. In addition, subsequent researchers are suggested to bring interview contents and faculties observations of working status to their studies, so as to provide support for the research findings and give an overall judgment.

Prior studies have verified the relationship between time management behaviors and perceived time control. With time management behaviors, employees can perceive that they can deal with time-related conditions, increase the self-efficacy for use of work time. As a result, time management training is a good start point in order to develop the perceived time control into one of individual resources. Therefore, it is recommended in this study that managers can make good use of the important situational effect of the perceived time control on work time characteristic and its stress to increase the time-related individual resources of teachers and generate subsequent buffering and positive effects.

Limitations

Data of different sources were used in this study to avoid the possibility of common method variance (Podsakoff et al., 2003). However, such data were collected at a single time point, making this study a cross-sectional one that has causal inferences among research variables to be clarified (Spector, 2006). The research framework of this study is based on JD-R model's health impairment process: job demand-stress-job outcome. A variety of study designs such as longitudinal design (Hakanen et al., 2008; Bakker et al., 2014) and diary method design (Simbula, 2010) have been created to establish the casual inference of this theoretical model. Future studies are suggested to verify the health impairment process from the time perspective using the diary method or longitudinal design based on the results of this study.

The tightness orientation is considered as a major variable in this study, but there are many other orientations such as synchronization orientation and future orientation to be clarified.

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All these orientations can cause time pressure on employees and thus produce the health impairment process. Furthermore, in this study, employees are invited to make an overall assessment on their work time, but we suggest researchers should discuss other better time-related individual resources (e.g., multi-task capability and care for future outcomes) for specific work time characteristics.

This study is a general staff survey and is limited to the education service industry, since we expect it is applicable to the faculty in most universities. Only 347 valid questionnaires were obtained in this study. Although specific sampling conditions are proposed during sampling to ensure the sample validity, more samples will contribute to valuable insights and generalization for research results and theories. We suggest researchers expand the sampling size to reach more diversified and enriched conclusions. However, previous studies demonstrated that different work groups may have specific job demands and resources, which have a greater effect on their psychological health impairment. Therefore, different types of teachers such as senior high school teachers or junior high school teachers can be taken into account in future studies to clarify work and/or individual resources that benefit them the most.

DATA AVAILABILITY STATEMENT

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

ETHICS STATEMENT

The studies involving human participants were reviewed and approved by Academic Committee of School of Economics and Management, Xiangnan University. The patients/participants provided their written informed consent to participate in this study.

AUTHOR CONTRIBUTIONS

ZC and TL contributed to conception and design of the study. TL organized the database. XY performed the statistical analysis. ZC and XY wrote the first draft of the manuscript. ZC, XY, TL, and JW wrote sections of the manuscript. All authors contributed to manuscript revision, read, and approved the submitted version.

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